

COMPLEX ANALYSIS AND RIEMANN SURFACES I/II

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ABSTRACT. Lecture notes for MATH6176GR.

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¹Based on lectures by Duong Phong [3].

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²Based on lectures by Duong Phong [3] .

³Based on lectures by Mu-Tao Wang, Aaron Chow [4] [6].

Part 1. The Line-Bundle Interpretation⁴

1. PRELIMINARIES AND NOTATION

1.1. Complex coordinates, frames, and indices.

Definition 1.1.1 (Local holomorphic coordinates). Let X be a complex manifold of complex dimension n . Fix

$$U \xrightarrow{\sim} \mathbb{C}^n, \quad z = (z^1, \dots, z^n).$$

For every $j \in \{1, \dots, n\}$, write $z^j = x^j + i \cdot y^j$. Then

$$\frac{\partial}{\partial z^j} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x^j} - i \cdot \frac{\partial}{\partial y^j} \right), \quad \frac{\partial}{\partial \bar{z}^j} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x^j} + i \cdot \frac{\partial}{\partial y^j} \right).$$

Definition 1.1.2 (Index conventions). Throughout,

$$i, j, k, \ell, m, p, q, r, s \in \{1, \dots, n\}, \quad \bar{i}, \bar{j}, \bar{k}, \bar{\ell}, \bar{m}, \bar{p}, \bar{q}, \bar{r}, \bar{s} \in \{\bar{1}, \dots, \bar{n}\}.$$

A barred index denotes an anti-holomorphic slot; its underlying numerical index lies in $\{1, \dots, n\}$. Unless explicitly stated otherwise, every sum below displays its full index range.

Definition 1.1.3 (Holomorphic and anti-holomorphic frames). In local holomorphic coordinates, define

$$e_j = \frac{\partial}{\partial z^j}, \quad \bar{e}_k = \frac{\partial}{\partial \bar{z}^k}.$$

For vector fields $V \in \Gamma(U, T^{1,0}X)$ and $\bar{W} \in \Gamma(U, T^{0,1}X)$, write

$$V = \sum_{m=1}^n V^m \cdot e_m, \quad [V]_e = \begin{bmatrix} V^1 \\ \vdots \\ V^n \end{bmatrix}, \quad \bar{W} = \sum_{k=1}^n W^{\bar{k}} \cdot \bar{e}_k, \quad [\bar{W}]_{\bar{e}} = \begin{bmatrix} W^{\bar{1}} \\ \vdots \\ W^{\bar{n}} \end{bmatrix}, \quad V^m, W^{\bar{k}} \in C^\infty(U).$$

Definition 1.1.4 (Dual coframes). The dual coframes are

$$e^j = dz^j, \quad \bar{e}^k = d\bar{z}^k,$$

with

$$e^j(e_k) = \delta_k^j, \quad \bar{e}^k(\bar{e}_\ell) = \delta_\ell^k, \quad e^j(\bar{e}_k) = \bar{e}^k(e_j) = 0.$$

Thus every $(1,0)$ -form α and $(0,1)$ -form β has the expansion

$$\alpha = \sum_{j=1}^n \alpha_j \cdot dz^j, \quad \alpha = [\alpha]_{e^*} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix},$$

$$\beta = \sum_{k=1}^n \beta_{\bar{k}} \cdot d\bar{z}^k, \quad \beta = [\beta]_{\bar{e}^*} = \begin{bmatrix} \beta_{\bar{1}} \\ \vdots \\ \beta_{\bar{n}} \end{bmatrix}, \quad \alpha_j, \beta_{\bar{k}} \in C^\infty(U).$$

⁴Based on lectures by Duong Phong [3].

1.2. Hermitian metrics and matrix identities.

Definition 1.2.1 (Hermitian metric). A Hermitian metric g on $T^{1,0}X$ is complex-linear in its first slot and conjugate-linear in its second slot. Set

$$g_{j\bar{k}} = g(e_j, e_k).$$

Thus the bar on $\bar{k}$ records the conjugate-linear slot; equivalently, if the underlying real metric is $g_{\mathbb{R}}$ and $g_{\mathbb{R},\mathbb{C}}$ is its complex-bilinear extension, our normalization is

$$g_{j\bar{k}} = 2 \cdot g_{\mathbb{R},\mathbb{C}}(e_j, \bar{e}_k).$$

This normalization produces the factor $i/2$ in the associated $(1,1)$ -form. The coefficients satisfy

$$g_{j\bar{k}} = \overline{g_{k\bar{j}}}, \quad \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot \xi^j \cdot \overline{\xi^k} > 0 \quad \text{for every } 0 \neq \xi \in \mathbb{C}^n.$$

We index the metric matrix by barred rows and holomorphic columns,

$$G = [g_{j\bar{k}}]_{\bar{k},j} = \begin{bmatrix} g_{1\bar{1}} & g_{2\bar{1}} & \cdots & g_{n\bar{1}} \\ g_{1\bar{2}} & g_{2\bar{2}} & \cdots & g_{n\bar{2}} \\ \vdots & \vdots & \ddots & \vdots \\ g_{1\bar{n}} & g_{2\bar{n}} & \cdots & g_{n\bar{n}} \end{bmatrix} \succ 0,$$

and, for coordinate columns,

$$g(V, W) = [W]_e^* \cdot G \cdot [V]_e = \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot V^j \cdot \overline{W^k}.$$

and write the Hermitian tensor as

$$g = \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot dz^j \otimes \overline{dz^k}.$$

Here the second tensor factor denotes the conjugate-linear covector slot, and $\overline{dz^k} = d\bar{z}^k$. The Hermitian symmetry is precisely $G = G^*$ under this convention.

Definition 1.2.2 (Inverse metric matrix). Let

$$G^{-1} = [g^{m\bar{p}}]_{m,\bar{p}} = \begin{bmatrix} g^{1\bar{1}} & g^{1\bar{2}} & \cdots & g^{1\bar{n}} \\ g^{2\bar{1}} & g^{2\bar{2}} & \cdots & g^{2\bar{n}} \\ \vdots & \vdots & \ddots & \vdots \\ g^{n\bar{1}} & g^{n\bar{2}} & \cdots & g^{n\bar{n}} \end{bmatrix}.$$

$$(G = G^* \succ 0) \xrightarrow{1.2.1} G^{-1} \cdot G = I_n, \quad G \cdot G^{-1} = I_n,$$

or, entry by entry,

$$\sum_{p=1}^n g^{m\bar{p}} \cdot g_{j\bar{p}} = \delta_j^m, \quad \sum_{m=1}^n g_{m\bar{k}} \cdot g^{m\bar{p}} = \delta_k^{\bar{p}}.$$

For $n = 2$, put

$$D = g_{1\bar{1}} \cdot g_{2\bar{2}} - g_{2\bar{1}} \cdot g_{1\bar{2}} = \det(G).$$

Then

$$\begin{bmatrix} g_{1\bar{1}} & g_{2\bar{1}} \\ g_{1\bar{2}} & g_{2\bar{2}} \end{bmatrix}^{-1} = \frac{1}{D} \cdot \begin{bmatrix} g_{2\bar{2}} & -g_{2\bar{1}} \\ -g_{1\bar{2}} & g_{1\bar{1}} \end{bmatrix} = \begin{bmatrix} g^{1\bar{1}} & g^{1\bar{2}} \\ g^{2\bar{1}} & g^{2\bar{2}} \end{bmatrix}.$$

Moreover, $G^{-1} = (G^{-1})^*$, so $g^{m\bar{p}} = \overline{g^{p\bar{m}}}$.

Definition 1.2.3 (Raising and lowering indices). For $V = \sum_{m=1}^n V^m \cdot e_m$, define

$$V_{\bar{k}} = \sum_{m=1}^n g_{m\bar{k}} \cdot V^m, \quad V^{\flat} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_{\bar{n}} \end{bmatrix} = \begin{bmatrix} g_{1\bar{1}} & g_{2\bar{1}} & \cdots & g_{n\bar{1}} \\ g_{1\bar{2}} & g_{2\bar{2}} & \cdots & g_{n\bar{2}} \\ \vdots & \vdots & \ddots & \vdots \\ g_{1\bar{n}} & g_{2\bar{n}} & \cdots & g_{n\bar{n}} \end{bmatrix} \cdot \begin{bmatrix} V^1 \\ V^2 \\ \vdots \\ V^n \end{bmatrix} = G \cdot [V]_e.$$

For a $(0, 1)$ -covector with components $\alpha_{\bar{p}}$, define

$$\alpha^m = \sum_{p=1}^n g^{m\bar{p}} \cdot \alpha_{\bar{p}}, \quad \alpha^{\sharp} = \begin{bmatrix} \alpha^1 \\ \alpha^2 \\ \vdots \\ \alpha^n \end{bmatrix} = G^{-1} \cdot \begin{bmatrix} \alpha_{\bar{1}} \\ \alpha_{\bar{2}} \\ \vdots \\ \alpha_{\bar{n}} \end{bmatrix}.$$

$$G \cdot G^{-1} = I_n \xrightarrow{1.2.2} \sum_{m=1}^n g_{m\bar{k}} \cdot \alpha^m = \sum_{m=1}^n g_{m\bar{k}} \cdot \left(\sum_{p=1}^n g^{m\bar{p}} \cdot \alpha_{\bar{p}} \right) = \sum_{p=1}^n \left(\sum_{m=1}^n g_{m\bar{k}} \cdot g^{m\bar{p}} \right) \cdot \alpha_{\bar{p}} \xrightarrow{1.2.2} \sum_{p=1}^n \delta_k^{\bar{p}} \cdot \alpha_{\bar{p}} = \alpha_{\bar{k}}.$$

Example 1.2.4 (A complete rank-two index computation). Let

$$G = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}, \quad \det(G) = 4 \cdot 3 - 2 \cdot 2 = 12 - 4 = 8, \quad G^{-1} = \frac{1}{8} \cdot \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix}.$$

Then every entry of the inverse check is

$$\begin{aligned} G^{-1} \cdot G &= \frac{1}{8} \cdot \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} = \frac{1}{8} \cdot \begin{bmatrix} 3 \cdot 4 + (-2) \cdot 2 & 3 \cdot 2 + (-2) \cdot 3 \\ (-2) \cdot 4 + 4 \cdot 2 & (-2) \cdot 2 + 4 \cdot 3 \end{bmatrix} \\ &= \frac{1}{8} \cdot \begin{bmatrix} 12 - 4 & 6 - 6 \\ -8 + 8 & -4 + 12 \end{bmatrix} = \frac{1}{8} \cdot \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \end{aligned}$$

For $[V]_e = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$,

$$\begin{aligned} [V]_e^{\flat} &= G \cdot [V]_e = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \cdot 4 + 2 \cdot 3 \\ 2 \cdot 4 + 3 \cdot 3 \end{bmatrix} = \begin{bmatrix} 16 + 6 \\ 8 + 9 \end{bmatrix} = \begin{bmatrix} 22 \\ 17 \end{bmatrix}, \\ G^{-1} \cdot [V]_e^{\flat} &= \frac{1}{8} \cdot \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 22 \\ 17 \end{bmatrix} = \frac{1}{8} \cdot \begin{bmatrix} 3 \cdot 22 - 2 \cdot 17 \\ -2 \cdot 22 + 4 \cdot 17 \end{bmatrix} = \frac{1}{8} \cdot \begin{bmatrix} 66 - 34 \\ -44 + 68 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}. \end{aligned}$$

1.3. Differential operators.

Definition 1.3.1 (Coordinate derivatives and type decomposition). In local holomorphic coordinates, define

$$\partial_j = \frac{\partial}{\partial z^j}, \quad \bar{\partial}_{\bar{k}} = \frac{\partial}{\partial \bar{z}^{\bar{k}}}.$$

The exterior derivative splits as $d = \partial + \bar{\partial}$. For $u \in C^\infty(U)$,

$$\partial(u) = \sum_{j=1}^n \partial_j(u) \cdot dz^j = \begin{bmatrix} \partial_1(u) & \cdots & \partial_n(u) \end{bmatrix} \cdot \begin{bmatrix} dz^1 \\ \vdots \\ dz^n \end{bmatrix}, \quad \bar{\partial}(u) = \sum_{k=1}^n \bar{\partial}_{\bar{k}}(u) \cdot d\bar{z}^{\bar{k}}, \quad d(u) = \partial(u) + \bar{\partial}(u).$$

Since the summands below have types $(2, 0)$, $(1, 1)$, and $(0, 2)$,

$$0 = d^2 = (\partial + \bar{\partial})^2 = \partial^2 + (\partial \bar{\partial} + \bar{\partial} \partial) + \bar{\partial}^2 \xrightarrow{1.3.1} \partial^2 = 0, \quad \partial \bar{\partial} + \bar{\partial} \partial = 0, \quad \bar{\partial}^2 = 0.$$

1.4. Kähler forms.

Definition 1.4.1 (Kähler form). For a Hermitian metric g , define the associated real $(1, 1)$ -form

$$\omega = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k.$$

It is real because

$$\begin{aligned} \bar{\omega} &= -\frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n \overline{g_{j\bar{k}}} \cdot d\bar{z}^j \wedge dz^k \xrightarrow{1.2.1} -\frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{k\bar{j}} \cdot d\bar{z}^j \wedge dz^k \\ &\stackrel{j \leftrightarrow k}{=} -\frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot d\bar{z}^k \wedge dz^j = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k = \omega. \end{aligned}$$

Definition 1.4.2 (Kähler metric). The metric g and its associated form ω are *Kähler* if $d(\omega) = 0$.

Proposition 1.4.3 (Coordinate characterization of the Kähler condition). *The metric g is Kähler if and only if*

$$\partial_j(g_{p\bar{q}}) = \partial_p(g_{j\bar{q}}), \quad \partial_{\bar{k}}(g_{p\bar{q}}) = \partial_{\bar{q}}(g_{p\bar{k}}).$$

Proof. By $d = \partial + \bar{\partial}$, $\omega \in \Omega^{1,1}(X)$, and $d(\omega) = \partial(\omega) + \bar{\partial}(\omega) \in \Omega^{2,1}(X) \oplus \Omega^{1,2}(X)$,

$$d(\omega) = 0 \xLeftrightarrow{1.3.1} \partial(\omega) = 0 = \bar{\partial}(\omega).$$

$$\partial(\omega) = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{p=1}^n \sum_{q=1}^n (\partial_j(g_{p\bar{q}})) \cdot dz^j \wedge dz^p \wedge d\bar{z}^q = \frac{i}{4} \cdot \sum_{j=1}^n \sum_{p=1}^n \sum_{q=1}^n (\partial_j(g_{p\bar{q}}) - \partial_p(g_{j\bar{q}})) \cdot dz^j \wedge dz^p \wedge d\bar{z}^q,$$

$$\partial(\omega) = 0 \iff \partial_j(g_{p\bar{q}}) = \partial_p(g_{j\bar{q}}) \quad \text{for all } j, p, q \in \{1, \dots, n\}.$$

$$\bar{\partial}(\omega) = -\frac{i}{4} \cdot \sum_{k=1}^n \sum_{p=1}^n \sum_{q=1}^n (\partial_{\bar{k}}(g_{p\bar{q}}) - \partial_{\bar{q}}(g_{p\bar{k}})) \cdot dz^p \wedge d\bar{z}^k \wedge d\bar{z}^q.$$

$$\bar{\partial}(\omega) = 0 \iff \partial_{\bar{k}}(g_{p\bar{q}}) = \partial_{\bar{q}}(g_{p\bar{k}}) \quad \text{for all } k, p, q \in \{1, \dots, n\}.$$

Thus both derivative symmetries are equivalent to $d(\omega) = 0$. □

1.5. Connection matrices.

Definition 1.5.1 (Connection matrix in a holomorphic frame). Let ∇ be a connection on $T^{1,0}X$ and fix $\{e_p\}_{p=1}^n$. Define

$$A = [A^m_p]_{m,p} = \begin{bmatrix} A^1_1 & A^1_2 & \cdots & A^1_n \\ A^2_1 & A^2_2 & \cdots & A^2_n \\ \vdots & \vdots & \ddots & \vdots \\ A^n_1 & A^n_2 & \cdots & A^n_n \end{bmatrix}$$

by

$$\nabla(e_p) = \sum_{m=1}^n A^m_p \otimes e_m.$$

Thus the p -th column records $\nabla(e_p)$. Before any type condition is imposed,

$$A^m_p = \sum_{j=1}^n A^m_{jp} \cdot dz^j + \sum_{k=1}^n A^m_{\bar{k}p} \cdot d\bar{z}^k,$$

$$A = \sum_{j=1}^n A_j \cdot dz^j + \sum_{k=1}^n A_{\bar{k}} \cdot d\bar{z}^k, \quad A_j = [A^m_{jp}]_{m,p} = \begin{bmatrix} A^1_{j1} & \cdots & A^1_{jn} \\ \vdots & \ddots & \vdots \\ A^n_{j1} & \cdots & A^n_{jn} \end{bmatrix}.$$

Definition 1.5.2 (Matrix wedge product). For matrix-valued differential forms $B = [B^m_p]$ and $C = [C^m_p]$,

$$(B \wedge C)^m_p = \sum_{q=1}^n B^m_q \wedge C^q_p.$$

If $A = \sum_{j=1}^n A_j \cdot dz^j$ has type $(1,0)$, then

$$\begin{aligned} (A \wedge A)^m_p &= \sum_{q=1}^n A^m_q \wedge A^q_p = \sum_{j=1}^n \sum_{\ell=1}^n \sum_{q=1}^n A^m_{jq} \cdot A^q_{\ell p} \cdot dz^j \wedge dz^\ell, \\ A \wedge A &= \sum_{j=1}^n \sum_{\ell=1}^n (A_j \cdot A_\ell) \cdot dz^j \wedge dz^\ell \\ &= \sum_{j=1}^{n-1} \sum_{\ell=j+1}^n (A_j \cdot A_\ell - A_\ell \cdot A_j) \cdot dz^j \wedge dz^\ell \in \Omega^{2,0}(\text{End}(T^{1,0}X)). \end{aligned}$$

1.6. The Chern connection.

Theorem 1.6.1 (Chern connection). *Let (X, g) be a Hermitian manifold. There exists a unique connection ∇ on $T^{1,0}X$ satisfying*

$$(1.6.1) \quad (i) \quad \nabla^{0,1} = \bar{\partial} \quad \text{on } T^{1,0}X,$$

$$(1.6.2) \quad (ii) \quad \nabla(g) = 0.$$

Proof. By $e = (e_1, \dots, e_n)$, $V = \sum_{p=1}^n V^p \cdot e_p$, and $\nabla(e) = e \cdot A$,

$$\nabla^{0,1}(V) - \bar{\partial}(V) = \sum_{m=1}^n \left(\sum_{p=1}^n (A^{0,1})^m_p \cdot V^p \right) \otimes e_m.$$

$$\nabla^{0,1}(V) = \bar{\partial}(V) \iff (A^{0,1})^m_p = 0 \quad \text{for all } m, p \in \{1, \dots, n\}, \quad A = A^{1,0}.$$

$$d(g_{p\bar{q}}) = \sum_{m=1}^n A^m_p \cdot g_{m\bar{q}} + \sum_{\ell=1}^n g_{p\bar{\ell}} \cdot \overline{A^\ell_q}.$$

$$d(G) = G \cdot A + A^* \cdot G \xrightarrow{1.3.1} \partial(G) = G \cdot A, \quad \bar{\partial}(G) = A^* \cdot G.$$

$$A = G^{-1} \cdot \partial(G), \quad A^m_p = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial(g_{p\bar{q}}) = \sum_{j=1}^n \left(\sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}}) \right) \cdot dz^j, \quad A^m_{jp} = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}}).$$

$$(G \cdot A)_{\bar{q},p} = \sum_{m=1}^n g_{m\bar{q}} \cdot A^m_p = \sum_{m=1}^n \sum_{\ell=1}^n g_{m\bar{q}} \cdot g^{m\bar{\ell}} \cdot \partial(g_{p\bar{\ell}}) \xrightarrow{1.2.2} \sum_{\ell=1}^n \delta_{\bar{q}}^{\bar{\ell}} \cdot \partial(g_{p\bar{\ell}}) = \partial(g_{p\bar{q}}).$$

$$A^* = (G^{-1} \cdot \partial(G))^* = \bar{\partial}(G) \cdot G^{-1}, \quad G \cdot A = \partial(G), \quad A^* \cdot G = \bar{\partial}(G),$$

$$d(G) = \partial(G) + \bar{\partial}(G) = G \cdot A + A^* \cdot G.$$

On an overlap let $e' = e \cdot T$, where T is holomorphic.

$$G' = T^* \cdot G \cdot T, \quad (G')^{-1} = T^{-1} \cdot G^{-1} \cdot (T^*)^{-1}, \quad \partial(T^*) = 0.$$

$$\partial(T^* \cdot G \cdot T) = \partial(T^*) \cdot G \cdot T + T^* \cdot \partial(G) \cdot T + T^* \cdot G \cdot \partial(T) = T^* \cdot \partial(G) \cdot T + T^* \cdot G \cdot \partial(T).$$

$$(G')^{-1} \cdot \partial(G') = T^{-1} \cdot G^{-1} \cdot (T^*)^{-1} \cdot (T^* \cdot \partial(G) \cdot T + T^* \cdot G \cdot \partial(T)).$$

$$(G')^{-1} \cdot \partial(G') = T^{-1} \cdot G^{-1} \cdot ((T^*)^{-1} \cdot T^*) \cdot \partial(G) \cdot T + T^{-1} \cdot G^{-1} \cdot ((T^*)^{-1} \cdot T^*) \cdot G \cdot \partial(T)$$

$$(G')^{-1} \cdot \partial(G') = T^{-1} \cdot G^{-1} \cdot \partial(G) \cdot T + T^{-1} \cdot (G^{-1} \cdot G) \cdot \partial(T) = T^{-1} \cdot A \cdot T + T^{-1} \cdot \partial(T).$$

$$(1.6.3) \quad A' = T^{-1} \cdot A \cdot T + T^{-1} \cdot \partial(T).$$

$$\nabla(e \cdot T) = (\nabla(e)) \cdot T + e \cdot d(T) = (e \cdot T) \cdot (T^{-1} \cdot A \cdot T + T^{-1} \cdot \partial(T)).$$

The frame-change law gives existence, while $\partial(G) = G \cdot A$ forces uniqueness. $\square$

Definition 1.6.2 (Chern connection coefficients). For the Chern connection in a local holomorphic frame,

$$\nabla_{\bar{k}}(e_p) = 0, \quad \nabla_j(e_p) = \sum_{m=1}^n A^m_{jp} \cdot e_m, \quad A^m_{jp} = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}}).$$

For each fixed $j \in \{1, \dots, n\}$,

$$A_j = [A^m_{jp}]_{m,p} = \begin{bmatrix} \sum_{q=1}^n g^{1\bar{q}} \cdot \partial_j(g_{1\bar{q}}) & \cdots & \sum_{q=1}^n g^{1\bar{q}} \cdot \partial_j(g_{n\bar{q}}) \\ \vdots & \ddots & \vdots \\ \sum_{q=1}^n g^{n\bar{q}} \cdot \partial_j(g_{1\bar{q}}) & \cdots & \sum_{q=1}^n g^{n\bar{q}} \cdot \partial_j(g_{n\bar{q}}) \end{bmatrix} = G^{-1} \cdot \partial_j(G).$$

Proposition 1.6.3 (Action on components). For $V = \sum_{p=1}^n V^p \cdot e_p$, the Chern connection satisfies

$$(\nabla_{\bar{k}}(V))^m = \partial_{\bar{k}}(V^m), \quad (\nabla_j(V))^m = \partial_j(V^m) + \sum_{p=1}^n A^m_{jp} \cdot V^p.$$

Equivalently,

$$[\nabla_j(V)]_e = \partial_j \left(\begin{bmatrix} V^1 \\ \vdots \\ V^n \end{bmatrix} \right) + \begin{bmatrix} A^1_{j1} & \cdots & A^1_{jn} \\ \vdots & \ddots & \vdots \\ A^n_{j1} & \cdots & A^n_{jn} \end{bmatrix} \cdot \begin{bmatrix} V^1 \\ \vdots \\ V^n \end{bmatrix} = \partial_j([V]_e) + A_j \cdot [V]_e.$$

Proof. By $\nabla_{\bar{k}}(e_p) = 0$, $\nabla_j(e_p) = \sum_{m=1}^n A^m_{jp} \cdot e_m$, and $\nabla_j(f \cdot V) = \partial_j(f) \cdot V + f \cdot \nabla_j(V)$,

$$\nabla_{\bar{k}}(V) = \sum_{p=1}^n \partial_{\bar{k}}(V^p) \cdot e_p.$$

$$\nabla_j(V) = \sum_{p=1}^n \partial_j(V^p) \cdot e_p + \sum_{p=1}^n \sum_{m=1}^n V^p \cdot A^m_{jp} \cdot e_m = \sum_{m=1}^n \left(\partial_j(V^m) + \sum_{p=1}^n A^m_{jp} \cdot V^p \right) \cdot e_m.$$

Thus the displayed component and column formulas agree. $\square$

Proposition 1.6.4 (Metric compatibility and the connection matrix). The Chern connection matrix and Hermitian metric satisfy

$$A = G^{-1} \cdot \partial(G), \quad A^m_{jp} = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}}).$$

For every fixed $j \in \{1, \dots, n\}$,

$$A_j = \begin{bmatrix} g^{1\bar{1}} & \cdots & g^{1\bar{n}} \\ \vdots & \ddots & \vdots \\ g^{n\bar{1}} & \cdots & g^{n\bar{n}} \end{bmatrix} \cdot \begin{bmatrix} \partial_j(g_{1\bar{1}}) & \cdots & \partial_j(g_{n\bar{1}}) \\ \vdots & \ddots & \vdots \\ \partial_j(g_{1\bar{n}}) & \cdots & \partial_j(g_{n\bar{n}}) \end{bmatrix} = G^{-1} \cdot \partial_j(G).$$

Proof. By $\nabla(g) = 0$, $G^{-1} \cdot G = I_n$, and $\partial(G) = G \cdot A$,

$$\nabla(g) = 0 \xrightarrow{1.6.2} \partial(G) = G \cdot A \xrightarrow{1.2.2} A = G^{-1} \cdot (G \cdot A) = G^{-1} \cdot \partial(G).$$

For each $m, p, j \in \{1, \dots, n\}$,

$$(G^{-1} \cdot \partial_j(G))^m_p = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}}) = A^m_{jp}.$$

Thus the matrix identity and its component form coincide. $\square$

Corollary 1.6.5 (Kähler symmetry of Christoffel-type coefficients). *If g is Kähler, then*

$$A^m_{jp} = A^m_{pj}.$$

Proof. By g Kähler, $A^m_{jp} = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}})$, and $\partial_j(g_{p\bar{q}}) = \partial_p(g_{j\bar{q}})$,

$$A^m_{jp} = \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{p\bar{q}}) \xrightarrow{1.4.3} \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_p(g_{j\bar{q}}) = A^m_{pj}.$$

□

1.7. Chern curvature and its components.

Definition 1.7.1 (Curvature matrix). Let A be the Chern connection matrix. Its curvature is

$$F = d(A) + A \wedge A, \quad F = [F^m_p]_{m,p} = \begin{bmatrix} F^1_1 & \cdots & F^1_n \\ \vdots & \ddots & \vdots \\ F^n_1 & \cdots & F^n_n \end{bmatrix}.$$

Since $A = A^{1,0}$,

$$\begin{aligned} 0 &= \partial(I_n) = \partial(G^{-1} \cdot G) = \partial(G^{-1}) \cdot G + G^{-1} \cdot \partial(G) \implies \partial(G^{-1}) = -G^{-1} \cdot \partial(G) \cdot G^{-1}. \\ \partial(A) &= \partial(G^{-1} \cdot \partial(G)) = \partial(G^{-1}) \wedge \partial(G) + G^{-1} \cdot \partial^2(G) = \partial(G^{-1}) \wedge \partial(G) + G^{-1} \cdot 0. \\ \partial(A) &= -(G^{-1} \cdot \partial(G) \cdot G^{-1}) \wedge \partial(G) = -(G^{-1} \cdot \partial(G)) \wedge (G^{-1} \cdot \partial(G)) = -A \wedge A. \\ F^{2,0} &= \partial(A) + A \wedge A = 0, \quad F = F^{1,1} = \bar{\partial}(A). \end{aligned}$$

Definition 1.7.2 (Type decomposition of curvature entries). Using the local coframe and the curvature matrix, write

$$F^m_p = \sum_{k=1}^n \sum_{j=1}^n F^m_{\bar{k}jp} \cdot d\bar{z}^k \wedge dz^j.$$

Equivalently,

$$F = \sum_{k=1}^n \sum_{j=1}^n \begin{bmatrix} F^1_{\bar{k}j1} & \cdots & F^1_{\bar{k}jn} \\ \vdots & \ddots & \vdots \\ F^n_{\bar{k}j1} & \cdots & F^n_{\bar{k}jn} \end{bmatrix} \cdot d\bar{z}^k \wedge dz^j.$$

Definition 1.7.3 (Curvature coefficients as commutators). For $V = \sum_{p=1}^n V^p \cdot e_p$, define

$$\begin{aligned} (\nabla_j \circ \nabla_k - \nabla_k \circ \nabla_j)(V) &= \sum_{m=1}^n \left(\sum_{p=1}^n R_{jk}^m{}_p \cdot V^p \right) \cdot e_m. \\ (\nabla_j \circ \nabla_{\bar{k}} - \nabla_{\bar{k}} \circ \nabla_j)(V) &= \sum_{m=1}^n \left(\sum_{p=1}^n R_{\bar{k}j}^m{}_p \cdot V^p \right) \cdot e_m. \end{aligned}$$

Proposition 1.7.4 (The $(1,1)$ -curvature coefficients). *For the Chern connection,*

$$R_{\bar{k}j}^m{}_p = -\partial_{\bar{k}}(A^m_{jp}), \quad F^m_{\bar{k}jp} = \partial_{\bar{k}}(A^m_{jp}).$$

For every $j, k \in \{1, \dots, n\}$,

$$(F^m_{\bar{k}jp})_{m,p} = \begin{bmatrix} \partial_{\bar{k}}(A^1_{j1}) & \cdots & \partial_{\bar{k}}(A^1_{jn}) \\ \vdots & \ddots & \vdots \\ \partial_{\bar{k}}(A^n_{j1}) & \cdots & \partial_{\bar{k}}(A^n_{jn}) \end{bmatrix} = \partial_{\bar{k}}(A_j) = \partial_{\bar{k}}(G^{-1} \cdot \partial_j(G)).$$

Proof. By $F = \bar{\partial}(A)$, $A^m_p = \sum_{j=1}^n A^m_{jp} \cdot dz^j$, and $F^m_p = \sum_{k=1}^n \sum_{j=1}^n F^m_{\bar{k}jp} \cdot d\bar{z}^k \wedge dz^j$,

$$\bar{\partial}(A^m_p) = \bar{\partial}\left(\sum_{j=1}^n A^m_{jp} \cdot dz^j\right) = \sum_{j=1}^n \sum_{k=1}^n (\partial_{\bar{k}}(A^m_{jp})) \cdot d\bar{z}^k \wedge dz^j = F^m_p \xrightarrow{1.7.2} F^m_{\bar{k}jp} = \partial_{\bar{k}}(A^m_{jp}).$$

$$(\nabla_{\bar{k}}(\nabla_j(V)))^m = \partial_{\bar{k}}(\partial_j(V^m)) + \sum_{p=1}^n \partial_{\bar{k}}(A^m_{jp}) \cdot V^p + \sum_{p=1}^n A^m_{jp} \cdot \partial_{\bar{k}}(V^p),$$

$$(\nabla_j(\nabla_{\bar{k}}(V)))^m = \partial_j(\partial_{\bar{k}}(V^m)) + \sum_{p=1}^n A^m_{jp} \cdot \partial_{\bar{k}}(V^p).$$

$$\left((\nabla_j \circ \nabla_{\bar{k}} - \nabla_{\bar{k}} \circ \nabla_j)(V)\right)^m = - \sum_{p=1}^n \partial_{\bar{k}}(A^m_{jp}) \cdot V^p \xrightarrow{1.7.3} R_{\bar{k}j}^m{}_p = -\partial_{\bar{k}}(A^m_{jp}) = -F^m_{\bar{k}jp}.$$

Thus the commutator coefficient is the negative of the displayed form coefficient. $\square$

1.8. Ricci curvature.

Definition 1.8.1 (Ricci tensor). Trace the endomorphism indices over $m = p$ and define

$$R_{\bar{k}j} = \sum_{m=1}^n R_{\bar{k}j}^m{}_m = - \sum_{m=1}^n \partial_{\bar{k}}(A^m_{jm}).$$

Its coordinate matrix is

$$[R_{\bar{k}j}]_{k,j} = \begin{bmatrix} R_{\bar{1}1} & R_{\bar{1}2} & \cdots & R_{\bar{1}n} \\ R_{\bar{2}1} & R_{\bar{2}2} & \cdots & R_{\bar{2}n} \\ \vdots & \vdots & \ddots & \vdots \\ R_{\bar{n}1} & R_{\bar{n}2} & \cdots & R_{\bar{n}n} \end{bmatrix}.$$

Proposition 1.8.2 (Ricci log-determinant formula). *In local holomorphic coordinates,*

$$R_{\bar{k}j} = -\partial_{\bar{k}}(\partial_j(\log(\det(G)))).$$

Proof. By $A_j = G^{-1} \cdot \partial_j(G)$, $R_{\bar{k}j} = -\partial_{\bar{k}}(\text{Tr}(A_j))$, and $\partial_j(\det(G)) = \det(G) \cdot \text{Tr}(G^{-1} \cdot \partial_j(G))$,

$$\partial_j(\det(G)) = \det(G) \cdot \text{Tr}(G^{-1} \cdot \partial_j(G)).$$

$$\partial_j(\log(\det(G))) = \frac{1}{\det(G)} \cdot \partial_j(\det(G)) = \text{Tr}(G^{-1} \cdot \partial_j(G)) = \sum_{m=1}^n (G^{-1} \cdot \partial_j(G))^m{}_m = \sum_{m=1}^n \sum_{q=1}^n g^{m\bar{q}} \cdot \partial_j(g_{m\bar{q}}).$$

$$R_{\bar{k}j} = -\partial_{\bar{k}}\left(\sum_{m=1}^n A^m_{jm}\right) \xrightarrow{1.6.4} -\partial_{\bar{k}}(\text{Tr}(G^{-1} \cdot \partial_j(G))) = -\partial_{\bar{k}}(\partial_j(\log(\det(G)))).$$

This is the Chern–Ricci tensor, equal to the usual Ricci tensor when g is Kähler. $\square$

Definition 1.8.3 (Ricci form). Define the Chern–Ricci form by

$$\rho(\omega) = i \cdot \sum_{k=1}^n \sum_{j=1}^n R_{\bar{k}j} \cdot dz^j \wedge d\bar{z}^k \xrightarrow{1.8.2} -i \cdot \partial(\bar{\partial}(\log(\det(G)))).$$

Since $F^m_{\bar{k}jp} = \partial_{\bar{k}}(A^m_{jp})$ and $d\bar{z}^k \wedge dz^j = -dz^j \wedge d\bar{z}^k$,

$$i \cdot \text{Tr}(F) = -i \cdot \sum_{k=1}^n \sum_{j=1}^n \partial_{\bar{k}}(\text{Tr}(A_j)) \cdot dz^j \wedge d\bar{z}^k = \rho(\omega).$$

For a complete row–column representation, set

$$\begin{aligned} \eta &= \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_n \end{bmatrix} = [R_{\bar{k}j}]_{k,j}^T \cdot \begin{bmatrix} d\bar{z}^1 \\ \vdots \\ d\bar{z}^n \end{bmatrix} \\ &= \begin{bmatrix} R_{\bar{1}1} & R_{\bar{2}1} & \cdots & R_{\bar{n}1} \\ R_{\bar{1}2} & R_{\bar{2}2} & \cdots & R_{\bar{n}2} \\ \vdots & \vdots & \ddots & \vdots \\ R_{\bar{1}n} & R_{\bar{2}n} & \cdots & R_{\bar{n}n} \end{bmatrix} \cdot \begin{bmatrix} d\bar{z}^1 \\ d\bar{z}^2 \\ \vdots \\ d\bar{z}^n \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^n R_{\bar{k}1} \cdot d\bar{z}^k \\ \sum_{k=1}^n R_{\bar{k}2} \cdot d\bar{z}^k \\ \vdots \\ \sum_{k=1}^n R_{\bar{k}n} \cdot d\bar{z}^k \end{bmatrix}. \\ \rho(\omega) &= i \cdot \sum_{j=1}^n dz^j \wedge \eta_j = i \cdot [dz^1 \quad \cdots \quad dz^n] \wedge \eta, \end{aligned}$$

where the last product means $\sum_{j=1}^n dz^j \wedge \eta_j$. Finally,

$$d(\rho(\omega)) = -i \cdot \partial(\partial(\bar{\partial}(\log(\det(G)))))) - i \cdot \bar{\partial}(\partial(\bar{\partial}(\log(\det(G)))))) \xrightarrow{1.3.1} i \cdot \partial(\bar{\partial}^2(\log(\det(G)))) = 0.$$

Since $\log(\det(G)) \in \mathbb{R}$, one also has $\overline{\rho(\omega)} = \rho(\omega)$.

2. THE CANONICAL LINE BUNDLE AND RICCI CURVATURE

The line-bundle interpretation begins with rank-one Chern geometry. A Hermitian holomorphic line bundle produces a scalar connection and curvature. Applied to K_X and K_X^{-1} , this construction produces the Ricci form and the volume-ratio identity that leads to the complex Monge–Ampère equation.

2.1. Hermitian holomorphic line bundles.

Definition 2.1.1 (Holomorphic line bundle). Let $L \xrightarrow{\pi} X$ be a holomorphic line bundle. Choose an open cover $\{U_\mu\}_{\mu \in I}$ and nowhere-vanishing holomorphic frames $s_\mu \in H^0(U_\mu, L)$. The transition functions satisfy

$$s_\nu = s_\mu \cdot \lambda_{\mu\nu}, \quad \lambda_{\mu\nu} \in \mathcal{O}^*(U_\mu \cap U_\nu), \quad \lambda_{\mu\mu} = 1, \quad \lambda_{\mu\nu} \cdot \lambda_{\nu\mu} = 1, \quad \lambda_{\mu\nu} \cdot \lambda_{\nu\kappa} = \lambda_{\mu\kappa}.$$

Definition 2.1.2 (Hermitian line metric). A Hermitian metric h on L is determined by the positive local weights

$$h_\mu = \|s_\mu\|_h^2 > 0, \quad h_\nu = |\lambda_{\mu\nu}|^2 \cdot h_\mu, \quad \sigma = s_\mu \cdot f_\mu \implies \|\sigma\|_h^2 = |f_\mu|^2 \cdot h_\mu, \quad f_\nu = \lambda_{\mu\nu}^{-1} \cdot f_\mu.$$

Proposition 2.1.3 (Chern connection and curvature). *The Chern connection of (L, h) has local connection form*

$$A_{h,\mu} = h_\mu^{-1} \cdot \partial(h_\mu) = \partial(\log(h_\mu)), \quad A_{h,\nu} = A_{h,\mu} + \lambda_{\mu\nu}^{-1} \cdot \partial(\lambda_{\mu\nu}).$$

Its curvature is a global closed $(1, 1)$ -form, and $i \cdot F_h$ is real. Moreover,

$$(2.1.1) \quad F_h = \bar{\partial}(A_{h,\mu}) = -\partial(\bar{\partial}(\log(h_\mu))).$$

The closed class

$$c_1(L) = \left[\frac{i}{2 \cdot \pi} \cdot F_h \right]$$

is independent of h .

Proof. By $s_\nu = s_\mu \cdot \lambda_{\mu\nu}$, $h_\nu = |\lambda_{\mu\nu}|^2 \cdot h_\mu$, and $\bar{\partial}(\lambda_{\mu\nu}) = 0$,

$$A_{h,\nu} = \partial(\log(h_\nu)) = \partial(\log(h_\mu)) + \partial(\log(|\lambda_{\mu\nu}|^2)) = A_{h,\mu} + \lambda_{\mu\nu}^{-1} \cdot \partial(\lambda_{\mu\nu}), \quad F_{h,\nu} - F_{h,\mu} = \bar{\partial}(\lambda_{\mu\nu}^{-1} \cdot \partial(\lambda_{\mu\nu})) = 0.$$

$$\partial(A_{h,\mu}) = \partial^2(\log(h_\mu)) = 0, \quad A_{h,\mu} \wedge A_{h,\mu} = 0.$$

$$F_h = d(A_{h,\mu}) + A_{h,\mu} \wedge A_{h,\mu} = \partial(A_{h,\mu}) + \bar{\partial}(A_{h,\mu}) + A_{h,\mu} \wedge A_{h,\mu} = \bar{\partial}(A_{h,\mu}) = \bar{\partial}(\partial(\log(h_\mu))) \xrightarrow{1.3.1} -\partial(\bar{\partial}(\log(h_\mu))).$$

$$d(F_h) = -\partial(\partial(\bar{\partial}(\log(h_\mu)))) - \bar{\partial}(\partial(\bar{\partial}(\log(h_\mu)))) = 0, \quad \bar{F}_h = -F_h, \quad \bar{i} \cdot \bar{F}_h = i \cdot F_h.$$

$$\psi \in C^\infty(X, \mathbb{R}), \quad \tilde{h} = \exp(\psi) \cdot h \implies F_{\tilde{h}} - F_h = -\partial(\bar{\partial}(\psi)) = -d(\bar{\partial}(\psi)) \implies \left[\frac{i}{2 \cdot \pi} \cdot F_{\tilde{h}} \right] = \left[\frac{i}{2 \cdot \pi} \cdot F_h \right].$$

If $\tilde{A}_{h,\mu}$ is another compatible $(1,0)$ connection form, then

$$\partial(h_\mu) = h_\mu \cdot A_{h,\mu} = h_\mu \cdot \tilde{A}_{h,\mu} \implies A_{h,\mu} = \tilde{A}_{h,\mu}.$$

Thus the curvature and its first Chern class are global and metric-independent. $\square$

2.2. Canonical and anticanonical bundles. Let (X^n, ω) be Kähler. The metric matrix, Kähler form, and canonical line bundles are

$$G = [g_{j\bar{k}}]_{\bar{k},j}, \quad \omega = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k, \quad K_X = \Lambda^{n,0} T^* X = (\det(T^{1,0} X))^*, \quad K_X^{-1} = \det(T^{1,0} X).$$

On a holomorphic coordinate chart write

$$\theta = dz^1 \wedge \cdots \wedge dz^n, \quad s = e_1 \wedge \cdots \wedge e_n, \quad \theta(s) = 1.$$

Proposition 2.2.1 (Canonical curvature and the Ricci form). *The induced metrics and curvatures satisfy*

$$\|s\|_\omega^2 = \det(G), \quad \|\theta\|_\omega^2 = (\det(G))^{-1}, \quad F_{K_X^{-1}} = -\partial(\bar{\partial}(\log(\det(G)))), \quad F_{K_X} = \partial(\bar{\partial}(\log(\det(G)))).$$

Consequently,

$$(2.2.1) \quad \rho(\omega) = -i \cdot F_{K_X} = i \cdot F_{K_X^{-1}}, \quad \frac{1}{2 \cdot \pi} [\rho(\omega)] = c_1(K_X^{-1}) = c_1(X) = -c_1(K_X).$$

Proof. By $\theta(s) = 1$, $G = [g_{j\bar{k}}]_{\bar{k},j}$, and $\rho(\omega) = -i \cdot \partial(\bar{\partial}(\log(\det(G))))$,

$$\|s\|_\omega^2 = \det\left(\langle [e_j, e_k]_\omega \rangle_{\bar{k},j}\right) \xrightarrow{1.2.1} \|s\|_\omega^2 = \det(G), \quad \|\theta\|_\omega^2 = \|s\|_\omega^{-2} = (\det(G))^{-1}.$$

$$h_{K_X^{-1}} = \det(G) \xrightarrow{2.1.3} F_{K_X^{-1}} = -\partial(\bar{\partial}(\log(\det(G)))) , \quad h_{K_X} = (\det(G))^{-1} \xrightarrow{2.1.3} F_{K_X} = \partial(\bar{\partial}(\log(\det(G)))).$$

$$\rho(\omega) \xrightarrow{1.8.2} \rho(\omega) = -i \cdot \partial(\bar{\partial}(\log(\det(G)))) = -i \cdot F_{K_X} = i \cdot F_{K_X^{-1}}.$$

$$c_1(K_X^{-1}) = \left[\frac{i}{2 \cdot \pi} \cdot F_{K_X^{-1}} \right] = \frac{1}{2 \cdot \pi} [\rho(\omega)] = -c_1(K_X), \quad c_1(X) = c_1(T^{1,0} X) = c_1(\det(T^{1,0} X)) = c_1(K_X^{-1}).$$

Thus Ricci curvature is the curvature of the canonical line bundle up to $-i$. $\square$

2.3. Divisor currents and the Poincaré–Lelong formula.

Definition 2.3.1 (Divisor current). A $(1, 1)$ -current acts continuously on compactly supported smooth $(n-1, n-1)$ -forms. For an irreducible hypersurface $V \subset U$, its integration current is

$$[V](\eta) = \int_{V_{\text{reg}}} \eta.$$

For a nonzero meromorphic function f define

$$\text{div}(f) = \sum_{\substack{V \subset U \\ V \text{ irreducible hypersurface}}} \text{ord}_V(f) \cdot V, \quad [\text{div}(f)] = \sum_{\substack{V \subset U \\ V \text{ irreducible hypersurface}}} \text{ord}_V(f) \cdot [V].$$

Fact 2.3.2 (Dimension principle for divisor currents). *If a closed order-zero $(1, 1)$ -current is supported on an analytic subset of complex codimension at least two, then it vanishes.*

Theorem 2.3.3 (Poincaré–Lelong formula). *For every nonzero meromorphic function f ,*

$$(2.3.1) \quad \log(|f|^2) \in L^1_{\text{loc}}(U), \quad \frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|f|^2))) = [\text{div}(f)]$$

as $(1, 1)$ -currents.

Proof. By $f = g/h$ locally with $g, h \in \mathcal{O}(U) \setminus \{0\}$ having no common divisor, $D = \text{div}(f) = \sum_{\alpha=1}^N m_{\alpha} \cdot V_{\alpha}$, and $T = \frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|f|^2))) - [D]$,

$$\log(|g|^2), \log(|h|^2) \in \text{PSH}_{\text{loc}}(U), \quad i \cdot \partial(\bar{\partial}(\log(|g|^2))) \geq 0, \quad i \cdot \partial(\bar{\partial}(\log(|h|^2))) \geq 0.$$

$$\log(|f|^2) = \log(|g|^2) - \log(|h|^2) \in L^1_{\text{loc}}(U), \quad \text{ord}(T) = 0, \quad d(T) = 0.$$

For $p \in V_{\alpha, \text{reg}}$ away from every other component and the indeterminacy set, choose coordinates with $V_{\alpha} \cap U = \{z^1 = 0\}$.

$$f = u \cdot (z^1)^{m_{\alpha}}, \quad u \in \mathcal{O}^*(U) \implies \log(|f|^2) = m_{\alpha} \cdot \log(|z^1|^2) + \log(|u|^2), \quad \partial(\bar{\partial}(\log(|u|^2))) = 0.$$

$$\Delta_{\mathbb{R}^2}(\log(|z|^2)) = 4 \cdot \pi \cdot \delta_{\{0\}}, \quad \frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|z|^2))) = [\{0\}].$$

$$\left\langle \frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|z^1|^2))), \eta \right\rangle = \int_{\{z^1=0\}} \eta = [\{z^1=0\}](\eta) \implies \frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|z^1|^2))) = [\{z^1=0\}] = [V_{\alpha}].$$

$$\frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|f|^2))) = m_{\alpha} \cdot [V_{\alpha}] = \text{ord}_{V_{\alpha}}(f) \cdot [V_{\alpha}].$$

$$I(f) = \{x \in U \mid g(x) = h(x) = 0\}, \quad Z = I(f) \cup \bigcup_{\alpha=1}^N V_{\alpha, \text{sing}} \cup \bigcup_{\alpha=1}^N \bigcup_{\substack{\beta=1 \\ \beta \neq \alpha}}^N (V_{\alpha} \cap V_{\beta}),$$

$$\text{supp}(T) \subset Z, \quad d(T) = 0, \quad \text{ord}(T) = 0, \quad \text{codim}_{\mathbb{C}}(Z) \geq 2 \xrightarrow{2.3.2} T = 0.$$

$$T = 0 \implies \frac{i}{2 \cdot \pi} \cdot \partial(\bar{\partial}(\log(|f|^2))) = [\text{div}(f)].$$

Thus the local divisor identity extends across the codimension-two locus. $\square$

2.4. Meromorphic canonical forms.

Proposition 2.4.1 (Meromorphic Ricci formula). *Let $\Omega = f \cdot \theta$ be a nonzero meromorphic section of K_X on U and let $D = \text{div}(\Omega) = \text{div}(f)$. Then*

$$\|\Omega\|_\omega^2 = \frac{|f|^2}{\det(G)}, \quad \rho(\omega) = i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))) \quad \text{on } U \setminus \text{supp}(D).$$

As currents on U ,

$$(2.4.1) \quad \rho(\omega) = i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))) - 2 \cdot \pi \cdot [D].$$

Proof. By $\Omega = f \cdot \theta$, $D = \text{div}(f)$, and $\|\theta\|_\omega^2 = (\det(G))^{-1}$,

$$\Omega = f \cdot \theta \xrightarrow{2.2.1} \|\Omega\|_\omega^2 = |f|^2 \cdot (\det(G))^{-1}, \quad \log(\|\Omega\|_\omega^2) = \log(|f|^2) - \log(\det(G)).$$

$$i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))) = i \cdot \partial(\bar{\partial}(\log(|f|^2))) - i \cdot \partial(\bar{\partial}(\log(\det(G)))) \xrightarrow{1.8.2} i \cdot \partial(\bar{\partial}(\log(|f|^2))) + \rho(\omega).$$

$$(D = \text{div}(f), f \neq 0) \xrightarrow{2.3.3} i \cdot \partial(\bar{\partial}(\log(|f|^2))) = 2 \cdot \pi \cdot [D] \implies \rho(\omega) = i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))) - 2 \cdot \pi \cdot [D].$$

This proves both the smooth and current identities. $\square$

Corollary 2.4.2 (Holomorphic volume form). *If $\Omega \in H^0(U, K_X)$ is nowhere vanishing, then*

$$\rho(\omega) = i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))).$$

Proof. By $\Omega \in H^0(U, K_X)$, $\Omega(x) \neq 0$ for every $x \in U$, and $D = \text{div}(\Omega)$,

$$\Omega(x) \neq 0 \implies D = 0 \implies [D] = 0 \xrightarrow{2.4.1} \rho(\omega) = i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))).$$

Thus the divisor contribution vanishes. $\square$

2.5. Volume normalization.

Proposition 2.5.1 (Volume normalization and Ricci curvature). *Let $\Omega = f \cdot \theta$ be a nonzero meromorphic section of K_X , let $D = \text{div}(\Omega)$, and suppose*

$$\omega^n = \exp(\Phi) \cdot i^{n^2} \cdot \Omega \wedge \bar{\Omega}$$

on $U \setminus \text{supp}(D)$. Then

$$\Phi = \log(\det(G)) - \log(|f|^2) + \log\left(\frac{n!}{2^n}\right) = -\log(\|\Omega\|_\omega^2) + \log\left(\frac{n!}{2^n}\right).$$

The right-hand side gives the locally integrable extension of Φ across D , and

$$(2.5.1) \quad \rho(\omega) = -i \cdot \partial(\bar{\partial}(\Phi)) - 2 \cdot \pi \cdot [D]$$

as currents. If Ω is holomorphic and nowhere vanishing, then

$$\rho(\omega) = -i \cdot \partial(\bar{\partial}(\Phi)).$$

Proof. By $\Omega = f \cdot \theta$, $D = \text{div}(\Omega)$, and $\omega^n = \exp(\Phi) \cdot i^{n^2} \cdot \Omega \wedge \bar{\Omega}$,

$$\omega^n \xrightarrow{1.4.1} \omega^n = n! \cdot \left(\frac{i}{2}\right)^n \cdot \det(G) \cdot dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n = \frac{n!}{2^n} \cdot i^{n^2} \cdot \det(G) \cdot \theta \wedge \bar{\theta}.$$

$$\Omega \wedge \bar{\Omega} = |f|^2 \cdot \theta \wedge \bar{\theta} \implies \exp(\Phi) = \frac{n! \cdot \det(G)}{2^n \cdot |f|^2}.$$

$$\Phi = \log(\det(G)) - \log(|f|^2) + \log\left(\frac{n!}{2^n}\right) = -\log(\|\Omega\|_\omega^2) + \log\left(\frac{n!}{2^n}\right), \quad \log(|f|^2) \in L_{\text{loc}}^1(U) \xrightarrow{2.3.3} \Phi \in L_{\text{loc}}^1(U).$$

$$-i \cdot \partial(\bar{\partial}(\Phi)) = i \cdot \partial(\bar{\partial}(\log(\|\Omega\|_\omega^2))) \xrightarrow{2.4.1} -i \cdot \partial(\bar{\partial}(\Phi)) = \rho(\omega) + 2 \cdot \pi \cdot [D].$$

$$\Omega \in H^0(U, K_X), \quad \Omega(x) \neq 0 \text{ for every } x \in U \implies D = 0 \implies [D] = 0 \xrightarrow{2.5.1} \rho(\omega) = -i \cdot \partial(\bar{\partial}(\Phi)).$$

Thus the volume potential determines the Ricci current. $\square$

Example 2.5.2 (A two-dimensional volume coefficient). Let $n = 2$, $G = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$, and $f = 2$. Then

$$\det(G) = 4 \cdot 3 - 2 \cdot 2 = 12 - 4 = 8, \quad |f|^2 = 2 \cdot 2 = 4,$$

$$\exp(\Phi) = \frac{2! \cdot \det(G)}{2^2 \cdot |f|^2} = \frac{2 \cdot 8}{(2 \cdot 2) \cdot (2 \cdot 2)} = \frac{16}{4 \cdot 4} = \frac{16}{16} = 1, \quad \Phi = \log(1) = 0.$$

Corollary 2.5.3 (Ricci variation by a volume ratio). *For Kähler forms ω_0 and ω_1 on X ,*

$$(2.5.2) \quad \rho(\omega_1) - \rho(\omega_0) = -i \cdot \partial \left(\bar{\partial} \left(\log \left(\frac{\omega_1^n}{\omega_0^n} \right) \right) \right).$$

If $F \in C^\infty(X, \mathbb{R})$, then

$$\omega_1^n = \exp(F) \cdot \omega_0^n \xrightarrow{2.5.2} \rho(\omega_1) = \rho(\omega_0) - i \cdot \partial(\bar{\partial}(F)).$$

Proof. By $G_a = [g_{a,j\bar{k}}]_{\bar{k},j}$, $\omega_a = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{a,j\bar{k}} \cdot dz^j \wedge d\bar{z}^k$, and $a \in \{0, 1\}$, $F \in C^\infty(X, \mathbb{R})$,

$$\frac{\omega_1^n}{\omega_0^n} = \frac{\det(G_1)}{\det(G_0)}, \quad \log \left(\frac{\omega_1^n}{\omega_0^n} \right) = \log(\det(G_1)) - \log(\det(G_0)) \xrightarrow{1.8.2} \rho(\omega_1) - \rho(\omega_0) = -i \cdot \partial \left(\bar{\partial} \left(\log \left(\frac{\omega_1^n}{\omega_0^n} \right) \right) \right).$$

$$\omega_1^n = \exp(F) \cdot \omega_0^n \implies \log \left(\frac{\omega_1^n}{\omega_0^n} \right) = F \implies \rho(\omega_1) = \rho(\omega_0) - i \cdot \partial(\bar{\partial}(F)).$$

Thus the volume ratio is the Ricci potential. $\square$

3. THE COMPLEX MONGE–AMPÈRE EQUATION AND THE CONTINUITY METHOD

The line-bundle formula converts prescribed Ricci curvature into a scalar volume equation. We formulate that equation, compute its linearization, prove uniqueness, and establish openness of the continuity path.

3.1. Kähler potentials and prescribed Ricci forms. Fix a connected compact Kähler manifold (X^n, ω_0) and define

$$\mathcal{H}_{\omega_0} = \left\{ \phi \in C^\infty(X, \mathbb{R}) \mid \omega_\phi = \omega_0 + \frac{i}{2} \cdot \partial(\bar{\partial}(\phi)) > 0 \right\}.$$

Proposition 3.1.1 (Kähler potentials and volume invariance). *If ω is Kähler and $[\omega] = [\omega_0]$, then $\omega = \omega_\phi$ for some $\phi \in \mathcal{H}_{\omega_0}$, unique modulo constants. Moreover,*

$$\int_X \omega_\phi^n = \int_X \omega_0^n \quad (\phi \in \mathcal{H}_{\omega_0}).$$

Proof. By $[\omega] = [\omega_0]$, $d(\omega - \omega_0) = 0$, and $[\omega - \omega_0] = 0$,

$$\alpha = \omega - \omega_0, \quad d(\alpha) = 0, \quad [\alpha] = 0 \xrightarrow{A.2.3} \alpha = \frac{i}{2} \cdot \partial(\bar{\partial}(\phi)), \quad \omega = \omega_\phi.$$

$$\omega_\phi = \omega_\psi \implies \partial(\bar{\partial}(\phi - \psi)) = 0 \implies \Delta_{\omega_0}(\phi - \psi) = 0 \xrightarrow{3.5.2} \phi - \psi \in \mathbb{R}.$$

$$\Theta_\phi = \sum_{q=0}^{n-1} \omega_\phi^q \wedge \omega_0^{n-1-q}, \quad d(\omega_\phi) = d(\omega_0) = 0 \xrightarrow{1.4.1} d(\Theta_\phi) = 0.$$

$$(3.1.1) \quad \omega_\phi^n - \omega_0^n = \frac{i}{2} \cdot \partial(\bar{\partial}(\phi)) \wedge \Theta_\phi = d \left(\frac{i}{2} \cdot \bar{\partial}(\phi) \wedge \Theta_\phi \right).$$

$$\int_X (\omega_\phi^n - \omega_0^n) \xrightarrow{3.1.1} \int_X d \left(\frac{i}{2} \cdot \bar{\partial}(\phi) \wedge \Theta_\phi \right) = 0.$$

Thus the potential is unique modulo constants and its volume is fixed by $[\omega_0]$. $\square$

Proposition 3.1.2 (Prescribed Ricci reduction). *Let η be a real d -closed $(1, 1)$ -form satisfying*

$$[\eta] = 2 \cdot \pi \cdot c_1(X).$$

There is a unique $F \in C^\infty(X, \mathbb{R})$ satisfying

$$\rho(\omega_0) - \eta = i \cdot \partial(\bar{\partial}(F)), \quad \int_X e^F \cdot \omega_0^n = \int_X \omega_0^n.$$

For every $\phi \in \mathcal{H}_{\omega_0}$,

$$(3.1.2) \quad \rho(\omega_\phi) = \eta \iff \omega_\phi^n = e^F \cdot \omega_0^n.$$

Proof. By $[\eta] = 2 \cdot \pi \cdot c_1(X)$, $[\rho(\omega_0)] = 2 \cdot \pi \cdot c_1(X)$, and X is connected compact Kähler,

$$[\eta] = 2 \cdot \pi \cdot c_1(X) \xrightarrow{2.2.1} [\rho(\omega_0) - \eta] = 0 \xrightarrow{A.2.3} \rho(\omega_0) - \eta = \frac{i}{2} \cdot \partial(\bar{\partial}(u)) = i \cdot \partial(\bar{\partial}(f)), \quad f = \frac{u}{2}.$$

$$V_0 = \int_X \omega_0^n, \quad c = \log\left(\frac{V_0}{\int_X e^f \cdot \omega_0^n}\right), \quad F = f + c \implies \int_X e^F \cdot \omega_0^n = V_0.$$

$$\omega_\phi^n = e^F \cdot \omega_0^n \xrightarrow{2.5.2} \rho(\omega_\phi) = \rho(\omega_0) - i \cdot \partial(\bar{\partial}(F)) = \eta.$$

Conversely, let $h = \log(\omega_\phi^n / \omega_0^n) - F$.

$$\rho(\omega_\phi) = \eta \xrightarrow{2.5.2} \partial(\bar{\partial}(h)) = 0 \implies \Delta_{\omega_0}(h) = 0 \xrightarrow{3.5.2} h = a \in \mathbb{R}.$$

$$\phi \in \mathcal{H}_{\omega_0} \xrightarrow{3.1.1} V_0 = \int_X \omega_\phi^n = \int_X e^{F+a} \cdot \omega_0^n = e^a \cdot V_0 \implies a = 0 \implies \omega_\phi^n = e^F \cdot \omega_0^n.$$

$$i \cdot \partial(\bar{\partial}(F_1 - F_2)) = 0 \implies \Delta_{\omega_0}(F_1 - F_2) = 0 \xrightarrow{3.5.2} F_1 - F_2 = b \in \mathbb{R}, \quad V_0 = e^b \cdot V_0 \implies b = 0.$$

Thus the normalized Ricci potential is unique. $\square$

3.2. The complex Monge–Ampère equation. For $\phi \in \mathcal{H}_{\omega_0}$ write

$$H_\phi = \begin{bmatrix} \phi_{1\bar{1}} & \cdots & \phi_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ \phi_{1\bar{n}} & \cdots & \phi_{n\bar{n}} \end{bmatrix}, \quad \phi_{j\bar{k}} = \frac{\partial^2 \phi}{\partial z^j \partial \bar{z}^k}, \quad G_\phi = G_0 + H_\phi.$$

The complex Monge–Ampère equation with datum $F \in C^\infty(X, \mathbb{R})$ is

$$(3.2.1) \quad \omega_\phi^n = e^F \cdot \omega_0^n, \quad \omega_\phi > 0, \quad \int_X \phi \cdot d\mu_{\omega_0} = 0.$$

Equivalently,

$$\frac{\omega_\phi^n}{\omega_0^n} = \frac{\det(G_\phi)}{\det(G_0)} = e^F, \quad G_\phi = G_0 + H_\phi > 0.$$

Corollary 3.2.1 (Volume normalization). *The complex Monge–Ampère equation can hold only if*

$$(3.2.2) \quad \int_X e^F \cdot \omega_0^n = \int_X \omega_0^n.$$

Proof. By $\phi \in \mathcal{H}_{\omega_0}$, $\omega_\phi^n = e^F \cdot \omega_0^n$, and $\int_X \omega_\phi^n = \int_X \omega_0^n$,

$$\omega_\phi^n = e^F \cdot \omega_0^n \xrightarrow{3.1.1} \int_X e^F \cdot \omega_0^n = \int_X \omega_\phi^n = \int_X \omega_0^n.$$

Thus the volume constraint is necessary. $\square$

Example 3.2.2 (Complex dimension one). If $n = 1$, then

$$G_0 = [g_{1\bar{1}}], \quad H_\phi = [\phi_{1\bar{1}}], \quad \frac{\det(G_0 + H_\phi)}{\det(G_0)} = \frac{\det([g_{1\bar{1}} + \phi_{1\bar{1}}])}{\det([g_{1\bar{1}}])} = 1 + g^{1\bar{1}} \cdot \phi_{1\bar{1}} \xrightarrow{3.4.1} 1 + \Delta_{\omega_0}(\phi).$$

Hence

$$\omega_\phi = e^F \cdot \omega_0 \iff \Delta_{\omega_0}(\phi) = e^F - 1, \quad \omega_\phi = e^F \cdot \omega_0 \xrightarrow{3.2.1} \int_X (e^F - 1) \cdot \omega_0 = 0.$$

Example 3.2.3 (Complex dimension two). At a point write

$$\begin{aligned} G_0 &= \begin{bmatrix} a & b \\ \bar{b} & c \end{bmatrix}, \quad H_\phi = \begin{bmatrix} p & q \\ \bar{q} & r \end{bmatrix}, \quad \det(G_0) = a \cdot c - |b|^2 > 0 \xrightarrow{1.2.2} G_0^{-1} = \frac{1}{a \cdot c - |b|^2} \cdot \begin{bmatrix} c & -b \\ -\bar{b} & a \end{bmatrix}. \\ G_0^{-1} \cdot H_\phi &= \frac{1}{\det(G_0)} \cdot \begin{bmatrix} c & -b \\ -\bar{b} & a \end{bmatrix} \cdot \begin{bmatrix} p & q \\ \bar{q} & r \end{bmatrix}. \\ (G_0^{-1} \cdot H_\phi)^1_1 &= \frac{c \cdot p + (-b) \cdot \bar{q}}{\det(G_0)} = \frac{c \cdot p - b \cdot \bar{q}}{\det(G_0)}, \quad (G_0^{-1} \cdot H_\phi)^1_2 = \frac{c \cdot q + (-b) \cdot r}{\det(G_0)} = \frac{c \cdot q - b \cdot r}{\det(G_0)}. \\ (G_0^{-1} \cdot H_\phi)^2_1 &= \frac{(-\bar{b}) \cdot p + a \cdot \bar{q}}{\det(G_0)} = \frac{-\bar{b} \cdot p + a \cdot \bar{q}}{\det(G_0)}, \quad (G_0^{-1} \cdot H_\phi)^2_2 = \frac{(-\bar{b}) \cdot q + a \cdot r}{\det(G_0)} = \frac{-\bar{b} \cdot q + a \cdot r}{\det(G_0)}. \\ G_0^{-1} \cdot H_\phi &= \frac{1}{\det(G_0)} \cdot \begin{bmatrix} c \cdot p - b \cdot \bar{q} & c \cdot q - b \cdot r \\ -\bar{b} \cdot p + a \cdot \bar{q} & -\bar{b} \cdot q + a \cdot r \end{bmatrix}. \\ \det(G_0 + H_\phi) &= (a + p) \cdot (c + r) - (b + q) \cdot (\bar{b} + \bar{q}) \\ &= a \cdot c + a \cdot r + p \cdot c + p \cdot r - |b|^2 - b \cdot \bar{q} - q \cdot \bar{b} - |q|^2 \\ &= \det(G_0) + c \cdot p + a \cdot r - b \cdot \bar{q} - \bar{b} \cdot q + \det(H_\phi). \end{aligned}$$

Therefore

$$\frac{\det(G_0 + H_\phi)}{\det(G_0)} = 1 + \text{Tr}(G_0^{-1} \cdot H_\phi) + \frac{\det(H_\phi)}{\det(G_0)} \xrightarrow{3.4.1} 1 + \Delta_{\omega_0}(\phi) + \frac{\det(H_\phi)}{\det(G_0)} = e^F.$$

Theorem 3.2.4 (Calabi–Yau). Let (X, ω_0) be compact Kähler and let $F \in C^\infty(X, \mathbb{R})$ satisfy

$$\int_X e^F \cdot \omega_0^n = \int_X \omega_0^n \implies \exists! \phi \in \mathcal{H}_{\omega_0}, \quad \int_X \phi \cdot d\mu_{\omega_0} = 0, \quad \omega_\phi^n = e^F \cdot \omega_0^n.$$

3.3. Hölder spaces and analytic input.

Definition 3.3.1 (Hölder spaces). For a bounded domain $\Omega \subset \mathbb{R}^m$, $\alpha \in (0, 1)$, and $\phi \in C^0(\Omega)$, define

$$[\phi]_{C^{0,\alpha}(\Omega)} = \sup_{\substack{x, y \in \Omega \\ x \neq y}} \frac{|\phi(x) - \phi(y)|}{|x - y|^\alpha}, \quad \|\phi\|_{C^{0,\alpha}(\Omega)} = \sup_{x \in \Omega} |\phi(x)| + [\phi]_{C^{0,\alpha}(\Omega)}.$$

For $k \in \mathbb{N}$ set

$$\begin{aligned} C^{k,\alpha}(\Omega) &= \left\{ \phi \in C^k(\Omega) \mid [\partial^\beta(\phi)]_{C^{0,\alpha}(\Omega)} < \infty \quad (|\beta| = k) \right\}, \\ \|\phi\|_{C^{k,\alpha}(\Omega)} &= \sum_{\substack{\beta \in \mathbb{N}^m \\ 0 \leq |\beta| \leq k}} \|\partial^\beta(\phi)\|_{C^0(\Omega)} + \sum_{\substack{\beta \in \mathbb{N}^m \\ |\beta| = k}} [\partial^\beta(\phi)]_{C^{0,\alpha}(\Omega)}. \end{aligned}$$

The manifold norm is obtained from a fixed finite atlas and partition of unity.

Lemma 3.3.2. The normed space $C^{k,\alpha}(X)$ is Banach.

Proof. By $(\phi_\nu)_{\nu=1}^\infty$ is Cauchy in $C^{k,\alpha}(X)$, $0 \leq |\beta| \leq k$, and $\|\partial^\beta(\phi_\nu - \phi_\mu)\|_{C^0} \leq \|\phi_\nu - \phi_\mu\|_{C^{k,\alpha}}$,

$$\|\phi_\nu - \phi_\mu\|_{C^{k,\alpha}} \longrightarrow 0 \xrightarrow{3.3.1} \|\partial^\beta(\phi_\nu) - \partial^\beta(\phi_\mu)\|_{C^0} \leq \|\phi_\nu - \phi_\mu\|_{C^{k,\alpha}} \longrightarrow 0 \implies \partial^\beta(\phi_\nu) \xrightarrow{C^0} u_\beta \quad (0 \leq |\beta| \leq k).$$

For every test function ψ supported in a coordinate chart U with Euclidean density dx ,

$$\int_U u_\beta \cdot \psi \cdot dx = \lim_{\nu \rightarrow \infty} \int_U \partial^\beta(\phi_\nu) \cdot \psi \cdot dx = (-1)^{|\beta|} \cdot \lim_{\nu \rightarrow \infty} \int_U \phi_\nu \cdot \partial^\beta(\psi) \cdot dx = (-1)^{|\beta|} \cdot \int_U u_0 \cdot \partial^\beta(\psi) \cdot dx.$$

$$u_\beta = \partial^\beta(u_0), \quad [\partial^\beta(\phi_\nu) - u_\beta]_{C^{0,\alpha}} \leq \liminf_{\mu \rightarrow \infty} [\partial^\beta(\phi_\nu) - \partial^\beta(\phi_\mu)]_{C^{0,\alpha}} \longrightarrow 0 \quad (|\beta| = k),$$

$$\partial^\beta(\phi_\nu) \xrightarrow{C^0} \partial^\beta(u_0), \quad [\partial^\beta(\phi_\nu) - \partial^\beta(u_0)]_{C^{0,\alpha}} \longrightarrow 0 \xrightarrow{3.3.1} \|\phi_\nu - u_0\|_{C^{k,\alpha}} \longrightarrow 0.$$

Thus every Cauchy sequence converges in $C^{k,\alpha}(X)$. $\square$

Definition 3.3.3 (Banach spaces for the Monge–Ampère map). Fix $N \in \mathbb{N}$ and $\alpha \in (0, 1)$. Define

$$\mathcal{B}_1 = C^{N+2,\alpha}(X), \quad \mathcal{B}_2 = C^{N,\alpha}(X), \quad \mathcal{B}_a^0 = \left\{ u \in \mathcal{B}_a \mid \int_X u \cdot \omega_0^n = 0 \right\} \quad (a \in \{1, 2\}).$$

Then $u \mapsto H_u$ is bounded from $\mathcal{B}_1$ to the $C^{N,\alpha}$ Hermitian-matrix fields.

Theorem 3.3.4 (Banach implicit function theorem). Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be Banach spaces, let $\mathcal{O} \subset \mathcal{X} \times \mathcal{Y}$ be open, and let $\Psi \in C^1(\mathcal{O}, \mathcal{Z})$. If

$$\Psi(x_0, y_0) = 0, \quad D_y \Psi(x_0, y_0) \in \text{Iso}(\mathcal{Y}, \mathcal{Z}),$$

then $\Psi(x, f(x)) = 0$ on a neighborhood of x_0 for a unique local C^1 map f satisfying $f(x_0) = y_0$.

We also use the bounded inverse theorem, the compact embedding $H^1(X) \hookrightarrow L^2(X)$, and elliptic Schauder regularity.

3.4. The complex Laplacian.

Definition 3.4.1 (Complex Laplacian). Let (X, ω) be compact Kähler. For $u \in C^\infty(X, \mathbb{R})$ define

$$H_u = \begin{bmatrix} u_{1\bar{1}} & \cdots & u_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ u_{1\bar{n}} & \cdots & u_{n\bar{n}} \end{bmatrix}, \quad \Delta_\omega(u) = \text{Tr}(G^{-1} \cdot H_u) = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot u_{j\bar{k}}, \quad d\mu_\omega = \frac{\omega^n}{n!}.$$

Explicitly,

$$G^{-1} \cdot H_u = \begin{bmatrix} g^{1\bar{1}} & \cdots & g^{1\bar{n}} \\ \vdots & \ddots & \vdots \\ g^{n\bar{1}} & \cdots & g^{n\bar{n}} \end{bmatrix} \cdot \begin{bmatrix} u_{1\bar{1}} & \cdots & u_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ u_{1\bar{n}} & \cdots & u_{n\bar{n}} \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^n g^{1\bar{k}} \cdot u_{1\bar{k}} & \cdots & \sum_{k=1}^n g^{1\bar{k}} \cdot u_{n\bar{k}} \\ \vdots & \ddots & \vdots \\ \sum_{k=1}^n g^{n\bar{k}} \cdot u_{1\bar{k}} & \cdots & \sum_{k=1}^n g^{n\bar{k}} \cdot u_{n\bar{k}} \end{bmatrix},$$

$$\Delta_\omega(u) = \sum_{j=1}^n (G^{-1} \cdot H_u)^j_j = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot u_{j\bar{k}}.$$

Example 3.4.2 (Flat complex Laplacian). For $z^j = x^j + i \cdot y^j$ and $\omega_{\text{std}} = (i/2) \cdot \sum_{j=1}^n dz^j \wedge d\bar{z}^j$,

$$z^j = x^j + i \cdot y^j \xrightarrow{1.1.1} \frac{\partial}{\partial z^j} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x^j} - i \cdot \frac{\partial}{\partial y^j} \right), \quad \frac{\partial}{\partial \bar{z}^j} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x^j} + i \cdot \frac{\partial}{\partial y^j} \right).$$

$$u_{j\bar{j}} = \frac{1}{4} \cdot \left(\frac{\partial^2 u}{\partial (x^j)^2} + \frac{\partial^2 u}{\partial (y^j)^2} \right) \xrightarrow{3.4.1} \Delta_{\omega_{\text{std}}}(u) = \frac{1}{4} \cdot \Delta_{\mathbb{R}^{2n}}(u).$$

Lemma 3.4.3 (Trace–wedge identity). For $u \in C^\infty(X, \mathbb{R})$,

$$(3.4.1) \quad \frac{i}{2} \cdot \partial(\bar{\partial}(u)) \wedge \frac{\omega^{n-1}}{(n-1)!} = (\Delta_\omega(u)) \cdot d\mu_\omega.$$

Proof. By $p \in X$, $G(p) = I_n$, and $\alpha_j = (i/2) \cdot dz^j \wedge d\bar{z}^j$,

$$\begin{aligned}\widehat{\alpha}_m &= \alpha_1 \wedge \cdots \wedge \widehat{\alpha_m} \wedge \cdots \wedge \alpha_n, & \omega^{n-1}(p) &= (n-1)! \cdot \sum_{m=1}^n \widehat{\alpha}_m. \\ \frac{i}{2} \cdot \partial(\bar{\partial}(u)) \wedge \widehat{\alpha}_m &= u_{m\bar{m}} \cdot \alpha_1 \wedge \cdots \wedge \alpha_n. \\ \frac{i}{2} \cdot \partial(\bar{\partial}(u)) \wedge \frac{\omega^{n-1}}{(n-1)!} &= \sum_{m=1}^n u_{m\bar{m}} \cdot d\mu_\omega = \text{Tr}(I_n \cdot H_u) \cdot d\mu_\omega \xrightarrow{3.4.1} = (\Delta_\omega(u)) \cdot d\mu_\omega.\end{aligned}$$

Since p was arbitrary, the identity is global. $\square$

3.5. Analysis of the Laplacian.

Lemma 3.5.1 (Integration by parts). *For $u, v \in C^\infty(X, \mathbb{R})$,*

$$(3.5.1) \quad \int_X (\Delta_\omega(u)) \cdot v \cdot d\mu_\omega = - \int_X \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot u_j \cdot v_{\bar{k}} \cdot d\mu_\omega.$$

Consequently $\int_X \Delta_\omega(u) \cdot d\mu_\omega = 0$ and Δ_ω is self-adjoint.

Proof. By $u, v \in C^\infty(X, \mathbb{R})$, $d(\omega) = 0$, and $\omega^{n-1}/(n-1)! \in \Omega^{n-1, n-1}(X)$,

$$\begin{aligned}d(\omega) &= 0 \xrightarrow{1.4.1} d\left(\frac{\omega^{n-1}}{(n-1)!}\right) = 0, \\ \int_X (\Delta_\omega(u)) \cdot v \cdot d\mu_\omega &\xrightarrow{3.4.3} \int_X \frac{i}{2} \cdot v \cdot \partial(\bar{\partial}(u)) \wedge \frac{\omega^{n-1}}{(n-1)!}. \\ d\left(\frac{i}{2} \cdot v \cdot \partial(u) \wedge \frac{\omega^{n-1}}{(n-1)!}\right) &= -\frac{i}{2} \cdot \partial(u) \wedge \bar{\partial}(v) \wedge \frac{\omega^{n-1}}{(n-1)!} - \frac{i}{2} \cdot v \cdot \partial(\bar{\partial}(u)) \wedge \frac{\omega^{n-1}}{(n-1)!}. \\ \int_X (\Delta_\omega(u)) \cdot v \cdot d\mu_\omega &= - \int_X \frac{i}{2} \cdot \partial(u) \wedge \bar{\partial}(v) \wedge \frac{\omega^{n-1}}{(n-1)!} = - \int_X \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot u_j \cdot v_{\bar{k}} \cdot d\mu_\omega. \\ v = 1 &\xrightarrow{3.5.1} \int_X \Delta_\omega(u) \cdot d\mu_\omega = 0, \quad g^{j\bar{k}} = \overline{g^{k\bar{j}}}, \quad u, v \in \mathbb{R} \xrightarrow{3.5.1} \int_X (\Delta_\omega(u)) \cdot v \cdot d\mu_\omega = \int_X u \cdot (\Delta_\omega(v)) \cdot d\mu_\omega.\end{aligned}$$

Thus the energy identity gives both conclusions. $\square$

Corollary 3.5.2 (Kernel of the complex Laplacian). *If X is connected, then $\Delta_\omega(u) = 0$ if and only if u is constant.*

Proof. By $u \in C^\infty(X, \mathbb{R})$, $\Delta_\omega(u) = 0$, and $(G^{-1})^* = G^{-1} \succ 0$,

$$u_{\bar{z}} = \begin{bmatrix} u_{\bar{1}} \\ \vdots \\ u_{\bar{n}} \end{bmatrix}, \quad \Delta_\omega(u) = 0 \xrightarrow{3.5.1} 0 = - \int_X u_{\bar{z}}^* \cdot G^{-1} \cdot u_{\bar{z}} \cdot d\mu_\omega \implies u_{\bar{z}} = 0.$$

$$\bar{\partial}(u) = 0, \quad u \in \mathbb{R} \implies \partial(u) = 0 \xrightarrow{1.3.1} d(u) = \partial(u) + \bar{\partial}(u) = 0 \implies u \in \mathbb{R} \cdot 1_X.$$

The converse follows by direct differentiation. $\square$

Lemma 3.5.3 (Mean-zero Poincaré inequality). *There is $C_\omega > 0$ such that*

$$\int_X u^2 \cdot d\mu_\omega \leq C_\omega \cdot \int_X u_{\bar{z}}^* \cdot G^{-1} \cdot u_{\bar{z}} \cdot d\mu_\omega$$

whenever $u \in H^1(X, \mathbb{R})$ and $\int_X u \cdot d\mu_\omega = 0$.

Proof. By $(u_v)_{v=1}^\infty \subset H^1(X, \mathbb{R})$, $\int_X u_v \cdot d\mu_\omega = 0$, $\|u_v\|_{L^2} = 1$, and $\int_X u_{v,\bar{z}}^* \cdot G^{-1} \cdot u_{v,\bar{z}} \cdot d\mu_\omega \rightarrow 0$,

$$\int_X u_{v,\bar{z}}^* \cdot G^{-1} \cdot u_{v,\bar{z}} \cdot d\mu_\omega \rightarrow 0.$$

$$G^{-1} = (G^{-1})^* \succ 0, \quad X \text{ compact} \xrightarrow{1.2.2} c \cdot I_n \preceq G^{-1} \preceq C \cdot I_n.$$

$$c \cdot \|d(u_v)\|_{L^2}^2 \leq \int_X u_{v,\bar{z}}^* \cdot G^{-1} \cdot u_{v,\bar{z}} \cdot d\mu_\omega \leq C \cdot \|d(u_v)\|_{L^2}^2 \implies \|d(u_v)\|_{L^2} \rightarrow 0.$$

Compactness of $H^1(X) \hookrightarrow L^2(X)$ gives a subsequence with

$$u_v \xrightarrow{L^2} u, \quad d(u) = 0, \quad \int_X u \cdot d\mu_\omega = 0, \quad \|u\|_{L^2} = 1.$$

The conditions $d(u) = 0$ and $\int_X u \cdot d\mu_\omega = 0$ force $u = 0$, contradicting $\|u\|_{L^2} = 1$. $\square$

Theorem 3.5.4 (Lax–Milgram). *Let $\mathcal{H}$ be a real Hilbert space and let $B \in \mathcal{L}(\mathcal{H} \times \mathcal{H}, \mathbb{R})$ satisfy*

$$|B(u, v)| \leq M \cdot \|u\| \cdot \|v\|, \quad B(u, u) \geq c \cdot \|u\|^2$$

for $M, c > 0$. For every $\ell \in \mathcal{H}^$ there is a unique $u \in \mathcal{H}$ such that $B(u, v) = \ell(v)$ for every $v \in \mathcal{H}$.*

Proof. By $B(u, v) = \langle T(u), v \rangle$, $\ell(v) = \langle f, v \rangle$, and $B(u, u) \geq c \cdot \|u\|^2$,

$$c \cdot \|u\|^2 \leq \langle T(u), u \rangle \leq \|T(u)\| \cdot \|u\| \implies \|T(u)\| \geq c \cdot \|u\|.$$

$$T(u_v) \rightarrow y \implies \|u_v - u_\mu\| \leq \frac{1}{c} \cdot \|T(u_v) - T(u_\mu)\| \rightarrow 0$$

$$\implies u_v \rightarrow u, \quad y = \lim_{v \rightarrow \infty} T(u_v) = T(u) \implies \overline{\text{ran}(T)} = \text{ran}(T).$$

$$\ker(T) = \{0\}, \quad w \perp \text{ran}(T) \implies 0 = \langle T(w), w \rangle \geq c \cdot \|w\|^2 \implies w = 0.$$

Thus $\text{ran}(T)$ is closed and dense, so T is bijective and $T(u) = f$ has a unique solution. $\square$

Theorem 3.5.5 (Solvability of the complex Laplacian). *For $v \in C^{N,\alpha}(X, \mathbb{R})$, the equation $\Delta_\omega(u) = v$ has a solution $u \in C^{N+2,\alpha}(X, \mathbb{R})$ if and only if*

$$\int_X v \cdot d\mu_\omega = 0.$$

The solution is unique after imposing $\int_X u \cdot d\mu_\omega = 0$. Smooth data produce a smooth solution.

Proof. By $u \in C^{N+2,\alpha}(X, \mathbb{R})$, $v \in C^{N,\alpha}(X, \mathbb{R})$, and $\Delta_\omega(u) = v$,

$$\Delta_\omega(u) = v \xrightarrow{3.5.1} \int_X v \cdot d\mu_\omega = \int_X \Delta_\omega(u) \cdot d\mu_\omega = 0.$$

Conversely, let $\int_X v \cdot d\mu_\omega = 0$ and define on $H_0^1(X) = \{u \in H^1(X, \mathbb{R}) \mid \int_X u \cdot d\mu_\omega = 0\}$

$$B(u, w) = \text{Re} \left(\int_X u_{\bar{z}}^* \cdot G^{-1} \cdot w_{\bar{z}} \cdot d\mu_\omega \right), \quad \ell(w) = - \int_X v \cdot w \cdot d\mu_\omega.$$

$$|B(u, w)| \leq C \cdot \|u\|_{H^1} \cdot \|w\|_{H^1}, \quad B(u, u) \geq c \cdot \|d(u)\|_{L^2}^2 \xrightarrow{3.5.3} B(u, u) \geq c' \cdot \|u\|_{H^1}^2.$$

$$B \in \mathcal{L}(H_0^1(X) \times H_0^1(X), \mathbb{R}), \quad B(u, u) \geq c' \cdot \|u\|_{H^1}^2 \xrightarrow{3.5.4} \exists! u \in H_0^1(X) \quad B(u, w) = \ell(w) \quad (w \in H_0^1(X)).$$

$$B(u, w) = \ell(w) \xrightarrow{3.5.1} - \int_X (\Delta_\omega(u)) \cdot w \cdot d\mu_\omega = - \int_X v \cdot w \cdot d\mu_\omega \quad (w \in H_0^1(X)).$$

$$\tilde{w} = w - \frac{\int_X w \cdot d\mu_\omega}{\int_X d\mu_\omega} \implies \tilde{w} \in H_0^1(X), \quad \int_X v \cdot (w - \tilde{w}) \cdot d\mu_\omega = 0, \quad B(u, w) = B(u, \tilde{w}) \implies \Delta_\omega(u) = v \quad (H^{-1}(X)),$$

$$\Delta_\omega(u) = v \quad (H^{-1}(X)) \xrightarrow{A.3.2} u \in C^{N+2,\alpha}(X), \quad v \in C^\infty(X) \implies u \in C^\infty(X).$$

$$\Delta_\omega(u_1 - u_2) = 0 \xrightarrow{3.5.2} u_1 - u_2 \in \mathbb{R}.$$

Thus the mean-zero solution exists and is unique. $\square$

Remark 3.5.6. Self-adjointness and the kernel computation give

$$\Delta_\omega^* = \Delta_\omega \xrightarrow{3.5.1} (\text{ran}(\Delta_\omega))^\perp = \ker(\Delta_\omega), \quad \ker(\Delta_\omega) = \mathbb{R} \xrightarrow{3.5.2} \text{ran}(\Delta_\omega) = \left\{ v \in C^\infty(X, \mathbb{R}) \mid \int_X v \cdot d\mu_\omega = 0 \right\}.$$

3.6. The Monge–Ampère map. Let $J \subset \mathbb{R}$ be an open interval containing $[0, 1]$ and let $H \in C^\infty(J \times X, \mathbb{R})$ satisfy

$$\int_X e^{H(t)} \cdot \omega_0^n = \int_X \omega_0^n \quad (t \in J).$$

Definition 3.6.1 (Monge–Ampère map). On the open set

$$\mathcal{U} = \{(t, \phi) \in J \times \mathcal{B}_1^0 \mid \omega_\phi > 0\}$$

define $\mathcal{U} \xrightarrow{\Phi} \mathcal{B}_2^0$ by

$$\Phi(t, \phi) = \frac{\omega_\phi^n}{\omega_0^n} - e^{H(t)} = \frac{\det(G_\phi)}{\det(G_0)} - e^{H(t)}.$$

Indeed,

$$\phi \in \mathcal{H}_{\omega_0} \xrightarrow{3.1.1} \int_X \Phi(t, \phi) \cdot \omega_0^n = \int_X \omega_\phi^n - \int_X e^{H(t)} \cdot \omega_0^n = 0.$$

Lemma 3.6.2. The map $\mathcal{U} \xrightarrow{\Phi} \mathcal{B}_2^0$ is continuously differentiable.

Proof. By $(t, \phi) \in \mathcal{U}$, $u \in \mathcal{B}_1^0$, and $G_{\phi+s \cdot u} = G_\phi + s \cdot H_u$,

$$G_{\phi+s \cdot u} = G_\phi + s \cdot H_u = G_\phi \cdot (I_n + s \cdot G_\phi^{-1} \cdot H_u),$$

$$\det(G_{\phi+s \cdot u}) = \det(G_\phi) \cdot \det(I_n + s \cdot G_\phi^{-1} \cdot H_u) \xrightarrow{??} = \det(G_\phi) \cdot (1 + s \cdot \text{Tr}(G_\phi^{-1} \cdot H_u) + O(s^2)).$$

$$\|u\|_{C^{N+2, \alpha}} \leq \delta \implies \left\| \det(G_\phi + H_u) - \det(G_\phi) - \det(G_\phi) \cdot \text{Tr}(G_\phi^{-1} \cdot H_u) \right\|_{C^{N, \alpha}} \leq C_\delta \cdot \|u\|_{C^{N+2, \alpha}}^2.$$

$$u \in \mathcal{B}_1 \xrightarrow{3.3.3} H_u \in C^{N, \alpha}(X, \text{Herm}(n)), \quad \|H_u\|_{C^{N, \alpha}} \leq C \cdot \|u\|_{C^{N+2, \alpha}}.$$

The space $C^{N, \alpha}(X)$ is a Banach algebra, while determinant, matrix inverse, and exponential are smooth on their open domains. Thus Φ is continuously differentiable. $\square$

Proposition 3.6.3 (Linearization of the Monge–Ampère map). For $(t, \phi) \in \mathcal{U}$ and $u \in \mathcal{B}_1^0$,

$$(3.6.1) \quad D_\phi \Phi(t, \phi)[u] = \Delta_{\omega_\phi}(u) \cdot \frac{\omega_\phi^n}{\omega_0^n}.$$

If $\Phi(t, \phi) = 0$, then

$$D_\phi \Phi(t, \phi)[u] = e^{H(t)} \cdot \Delta_{\omega_\phi}(u).$$

Proof. By $\phi_s = \phi + s \cdot u$, $G_s = G_\phi + s \cdot H_u$, and $\frac{d}{ds}(G_s) = H_u$,

$$\frac{d}{ds}(\det(G_s)) \xrightarrow{??} = \det(G_s) \cdot \text{Tr}(G_s^{-1} \cdot H_u) \implies \frac{d}{ds} \bigg|_{s=0} \left(\frac{\det(G_s)}{\det(G_0)} \right) \xrightarrow{3.4.1} = \Delta_{\omega_\phi}(u) \cdot \frac{\det(G_\phi)}{\det(G_0)}.$$

$$\frac{\det(G_\phi)}{\det(G_0)} = \frac{\omega_\phi^n}{\omega_0^n} \implies D_\phi \Phi(t, \phi)[u] = \Delta_{\omega_\phi}(u) \cdot \frac{\omega_\phi^n}{\omega_0^n}, \quad \Phi(t, \phi) = 0 \xrightarrow{3.6.1} \frac{\omega_\phi^n}{\omega_0^n} = e^{H(t)}.$$

Thus the linearization is the weighted complex Laplacian. $\square$

3.7. Calabi uniqueness.

Lemma 3.7.1 (Positivity of intermediate wedge forms). *For Kähler forms ω_ϕ and ω_ψ , define*

$$\Theta_{\phi,\psi} = \sum_{q=0}^{n-1} \omega_\phi^q \wedge \omega_\psi^{n-1-q}, \quad \frac{i}{2} \cdot \partial(u) \wedge \bar{\partial}(u) \wedge \Theta_{\phi,\psi} \geq 0,$$

where $\Theta_{\phi,\psi}$ is strictly positive, $u \in C^\infty(X, \mathbb{R})$ is arbitrary, and equality holds at a point if and only if $\partial(u)$ vanishes there.

Proof. By $p \in X$, $G_\phi(p) = I_n$, and $G_\psi(p) = \text{diag}(\mu_1, \dots, \mu_n)$,

$$G_\phi(p) = I_n, \quad G_\psi(p) = \begin{bmatrix} \mu_1 & & 0 \\ & \ddots & \\ 0 & & \mu_n \end{bmatrix}, \quad \mu_j > 0, \quad \alpha_j = \frac{i}{2} \cdot dz^j \wedge d\bar{z}^j.$$

For $I = \{i_1 < \dots < i_q\} \subset \{1, \dots, n\}$ write $\alpha_I = \alpha_{i_1} \wedge \dots \wedge \alpha_{i_q}$.

$$\omega_\phi^q = q! \cdot \sum_{\substack{I \subset \{1, \dots, n\} \\ |I|=q}} \alpha_I, \quad \omega_\psi^{n-1-q} = (n-1-q)! \cdot \sum_{\substack{J \subset \{1, \dots, n\} \\ |J|=n-1-q}} \left(\prod_{j \in J} \mu_j \right) \cdot \alpha_J.$$

$$\Theta_{\phi,\psi}(p) = \sum_{m=1}^n c_m \cdot \alpha_1 \wedge \dots \wedge \widehat{\alpha_m} \wedge \dots \wedge \alpha_n,$$

$$c_m = \sum_{q=0}^{n-1} q! \cdot (n-1-q)! \cdot \sum_{\substack{J \subset \{1, \dots, n\} \setminus \{m\} \\ |J|=n-1-q}} \prod_{j \in J} \mu_j > 0.$$

$$\partial(u)(p) = \sum_{m=1}^n u_m \cdot dz^m \implies \frac{i}{2} \cdot \partial(u) \wedge \bar{\partial}(u) \wedge \Theta_{\phi,\psi} = \left(\sum_{m=1}^n c_m \cdot |u_m|^2 \right) \cdot \alpha_1 \wedge \dots \wedge \alpha_n.$$

Thus positivity is strict precisely when $\partial(u)(p) \neq 0$. □

Lemma 3.7.2 (Monge–Ampère difference factorization). *For $\phi, \psi \in \mathcal{H}_{\omega_0}$,*

$$(3.7.1) \quad \omega_\phi^n - \omega_\psi^n = \frac{i}{2} \cdot \partial(\bar{\partial}(\phi - \psi)) \wedge \Theta_{\phi,\psi}.$$

Proof. By $A = \omega_\phi$, $B = \omega_\psi$, and $A - B = (i/2) \cdot \partial(\bar{\partial}(\phi - \psi))$,

$$A^n - B^n = (A - B) \wedge \sum_{q=0}^{n-1} A^q \wedge B^{n-1-q}, \quad A - B = \frac{i}{2} \cdot \partial(\bar{\partial}(\phi - \psi)).$$

Substitution gives the claimed factorization. □

Theorem 3.7.3 (Calabi uniqueness). *If $\phi, \psi \in \mathcal{H}_{\omega_0}$ and $\omega_\phi^n = \omega_\psi^n$, then $\phi - \psi$ is constant. If both potentials have zero ω_0 -mean, then $\phi = \psi$.*

Proof. By $u = \phi - \psi$, $\omega_\phi^n = \omega_\psi^n$, and $\phi, \psi \in \mathcal{H}_{\omega_0}$,

$$d(\omega_\phi) = d(\omega_\psi) = 0 \xrightarrow{1.4.1} d(\Theta_{\phi,\psi}) = 0,$$

$$\omega_\phi^n = \omega_\psi^n \xrightarrow{3.7.2} \frac{i}{2} \cdot \partial(\bar{\partial}(u)) \wedge \Theta_{\phi,\psi} = 0.$$

$$0 = \int_X d\left(\frac{i}{2} \cdot u \cdot \bar{\partial}(u) \wedge \Theta_{\phi,\psi}\right) = \int_X \frac{i}{2} \cdot \partial(u) \wedge \bar{\partial}(u) \wedge \Theta_{\phi,\psi} + \int_X \frac{i}{2} \cdot u \cdot \partial(\bar{\partial}(u)) \wedge \Theta_{\phi,\psi}.$$

$$\begin{aligned} \frac{i}{2} \cdot \partial(\bar{\partial}(u)) \wedge \Theta_{\phi, \psi} &= 0 \xrightarrow{3.7.1} \int_X \frac{i}{2} \cdot u \cdot \partial(\bar{\partial}(u)) \wedge \Theta_{\phi, \psi} = 0 \implies \int_X \frac{i}{2} \cdot \partial(u) \wedge \bar{\partial}(u) \wedge \Theta_{\phi, \psi} = 0 \xrightarrow{3.7.1} \partial(u) = 0. \\ \partial(u) &= 0, \quad u \in C^\infty(X, \mathbb{R}) \xrightarrow{3.4.1} \Delta_{\omega_0}(u) = 0 \xrightarrow{3.5.2} u \in \mathbb{R}. \end{aligned}$$

Thus the mean normalization makes that constant zero. $\square$

Theorem 3.7.4 (Comparison principle). *If $\phi, \psi \in \mathcal{H}_{\omega_0}$ satisfy $\omega_\phi^n \geq \omega_\psi^n$ and both have zero ω_0 -mean, then $\phi = \psi$.*

Proof. By $\phi, \psi \in \mathcal{H}_{\omega_0}$, $\omega_\phi^n - \omega_\psi^n \geq 0$, and $\int_X (\phi - \psi) \cdot \omega_0^n = 0$,

$$0 \leq \int_X (\omega_\phi^n - \omega_\psi^n) \xrightarrow{3.1.1} \int_X (\omega_\phi^n - \omega_\psi^n) = 0 \implies \omega_\phi^n = \omega_\psi^n \xrightarrow{3.7.3} \phi - \psi \in \mathbb{R}, \quad \int_X (\phi - \psi) \cdot \omega_0^n = 0 \implies \phi = \psi.$$

Thus the ordered normalized potentials coincide. $\square$

3.8. The continuity path. Assume from now on that F satisfies the volume normalization above.

Definition 3.8.1 (Continuity family). For $t \in \mathbb{R}$ choose $c_t \in \mathbb{R}$ so that $\int_X e^{t \cdot F + c_t} \cdot \omega_0^n = \int_X \omega_0^n$. The continuity family is

$$(3.8.1) \quad \omega_\phi^n = e^{t \cdot F + c_t} \cdot \omega_0^n, \quad \omega_\phi > 0, \quad \int_X \phi \cdot \omega_0^n = 0.$$

The solution set is

$$I = \{t \in [0, 1] \mid \text{there is } \phi \in \mathcal{H}_{\omega_0} \cap \mathcal{B}_1^0 \text{ satisfying the displayed equation}\}.$$

Lemma 3.8.2 (Normalization constant and nonemptiness). *The normalization constant is*

$$c_t = \log \left(\frac{\int_X \omega_0^n}{\int_X e^{t \cdot F} \cdot \omega_0^n} \right), \quad c_0 = c_1 = 0, \quad 0 \in I.$$

Proof. By $c_t \in \mathbb{R}$, $\int_X e^{t \cdot F + c_t} \cdot \omega_0^n = \int_X \omega_0^n$, and $\int_X e^F \cdot \omega_0^n = \int_X \omega_0^n$,

$$e^{c_t} \cdot \int_X e^{t \cdot F} \cdot \omega_0^n = \int_X \omega_0^n \implies c_t = \log \left(\frac{\int_X \omega_0^n}{\int_X e^{t \cdot F} \cdot \omega_0^n} \right), \quad t = 0 \implies c_0 = 0, \quad \phi_0 = 0, \quad t = 1 \xrightarrow{3.2.2} c_1 = 0.$$

$$t = 0, \quad c_0 = 0, \quad \phi_0 = 0 \xrightarrow{3.8.1} 0 \in I.$$

Thus the path begins at the reference metric and ends at the target equation. $\square$

3.9. Openness.

Proposition 3.9.1 (Openness of the continuity set). *For every $t_0 \in I$ there is $\varepsilon > 0$ such that*

$$(t_0 - \varepsilon, t_0 + \varepsilon) \cap [0, 1] \subset I.$$

Proof. By $t_0 \in I$, $H(t) = t \cdot F + c_t$, $H_0 = t_0 \cdot F + c_{t_0}$, and $\mathcal{U} \xrightarrow{\Phi} \mathcal{B}_2^0$,

$$c_t = \log \left(\frac{\int_X \omega_0^n}{\int_X e^{t \cdot F} \cdot \omega_0^n} \right) \xrightarrow{3.8.2} c_t \in C^\infty(\mathbb{R}), \quad H \in C^\infty(\mathbb{R} \times X).$$

$$t_0 \in I \xrightarrow{3.8.1} \exists \phi_0 \in \mathcal{H}_{\omega_0} \cap \mathcal{B}_1^0, \quad \omega' = \omega_{\phi_0}, \quad \Phi(t_0, \phi_0) = 0.$$

$$\Phi(t_0, \phi_0) = 0, \quad \frac{\omega'^n}{\omega_0^n} = e^{H_0} \xrightarrow{3.6.3} D_\phi \Phi(t_0, \phi_0)[u] = e^{H_0} \cdot \Delta_{\omega'}(u).$$

$$\int_X D_\phi \Phi(t_0, \phi_0)[u] \cdot \omega_0^n = \int_X e^{H_0} \cdot \Delta_{\omega'}(u) \cdot \omega_0^n = \int_X \Delta_{\omega'}(u) \cdot \omega'^n \xrightarrow{3.5.1} 0.$$

$$v \in \mathcal{B}_2^0, \quad q = e^{-H_0} \cdot v, \quad \int_X q \cdot \omega'^n = \int_X e^{-H_0} \cdot v \cdot e^{H_0} \cdot \omega_0^n = \int_X v \cdot \omega_0^n = 0 \xrightarrow{3.5.5} \exists u \in \mathcal{B}_1 \quad \Delta_{\omega'}(u) = q.$$

$$\begin{aligned}
\hat{u} &= u - \frac{\int_X u \cdot \omega_0^n}{\int_X \omega_0^n} \implies \hat{u} \in \mathcal{B}_1^0, \quad \Delta_{\omega'}(\hat{u}) = q, \quad D_\phi \Phi(t_0, \phi_0)[\hat{u}] = v. \\
D_\phi \Phi(t_0, \phi_0)[u] &= 0 \implies \Delta_{\omega'}(u) = 0 \xrightarrow{3.5.2} u \in \mathbb{R}, \quad u \in \mathcal{B}_1^0 \implies u = 0. \\
\Phi \in C^1(\mathcal{U}, \mathcal{B}_2^0) &\xrightarrow{3.6.2} D_\phi \Phi(t_0, \phi_0) \in \mathcal{L}(\mathcal{B}_1^0, \mathcal{B}_2^0), \quad \ker(D_\phi \Phi(t_0, \phi_0)) = \{0\}. \\
\text{ran}(D_\phi \Phi(t_0, \phi_0)) &= \mathcal{B}_2^0 \implies D_\phi \Phi(t_0, \phi_0) \in \text{Iso}(\mathcal{B}_1^0, \mathcal{B}_2^0). \\
\Phi(t_0, \phi_0) &= 0, \quad \Phi \in C^1(\mathcal{U}, \mathcal{B}_2^0), \quad D_\phi \Phi(t_0, \phi_0) \in \text{Iso}(\mathcal{B}_1^0, \mathcal{B}_2^0) \xrightarrow{3.3.4} \exists \varepsilon > 0, \quad \exists \phi \in C^1((t_0 - \varepsilon, t_0 + \varepsilon), \mathcal{B}_1^0), \\
\Phi(t, \phi(t)) &= 0, \quad |t - t_0| < \varepsilon, \quad \phi(t) \in \mathcal{B}_1^0, \quad \omega_{\phi(t)} > 0 \xrightarrow{??} \phi(t) \in \mathcal{B}_1^0 \cap \mathcal{H}_{\omega_0}. \\
\Phi(t, \phi(t)) &= 0, \quad \phi(t) \in \mathcal{B}_1^0 \cap \mathcal{H}_{\omega_0}, \quad t \in (t_0 - \varepsilon, t_0 + \varepsilon) \cap [0, 1] \xrightarrow{3.8.1} t \in I.
\end{aligned}$$

Thus every $t_0 \in I$ is interior relative to $[0, 1]$. $\square$

4. CLOSEDNESS THROUGH COMPACTNESS AND EIGENVALUE CONTROL

Openness continues one solution locally in the path parameter. Closedness requires a sequence of solutions to converge while preserving the Monge–Ampère equation and strict positivity. The compactness and eigenvalue mechanisms are isolated here.

$$\begin{aligned}
t_0 \in I &\xrightarrow{3.9.1} t_0 \in \text{int}_{[0,1]}(I). \\
\|\phi_v\|_{C^{2,\alpha}} &\leq C \xrightarrow{4.1.3} \phi_{v_\ell} \xrightarrow{C^{2,\beta}} \phi_\infty. \\
\left(\text{Tr}(h_v) \leq C, \left| \log \left(\frac{\omega_v^n}{\omega_0^n} \right) \right| \leq C \right) &\xrightarrow{4.2.6} \omega_\infty > 0.
\end{aligned}$$

Throughout this section, C is a positive constant depending only on fixed background data and may increase from line to line. Reused constants carry descriptive subscripts.

4.1. Equicontinuity and Hölder compactness.

Definition 4.1.1 (Equicontinuity). Let $K \subset \mathbb{R}^m$ be compact and let $\mathcal{F} \subset C^0(K)$. The family $\mathcal{F}$ is equicontinuous if

$$\varepsilon > 0 \implies \exists \delta > 0 \quad \|x - y\| < \delta \implies |u(x) - u(y)| < \varepsilon \quad (u \in \mathcal{F}, x, y \in K).$$

In particular,

$$[u]_{C^{0,\alpha}(K)} \leq C, \xrightarrow{3.3.1} |u(x) - u(y)| \leq C \cdot \|x - y\|^\alpha, \quad \delta = \left(\frac{\varepsilon}{C} \right)^{1/\alpha} \implies |u(x) - u(y)| < \varepsilon.$$

Theorem 4.1.2 (Arzelà–Ascoli). Every uniformly bounded equicontinuous sequence $(u_v)_{v=1}^\infty \subset C^0(K)$ has a uniformly convergent subsequence.

Proof. By $K \Subset \mathbb{R}^m$, $\sup_{v \geq 1} \|u_v\|_{C^0(K)} < \infty$, and $(u_v)_{v=1}^\infty$ equicontinuous,

$$\begin{aligned}
\{x_r\}_{r=1}^\infty &\subset K \text{ countable and dense,} \quad u_{v_\ell}(x_r) \longrightarrow a_r \quad (r \in \mathbb{N}). \\
\mathcal{F} \text{ equicontinuous} &\xrightarrow{4.1.1} \sup_{\substack{u \in \mathcal{F} \\ \|y-z\| < \delta}} |u(y) - u(z)| < \frac{\varepsilon}{3}. \\
\|x - x_r\| < \delta, \quad \ell, q \geq L &\implies |u_{v_\ell}(x) - u_{v_q}(x)| \leq |u_{v_\ell}(x) - u_{v_\ell}(x_r)| + |u_{v_\ell}(x_r) - u_{v_q}(x_r)| \\
&\quad + |u_{v_q}(x_r) - u_{v_q}(x)| \leq |u_{v_\ell}(x_r) - u_{v_q}(x_r)| + |u_{v_q}(x_r) - u_{v_q}(x)|. \\
|u_{v_\ell}(x) - u_{v_q}(x)| &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \\
K \subset \bigcup_{r=1}^R B_\delta(x_r) &\implies \ell, q \geq L \implies \sup_{x \in K} |u_{v_\ell}(x) - u_{v_q}(x)| < \varepsilon.
\end{aligned}$$

Thus the extracted subsequence is Cauchy in $C^0(K)$ and converges uniformly. $\square$

Corollary 4.1.3 (Compact Hölder embedding). *For $0 < \beta < \alpha < 1$,*

$$C^{2,\alpha}(X) \subseteq C^{2,\beta}(X).$$

Consequently, every $C^{2,\alpha}$ -bounded sequence has a subsequence converging in $C^{2,\beta}(X)$.

Proof. By $0 < \beta < \alpha < 1$, $\|u_v\|_{C^{2,\alpha}(X)} \leq C$, and X covered by a fixed finite atlas,

$$\|D^\gamma(u_v)\|_{C^0} + [D^\gamma(u_v)]_{C^{0,\alpha}} \leq C \quad (0 \leq |\gamma| \leq 2) \xrightarrow{3.3.1} |D^\gamma(u_v)(x) - D^\gamma(u_v)(y)| \leq C \cdot \|x - y\|^\alpha.$$

$$|D^\gamma(u_v)(x) - D^\gamma(u_v)(y)| \leq C \cdot \|x - y\|^\alpha \xrightarrow{4.1.1} \{D^\gamma(u_v)\}_{v=1}^\infty \text{ uniformly bounded and equicontinuous.}$$

$$\{D^\gamma(u_v)\}_{v=1}^\infty \text{ uniformly bounded and equicontinuous} \xrightarrow{4.1.2} D^\gamma(u_{v_\ell}) \xrightarrow{C^0} D^\gamma(u_\infty).$$

$$f_\ell = D^\gamma(u_{v_\ell} - u_\infty), \quad [f_\ell]_{C^{0,\beta}} \leq 2^{1-\beta/\alpha} \cdot \|f_\ell\|_{C^0}^{1-\beta/\alpha} \cdot [f_\ell]_{C^{0,\alpha}}^{\beta/\alpha} \rightarrow 0.$$

The finite atlas and the finitely many derivatives give convergence in $C^{2,\beta}(X)$. $\square$

4.2. Relative metrics and scalar identities.

Definition 4.2.1 (Relative eigenvalues). Let

$$\omega = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k, \quad \eta = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n h_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k.$$

With barred rows and holomorphic columns, set

$$G = [g_{j\bar{k}}]_{\bar{k},j}, \quad H = [h_{j\bar{k}}]_{\bar{k},j}, \quad A = G^{-1} \cdot H = \begin{bmatrix} g^{1\bar{1}} & \cdots & g^{1\bar{n}} \\ \vdots & \ddots & \vdots \\ g^{n\bar{1}} & \cdots & g^{n\bar{n}} \end{bmatrix} \cdot \begin{bmatrix} h_{1\bar{1}} & \cdots & h_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ h_{1\bar{n}} & \cdots & h_{n\bar{n}} \end{bmatrix}.$$

If $\omega, \eta > 0$, then

$$\omega, \eta > 0 \xrightarrow{1.2.1} G, H > 0, \quad B = G^{-1/2} \cdot H \cdot G^{-1/2} > 0, \quad A = G^{-1/2} \cdot B \cdot G^{1/2}, \quad B = Q \cdot \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \cdot Q^*.$$

The positive numbers $\lambda_1, \dots, \lambda_n$ are the eigenvalues of η relative to ω . Equivalently,

$$A \cdot v_a = \lambda_a \cdot v_a \quad (1 \leq a \leq n).$$

Lemma 4.2.2 (Perturbed metric matrix). *For $\phi \in \mathcal{H}_{\omega_0}$,*

$$(4.2.1) \quad G_\phi = G_0 + H_\phi, \quad H_\phi = [\phi_{j\bar{k}}]_{\bar{k},j}, \quad h = G_0^{-1} \cdot G_\phi = I_n + G_0^{-1} \cdot H_\phi.$$

Moreover, $\omega_\phi > 0$ if and only if $G_\phi > 0$.

Proof. By $\phi \in \mathcal{H}_{\omega_0}$, $V = \begin{bmatrix} V^1 \\ \vdots \\ V^n \end{bmatrix} \neq 0$, and $[G_\phi]_{\bar{k},j}$ in barred-row convention,

$$\phi \in \mathcal{H}_{\omega_0} \xrightarrow{3.1.1} G_\phi = \begin{bmatrix} g_{0,1\bar{1}} + \phi_{1\bar{1}} & \cdots & g_{0,n\bar{1}} + \phi_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ g_{0,1\bar{n}} + \phi_{1\bar{n}} & \cdots & g_{0,n\bar{n}} + \phi_{n\bar{n}} \end{bmatrix}, \quad -2 \cdot i \cdot \omega_\phi(V, \bar{V}) = \bar{V}^\top \cdot G_\phi \cdot V.$$

$$\bar{V}^\top \cdot G_\phi \cdot V = -2 \cdot i \cdot \omega_\phi(V, \bar{V}) \xrightarrow{1.2.1} \omega_\phi > 0 \iff \bar{V}^\top \cdot G_\phi \cdot V > 0 \quad (V \neq 0) \iff G_\phi > 0.$$

$$h_p^j = \sum_{k=1}^n g_0^{j\bar{k}} \cdot (g_{0,p\bar{k}} + \phi_{p\bar{k}}) = \delta_p^j + \sum_{k=1}^n g_0^{j\bar{k}} \cdot \phi_{p\bar{k}} \xrightarrow{4.2.1} \text{spec}(h) = \{\lambda_1, \dots, \lambda_n\}.$$

Thus positivity and the relative spectrum are encoded by the displayed matrices. $\square$

Lemma 4.2.3 (Determinant and trace identities). *Let $\lambda_1, \dots, \lambda_n > 0$ be the eigenvalues of ω_ϕ relative to ω_0 . Then*

$$(4.2.2) \quad \frac{\omega_\phi^n}{\omega_0^n} = \frac{\det(G_\phi)}{\det(G_0)} = \det(h) = \prod_{a=1}^n \lambda_a.$$

$$\text{Tr}(h) = \sum_{a=1}^n \lambda_a = n + \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}} \cdot \phi_{j\bar{k}} = n + \Delta_{\omega_0}(\phi).$$

$$\text{Tr}(h^{-1}) = \sum_{a=1}^n \lambda_a^{-1} = n - \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot \phi_{j\bar{k}} = n - \Delta_{\omega_\phi}(\phi).$$

Proof. By $\omega_\phi > 0$, $\lambda_1, \dots, \lambda_n > 0$, and $p \in X$,

$$G_0(p) = I_n, \quad G_\phi(p) = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad G_\phi(p)^{-1} = \begin{bmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{bmatrix}.$$

$$\omega_0, \omega_\phi \xrightarrow{1.4.1} \frac{\omega_\phi^n}{\omega_0^n} = \frac{\det(G_\phi)}{\det(G_0)} = \det(h) = \prod_{a=1}^n \lambda_a.$$

$$\text{Tr}(h) = \text{Tr}(G_0^{-1} \cdot G_\phi) = \sum_{a=1}^n \lambda_a, \quad \text{Tr}(h^{-1}) = \text{Tr}(G_\phi^{-1} \cdot G_0) = \sum_{a=1}^n \lambda_a^{-1}.$$

$$h = I_n + G_0^{-1} \cdot H_\phi \xrightarrow{4.2.1} \text{Tr}(h) = \text{Tr}(I_n + G_0^{-1} \cdot H_\phi) = n + \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}} \cdot \phi_{j\bar{k}} \xrightarrow{3.4.1} \text{Tr}(h) = n + \Delta_{\omega_0}(\phi).$$

$$\text{Tr}(h^{-1}) = \text{Tr}(I_n - G_\phi^{-1} \cdot H_\phi) = n - \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot \phi_{j\bar{k}} \xrightarrow{3.4.1} \text{Tr}(h^{-1}) = n - \Delta_{\omega_\phi}(\phi).$$

Thus determinant and both metric traces are the elementary spectral data. $\square$

Lemma 4.2.4 (Uniform right-hand-side bounds). *Set*

$$m_F = \min_X(F), \quad M_F = \max_X(F), \quad \text{osc}_X(F) = M_F - m_F, \quad V_0 = \int_X \omega_0^n.$$

For $0 \leq t \leq 1$,

$$(4.2.3) \quad -t \cdot M_F \leq c_t \leq -t \cdot m_F, \quad -t \cdot \text{osc}_X(F) \leq t \cdot F + c_t \leq t \cdot \text{osc}_X(F).$$

Hence every solution of the continuity equation satisfies

$$e^{-\text{osc}_X(F)} \leq \frac{\omega_t^n}{\omega_0^n} \leq e^{\text{osc}_X(F)}.$$

Set $H_t = t \cdot F + c_t$. For every integer $r \geq 0$ and $0 < \alpha < 1$,

$$\sup_{0 \leq t \leq 1} \|H_t\|_{C^{r,\alpha}(X)} \leq \text{osc}_X(F) + \|F\|_{C^{r,\alpha}(X)}.$$

Moreover,

$$D_* = \sup_{\substack{0 \leq t \leq 1 \\ x \in X}} \left| \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}}(x) \cdot (H_t)_{j\bar{k}}(x) - S(\omega_0)(x) \right| \leq \left\| \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}} \cdot F_{j\bar{k}} \right\|_{C^0(X)} + \|S(\omega_0)\|_{C^0(X)}.$$

Proof. By $0 \leq t \leq 1$, $m_F \leq F \leq M_F$, and $e^{c_t} \cdot \int_X e^{t \cdot F} \cdot \omega_0^n = V_0$,

$$\omega_t^n = e^{t \cdot F + c_t} \cdot \omega_0^n \xrightarrow{3.8.1} e^{c_t} \cdot \int_X e^{t \cdot F} \cdot \omega_0^n = V_0, \quad e^{t \cdot m_F} \cdot V_0 \leq \int_X e^{t \cdot F} \cdot \omega_0^n \leq e^{t \cdot M_F} \cdot V_0 \xrightarrow{3.8.2} -t \cdot M_F \leq c_t \leq -t \cdot m_F.$$

$$t \cdot m_F - t \cdot M_F \leq t \cdot F + c_t \leq t \cdot M_F - t \cdot m_F \xrightarrow{3.8.1} e^{-\text{osc}_X(F)} \leq \frac{\omega_t^n}{\omega_0^n} \leq e^{\text{osc}_X(F)}.$$

$$(H_t)_{j\bar{k}} = t \cdot F_{j\bar{k}}, \quad \partial^\gamma(H_t) = t \cdot \partial^\gamma(F) \quad (1 \leq |\gamma| \leq r) \implies \sup_{0 \leq t \leq 1} \|H_t\|_{C^{r,\alpha}} \leq \text{osc}_X(F) + \|F\|_{C^{r,\alpha}}.$$

$$\left| \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}} \cdot (H_t)_{j\bar{k}} - S(\omega_0) \right| \leq \left| \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}} \cdot F_{j\bar{k}} \right| + |S(\omega_0)| \implies D_* \leq \left\| \sum_{j=1}^n \sum_{k=1}^n g_0^{j\bar{k}} \cdot F_{j\bar{k}} \right\|_{C^0} + \|S(\omega_0)\|_{C^0}.$$

Thus the density and all spatial derivatives of its logarithm are uniformly bounded. $\square$

Lemma 4.2.5 (Trace comparison). *For $n \geq 2$,*

$$(4.2.4) \quad \text{Tr}(h) \leq \frac{1}{(n-1)!} \cdot (\text{Tr}(h^{-1}))^{n-1} \cdot \frac{\omega_\phi^n}{\omega_0^n}.$$

For $n = 1$ the corresponding identity is immediate.

Proof. By $n \geq 2$, $\lambda_1, \dots, \lambda_n > 0$, and $\omega_\phi^n / \omega_0^n = \prod_{a=1}^n \lambda_a$,

$$\left(\sum_{a=1}^n \lambda_a^{-1} \right)^{n-1} = \sum_{(a_1, \dots, a_{n-1}) \in \{1, \dots, n\}^{n-1}} \prod_{q=1}^{n-1} \lambda_{a_q}^{-1} \geq (n-1)! \cdot \sum_{a=1}^n \prod_{\substack{1 \leq b \leq n \\ b \neq a}} \lambda_b^{-1}.$$

$$\frac{1}{(n-1)!} \cdot \left(\sum_{a=1}^n \lambda_a^{-1} \right)^{n-1} \cdot \prod_{a=1}^n \lambda_a \geq \sum_{a=1}^n \lambda_a \xrightarrow{4.2.3} \frac{1}{(n-1)!} \cdot (\text{Tr}(h^{-1}))^{n-1} \cdot \frac{\omega_\phi^n}{\omega_0^n} \geq \text{Tr}(h).$$

This is the standard spectral trace comparison. $\square$

Corollary 4.2.6 (Trace–determinant eigenvalue bounds). *Suppose*

$$\text{Tr}(h) \leq C_{\text{tr}}, \quad \frac{\omega_\phi^n}{\omega_0^n} = e^H, \quad \|H\|_{C^0(X)} \leq C_{\text{vol}}.$$

Then every relative eigenvalue satisfies

$$(4.2.5) \quad e^{-C_{\text{vol}}} \cdot C_{\text{tr}}^{1-n} \leq \lambda_a \leq C_{\text{tr}} \quad (1 \leq a \leq n).$$

Proof. By $\sum_{a=1}^n \lambda_a \leq C_{\text{tr}}$, $\prod_{a=1}^n \lambda_a = e^H$, and $|H| \leq C_{\text{vol}}$,

$$\text{Tr}(h) \leq C_{\text{tr}} \xrightarrow{4.2.3} \sum_{b=1}^n \lambda_b \leq C_{\text{tr}} \implies 0 < \lambda_a \leq \sum_{b=1}^n \lambda_b \leq C_{\text{tr}}.$$

$$\frac{\omega_\phi^n}{\omega_0^n} = e^H \xrightarrow{4.2.2} \prod_{b=1}^n \lambda_b = e^H.$$

$$|H| \leq C_{\text{vol}} \implies e^{-C_{\text{vol}}} \leq e^H = \lambda_a \cdot \prod_{\substack{1 \leq b \leq n \\ b \neq a}} \lambda_b \leq \lambda_a \cdot C_{\text{tr}}^{n-1}.$$

Rearrangement gives the claimed lower bound. $\square$

Theorem 4.2.7 (Explicit eigenvalue bounds from scalar data). *Let $\phi_j \in \mathcal{H}_{\omega_0}$ be normalized solutions, and set*

$$\omega_j = \omega_{\phi_j}, \quad G_j = G_0 + [\partial_p(\partial_{\bar{q}}(\phi_j))]_{\bar{q},p}, \quad h_j = G_0^{-1} \cdot G_j.$$

Suppose

$$\omega_j^n = e^{F_{t_j}} \cdot \omega_0^n, \quad \int_X \phi_j \cdot \omega_0^n = 0, \quad \|\phi_j\|_{C^0(X)} \leq C_0, \quad \text{Tr}(h_j) \leq C_2, \quad \|F_{t_j}\|_{C^0(X)} \leq C_F.$$

Then

$$(4.2.6) \quad e^{-C_F} \cdot C_2^{1-n} \cdot \omega_0 \leq \omega_j \leq C_2 \cdot \omega_0.$$

Proof. By $\text{Tr}(h_j) \leq C_2$, $\det(h_j) = e^{F_{t_j}}$, and $\|F_{t_j}\|_{C^0(X)} \leq C_F$,

$$G_j = G_0 + [\partial_p(\partial_{\bar{q}}(\phi_j))]_{\bar{q},p}, \quad h_j = G_0^{-1} \cdot G_j \xrightarrow{4.2.1} \text{spec}(h_j) = \{\lambda_1^{(j)}, \dots, \lambda_n^{(j)}\}.$$

$$\text{Tr}(h_j) \leq C_2 \xrightarrow{4.2.3} \sum_{a=1}^n \lambda_a^{(j)} = \text{Tr}(h_j) \leq C_2.$$

$$\det(h_j) = e^{F_{t_j}} \xrightarrow{4.2.3} \prod_{a=1}^n \lambda_a^{(j)} = \det(h_j) = e^{F_{t_j}} \geq e^{-C_F}.$$

$$0 < \lambda_a^{(j)} \leq C_2, \quad e^{-C_F} \leq \lambda_a^{(j)} \cdot \prod_{\substack{1 \leq b \leq n \\ b \neq a}} \lambda_b^{(j)} \leq \lambda_a^{(j)} \cdot C_2^{n-1}.$$

$$e^{-C_F} \cdot C_2^{1-n} \leq \lambda_a^{(j)} \leq C_2 \xrightarrow{4.2.1} e^{-C_F} \cdot C_2^{1-n} \cdot \omega_0 \leq \omega_j \leq C_2 \cdot \omega_0.$$

□

4.3. Log-volume linearization and uniform ellipticity.

Proposition 4.3.1 (Log-volume linearization). *For*

$$\mathcal{M}(\phi) = \log\left(\frac{\omega_\phi^n}{\omega_0^n}\right), \quad u \in C^\infty(X, \mathbb{R}),$$

one has

$$(4.3.1) \quad D\mathcal{M}_\phi[u] = \text{Tr}(G_\phi^{-1} \cdot H_u) = \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot u_{j\bar{k}}.$$

Proof. By $\phi_s = \phi + s \cdot u$, $H_u = [u_{j\bar{k}}]_{\bar{k},j}$, and $|s| \ll 1 \implies \omega_{\phi_s} > 0$,

$$\frac{\omega_{\phi_s}^n}{\omega_0^n} = \frac{\det(G_{\phi_s})}{\det(G_0)} \xrightarrow{4.2.3} \mathcal{M}(\phi_s) = \log(\det(G_{\phi_s})) - \log(\det(G_0)).$$

$$\frac{d}{ds} \Big|_{s=0} \mathcal{M}(\phi_s) = \frac{\omega_0^n}{\omega_\phi^n} \cdot \frac{d}{ds} \Big|_{s=0} \frac{\omega_{\phi_s}^n}{\omega_0^n} \xrightarrow{3.6.3} \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot u_{j\bar{k}}.$$

$$\sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot u_{j\bar{k}} \xrightarrow{3.4.1} \Delta_{\omega_\phi}(u) = \text{Tr}(G_\phi^{-1} \cdot H_u) \xrightarrow{1.2.2} \text{Tr}\left(\begin{bmatrix} g_\phi^{1\bar{1}} & \cdots & g_\phi^{1\bar{n}} \\ \vdots & \ddots & \vdots \\ g_\phi^{n\bar{1}} & \cdots & g_\phi^{n\bar{n}} \end{bmatrix} \cdot \begin{bmatrix} u_{1\bar{1}} & \cdots & u_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ u_{1\bar{n}} & \cdots & u_{n\bar{n}} \end{bmatrix} \right).$$

Thus the logarithmic linearization is the complex Laplacian.

□

Proposition 4.3.2 (Explicit uniform ellipticity). *Suppose*

$$\mathrm{Tr}(h) \leq C_{\mathrm{tr}}, \quad \frac{\omega_\phi^n}{\omega_0^n} = e^H, \quad \|H\|_{C^0(X)} \leq C_{\mathrm{vol}}.$$

Set $\tilde{h} = G_0^{1/2} \cdot h \cdot G_0^{-1/2} = G_0^{-1/2} \cdot G_\phi \cdot G_0^{-1/2}$. Then

$$(4.3.2) \quad e^{-C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{1-n} \cdot \omega_0 \leq \omega_\phi \leq C_{\mathrm{tr}} \cdot \omega_0, \quad e^{-C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{1-n} \cdot I_n \leq \tilde{h} \leq C_{\mathrm{tr}} \cdot I_n.$$

Moreover,

$$C_{\mathrm{tr}}^{-1} \cdot G_0^{-1} \leq G_\phi^{-1} \leq e^{C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{n-1} \cdot G_0^{-1},$$

and, for $\xi = [\xi_1 \ \cdots \ \xi_n]^\top \neq 0$,

$$C_{\mathrm{tr}}^{-1} \cdot \bar{\xi}^\top \cdot G_0^{-1} \cdot \xi \leq \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot \xi_j \cdot \bar{\xi}_k \leq e^{C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{n-1} \cdot \bar{\xi}^\top \cdot G_0^{-1} \cdot \xi.$$

Thus the complete linearized elliptic equation is

$$D\mathcal{M}_\phi[u] = \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot \frac{\partial^2 u}{\partial z^j \partial \bar{z}^k}.$$

Proof. By $\mathrm{Tr}(h) \leq C_{\mathrm{tr}}$, $\omega_\phi^n / \omega_0^n = e^H$, and $\|H\|_{C^0(X)} \leq C_{\mathrm{vol}}$,

$$\left(\mathrm{Tr}(h) \leq C_{\mathrm{tr}}, \quad \frac{\omega_\phi^n}{\omega_0^n} = e^H, \quad \|H\|_{C^0(X)} \leq C_{\mathrm{vol}} \right) \xrightarrow{4.2.6} e^{-C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{1-n} \leq \lambda_a \leq C_{\mathrm{tr}} \quad (1 \leq a \leq n).$$

$$e^{-C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{1-n} \leq \lambda_a \leq C_{\mathrm{tr}} \quad (1 \leq a \leq n) \xrightarrow{4.2.1} e^{-C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{1-n} \cdot I_n \leq \tilde{h} \leq C_{\mathrm{tr}} \cdot I_n.$$

$$G_\phi = G_0^{1/2} \cdot \tilde{h} \cdot G_0^{1/2} \xrightarrow{4.2.1} e^{-C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{1-n} \cdot G_0 \leq G_\phi \leq C_{\mathrm{tr}} \cdot G_0 \xrightarrow{1.2.2} C_{\mathrm{tr}}^{-1} \cdot G_0^{-1} \leq G_\phi^{-1} \leq e^{C_{\mathrm{vol}}} \cdot C_{\mathrm{tr}}^{n-1} \cdot G_0^{-1}.$$

$$\bar{\xi}^\top \cdot G_\phi^{-1} \cdot \xi = \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot \xi_j \cdot \bar{\xi}_k.$$

$$(\phi \in \mathcal{H}_{\omega_0}, \quad u \in C^\infty(X, \mathbb{R})) \xrightarrow{4.3.1} D\mathcal{M}_\phi[u] = \sum_{j=1}^n \sum_{k=1}^n g_\phi^{j\bar{k}} \cdot \frac{\partial^2 u}{\partial z^j \partial \bar{z}^k}.$$

Thus the displayed lower and upper coefficients are the ellipticity constants. $\square$

4.4. Computational eigenvalue experiments.

Example 4.4.1 (Fixed determinant without uniform ellipticity). At one point take $G_0 = I_2$. For $N \geq 1$ set

$$Q_2 = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = [v_+ \ v_-], \quad D_N = \begin{bmatrix} N & 0 \\ 0 & N^{-1} \end{bmatrix}, \quad v_+ = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_- = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

$$Q_2^* \cdot Q_2 = \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot (-1) + 1 \cdot 1 \\ (-1) \cdot 1 + 1 \cdot 1 & (-1) \cdot (-1) + 1 \cdot 1 \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I_2.$$

The full relative metric is

$$\begin{aligned} G_N = h_N &= \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} N & 0 \\ 0 & N^{-1} \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} N & -N^{-1} \\ N & N^{-1} \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\ &= \frac{1}{2} \cdot \begin{bmatrix} N \cdot 1 + (-N^{-1}) \cdot (-1) & N \cdot 1 + (-N^{-1}) \cdot 1 \\ N \cdot 1 + N^{-1} \cdot (-1) & N \cdot 1 + N^{-1} \cdot 1 \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} N + N^{-1} & N - N^{-1} \\ N - N^{-1} & N + N^{-1} \end{bmatrix}. \\ Q_2^* \cdot Q_2 &= I_2, \quad h_N = Q_2 \cdot D_N \cdot Q_2^* \xrightarrow{4.2.1} h_N \cdot v_+ = N \cdot v_+, \quad h_N \cdot v_- = N^{-1} \cdot v_-. \end{aligned}$$

Let ω_N be the associated Hermitian form. For $N = 100$, all eigenvectors are checked simultaneously by

$$\begin{aligned} & \begin{bmatrix} 50.005 & 49.995 \\ 49.995 & 50.005 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 50.005 \cdot 1 + 49.995 \cdot 1 & 50.005 \cdot (-1) + 49.995 \cdot 1 \\ 49.995 \cdot 1 + 50.005 \cdot 1 & 49.995 \cdot (-1) + 50.005 \cdot 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 100 & -0.01 \\ 100 & 0.01 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 100 & 0 \\ 0 & 0.01 \end{bmatrix}. \end{aligned}$$

The deterministic sweep $N_r = 10^r$ for $0 \leq r \leq 6$ gives

$$\text{spec}(h_{N_r}) = \{10^r, 10^{-r}\} \xrightarrow{4.2.1} (\lambda_{\min}(h_{N_r}), \lambda_{\max}(h_{N_r}), \kappa_2(h_{N_r})) = (10^{-r}, 10^r, 10^{2r}).$$

$$\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}, \quad \det(h_{N_r}) = 1 \xrightarrow{4.2.3} \frac{\omega_{N_r}^2}{\omega_0^2} = 1.$$

Hence fixed volume does not prevent the ellipticity ratio from diverging.

Remark 4.4.2. The Monge–Ampère equation controls $\det(h) = \prod_{a=1}^n \lambda_a$, but a fixed positive determinant does not bound the individual eigenvalues.

$$\omega_\phi > 0 \xrightarrow{4.2.3} \det(h) = \prod_{a=1}^n \lambda_a \xrightarrow{4.4.1} (\det(h_N) = 1, \quad \lambda_{\min}(h_N) = N^{-1}, \quad \lambda_{\max}(h_N) = N).$$

$$\det(h) \geq e^{-C_{\text{vol}}}, \quad \text{Tr}(h) \leq C_{\text{tr}} \xrightarrow{4.2.6} e^{-C_{\text{vol}}} \cdot C_{\text{tr}}^{1-n} \leq \lambda_a \leq C_{\text{tr}}.$$

Thus the determinant and trace together preserve strict positivity in a limit.

Example 4.4.3 (Determinant and trace control). At one point take $G_0 = I_3$ and set

$$\begin{aligned} Q_3 &= \frac{1}{3} \cdot \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix} = [v_1 \quad v_2 \quad v_3], \quad D_s = \begin{bmatrix} s & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & s^{-1} \end{bmatrix}. \\ Q_3^* \cdot Q_3 &= \frac{1}{9} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix} \\ &= \frac{1}{9} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{9} \cdot \begin{bmatrix} 2 \cdot 2 + 2 \cdot 2 + 1 \cdot 1 \\ (-2) \cdot 2 + 1 \cdot 2 + 2 \cdot 1 \\ 1 \cdot 2 + (-2) \cdot 2 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \\ &= \frac{1}{9} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} = \frac{1}{9} \cdot \begin{bmatrix} 2 \cdot (-2) + 2 \cdot 1 + 1 \cdot 2 \\ (-2) \cdot (-2) + 1 \cdot 1 + 2 \cdot 2 \\ 1 \cdot (-2) + (-2) \cdot 1 + 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \\ &= \frac{1}{9} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \frac{1}{9} \cdot \begin{bmatrix} 2 \cdot 1 + 2 \cdot (-2) + 1 \cdot 2 \\ (-2) \cdot 1 + 1 \cdot (-2) + 2 \cdot 2 \\ 1 \cdot 1 + (-2) \cdot (-2) + 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \\ Q_3^* \cdot Q_3 &= \frac{1}{9} \cdot \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = I_3. \\ v_1 &= \frac{1}{3} \cdot \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \frac{1}{3} \cdot \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}, \quad v_3 = \frac{1}{3} \cdot \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}. \end{aligned}$$

$$G_s = h_s = \frac{1}{3} \cdot \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} s & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & s^{-1} \end{bmatrix} \cdot \frac{1}{3} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}.$$

$$\mathcal{Q}_3^* \cdot \mathcal{Q}_3 = I_3, \quad h_s = \mathcal{Q}_3 \cdot D_s \cdot \mathcal{Q}_3^* \xrightarrow{4.2.1} \text{spec}(h_s) = \{s, 1, s^{-1}\}.$$

$$\text{spec}(h_s) = \{s, 1, s^{-1}\} \xrightarrow{4.2.3} \det(h_s) = s \cdot 1 \cdot s^{-1} = 1, \quad \text{Tr}(h_s) = s + 1 + s^{-1}.$$

For the deterministic grid $s_\ell = (\ell + 4)/10$ with $0 \leq \ell \leq 21$, direct evaluation gives

$$\det(h_{s_\ell}) = 1, \quad \max_{0 \leq \ell \leq 21} \text{Tr}(h_{s_\ell}) = \frac{39}{10}, \quad \min_{\substack{0 \leq \ell \leq 21 \\ 1 \leq a \leq 3}} \lambda_a = \frac{2}{5}, \quad \max_{\substack{0 \leq \ell \leq 21 \\ 1 \leq a \leq 3}} \lambda_a = \frac{5}{2}, \quad \max_{0 \leq \ell \leq 21} \kappa_2(h_{s_\ell}) = \frac{25}{4}.$$

The scalar estimate alone gives

$$\det(h_{s_\ell}) = 1, \quad \text{Tr}(h_{s_\ell}) \leq \frac{39}{10} \xrightarrow{4.2.6} \frac{100}{1521} \cdot I_3 \leq h_{s_\ell} \leq \frac{39}{10} \cdot I_3 \implies \frac{10}{39} \cdot I_3 \leq h_{s_\ell}^{-1} \leq \frac{1521}{100} \cdot I_3.$$

The explicit endpoint bounds give

$$\frac{100}{1521} \leq \lambda_{\min}(h_{s_\ell}) \leq \lambda_{\max}(h_{s_\ell}) \leq \frac{39}{10} \xrightarrow{4.3.2} \kappa_2(h_{s_\ell}) \leq \frac{39}{10} \cdot \frac{1521}{100} = \frac{59319}{1000}.$$

At $s = 2$ the complete multiplication is

$$\begin{aligned} h_2 &= \frac{1}{3} \cdot \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \cdot \frac{1}{3} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} = \frac{1}{9} \cdot \begin{bmatrix} 4 & -2 & \frac{1}{2} \\ 4 & 1 & -1 \\ 2 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \\ &= \frac{1}{9} \cdot \begin{bmatrix} 4 \cdot 2 + (-2) \cdot (-2) + \frac{1}{2} \cdot 1 & 4 \cdot 2 + (-2) \cdot 1 + \frac{1}{2} \cdot (-2) & 4 \cdot 1 + (-2) \cdot 2 + \frac{1}{2} \cdot 2 \\ 4 \cdot 2 + 1 \cdot (-2) + (-1) \cdot 1 & 4 \cdot 2 + 1 \cdot 1 + (-1) \cdot (-2) & 4 \cdot 1 + 1 \cdot 2 + (-1) \cdot 2 \\ 2 \cdot 2 + 2 \cdot (-2) + 1 \cdot 1 & 2 \cdot 2 + 2 \cdot 1 + 1 \cdot (-2) & 2 \cdot 1 + 2 \cdot 2 + 1 \cdot 2 \end{bmatrix} = \frac{1}{18} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix}. \end{aligned}$$

Each bold eigenvector is verified directly.

$$\begin{aligned} \frac{1}{18} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \frac{1}{3} \cdot \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} &= \frac{1}{54} \cdot \begin{bmatrix} 25 \cdot 2 + 10 \cdot 2 + 2 \cdot 1 \\ 10 \cdot 2 + 22 \cdot 2 + 8 \cdot 1 \\ 2 \cdot 2 + 8 \cdot 2 + 16 \cdot 1 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix} = 2 \cdot v_1. \\ \frac{1}{18} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \frac{1}{3} \cdot \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} &= \frac{1}{54} \cdot \begin{bmatrix} 25 \cdot (-2) + 10 \cdot 1 + 2 \cdot 2 \\ 10 \cdot (-2) + 22 \cdot 1 + 8 \cdot 2 \\ 2 \cdot (-2) + 8 \cdot 1 + 16 \cdot 2 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} = v_2. \\ \frac{1}{18} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \frac{1}{3} \cdot \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} &= \frac{1}{54} \cdot \begin{bmatrix} 25 \cdot 1 + 10 \cdot (-2) + 2 \cdot 2 \\ 10 \cdot 1 + 22 \cdot (-2) + 8 \cdot 2 \\ 2 \cdot 1 + 8 \cdot (-2) + 16 \cdot 2 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} \frac{1}{2} \\ -1 \\ 1 \end{bmatrix} = \frac{1}{2} \cdot v_3. \\ \left(h_2 \cdot v_1 = 2 \cdot v_1, \quad h_2 \cdot v_2 = v_2, \quad h_2 \cdot v_3 = \frac{1}{2} \cdot v_3 \right) &\xrightarrow{4.2.1} \text{spec}(h_2) = \left\{ 2, 1, \frac{1}{2} \right\}. \\ \text{spec}(h_2) = \left\{ 2, 1, \frac{1}{2} \right\} &\xrightarrow{4.2.3} \det(h_2) = 2 \cdot 1 \cdot \frac{1}{2} = 1, \quad \text{Tr}(h_2) = 2 + 1 + \frac{1}{2} = \frac{7}{2}. \end{aligned}$$

Finally,

$$\begin{aligned} h_2^{-1} &= \frac{1}{18} \cdot \begin{bmatrix} 16 & -8 & 2 \\ -8 & 22 & -10 \\ 2 & -10 & 25 \end{bmatrix}. \\ h_2 \cdot h_2^{-1} &= \frac{1}{18} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \frac{1}{18} \cdot \begin{bmatrix} 16 & -8 & 2 \\ -8 & 22 & -10 \\ 2 & -10 & 25 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
\frac{1}{324} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \begin{bmatrix} 16 \\ -8 \\ 2 \end{bmatrix} &= \frac{1}{324} \cdot \begin{bmatrix} 25 \cdot 16 + 10 \cdot (-8) + 2 \cdot 2 \\ 10 \cdot 16 + 22 \cdot (-8) + 8 \cdot 2 \\ 2 \cdot 16 + 8 \cdot (-8) + 16 \cdot 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \\
\frac{1}{324} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \begin{bmatrix} -8 \\ 22 \\ -10 \end{bmatrix} &= \frac{1}{324} \cdot \begin{bmatrix} 25 \cdot (-8) + 10 \cdot 22 + 2 \cdot (-10) \\ 10 \cdot (-8) + 22 \cdot 22 + 8 \cdot (-10) \\ 2 \cdot (-8) + 8 \cdot 22 + 16 \cdot (-10) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \\
\frac{1}{324} \cdot \begin{bmatrix} 25 & 10 & 2 \\ 10 & 22 & 8 \\ 2 & 8 & 16 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -10 \\ 25 \end{bmatrix} &= \frac{1}{324} \cdot \begin{bmatrix} 25 \cdot 2 + 10 \cdot (-10) + 2 \cdot 25 \\ 10 \cdot 2 + 22 \cdot (-10) + 8 \cdot 25 \\ 2 \cdot 2 + 8 \cdot (-10) + 16 \cdot 25 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \\
h_2 \cdot h_2^{-1} &= \frac{1}{324} \cdot \begin{bmatrix} 324 & 0 & 0 \\ 0 & 324 & 0 \\ 0 & 0 & 324 \end{bmatrix} = I_3.
\end{aligned}$$

Thus the trace and determinant controls produce fixed ellipticity constants.

4.5. Differentiation along the continuity path.

Lemma 4.5.1 (Differentiated continuity equation). *Let $t \mapsto \phi_t$ be a smooth branch of normalized solutions and write $\omega_t = \omega_{\phi_t}$ and $G_t^{-1} = [g_t^{j\bar{k}}]_{\bar{k},j}$. Then*

$$(4.5.1) \quad \sum_{j=1}^n \sum_{k=1}^n g_t^{j\bar{k}} \cdot \frac{\partial^2 \dot{\phi}_t}{\partial z^j \partial \bar{z}^k} = F + \dot{c}_t, \quad \dot{c}_t = -\frac{\int_X F \cdot \omega_t^n}{\int_X \omega_t^n}, \quad \int_X \dot{\phi}_t \cdot \omega_0^n = 0.$$

Proof. By $\omega_t^n = e^{t \cdot F + c_t} \cdot \omega_0^n$, $\int_X \phi_t \cdot \omega_0^n = 0$, and $G_t^{-1} = [g_t^{j\bar{k}}]_{\bar{k},j}$,

$$\begin{aligned}
\omega_t^n &= e^{t \cdot F + c_t} \cdot \omega_0^n \xrightarrow{3.8.1} \log \left(\frac{\omega_t^n}{\omega_0^n} \right) = t \cdot F + c_t \implies \frac{d}{dt} \log \left(\frac{\omega_t^n}{\omega_0^n} \right) = F + \dot{c}_t. \\
\frac{d}{dt} \log \left(\frac{\omega_t^n}{\omega_0^n} \right) &\xrightarrow{4.3.1} \sum_{j=1}^n \sum_{k=1}^n g_t^{j\bar{k}} \cdot \frac{\partial^2 \dot{\phi}_t}{\partial z^j \partial \bar{z}^k} = F + \dot{c}_t. \\
\int_X \phi_t \cdot \omega_0^n &= 0 \xrightarrow{3.8.1} \int_X \dot{\phi}_t \cdot \omega_0^n = 0. \\
\left(c_t = \log(V_0) - \log \left(\int_X e^{t \cdot F} \cdot \omega_0^n \right) \right) &\xrightarrow{3.8.2} \dot{c}_t = -\frac{\int_X F \cdot e^{t \cdot F} \cdot \omega_0^n}{\int_X e^{t \cdot F} \cdot \omega_0^n} = -\frac{\int_X F \cdot \omega_t^n}{\int_X \omega_t^n}. \\
\int_X (F + \dot{c}_t) \cdot \omega_t^n &= 0 \xrightarrow{3.5.5} \exists u_t \in C^\infty(X, \mathbb{R}) \quad \sum_{j=1}^n \sum_{k=1}^n g_t^{j\bar{k}} \cdot \frac{\partial^2 u_t}{\partial z^j \partial \bar{z}^k} = F + \dot{c}_t, \quad \int_X u_t \cdot \omega_t^n = 0. \\
\hat{u}_t &= u_t - \frac{\int_X u_t \cdot \omega_0^n}{V_0} \xrightarrow{3.4.1} \Delta_{\omega_t}(\hat{u}_t) = \Delta_{\omega_t}(u_t) = F + \dot{c}_t, \quad \int_X \hat{u}_t \cdot \omega_0^n = 0. \\
\Delta_{\omega_t}(\hat{u}_t) &= F + \dot{c}_t \xrightarrow{3.4.1} \sum_{j=1}^n \sum_{k=1}^n g_t^{j\bar{k}} \cdot \frac{\partial^2 \hat{u}_t}{\partial z^j \partial \bar{z}^k} = F + \dot{c}_t, \quad \int_X \hat{u}_t \cdot \omega_0^n = 0. \\
\sum_{j=1}^n \sum_{k=1}^n g_t^{j\bar{k}} \cdot \frac{\partial^2 (\dot{\phi}_t - \hat{u}_t)}{\partial z^j \partial \bar{z}^k} &= 0 \xrightarrow{3.5.2} \dot{\phi}_t - \hat{u}_t \in \mathbb{R}. \\
\int_X (\dot{\phi}_t - \hat{u}_t) \cdot \omega_0^n &= 0 \implies \dot{\phi}_t = \hat{u}_t.
\end{aligned}$$

Thus the tangent to the solution branch is the unique ω_0^n -normalized solution of the Poisson equation. $\square$

4.6. Conditional closedness.

Theorem 4.6.1 (Closedness under uniform $C^{2,\alpha}$ control). *Fix $0 < \beta < \alpha < 1$. Let $t_v \in I$ satisfy $t_v \rightarrow t_\infty \in [0, 1]$, and let ϕ_v be the normalized solution at t_v . If*

$$\sup_{v \geq 1} \|\phi_v\|_{C^{2,\alpha}(X)} \leq C,$$

then $t_\infty \in I$.

Proof. By $t_v \rightarrow t_\infty$, $\sup_{v \geq 1} \|\phi_v\|_{C^{2,\alpha}(X)} \leq C$, and $G_v = G_0 + H_{\phi_v}$,

$$\sup_{v \geq 1} \|\phi_v\|_{C^{2,\alpha}} \leq C \xrightarrow{4.1.3} \phi_{v_\ell} \xrightarrow{C^{2,\beta}} \phi_\infty, \quad G_v = G_0 + H_{\phi_v} \xrightarrow{4.2.2} G_{v_\ell} \xrightarrow{C^{0,\beta}} G_\infty = G_0 + H_{\phi_\infty}.$$

$$\omega_{\phi_{v_\ell}} > 0 \xrightarrow{4.2.2} G_{v_\ell} > 0 \implies G_\infty \geq 0.$$

$$t_{v_\ell} \in I \xrightarrow{3.8.1} \omega_{\phi_{v_\ell}}^n = e^{t_{v_\ell} \cdot F + c_{t_{v_\ell}}} \cdot \omega_0^n.$$

$$\omega_{\phi_{v_\ell}}^n = e^{t_{v_\ell} \cdot F + c_{t_{v_\ell}}} \cdot \omega_0^n \xrightarrow{1.4.1} \frac{\det(G_{v_\ell})}{\det(G_0)} = e^{t_{v_\ell} \cdot F + c_{t_{v_\ell}}} \implies \det(G_{v_\ell}) = e^{t_{v_\ell} \cdot F + c_{t_{v_\ell}}} \cdot \det(G_0).$$

$$t_{v_\ell} \rightarrow t_\infty \xrightarrow{3.8.2} c_{t_{v_\ell}} \rightarrow c_{t_\infty}, \quad G_{v_\ell} \rightarrow G_\infty \implies \det(G_\infty) = e^{t_\infty \cdot F + c_{t_\infty}} \cdot \det(G_0) > 0.$$

$$G_\infty \geq 0, \quad \det(G_\infty) > 0 \implies G_\infty > 0, \quad \omega_{\phi_\infty}^n = e^{t_\infty \cdot F + c_{t_\infty}} \cdot \omega_0^n, \quad \int_X \phi_{v_\ell} \cdot \omega_0^n = 0 \xrightarrow{3.8.1} \int_X \phi_\infty \cdot \omega_0^n = 0.$$

$$G_\infty > 0, \quad X \text{ compact}, \quad m_\infty = \min_{\substack{x \in X \\ 1 \leq a \leq n}} \lambda_a(G_0^{-1} \cdot G_\infty) > 0, \quad M_\infty = \max_{\substack{x \in X \\ 1 \leq a \leq n}} \lambda_a(G_0^{-1} \cdot G_\infty) < \infty.$$

$$0 < m_\infty \leq M_\infty \xrightarrow{4.2.1} m_\infty \cdot G_0 \leq G_\infty \leq M_\infty \cdot G_0, \quad \|\phi_\infty\|_{C^0} < \infty.$$

$$\left(\phi_\infty \in C^{2,\beta}(X), \quad m_\infty \cdot \omega_0 \leq \omega_{\phi_\infty} \leq M_\infty \cdot \omega_0, \quad \omega_{\phi_\infty}^n = e^{t_\infty \cdot F + c_{t_\infty}} \cdot \omega_0^n \right) \xrightarrow{5.8.1} \phi_\infty \in C^\infty(X) \xrightarrow{3.8.1} t_\infty \in I.$$

Thus every limit point of I belongs to I once the uniform estimate is known. $\square$

5. AUBIN–YAU ESTIMATES AND COMPLETION OF THE CONTINUITY METHOD

This section supplies the estimates required by the compactness argument. Fix a compact connected Kähler manifold $(X^n, \hat{\omega})$ and a normalized solution

$$\omega_\phi = \hat{\omega} + \frac{i}{2} \cdot \partial \bar{\partial} \phi > 0, \quad \omega_\phi^n = e^F \cdot \hat{\omega}^n, \quad \int_X \phi \cdot \hat{\omega}^n = 0.$$

The analytic chain is

$$\omega_\phi^n = e^F \cdot \hat{\omega}^n, \quad \int_X \phi \cdot \hat{\omega}^n = 0 \xrightarrow{5.3.1} \|\phi\|_{C^0} \leq C_0 \xrightarrow{5.6.1} \text{Tr}(h) \leq C_2.$$

$$\text{Tr}(h) \leq C_2 \xrightarrow{5.6.2} C^{-1} \cdot \hat{\omega} \leq \omega_\phi \leq C \cdot \hat{\omega} \xrightarrow{5.8.1} \|\phi\|_{C^{k,\alpha}} \leq C_{k,\alpha}.$$

5.1. Continuity data.

Definition 5.1.1 (Continuity notation). The continuity set I and its normalized solutions are the objects satisfying

$$t \in I \xrightarrow{3.8.1} \left(\omega_t = \hat{\omega} + \frac{i}{2} \cdot \partial \bar{\partial} \phi_t > 0, \quad \omega_t^n = e^{t \cdot F + c_t} \cdot \hat{\omega}^n, \quad \int_X \phi_t \cdot \hat{\omega}^n = 0 \right).$$

$$(t \in I, \quad H_t = t \cdot F + c_t) \xrightarrow{3.8.1} \omega_t^n = e^{H_t} \cdot \hat{\omega}^n.$$

Fact 5.1.2 (Uniform continuity density). Set $H_t = t \cdot F + c_t$. For $0 \leq t \leq 1$,

$$-\operatorname{osc}_X(F) \leq H_t \leq \operatorname{osc}_X(F), \quad e^{-\operatorname{osc}_X(F)} \leq e^{H_t} \leq e^{\operatorname{osc}_X(F)}.$$

For every integer $r \geq 0$ and $0 < \alpha < 1$,

$$\sup_{0 \leq t \leq 1} \|H_t\|_{C^{r,\alpha}(X)} \leq \operatorname{osc}_X(F) + \|F\|_{C^{r,\alpha}(X)},$$

$$D_* = \sup_{\substack{0 \leq t \leq 1 \\ x \in X}} \left| \sum_{j,k=1}^n \hat{g}^{j\bar{k}}(x) \cdot (H_t)_{j\bar{k}}(x) - \hat{S}(x) \right| \leq \left\| \sum_{j,k=1}^n \hat{g}^{j\bar{k}} \cdot F_{j\bar{k}} \right\|_{C^0(X)} + \|\hat{S}\|_{C^0(X)}.$$

Proof. By $0 \leq t \leq 1$, $m_F = \min_X(F)$, $M_F = \max_X(F)$, $V = \int_X \hat{\omega}^n$, and c_t normalized, apply the density estimates.

$$-\log\left(\frac{1}{V} \cdot \int_X e^{t \cdot F} \cdot \hat{\omega}^n\right) = c_t \xrightarrow{3.8.2} -t \cdot M_F \leq c_t \leq -t \cdot m_F.$$

$$-t \cdot M_F \leq c_t \leq -t \cdot m_F, \quad -t \cdot (M_F - m_F) \leq t \cdot F + c_t \leq t \cdot (M_F - m_F) \xrightarrow{4.2.4} e^{-\operatorname{osc}_X(F)} \leq e^{t \cdot F + c_t} \leq e^{\operatorname{osc}_X(F)}.$$

$$(H_t)_{j\bar{k}} = t \cdot F_{j\bar{k}}, \quad \sup_{0 \leq t \leq 1} \|H_t\|_{C^{r,\alpha}} \leq \operatorname{osc}_X(F) + \|F\|_{C^{r,\alpha}}, \quad D_* \leq \left\| \sum_{j,k=1}^n \hat{g}^{j\bar{k}} \cdot F_{j\bar{k}} \right\|_{C^0} + \|\hat{S}\|_{C^0}.$$

Thus every continuity density and every spatial derivative of its logarithm are uniformly controlled. $\square$

5.2. Logarithmic matrix calculus.

Theorem 5.2.1 (Jacobi formulas for the logarithmic determinant). Let $A(s)$ be a C^2 path of positive Hermitian $n \times n$ matrices. Then

$$(5.2.1) \quad \frac{d}{ds} \log(\det(A)) = \operatorname{Tr}(A^{-1} \cdot A'), \quad \frac{d^2}{ds^2} \log(\det(A)) = \operatorname{Tr}(A^{-1} \cdot A'') - \operatorname{Tr}(A^{-1} \cdot A' \cdot A^{-1} \cdot A').$$

Proof. By $A(s) > 0$, $A \in C^2$, and $A^{-1} \cdot A = I_n$, differentiate the determinant and the inverse identity.

$$\frac{d}{ds} \det(A) = \operatorname{Tr}(\operatorname{adj}(A) \cdot A') = \det(A) \cdot \operatorname{Tr}(A^{-1} \cdot A') \xrightarrow{??} \frac{d}{ds} \log(\det(A)) = \operatorname{Tr}(A^{-1} \cdot A').$$

$$0 = (A^{-1} \cdot A)' = (A^{-1})' \cdot A + A^{-1} \cdot A' \implies (A^{-1})' = -A^{-1} \cdot A' \cdot A^{-1}.$$

$$\frac{d^2}{ds^2} \log(\det(A)) = \operatorname{Tr}((A^{-1})' \cdot A' + A^{-1} \cdot A'') = \operatorname{Tr}(A^{-1} \cdot A'') - \operatorname{Tr}(A^{-1} \cdot A' \cdot A^{-1} \cdot A').$$

Thus both variations are obtained without choosing an s -dependent eigenbasis. $\square$

Theorem 5.2.2 (Concavity of the logarithmic determinant). If $A_s = A_0 + s \cdot B > 0$ and $B = B^*$, then

$$\frac{d^2}{ds^2} \log(\det(A_s)) = -\operatorname{Tr}([A_s^{-1/2} \cdot B \cdot A_s^{-1/2}]^2) = -\sum_{a=1}^n \mu_a(s)^2 \leq 0.$$

Consequently, $A \mapsto \log(\det(A))$ is concave on the positive Hermitian cone.

Proof. By $A'_s = B$, $A''_s = 0$, and $K_s = A_s^{-1/2} \cdot B \cdot A_s^{-1/2} = K_s^*$, apply the second Jacobi formula.

$$\frac{d^2}{ds^2} \log(\det(A_s)) \xrightarrow{5.2.1} -\text{Tr}(A_s^{-1} \cdot B \cdot A_s^{-1} \cdot B) = -\text{Tr}(K_s^2) = -\sum_{a=1}^n \mu_a(s)^2 \leq 0.$$

Thus every affine restriction has nonpositive second derivative. $\square$

Example 5.2.3 (Exact logarithmic Hessian computation). Take

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 2 \end{bmatrix}, \quad A^{-1} = \frac{1}{5} \cdot \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}.$$

The full product and the determinant polynomial are

$$A^{-1} \cdot B = \frac{1}{5} \cdot \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{5} \\ -\frac{1}{2} & \frac{13}{10} \end{bmatrix}, \quad \det(A + s \cdot B) = 5 + 9 \cdot s + \frac{7}{4} \cdot s^2.$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{3}{5} \\ -\frac{1}{2} & \frac{13}{10} \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} & -\frac{3}{5} \\ -\frac{1}{2} & \frac{13}{10} \end{bmatrix} = \begin{bmatrix} \frac{11}{20} & -\frac{27}{100} \\ -\frac{9}{10} & \frac{199}{100} \end{bmatrix}, \quad \text{Tr}([A^{-1} \cdot B]^2) = \frac{11}{20} + \frac{199}{100} = \frac{127}{50}.$$

Hence

$$\left. \frac{d}{ds} \right|_{s=0} \log(\det(A + s \cdot B)) = \frac{9}{5} \xrightarrow{5.2.1} \text{Tr}(A^{-1} \cdot B).$$

$$\left. \frac{d^2}{ds^2} \right|_{s=0} \log(\det(A + s \cdot B)) = -\frac{127}{50} \xrightarrow{5.2.1} -\text{Tr}\left(\begin{bmatrix} \frac{11}{20} & -\frac{27}{100} \\ -\frac{9}{10} & \frac{199}{100} \end{bmatrix}\right) < 0.$$

The computed derivatives satisfy

$$\left(\frac{9}{5}, -\frac{127}{50}\right) \xrightarrow{5.2.1} \left(\text{Tr}\left(\begin{bmatrix} \frac{1}{2} & -\frac{3}{5} \\ -\frac{1}{2} & \frac{13}{10} \end{bmatrix}\right), -\text{Tr}\left(\begin{bmatrix} \frac{11}{20} & -\frac{27}{100} \\ -\frac{9}{10} & \frac{199}{100} \end{bmatrix}\right)\right).$$

$$(A + s \cdot B > 0, \quad B = B^*) \xrightarrow{5.2.2} \frac{d^2}{ds^2} \log(\det(A + s \cdot B)) \leq 0.$$

5.3. The zeroth-order estimate.

Theorem 5.3.1 (Yau C^0 estimate). *Let $\phi \in C^\infty(X, \mathbb{R})$ satisfy*

$$\hat{\omega} + \frac{i}{2} \cdot \partial \bar{\partial} \phi > 0, \quad \left(\hat{\omega} + \frac{i}{2} \cdot \partial \bar{\partial} \phi\right)^n = e^F \cdot \hat{\omega}^n, \quad \int_X \phi \cdot \hat{\omega}^n = 0.$$

Then

$$(5.3.1) \quad \|\phi\|_{C^0(X)} \leq C(X, \hat{\omega}, \|F\|_{C^0(X)}).$$

Along the continuity path the right side depends only on $(X, \hat{\omega}, \text{osc}_X(F))$.

Proof. By $\omega_\phi > 0$, $\omega_\phi^n = e^F \cdot \hat{\omega}^n$, and $\int_X \phi \cdot \hat{\omega}^n = 0$, set $V = \int_X \hat{\omega}^n$, shift ϕ so that $\sup_X(\phi) = 0$, and write

$$\psi = -\phi \geq 0, \quad w = 1 + \psi \geq 1, \quad \Theta = \sum_{q=0}^{n-1} \omega_\phi^q \wedge \hat{\omega}^{n-1-q} > 0, \quad G_{\hat{\omega}}(x, y) \leq 0, \quad C_S > 0.$$

$$\sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot \phi_{j\bar{k}} = \text{Tr}(h) - n \xrightarrow{4.2.3} \geq -n, \quad \phi(x) - \frac{1}{V} \int_X \phi \cdot \hat{\omega}^n = \int_X G_{\hat{\omega}}(x, y) \cdot \sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot \phi_{j\bar{k}}(y) \cdot \hat{\omega}^n(y).$$

$$\phi(x_0) = \sup_X(\phi) = 0, \quad G_{\hat{\omega}} \leq 0 \implies -\frac{1}{V} \int_X \phi \cdot \hat{\omega}^n \leq n \cdot \sup_{x \in X} \int_X [-G_{\hat{\omega}}(x, y)] \cdot \hat{\omega}^n(y) \implies \|w\|_{L^1(\hat{\omega}^n)} \leq C_G.$$

$$p \geq 2, \quad d\Theta = 0 \implies \omega_\phi^n - \hat{\omega}^n \xrightarrow{3.7.2} -\frac{i}{2} \cdot \partial \bar{\partial} w \wedge \Theta \xrightarrow{3.2.1} (e^F - 1) \cdot \hat{\omega}^n.$$

$$(p \geq 2, \quad d\Theta = 0) \xrightarrow{3.5.1} (p-1) \cdot \int_X w^{p-2} \cdot \frac{i}{2} \cdot \partial w \wedge \bar{\partial} w \wedge \Theta = \int_X w^{p-1} \cdot (e^F - 1) \cdot \hat{\omega}^n \leq (1 + e^{\|F\|_{C^0}}) \cdot \int_X w^{p-1} \cdot \hat{\omega}^n.$$

$$\Theta = \hat{\omega}^{n-1} + \sum_{q=1}^{n-1} \omega_{\phi}^q \wedge \hat{\omega}^{n-1-q} \xrightarrow{3.7.1} \Theta \geq \hat{\omega}^{n-1}, \quad n \cdot \frac{i}{2} \cdot \partial v \wedge \bar{\partial} v \wedge \hat{\omega}^{n-1} = \left(\sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot v_j \cdot v_{\bar{k}} \right) \cdot \hat{\omega}^n, \quad v = w^{p/2},$$

$$\int_X \sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot \frac{\partial w^{p/2}}{\partial z^j} \cdot \frac{\partial w^{p/2}}{\partial \bar{z}^k} \cdot \hat{\omega}^n \leq \frac{n \cdot (1 + e^{\|F\|_{C^0}}) \cdot p^2}{4 \cdot (p-1)} \cdot \int_X w^{p-1} \cdot \hat{\omega}^n \leq \frac{n \cdot (1 + e^{\|F\|_{C^0}}) \cdot p}{2} \cdot \int_X w^p \cdot \hat{\omega}^n.$$

$$\gamma = \frac{n}{n-1} \quad (n \geq 2), \quad \gamma = 2 \quad (n = 1), \quad \|w\|_{L^{p\gamma}} \leq \left[C_S \cdot \left(1 + \frac{n \cdot (1 + e^{\|F\|_{C^0}}) \cdot p}{2} \right) \right]^{1/p} \cdot \|w\|_{L^p}.$$

$$p = 2, \quad \|w\|_{L^{2\gamma}} \leq C \cdot \|w\|_{L^2}, \quad \|w\|_{L^2} \leq \|w\|_{L^1}^{\frac{\gamma-1}{2\gamma-1}} \cdot \|w\|_{L^{2\gamma}}^{\frac{\gamma}{2\gamma-1}} \implies \|w\|_{L^2} \leq C.$$

$$p_r = 2 \cdot \gamma^r, \quad \|w\|_{L^{p_{r+1}}} \leq (C \cdot p_r)^{1/p_r} \cdot \|w\|_{L^{p_r}}, \quad \sum_{r=0}^{\infty} \frac{\log(C \cdot p_r)}{p_r} < \infty \implies \|w\|_{L^{\infty}} \leq C.$$

Thus $\text{osc}_X(\phi) \leq C$. Returning to $\int_X \phi \cdot \hat{\omega}^n = 0$ gives $\|\phi\|_{C^0} \leq C$. Along the path,

$$0 \leq t \leq 1 \xrightarrow{5.1.2} \|t \cdot F + c_t\|_{C^0(X)} \leq \text{osc}_X(F) \implies \sup_{t \in I} \|\phi_t\|_{C^0(X)} \leq C(X, \hat{\omega}, \text{osc}_X(F)).$$

This proves the uniform zeroth-order estimate. $\square$

5.4. Fixed-time matrices and curvature identities. Write

$$\hat{G} = [\hat{g}_{j\bar{k}}]_{\bar{k},j}, \quad H_{\phi} = [\phi_{j\bar{k}}]_{\bar{k},j}, \quad G = \hat{G} + H_{\phi} = [g_{j\bar{k}}]_{\bar{k},j}, \quad h = \hat{G}^{-1} \cdot G.$$

$$G = \hat{G} + H_{\phi} \xrightarrow{4.2.2} h = I_n + \hat{G}^{-1} \cdot H_{\phi}, \quad \omega_{\phi} > 0 \xLeftrightarrow{4.2.2} G > 0.$$

$$\text{Tr}(h) = \text{Tr}(\hat{G}^{-1} \cdot G) = \sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot g_{j\bar{k}}, \quad \text{Tr}(h^{-1}) = \text{Tr}(G^{-1} \cdot \hat{G}) = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \hat{g}_{j\bar{k}}.$$

Fact 5.4.1 (Determinant and Laplacian identities).

(5.4.1)

$$\det(h) = e^F, \quad \Delta_{\hat{\omega}}(\phi) = \sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot \phi_{j\bar{k}} = \text{Tr}(h) - n, \quad \Delta_{\omega_{\phi}}(\phi) = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \phi_{j\bar{k}} = n - \text{Tr}(h^{-1}).$$

Proof. By $G = \hat{G} + H_{\phi}$, $\omega_{\phi}^n = e^F \cdot \hat{\omega}^n$, and $h = \hat{G}^{-1} \cdot G$, expand both matrix traces.

$$\det(h) = \det(\hat{G}^{-1} \cdot G) = \frac{\det(G)}{\det(\hat{G})} \xrightarrow{4.2.3} \det(h) = \frac{\omega_{\phi}^n}{\hat{\omega}^n} \xrightarrow{3.2.1} \det(h) = e^F.$$

$$\text{Tr}(h) = \text{Tr}(\hat{G}^{-1} \cdot G) = \text{Tr}(I_n) + \text{Tr}(\hat{G}^{-1} \cdot H_{\phi}) = n + \sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot \phi_{j\bar{k}}, \quad \text{Tr}(h^{-1}) = \text{Tr}(G^{-1} \cdot \hat{G}) = n - \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \phi_{j\bar{k}}.$$

Thus the determinant and both traced elliptic operators are explicit. $\square$

Fact 5.4.2 (Trace arithmetic). *The relative traces satisfy*

$$\text{Tr}(h) \geq n \cdot e^{F/n}, \quad \text{Tr}(h^{-1}) \geq n \cdot e^{-F/n}, \quad \frac{1}{\text{Tr}(h)} \leq \frac{\text{Tr}(h^{-1})}{n^2}.$$

Proof. By $\lambda_a > 0$ for $1 \leq a \leq n$, $\text{Tr}(h) = \sum_{a=1}^n \lambda_a$, $\text{Tr}(h^{-1}) = \sum_{a=1}^n \lambda_a^{-1}$, and $\prod_{a=1}^n \lambda_a = e^F$, apply AM–GM and Cauchy–Schwarz.

$$\prod_{a=1}^n \lambda_a = e^F \xrightarrow{5.4.1} \frac{\text{Tr}(h)}{n} \geq \left(\prod_{a=1}^n \lambda_a \right)^{1/n} = e^{F/n}, \quad \prod_{a=1}^n \lambda_a^{-1} = e^{-F} \xrightarrow{5.4.1} \frac{\text{Tr}(h^{-1})}{n} \geq \left(\prod_{a=1}^n \lambda_a^{-1} \right)^{1/n} = e^{-F/n}.$$

$$n^2 = \left(\sum_{a=1}^n \lambda_a^{1/2} \cdot \lambda_a^{-1/2} \right)^2 \leq \left(\sum_{a=1}^n \lambda_a \right) \cdot \left(\sum_{a=1}^n \lambda_a^{-1} \right) \xrightarrow{4.2.3} n^2 \leq \text{Tr}(h) \cdot \text{Tr}(h^{-1}) \implies \frac{1}{\text{Tr}(h)} \leq \frac{\text{Tr}(h^{-1})}{n^2}.$$

Thus the two traces control one another once the determinant is fixed. $\square$

Fact 5.4.3 (Ricci identity from the Monge–Ampère equation). *One has*

$$(5.4.2) \quad \rho(\omega_\phi) - \rho(\hat{\omega}) = -i \cdot \partial \bar{\partial} F, \quad R_{\bar{k}j} - \hat{R}_{\bar{k}j} = -F_{j\bar{k}} \quad (1 \leq j, k \leq n).$$

Proof. By $\det(G) = e^F \cdot \det(\hat{G})$ and $\rho(\omega) = -i \cdot \partial \bar{\partial} \log(\det(G))$, subtract the two Ricci forms.

$$\begin{aligned} \rho(\omega_\phi) - \rho(\hat{\omega}) &\xrightarrow{1.8.2} -i \cdot \partial \bar{\partial} (\log(\det(G)) - \log(\det(\hat{G}))) = -i \cdot \partial \bar{\partial} F. \\ i \cdot \sum_{j=1}^n \sum_{k=1}^n (R_{\bar{k}j} - \hat{R}_{\bar{k}j}) \cdot dz^j \wedge d\bar{z}^k &= -i \cdot \sum_{j=1}^n \sum_{k=1}^n F_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k \implies R_{\bar{k}j} - \hat{R}_{\bar{k}j} = -F_{j\bar{k}}. \end{aligned}$$

Thus the form and component normalizations agree. $\square$

Fact 5.4.4 (Background-normal diagonal coordinates). *For every $p \in X$, there are holomorphic coordinates centered at p such that*

$$\hat{G}(p) = I_n, \quad \frac{\partial \hat{g}_{j\bar{k}}}{\partial z^\ell}(p) = 0, \quad G(p) = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad G(p)^{-1} = \begin{bmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{bmatrix}.$$

In these coordinates $\text{Tr}(h(p)) = \sum_{a=1}^n \lambda_a$ and $\text{Tr}(h^{-1}(p)) = \sum_{a=1}^n \lambda_a^{-1}$.

Proof. By $\hat{G}(p) > 0$, the Kähler symmetry, and $G(p) > 0$, first normalize the background metric, remove its first derivatives, and then apply a unitary diagonalization.

$$\begin{aligned} \hat{G}(p) > 0 &\implies \exists P \in \text{GL}(n, \mathbb{C}) \quad (P^*)^{-1} \cdot \hat{G}(p) \cdot P^{-1} = I_n. \\ u^m &= z^m + \frac{1}{2} \cdot \sum_{j=1}^n \sum_{\ell=1}^n \hat{\Gamma}_{j\bar{\ell}}^m(p) \cdot z^j \cdot \bar{z}^\ell \implies \tilde{\Gamma}_{j\bar{\ell}}^m(p) = 0 \implies \frac{\partial \tilde{g}_{j\bar{k}}}{\partial u^\ell}(p) = 0. \\ \hat{G}(p) = I_n, \quad G(p) = G(p)^* > 0 &\xrightarrow{4.2.1} \exists Q \in \mathbb{C}^{n \times n}, \quad Q^* \cdot Q = I_n. \\ Q^* \cdot G(p) \cdot Q &= \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad \lambda_a > 0 \quad (1 \leq a \leq n). \end{aligned}$$

Thus the same coordinates normalize $\hat{G}$ and diagonalize the relative metric. $\square$

5.5. The Aubin–Yau trace inequality.

Fact 5.5.1 (Laplacian of $\text{Tr}(h)$).

(5.5.1)

$$\Delta_{\omega_\phi}(\text{Tr}(h)) = \sum_{j,k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 \text{Tr}(h)}{\partial z^j \partial \bar{z}^k} = \sum_{p,q=1}^n \hat{g}^{p\bar{q}} \cdot F_{p\bar{q}} - \hat{S} + \sum_{j,k,p,m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}jp}^m \cdot h_p^m + \sum_{j,k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h)).$$

Proof. By $\hat{G}(p) = I_n$, $\partial \hat{G}(p) = 0$, $G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)$, and $\log(\det(G)) - \log(\det(\hat{G})) = F$, differentiate $\text{Tr}(h)$ twice at p .

$$(\hat{G}(p) = I_n, \quad \partial \hat{G}(p) = 0, \quad G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)) \xrightarrow{5.4.4} \text{Tr}(h) = \sum_{a,b=1}^n \hat{g}^{a\bar{b}} \cdot g_{a\bar{b}}.$$

$$\Delta_{\omega_\phi}(\text{Tr}(h)) = \sum_{j,k,a,b=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 (\hat{g}^{a\bar{b}} \cdot g_{a\bar{b}})}{\partial z^j \partial \bar{z}^k}.$$

$$\begin{aligned}
\frac{\partial^2(\hat{g}^{a\bar{b}} \cdot g_{a\bar{b}})}{\partial z^j \partial \bar{z}^k}(p) &= \hat{g}^{a\bar{b}} \cdot \frac{\partial^2 g_{a\bar{b}}}{\partial z^j \partial \bar{z}^k} + \frac{\partial^2 \hat{g}^{a\bar{b}}}{\partial z^j \partial \bar{z}^k} \cdot g_{a\bar{b}}. \\
G > 0 &\xrightarrow{5.2.1} \frac{\partial^2}{\partial z^p \partial \bar{z}^q} \log(\det(G)) = \sum_{j,k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 g_{j\bar{k}}}{\partial z^p \partial \bar{z}^q} - \sum_{j,k,a,b=1}^n g^{j\bar{k}} \cdot g^{a\bar{b}} \cdot \frac{\partial g_{j\bar{b}}}{\partial z^p} \cdot \frac{\partial g_{a\bar{k}}}{\partial \bar{z}^q}. \\
(\log(\det(G)) - \log(\det(\hat{G})) = F, \quad \hat{G}(p) = I_n) &\xrightarrow{5.4.3} \sum_{j,k,a,b=1}^n g^{j\bar{k}} \cdot \hat{g}^{a\bar{b}} \cdot \frac{\partial^2 g_{a\bar{b}}}{\partial z^j \partial \bar{z}^k} = \sum_{p,q=1}^n \hat{g}^{p\bar{q}} \cdot F_{p\bar{q}} - \hat{S} + \sum_{j,a,b=1}^n \frac{|\partial_j g_{b\bar{a}}|^2}{\lambda_j \cdot \lambda_b}. \\
G(p) = \text{diag}(\lambda_1, \dots, \lambda_n) &\xrightarrow{5.4.4} \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h)) = \sum_{j,a,b=1}^n \frac{|\partial_j g_{b\bar{a}}|^2}{\lambda_j \cdot \lambda_b}. \\
\frac{\partial^2 \hat{g}^{a\bar{b}}}{\partial z^j \partial \bar{z}^k}(p) = \hat{R}_{\bar{k}j}^a b(p) &\implies \sum_{j,k,a,b=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 \hat{g}^{a\bar{b}}}{\partial z^j \partial \bar{z}^k} \cdot g_{a\bar{b}} = \sum_{j=1}^n \sum_{p=1}^n \lambda_j^{-1} \cdot \lambda_p \cdot \hat{R}_{\bar{j}j}^p p. \\
\sum_{j,k,p,m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}jp}^m \cdot h^p_m &= \sum_{j=1}^n \sum_{p=1}^n \lambda_j^{-1} \cdot \lambda_p \cdot \hat{R}_{\bar{j}j}^p p.
\end{aligned}$$

Combining the two differentiated terms proves the claimed formula. $\square$

Theorem 5.5.2 (Aubin–Yau trace-gradient inequality).

$$(5.5.2) \quad \frac{\sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h))}{\text{Tr}(h)} - \frac{\sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot (\partial_j \text{Tr}(h)) \cdot (\partial_{\bar{k}} \text{Tr}(h))}{(\text{Tr}(h))^2} \geq 0.$$

Proof. By $h = \hat{G}^{-1} \cdot G$, $\hat{G}(p) = I_n$, $\partial \hat{G}(p) = 0$, $G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)$, and $g^{j\bar{k}}(p) = \lambda_j^{-1} \delta_{jk}$, apply Cauchy–Schwarz.

$$(\hat{G}(p) = I_n, \quad \partial \hat{G}(p) = 0, \quad G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)) \xrightarrow{5.4.4} h^a_b = g_{b\bar{a}} = \lambda_b \cdot \delta_{ab}, \quad (h^{-1})^b_c = \lambda_b^{-1} \cdot \delta_{bc}.$$

$$\begin{aligned}
(\partial_j h)^a_b &= \partial_j g_{b\bar{a}}, \quad \text{Tr}(h) = \sum_{a=1}^n \lambda_a. \\
\text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h)) &= \sum_{a,b,c=1}^n (\partial_j h)^a_b \cdot (h^{-1})^b_c \cdot (\partial_{\bar{k}} h)^c_a \\
&= \sum_{a,b,c=1}^n \partial_j g_{b\bar{a}} \cdot \lambda_b^{-1} \cdot \delta_{bc} \cdot \partial_{\bar{k}} g_{a\bar{c}} = \sum_{a,b=1}^n \lambda_b^{-1} \cdot \partial_j g_{b\bar{a}} \cdot \partial_{\bar{k}} g_{a\bar{b}}. \\
\sum_{j,k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h)) &= \sum_{j,a,b=1}^n \frac{|\partial_j g_{b\bar{a}}|^2}{\lambda_j \cdot \lambda_b}. \\
|\partial_j \text{Tr}(h)|^2 &= \left| \sum_{a=1}^n \lambda_a^{-1/2} \cdot \partial_j g_{a\bar{a}} \cdot \lambda_a^{1/2} \right|^2 \leq \text{Tr}(h) \cdot \sum_{a=1}^n \frac{|\partial_j g_{a\bar{a}}|^2}{\lambda_a}. \\
\sum_{j,k=1}^n g^{j\bar{k}} \cdot (\partial_j \text{Tr}(h)) \cdot (\partial_{\bar{k}} \text{Tr}(h)) &= \sum_{j=1}^n \lambda_j^{-1} \cdot |\partial_j \text{Tr}(h)|^2 \leq \text{Tr}(h) \cdot \sum_{j,a,b=1}^n \frac{|\partial_j g_{b\bar{a}}|^2}{\lambda_j \cdot \lambda_b}. \\
\sum_{j,k=1}^n g^{j\bar{k}} \cdot (\partial_j \text{Tr}(h)) \cdot (\partial_{\bar{k}} \text{Tr}(h)) &\leq \text{Tr}(h) \cdot \sum_{j,k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h)).
\end{aligned}$$

Thus the third-derivative term controls the gradient of the relative trace. $\square$

Fact 5.5.3 (Curvature term estimate). *There exists $C_{\hat{\omega}} > 0$ such that*

$$\left| \sum_{j,k,p,m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}jp}^m \cdot h^p_m \right| \leq C_{\hat{\omega}} \cdot (\text{Tr}(h^{-1})) \cdot (\text{Tr}(h)).$$

Proof. By $\hat{G}(p) = I_n$, $G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_a > 0$, and

$$C_{\hat{\omega}} = \sup_{\substack{x \in X \\ \|u\|_{\hat{\omega}}=1 \\ \|v\|_{\hat{\omega}}=1}} |\hat{R}(u, \bar{u}, v, \bar{v})| < \infty,$$

expand the complete curvature contraction.

$$\begin{aligned} (\hat{G}(p) = I_n, \quad G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)) &\xrightarrow{5.4.4} \left| \sum_{j,k,p,m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}jp}^m \cdot h^p_m \right| = \left| \sum_{j=1}^n \sum_{p=1}^n \lambda_j^{-1} \cdot \lambda_p \cdot \hat{R}_{j\bar{j}}^p \right| \\ &\leq C_{\hat{\omega}} \cdot \sum_{j=1}^n \sum_{p=1}^n \lambda_j^{-1} \cdot \lambda_p = C_{\hat{\omega}} \cdot (\text{Tr}(h^{-1})) \cdot (\text{Tr}(h)). \end{aligned}$$

Thus the background curvature contributes at most the product of the two relative traces. $\square$

Theorem 5.5.4 (Aubin–Yau differential inequality). *There exists $C_1 > 0$, depending only on $\hat{\omega}$ and $\|F\|_{C^2(\hat{\omega})}$, such that*

$$(5.5.3) \quad \Delta_{\omega_{\hat{\phi}}}(\log(\text{Tr}(h))) = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 \log(\text{Tr}(h))}{\partial z^j \partial \bar{z}^k} \geq -C_1 \cdot \text{Tr}(h^{-1}).$$

Proof. By $\text{Tr}(h) > 0$, $F \in C^2(X)$, and the complete elliptic identities below, set

$$\text{Tr}(h) \geq n \cdot e^{F/n} > 0 \xrightarrow{5.4.2} \log(\text{Tr}(h)) \text{ is defined.}$$

$$D_F = \sup_X \left| \sum_{p=1}^n \sum_{q=1}^n \hat{g}^{p\bar{q}} \cdot F_{p\bar{q}} - \hat{S} \right|, \quad C_1 = C_{\hat{\omega}} + \frac{D_F}{n^2},$$

differentiate the logarithm with every coefficient displayed.

$$\begin{aligned} \sum_{j,k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 \log(\text{Tr}(h))}{\partial z^j \partial \bar{z}^k} &= \frac{\sum_{j,k=1}^n g^{j\bar{k}} \cdot \partial_j \partial_{\bar{k}}(\text{Tr}(h))}{\text{Tr}(h)} - \frac{\sum_{j,k=1}^n g^{j\bar{k}} \cdot (\partial_j \text{Tr}(h)) \cdot (\partial_{\bar{k}} \text{Tr}(h))}{(\text{Tr}(h))^2} \\ &\xrightarrow{5.5.1} \frac{\sum_{p,q=1}^n \hat{g}^{p\bar{q}} \cdot F_{p\bar{q}} - \hat{S}}{\text{Tr}(h)} + \frac{\sum_{j,k,p,m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}jp}^m \cdot h^p_m}{\text{Tr}(h)} \\ &\quad + \frac{\sum_{j,k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h))}{\text{Tr}(h)} - \frac{\sum_{j,k=1}^n g^{j\bar{k}} \cdot (\partial_j \text{Tr}(h)) \cdot (\partial_{\bar{k}} \text{Tr}(h))}{(\text{Tr}(h))^2} \\ &\quad \left[\frac{\sum_{j,k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j h) \cdot h^{-1} \cdot (\partial_{\bar{k}} h))}{\text{Tr}(h)} - \frac{\sum_{j,k=1}^n g^{j\bar{k}} \cdot (\partial_j \text{Tr}(h)) \cdot (\partial_{\bar{k}} \text{Tr}(h))}{(\text{Tr}(h))^2} \right] \xrightarrow{5.5.2} \geq 0. \\ \frac{1}{\text{Tr}(h)} &\xrightarrow{5.4.2} \frac{\text{Tr}(h^{-1})}{n^2}, \quad \frac{1}{\text{Tr}(h)} \cdot \left| \sum_{j,k,p,m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}jp}^m \cdot h^p_m \right| \xrightarrow{5.5.3} C_{\hat{\omega}} \cdot \text{Tr}(h^{-1}). \\ \Delta_{\omega_{\hat{\phi}}}(\log(\text{Tr}(h))) &\geq -\frac{D_F}{n^2} \cdot \text{Tr}(h^{-1}) - C_{\hat{\omega}} \cdot \text{Tr}(h^{-1}) = -C_1 \cdot \text{Tr}(h^{-1}). \end{aligned}$$

Thus the taught Aubin–Yau inequality follows from the invariant trace-gradient term. $\square$

5.6. The second-order estimate and explicit metric bounds.

Theorem 5.6.1 (Yau second-order estimate). *Assume*

$$\omega_\phi^n = e^F \cdot \hat{\omega}^n, \quad \omega_\phi > 0, \quad \|\phi\|_{C^0(X)} \leq C_0.$$

Then there exists $C_2 > 0$, depending only on X , $\hat{\omega}$, C_0 , and $\|F\|_{C^2(\hat{\omega})}$, such that

$$(5.6.1) \quad \sup_X (\text{Tr}(h)) \leq C_2.$$

Proof. By $\|\phi\|_{C^0} \leq C_0$, $\Delta_{\omega_\phi}(\log(\text{Tr}(h))) \geq -C_1 \cdot \text{Tr}(h^{-1})$, $\Delta_{\omega_\phi}(\phi) = n - \text{Tr}(h^{-1})$, $A = C_1 + 1$, and $Q = \log(\text{Tr}(h)) - A \cdot \phi$,

$$(h = \hat{G}^{-1} \cdot G, \quad G = \hat{G} + H_\phi) \xrightarrow{5.4.1} \Delta_{\omega_\phi}(\phi) = n - \text{Tr}(h^{-1}).$$

$$\sum_{j,k=1}^n g^{j\bar{k}} \cdot Q_{j\bar{k}} = \sum_{j,k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 \log(\text{Tr}(h))}{\partial z^j \partial \bar{z}^k} - A \cdot \sum_{j,k=1}^n g^{j\bar{k}} \cdot \phi_{j\bar{k}}.$$

$$\sum_{j,k=1}^n g^{j\bar{k}} \cdot Q_{j\bar{k}} \xrightarrow{5.5.4} \geq -C_1 \cdot \text{Tr}(h^{-1}) - A \cdot (n - \text{Tr}(h^{-1})) = \text{Tr}(h^{-1}) - A \cdot n.$$

$$Q(x_0) = \max_X(Q) \implies 0 \geq \sum_{j,k=1}^n g^{j\bar{k}}(x_0) \cdot Q_{j\bar{k}}(x_0) \geq \text{Tr}(h(x_0)^{-1}) - A \cdot n \implies \text{Tr}(h(x_0)^{-1}) \leq A \cdot n.$$

$$\left(\det(h(x_0)) = e^{F(x_0)}, \quad \text{Tr}(h(x_0)^{-1}) \leq A \cdot n \right) \xrightarrow{4.2.5} \text{Tr}(h(x_0)) \leq \frac{e^{F(x_0)}}{(n-1)!} \cdot [\text{Tr}(h(x_0)^{-1})]^{n-1} \leq \frac{e^{\sup_X(F)}}{(n-1)!} \cdot (A \cdot n)^{n-1}.$$

$$Q(x) \leq Q(x_0) \implies \text{Tr}(h(x)) \leq \text{Tr}(h(x_0)) \cdot e^{A \cdot (\phi(x) - \phi(x_0))} \leq \frac{e^{\sup_X(F)}}{(n-1)!} \cdot (A \cdot n)^{n-1} \cdot e^{A \cdot \text{osc}_X(\phi)}.$$

$$C_2 = \frac{e^{\sup_X(F)}}{(n-1)!} \cdot (A \cdot n)^{n-1} \cdot e^{2 \cdot A \cdot C_0} \implies \sup_X (\text{Tr}(h)) \leq C_2.$$

Thus the trace bound follows with every maximum-principle coefficient displayed. $\square$

Corollary 5.6.2 (Uniform metric equivalence). *Assume $\text{Tr}(h) \leq C_2$ and $\det(h) = e^F$. There exists $C > 0$ such that*

$$(5.6.2) \quad C^{-1} \cdot \hat{G} \leq G \leq C \cdot \hat{G}, \quad C^{-1} \cdot \hat{\omega} \leq \omega_\phi \leq C \cdot \hat{\omega}.$$

Proof. By $\text{Tr}(h) = \sum_{a=1}^n \lambda_a \leq C_2$, $\prod_{a=1}^n \lambda_a = e^F$, and $\lambda_a > 0$, estimate each eigenvalue and then invert the matrices.

$$(\hat{\omega} > 0, \quad \omega_\phi > 0) \xrightarrow{4.2.1} \lambda_a > 0 \quad (1 \leq a \leq n).$$

$$\lambda_a \leq \sum_{b=1}^n \lambda_b \leq C_2, \quad \lambda_a = \frac{e^F}{\prod_{\substack{1 \leq b \leq n \\ b \neq a}} \lambda_b} \geq e^{\inf_X(F)} \cdot C_2^{1-n} \xrightarrow{4.2.1} e^{\inf_X(F)} \cdot C_2^{1-n} \cdot I_n \leq \hat{G}^{-1/2} \cdot G \cdot \hat{G}^{-1/2} \leq C_2 \cdot I_n.$$

$$e^{\inf_X(F)} \cdot C_2^{1-n} \cdot \hat{G} \leq G \leq C_2 \cdot \hat{G}, \quad C = \max\{C_2, e^{-\inf_X(F)} \cdot C_2^{n-1}\} \implies C^{-1} \cdot \hat{G} \leq G \leq C \cdot \hat{G}.$$

$$C^{-1} \cdot \hat{G} \leq G \leq C \cdot \hat{G} \implies C^{-1} \cdot \hat{G}^{-1} \leq G^{-1} \leq C \cdot \hat{G}^{-1}, \quad C^{-1} \cdot \hat{\omega} \leq \omega_\phi \leq C \cdot \hat{\omega}.$$

Thus the trace and determinant bounds give the exact coefficient interval. $\square$

Example 5.6.3 (Maximum-principle envelope in complex dimension two). For $F_q = -0.4 + 0.02 \cdot q$ with $0 \leq q \leq 40$, $x_r = -2 + 0.01 \cdot r$ with $0 \leq r \leq 400$, and $\hat{G} = I_2$, set

$$\lambda_1 = e^{x_r}, \quad \lambda_2 = e^{F_q - x_r}, \quad \text{Tr}(h_{q,r}) = \lambda_1 + \lambda_2, \quad \text{Tr}(h_{q,r}^{-1}) = \lambda_1^{-1} + \lambda_2^{-1}.$$

The deterministic grid points satisfying $\text{Tr}(h_{q,r}^{-1}) \leq 6$ give

$$\max(\text{Tr}(h_{q,r})) = 8.928617029566244 \quad \text{at} \quad (F_q, x_r) = (0.4, -1.77).$$

$$e^{F_q} = 1.4918246976412703, \quad \text{Tr}(h_{q,r}^{-1}) = 5.985030978293438.$$

$$\text{Tr}(h_{q,r}) = e^{F_q} \cdot \text{Tr}(h_{q,r}^{-1}) = 8.928617029566244.$$

The dimension-two identity and the maximum-principle envelope are

$$(n = 2, \quad \text{spec}(h) = \{\lambda_1, \lambda_2\}, \quad \det(h) = e^F) \xrightarrow{4.2.3} (\text{Tr}(h) = \lambda_1 + \lambda_2, \quad \text{Tr}(h^{-1}) = \lambda_1^{-1} + \lambda_2^{-1}, \quad e^F = \lambda_1 \cdot \lambda_2).$$

$$e^F \cdot \text{Tr}(h^{-1}) = \lambda_1 \cdot \lambda_2 \cdot (\lambda_1^{-1} + \lambda_2^{-1}) = \lambda_2 + \lambda_1 = \text{Tr}(h).$$

$$\text{Tr}(h_{q,r}) = e^{F_q} \cdot \text{Tr}(h_{q,r}^{-1}) = 8.928617029566244 \leq 6 \cdot e^{0.4} = 8.950948185847622.$$

With $Q_2 = 2^{-1/2} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, the full metric at the maximizing grid point is

$$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.1703329888 & 0 \\ 0 & 8.7582840407 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4.4643085148 & -4.2939755260 \\ -4.2939755260 & 4.4643085148 \end{bmatrix},$$

$$\begin{bmatrix} 4.4643085148 & -4.2939755260 \\ -4.2939755260 & 4.4643085148 \end{bmatrix}^{-1} = \begin{bmatrix} 2.9925154891 & 2.8783378722 \\ 2.8783378722 & 2.9925154891 \end{bmatrix}.$$

The experiment numerically reproduces the exact two-dimensional trace identity.

Corollary 5.6.4 (C^1 estimate). *Assume*

$$\omega_\phi = \hat{\omega} + \frac{i}{2} \cdot \partial \bar{\partial} \phi > 0, \quad \omega_\phi^n = e^F \cdot \hat{\omega}^n, \quad \int_X \phi \cdot \hat{\omega}^n = 0,$$

$$\|\phi\|_{C^0(X)} \leq C_0, \quad \text{Tr}(h) \leq C_2.$$

Then for every $0 < \alpha < 1$,

$$\|\phi\|_{C^{1,\alpha}(X, \hat{\omega})} \leq C(X, \hat{\omega}, \alpha, C_0, C_2).$$

Proof. By $\text{Tr}(h) \leq C_2$, $\int_X \phi \cdot \hat{\omega}^n = 0$, and $p > 2n/(1 - \alpha)$, solve the complete fixed-background Poisson equation.

$$\sum_{j=1}^n \sum_{k=1}^n \hat{g}^{j\bar{k}} \cdot \frac{\partial^2 \phi}{\partial \bar{z}^j \partial z^k} \xrightarrow{5.4.1} \Delta_{\hat{\omega}}(\phi) = \text{Tr}(h) - n, \quad \|\text{Tr}(h) - n\|_{L^p} \leq (C_2 + n) \cdot \text{Vol}_{\hat{\omega}}(X)^{1/p}.$$

$$\left(\Delta_{\hat{\omega}}(\phi) = \text{Tr}(h) - n, \quad \int_X \phi \cdot \hat{\omega}^n = 0 \right) \xrightarrow{3.5.5} \phi \text{ is the unique mean-zero solution.}$$

$$\|\phi\|_{W^{2,p}} \leq C_p \cdot \|\text{Tr}(h) - n\|_{L^p} \leq C_p, \quad p > \frac{2n}{1 - \alpha} \implies W^{2,p}(X) \hookrightarrow C^{1,\alpha}(X).$$

Thus the first derivative estimate follows from the trace estimate. $\square$

5.7. The classical Calabi refinement.

Definition 5.7.1 (Connection difference and Calabi quantity). Let ∇ and $\hat{\nabla}$ be the Chern connections of ω_ϕ and $\hat{\omega}$. Define

$$A_j^m{}_p = \Gamma_{jp}^m - \hat{\Gamma}_{jp}^m \xrightarrow{1.6.4} A_j^m{}_p = \sum_{q=1}^n g^{m\bar{q}} \cdot \hat{\nabla}_j g_{p\bar{q}},$$

$$S = |A|_{\omega_\phi}^2 = \sum_{j,k,p,q,m,\ell=1}^n g^{j\bar{k}} \cdot g^{p\bar{q}} \cdot g_{\ell\bar{m}} \cdot A_j^m{}_p \cdot \overline{A_k^\ell{}_q}.$$

At a background-normal point with $G = \text{diag}(\lambda_1, \dots, \lambda_n)$,

$$S = \sum_{j=1}^n \sum_{a=1}^n \sum_{b=1}^n \frac{|\partial_j g_{b\bar{a}}|^2}{\lambda_j \cdot \lambda_a \cdot \lambda_b}.$$

Fact 5.7.2 (Calabi differential inequalities). *Assume*

$$C^{-1} \cdot \hat{\omega} \leq \omega_\phi \leq C \cdot \hat{\omega}, \quad \omega_\phi^n = e^F \cdot \hat{\omega}^n, \quad \|F\|_{C^3(\hat{\omega})} \leq C_F.$$

Then constants $K_1, K_2, K_3, K_4 > 0$, depending only on $(X, \hat{\omega})$, C , and C_F , satisfy

$$(5.7.1) \quad \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot S_{j\bar{k}} \geq -K_1 \cdot S - K_2, \quad \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2 \text{Tr}(h)}{\partial z^j \partial \bar{z}^k} \geq K_3 \cdot S - K_4.$$

These inequalities are retained as the classical analytic input. The first is Calabi's connection-difference commutation formula, and the second is the quantified positive third-derivative term in the trace formula.

Theorem 5.7.3 (Calabi C^3 estimate). *Assume*

$$\omega_\phi^n = e^F \cdot \hat{\omega}^n, \quad \omega_\phi = \hat{\omega} + \frac{i}{2} \cdot \partial \bar{\partial} \phi > 0, \quad \|\phi\|_{C^0(X)} \leq C_0,$$

$$\text{Tr}(h) \leq C_2, \quad \|F\|_{C^3(\hat{\omega})} \leq C_F.$$

Then

$$\sup_X(S) \leq C(X, \hat{\omega}, C_0, C_2, C_F).$$

Proof. By $\text{Tr}(h) \leq C_2$, $\sum_{j,k=1}^n g^{j\bar{k}} \cdot S_{j\bar{k}} \geq -K_1 \cdot S - K_2$, $\sum_{j,k=1}^n g^{j\bar{k}} \cdot \partial_j \partial_{\bar{k}}(\text{Tr}(h)) \geq K_3 \cdot S - K_4$, $a = (K_1 + 1)/K_3$, and $Q = S + a \cdot \text{Tr}(h)$,

$$(\text{Tr}(h) \leq C_2, \quad \det(h) = e^F) \xrightarrow{5.6.2} C^{-1} \cdot \hat{\omega} \leq \omega_\phi \leq C \cdot \hat{\omega}.$$

$$\sum_{j,k=1}^n g^{j\bar{k}} \cdot Q_{j\bar{k}} = \sum_{j,k=1}^n g^{j\bar{k}} \cdot S_{j\bar{k}} + a \cdot \sum_{j,k=1}^n g^{j\bar{k}} \cdot \partial_j \partial_{\bar{k}}(\text{Tr}(h)) \xrightarrow{5.7.2} \geq -K_1 \cdot S - K_2 + a \cdot (K_3 \cdot S - K_4) = S - (K_2 + a \cdot K_4).$$

$$Q(x_0) = \max_X(Q) \implies 0 \geq \sum_{j,k=1}^n g^{j\bar{k}}(x_0) \cdot Q_{j\bar{k}}(x_0) \geq S(x_0) - (K_2 + a \cdot K_4) \implies S(x_0) \leq K_2 + a \cdot K_4.$$

$$S(x) + a \cdot \text{Tr}(h(x)) = Q(x) \leq Q(x_0) \implies S(x) \leq S(x_0) + a \cdot \text{Tr}(h(x_0)) \leq K_2 + a \cdot (K_4 + C_2).$$

Thus the classical third-order quantity is uniformly bounded. $\square$

5.8. Evans–Krylov and Schauder bootstrapping.

Theorem 5.8.1 (Evans–Krylov and Schauder bootstrapping). *Assume*

$$\|\phi\|_{C^0(X)} \leq C_0, \quad C^{-1} \cdot \hat{\omega} \leq \omega_\phi \leq C \cdot \hat{\omega}, \quad \omega_\phi^n = e^F \cdot \hat{\omega}^n.$$

Then there exists $\alpha_0 \in (0, 1)$ such that

$$\|\phi\|_{C^{2,\alpha_0}(X)} \leq C_{\alpha_0}(X, \hat{\omega}, C_0, C, \|F\|_{C^{\alpha_0}(X)}).$$

If $F \in C^\infty(X)$, then for every $r \geq 0$ and $0 < \alpha < 1$,

$$(5.8.1) \quad \|\phi\|_{C^{r+2,\alpha}(X)} \leq C_{r,\alpha}(X, \hat{\omega}, C_0, C, \|F\|_{C^{r,\alpha}(X)}).$$

Proof. By $\det(\hat{G} + H_\phi) = e^F \cdot \det(\hat{G})$, $C^{-1} \cdot \hat{G} \leq G \leq C \cdot \hat{G}$, and

$$A_s = A_0 + s \cdot B > 0 \xrightarrow{5.2.2} \frac{d^2}{ds^2} \log(\det(A_s)) = -\text{Tr}([A_s^{-1/2} \cdot B \cdot A_s^{-1/2}]^2) \leq 0,$$

write the nonlinear operator and its principal coefficients on a holomorphic coordinate ball.

$$\det \begin{pmatrix} \hat{g}_{1\bar{1}} + \phi_{1\bar{1}} & \cdots & \hat{g}_{n\bar{1}} + \phi_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ \hat{g}_{1\bar{n}} + \phi_{1\bar{n}} & \cdots & \hat{g}_{n\bar{n}} + \phi_{n\bar{n}} \end{pmatrix} = e^F \cdot \det \begin{pmatrix} \hat{g}_{1\bar{1}} & \cdots & \hat{g}_{n\bar{1}} \\ \vdots & \ddots & \vdots \\ \hat{g}_{1\bar{n}} & \cdots & \hat{g}_{n\bar{n}} \end{pmatrix}.$$

$$\mathcal{F}(H_\phi, x) = \log(\det(\hat{G} + H_\phi)) - \log(\det(\hat{G})) = F, \quad D\mathcal{F}_{H_\phi}[H_u] \xrightarrow{4.3.1} D\mathcal{F}_{H_\phi}[H_u] = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot u_{j\bar{k}}.$$

For every bold covector $\xi = [\xi_1 \ \cdots \ \xi_n]^\top$,

$$C^{-1} \cdot \hat{G} \leq G \leq C \cdot \hat{G} \implies C^{-1} \cdot \hat{G}^{-1} \leq G^{-1} \leq C \cdot \hat{G}^{-1}.$$

$$C^{-1} \cdot \bar{\xi}^\top \cdot \hat{G}^{-1} \cdot \xi \leq \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \xi_j \cdot \bar{\xi}_k \leq C \cdot \bar{\xi}^\top \cdot \hat{G}^{-1} \cdot \xi \xrightarrow{4.3.2} D\mathcal{F}_{H_\phi} \text{ is uniformly elliptic.}$$

The complex Evans–Krylov theorem on a finite coordinate cover gives one exponent $\alpha_0 \in (0, 1)$ and the stated C^{2,α_0} bound. For a real coordinate x^ℓ , differentiate the full equation.

$$\sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \frac{\partial^2}{\partial z^j \partial \bar{z}^k} \left(\frac{\partial \phi}{\partial x^\ell} \right) = \frac{\partial F}{\partial x^\ell} + \sum_{j=1}^n \sum_{k=1}^n (\hat{g}^{j\bar{k}} - g^{j\bar{k}}) \cdot \frac{\partial \hat{g}^{j\bar{k}}}{\partial x^\ell}.$$

$$g^{j\bar{k}} \in C^{\alpha_0}, \quad \frac{\partial F}{\partial x^\ell} + \sum_{j,k=1}^n (\hat{g}^{j\bar{k}} - g^{j\bar{k}}) \cdot \frac{\partial \hat{g}^{j\bar{k}}}{\partial x^\ell} \in C^{\alpha_0} \implies \left\| \frac{\partial \phi}{\partial x^\ell} \right\|_{C^{2,\alpha_0}} \leq C \implies \phi \in C^{3,\alpha_0}.$$

Thus G^{-1} is Lipschitz and $\phi \in C^{2,\alpha}$ for every $0 < \alpha < 1$. Repeated differentiation of the displayed linear equation followed by local Schauder estimates on the finite cover gives $\phi \in C^{r+2,\alpha}$ for every $r \geq 0$. $\square$

5.9. Completion of the continuity argument.

Theorem 5.9.1 (Continuity reduction). *Assume every normalized solution of the continuity equation satisfies*

$$t \in I, \quad h_t = \hat{G}^{-1} \cdot G_t, \quad \|\phi_t\|_{C^0(X)} \leq C_0, \quad \text{Tr}(h_t) \leq C_2.$$

Then $1 \in I$.

Proof. By $t \in I$, $\|\phi_t\|_{C^0} \leq C_0$, $\text{Tr}(h_t) \leq C_2$, $H_t = t \cdot F + c_t$, and $\omega_t^n = e^{H_t} \cdot \hat{\omega}^n$, evaluate the continuity chain.

$$\begin{aligned}
 t = 0 &\xrightarrow{3.8.2} c_0 = 0, \quad (t = 0, \quad c_0 = 0, \quad \phi_0 = 0) \xrightarrow{3.8.1} 0 \in I, \quad t_0 \in I \xrightarrow{3.9.1} t_0 \in \text{int}_{[0,1]} I. \\
 0 \leq t \leq 1 &\xrightarrow{5.1.2} |H_t| \leq \text{osc}_X(F), \quad \sup_{0 \leq t \leq 1} \|H_t\|_{C^{r,\alpha}} \leq \text{osc}_X(F) + \|F\|_{C^{r,\alpha}}. \\
 (\text{Tr}(h_t) \leq C_2, \quad \det(h_t) = e^{H_t}, \quad |H_t| \leq \text{osc}_X(F)) &\xrightarrow{5.6.2} C^{-1} \cdot \hat{\omega} \leq \omega_t \leq C \cdot \hat{\omega}. \\
 \left(\|\phi_t\|_{C^0} \leq C_0, \quad C^{-1} \cdot \hat{\omega} \leq \omega_t \leq C \cdot \hat{\omega}, \quad \|H_t\|_{C^{1,\alpha}(X)} < \infty \right) &\xrightarrow{5.8.1} \sup_{t \in I} \|\phi_t\|_{C^{2,\alpha}} < \infty \xrightarrow{4.6.1} I \text{ is closed in } [0, 1]. \\
 0 \in I, \quad I \text{ is open in } [0, 1], \quad I \text{ is closed in } [0, 1] &\implies I = [0, 1].
 \end{aligned}$$

Thus $1 \in I$. □

Corollary 5.9.2 (Completion of the continuity method). *The normalized complex Monge–Ampère equation at $t = 1$ has a unique smooth solution.*

Proof. By $H_t = t \cdot F + c_t$, $t \in I$, and the normalized continuity equation, apply the established zeroth- and second-order estimates.

$$\begin{aligned}
 0 \leq t \leq 1 &\xrightarrow{5.1.2} |H_t| \leq \text{osc}_X(F), \quad \sup_{0 \leq s \leq 1} \|H_s\|_{C^2(X)} < \infty. \\
 \left(t \in I, \quad \omega_t^n = e^{H_t} \cdot \hat{\omega}^n, \quad \|H_t\|_{C^0(X)} \leq \text{osc}_X(F) \right) &\xrightarrow{5.3.1} \|\phi_t\|_{C^0(X)} \leq C_0. \\
 \left(t \in I, \quad \omega_t^n = e^{H_t} \cdot \hat{\omega}^n, \quad \|\phi_t\|_{C^0(X)} \leq C_0, \quad \sup_{0 \leq s \leq 1} \|H_s\|_{C^2(X)} < \infty \right) &\xrightarrow{5.6.1} \text{Tr}(h_t) \leq C_2. \\
 \left(\sup_{t \in I} \|\phi_t\|_{C^0} \leq C_0, \quad \sup_{t \in I} \text{Tr}(h_t) \leq C_2 \right) &\xrightarrow{5.9.1} 1 \in I, \quad t = 1 \xrightarrow{3.8.2} c_1 = 0. \\
 (1 \in I, \quad c_1 = 0) &\implies \left(\omega_{\phi_1}^n = e^F \cdot \hat{\omega}^n, \quad \int_X \phi_1 \cdot \hat{\omega}^n = 0 \right) \xrightarrow{3.7.3} \phi_1 \text{ is unique.}
 \end{aligned}$$

Thus the line-bundle half closes with existence, regularity, and uniqueness. □

Part 2. The Vector-Bundle Interpretation⁵

6. HERMITIAN GEOMETRY ON HOLOMORPHIC VECTOR BUNDLES

We now pass from rank-one curvature to holomorphic vector bundles of arbitrary rank. The tangent bundle supplies the local model

$$E = T^{1,0}X, \quad H = G \xrightarrow{1.6.4} A_H = G^{-1} \cdot \partial G \xrightarrow{1.7.1} F_H = dA_H + A_H \wedge A_H \xrightarrow{6.4.3} F_H = \bar{\partial} A_H \xrightarrow{6.4.4} D_H(F_H) = 0.$$

Throughout,

$$j, k, \ell, m, p, q \in \{1, \dots, n\}, \quad \alpha, \beta, \gamma, \delta \in \{1, \dots, r\},$$

and every matrix acts from the left on a bold column vector.

6.1. Bundles and transition functions.

Definition 6.1.1 (Complex vector bundle). Let X be a smooth manifold with

$$X = \bigcup_{\mu \in \mathcal{I}} X_\mu.$$

A rank- r complex vector bundle $E \xrightarrow{\pi} X$ consists of local trivializations

$$E|_{X_\mu} \cong X_\mu \times \mathbb{C}^r$$

and smooth maps

$$X_\mu \cap X_\nu \xrightarrow{t_{\mu\nu}} \mathbf{GL}(r, \mathbb{C}), \quad t_{\mu\nu} = \begin{bmatrix} t_{\mu\nu}^1 1 & \cdots & t_{\mu\nu}^1 r \\ \vdots & \ddots & \vdots \\ t_{\mu\nu}^r 1 & \cdots & t_{\mu\nu}^r r \end{bmatrix},$$

such that

$$t_{\mu\mu} = I_r, \quad t_{\mu\nu} \cdot t_{\nu\rho} = t_{\mu\rho}, \quad t_{\nu\mu} = t_{\mu\nu}^{-1}.$$

For the local frame

$$e_\mu = (e_{\mu,1}, \dots, e_{\mu,r}), \quad e_\nu = e_\mu \cdot t_{\mu\nu},$$

a section $\varphi \in \Gamma(X, E)$ has local columns

$$\varphi_\mu = \begin{bmatrix} \varphi_\mu^1 \\ \vdots \\ \varphi_\mu^r \end{bmatrix}, \quad \varphi|_{X_\mu} = e_\mu \cdot \varphi_\mu,$$

with

$$\varphi_\mu = t_{\mu\nu} \cdot \varphi_\nu, \quad \varphi_\mu^\alpha = \sum_{\beta=1}^r t_{\mu\nu}^\alpha{}_\beta \cdot \varphi_\nu^\beta.$$

Example 6.1.2 (The holomorphic tangent bundle). For $E = T^{1,0}X$, set

$$e_{\mu,p} = \frac{\partial}{\partial z_\mu^p}, \quad e_{\nu,q} = \sum_{p=1}^n \frac{\partial z_\mu^p}{\partial z_\nu^q} \cdot e_{\mu,p}.$$

Thus

$$V = \sum_{p=1}^n V_\mu^p \cdot e_{\mu,p} = \sum_{q=1}^n V_\nu^q \cdot e_{\nu,q}, \quad e_{\nu,q} = \sum_{p=1}^n \frac{\partial z_\mu^p}{\partial z_\nu^q} \cdot e_{\mu,p} \xrightarrow{6.1.1} V_\mu = t_{\mu\nu} \cdot V_\nu,$$

⁵Based on lectures by Duong Phong [3].

$$t_{\mu\nu} = \begin{bmatrix} \frac{\partial z_\mu^1}{\partial z_\nu^1} & \cdots & \frac{\partial z_\mu^1}{\partial z_\nu^n} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_\mu^n}{\partial z_\nu^1} & \cdots & \frac{\partial z_\mu^n}{\partial z_\nu^n} \end{bmatrix}.$$

For

$$z_\mu^1 = 4 \cdot z_\nu^1 + z_\nu^2, \quad z_\mu^2 = 2 \cdot z_\nu^2,$$

one obtains

$$t_{\mu\nu} = \begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix}, \quad \det(t_{\mu\nu}) = 4 \cdot 2 - 1 \cdot 0 = 8 - 0 = 8.$$

For $V_\nu = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$,

$$\left(t_{\mu\nu} = \begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix}, \quad V_\nu = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right) \xrightarrow{6.1.1} V_\mu = \begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \cdot 4 + 1 \cdot 3 \\ 0 \cdot 4 + 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 16 + 3 \\ 0 + 6 \end{bmatrix} = \begin{bmatrix} 19 \\ 6 \end{bmatrix}.$$

Moreover,

$$t_{\mu\nu}^{-1} = \frac{1}{8} \cdot \begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix},$$

$$\begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix} \cdot \frac{1}{8} \cdot \begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix} = \frac{1}{8} \cdot \begin{bmatrix} 4 \cdot 2 + 1 \cdot 0 & 4 \cdot (-1) + 1 \cdot 4 \\ 0 \cdot 2 + 2 \cdot 0 & 0 \cdot (-1) + 2 \cdot 4 \end{bmatrix} = \frac{1}{8} \cdot \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} = I_2 \xrightarrow{6.1.1} t_{\nu\mu} = t_{\mu\nu}^{-1} = \frac{1}{8} \cdot \begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix}.$$

6.2. Holomorphic structures and Hermitian metrics.

Definition 6.2.1 (Holomorphic vector bundle). Let X be a complex manifold. A complex vector bundle $E \xrightarrow{\pi} X$ is holomorphic when

$$\bar{\partial} t_{\mu\nu} = 0, \quad \frac{\partial t_{\mu\nu}^\alpha{}_\beta}{\partial \bar{z}_\mu^\ell} = 0.$$

A section $\varphi \in \Gamma(X, E)$ is holomorphic when

$$\bar{\partial} \varphi_\mu = 0, \quad \frac{\partial \varphi_\mu^\alpha}{\partial \bar{z}_\mu^\ell} = 0.$$

Fact 6.2.2 (The $\bar{\partial}_E$ -operator). *The local formula*

$$(\bar{\partial}_E(\varphi))_\mu = \sum_{\alpha=1}^r \sum_{\ell=1}^n \frac{\partial \varphi_\mu^\alpha}{\partial \bar{z}_\mu^\ell} \cdot d\bar{z}_\mu^\ell \otimes e_{\mu,\alpha}$$

defines a global section of $\Lambda^{0,1} T^*X \otimes E$.

Proof. By $\varphi_\mu = t_{\mu\nu} \cdot \varphi_\nu$, $\bar{\partial} t_{\mu\nu} = 0$, and $\frac{\partial \varphi_\mu}{\partial \bar{z}_\mu^\ell} = \sum_{m=1}^n (\partial \bar{z}_\nu^m / \partial \bar{z}_\mu^\ell) \cdot \frac{\partial \varphi_\nu}{\partial \bar{z}_\nu^m}$,

$$\begin{aligned} \frac{\partial \varphi_\mu^\alpha}{\partial \bar{z}_\mu^\ell} &= \sum_{\beta=1}^r \frac{\partial t_{\mu\nu}^\alpha{}_\beta}{\partial \bar{z}_\mu^\ell} \cdot \varphi_\nu^\beta + \sum_{\beta=1}^r \sum_{m=1}^n t_{\mu\nu}^\alpha{}_\beta \cdot \frac{\partial \bar{z}_\nu^m}{\partial \bar{z}_\mu^\ell} \cdot \frac{\partial \varphi_\nu^\beta}{\partial \bar{z}_\nu^m}. \\ \bar{\partial} t_{\mu\nu} &= 0 \xrightarrow{6.2.1} \frac{\partial \varphi_\mu^\alpha}{\partial \bar{z}_\mu^\ell} = \sum_{\beta=1}^r \sum_{m=1}^n t_{\mu\nu}^\alpha{}_\beta \cdot \frac{\partial \bar{z}_\nu^m}{\partial \bar{z}_\mu^\ell} \cdot \frac{\partial \varphi_\nu^\beta}{\partial \bar{z}_\nu^m}. \\ \sum_{\ell=1}^n \frac{\partial \varphi_\mu^\alpha}{\partial \bar{z}_\mu^\ell} \cdot d\bar{z}_\mu^\ell &= \sum_{\beta=1}^r \sum_{m=1}^n \sum_{\ell=1}^n t_{\mu\nu}^\alpha{}_\beta \cdot \frac{\partial \bar{z}_\nu^m}{\partial \bar{z}_\mu^\ell} \cdot \frac{\partial \varphi_\nu^\beta}{\partial \bar{z}_\nu^m} \cdot d\bar{z}_\mu^\ell \xrightarrow{1.1.4} \sum_{\beta=1}^r \sum_{m=1}^n t_{\mu\nu}^\alpha{}_\beta \cdot \frac{\partial \varphi_\nu^\beta}{\partial \bar{z}_\nu^m} \cdot d\bar{z}_\nu^m. \end{aligned}$$

$$\sum_{\alpha=1}^r \sum_{\beta=1}^r \sum_{m=1}^n t_{\mu\nu}{}^{\alpha}{}_{\beta} \cdot \frac{\partial \phi_{\nu}^{\beta}}{\partial \bar{z}_{\nu}^m} \cdot d\bar{z}_{\nu}^m \otimes e_{\mu,\alpha} \xrightarrow{6.1.1} \sum_{\beta=1}^r \sum_{m=1}^n \frac{\partial \phi_{\nu}^{\beta}}{\partial \bar{z}_{\nu}^m} \cdot d\bar{z}_{\nu}^m \otimes e_{\nu,\beta}.$$

Thus the local $(0,1)$ -forms obey the section overlap law. $\square$

Definition 6.2.3 (Hermitian bundle metric). A Hermitian metric H is represented in the frame e_{μ} by

$$H_{\mu} = [H_{\mu,\beta\bar{\alpha}}]_{\bar{\alpha},\beta} = \begin{bmatrix} H_{\mu,1\bar{1}} & \cdots & H_{\mu,r\bar{1}} \\ \vdots & \ddots & \vdots \\ H_{\mu,1\bar{r}} & \cdots & H_{\mu,r\bar{r}} \end{bmatrix}, \quad H_{\mu} = H_{\mu}^* > 0,$$

$$H_{\mu}^{-1} = [H_{\mu}^{\alpha\bar{\gamma}}]_{\alpha,\bar{\gamma}}, \quad \sum_{\gamma=1}^r H_{\mu}^{\alpha\bar{\gamma}} \cdot H_{\mu,\beta\bar{\gamma}} = \delta^{\alpha}_{\beta}.$$

The pairing is linear in its first argument and conjugate-linear in its second argument

$$\langle V, W \rangle_H = W_{\mu}^* \cdot H_{\mu} \cdot V_{\mu} = \sum_{\alpha=1}^r \sum_{\beta=1}^r H_{\mu,\beta\bar{\alpha}} \cdot V_{\mu}^{\beta} \cdot \overline{W_{\mu}^{\alpha}}.$$

The metric matrices satisfy

$$(6.2.1) \quad H_{\nu} = t_{\mu\nu}^* \cdot H_{\mu} \cdot t_{\mu\nu},$$

$$H_{\nu,\delta\bar{\gamma}} = \sum_{\alpha=1}^r \sum_{\beta=1}^r \overline{t_{\mu\nu}^{\alpha\bar{\gamma}}} \cdot H_{\mu,\beta\bar{\alpha}} \cdot t_{\mu\nu}^{\beta\bar{\delta}}.$$

Example 6.2.4 (A rank-two metric transformation). Take

$$t_{\mu\nu} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad t_{\mu\nu}^* = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad H_{\mu} = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}.$$

Then

$$(t_{\mu\nu}, H_{\mu}) \xrightarrow{6.2.1} H_{\nu} = t_{\mu\nu}^* \cdot H_{\mu} \cdot t_{\mu\nu}.$$

$$t_{\mu\nu}^* \cdot H_{\mu} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 0 \cdot 1 & 1 \cdot 1 + 0 \cdot 2 \\ 1 \cdot 4 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 5 & 3 \end{bmatrix},$$

$$H_{\nu} = \begin{bmatrix} 4 & 1 \\ 5 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 \cdot 1 + 1 \cdot 0 & 4 \cdot 1 + 1 \cdot 1 \\ 5 \cdot 1 + 3 \cdot 0 & 5 \cdot 1 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 5 & 8 \end{bmatrix}.$$

For

$$V_{\nu} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad V_{\mu} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 2 \\ 0 \cdot 1 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

the two squared norms are

$$\langle V, V \rangle_H \xrightarrow{6.2.3} V_{\mu}^* \cdot H_{\mu} \cdot V_{\mu} = \begin{bmatrix} 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \cdot 3 + 1 \cdot 2 \\ 1 \cdot 3 + 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 14 \\ 7 \end{bmatrix} = 3 \cdot 14 + 2 \cdot 7 = 42 + 14 = 56,$$

$$\langle V, V \rangle_H \xrightarrow{6.2.3} V_{\nu}^* \cdot H_{\nu} \cdot V_{\nu} = \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \cdot 1 + 5 \cdot 2 \\ 5 \cdot 1 + 8 \cdot 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 14 \\ 21 \end{bmatrix} = 1 \cdot 14 + 2 \cdot 21 = 14 + 42 = 56.$$

6.3. The Chern connection.

Definition 6.3.1 (Chern unitary connection). For a holomorphic bundle $E \xrightarrow{\pi} X$ with Hermitian metric H , the Chern connection ∇_H is characterized by

$$\nabla_H^{0,1} = \bar{\partial}_E, \quad d(\langle V, W \rangle_H) = \langle \nabla_H(V), W \rangle_H + \langle V, \nabla_H(W) \rangle_H.$$

The pairing on bundle-valued one-forms conjugates the form coefficient in its second argument.

Proposition 6.3.2 (Local Chern connection formula). *The Chern connection exists and is unique. In a holomorphic frame,*

$$(6.3.1) \quad A_\mu = H_\mu^{-1} \cdot \partial H_\mu = \sum_{j=1}^n A_{\mu,j} \cdot dz_\mu^j,$$

$$A_{\mu,j}{}^\alpha{}_\beta = \sum_{\gamma=1}^r H_\mu^{\alpha\bar{\gamma}} \cdot \frac{\partial H_{\mu,\beta\bar{\gamma}}}{\partial z_\mu^j}, \quad A_{\mu,j} = \begin{bmatrix} A_{\mu,j}{}^1{}_1 & \cdots & A_{\mu,j}{}^1{}_r \\ \vdots & \ddots & \vdots \\ A_{\mu,j}{}^r{}_1 & \cdots & A_{\mu,j}{}^r{}_r \end{bmatrix}.$$

For

$$V_\mu = \begin{bmatrix} V_\mu^1 \\ \vdots \\ V_\mu^r \end{bmatrix},$$

$$\nabla_{H,\bar{k}}(V_\mu) = \frac{\partial V_\mu}{\partial \bar{z}_\mu^k}, \quad \nabla_{H,j}(V_\mu) = \frac{\partial V_\mu}{\partial z_\mu^j} + A_{\mu,j} \cdot V_\mu.$$

On overlaps,

$$(6.3.2) \quad A_\nu = t_{\mu\nu}^{-1} \cdot A_\mu \cdot t_{\mu\nu} + t_{\mu\nu}^{-1} \cdot \partial t_{\mu\nu}.$$

Proof. By $H_\nu = t_{\mu\nu}^* \cdot H_\mu \cdot t_{\mu\nu}$, $\partial(t_{\mu\nu}^*) = 0$, and $V_\mu = t_{\mu\nu} \cdot V_\nu$,

$$\begin{aligned} H_\nu &= t_{\mu\nu}^* \cdot H_\mu \cdot t_{\mu\nu} \xrightarrow{6.2.1} H_\nu^{-1} = (t_{\mu\nu}^* \cdot H_\mu \cdot t_{\mu\nu})^{-1} = t_{\mu\nu}^{-1} \cdot H_\mu^{-1} \cdot (t_{\mu\nu}^*)^{-1}. \\ \partial H_\nu &= 0 \cdot H_\mu \cdot t_{\mu\nu} + t_{\mu\nu}^* \cdot \partial H_\mu \cdot t_{\mu\nu} + t_{\mu\nu}^* \cdot H_\mu \cdot \partial t_{\mu\nu}. \\ (H_\nu^{-1}, \partial H_\nu) &\xrightarrow{6.3.1} A_\nu = H_\nu^{-1} \cdot \partial H_\nu = t_{\mu\nu}^{-1} \cdot H_\mu^{-1} \cdot \partial H_\mu \cdot t_{\mu\nu} + t_{\mu\nu}^{-1} \cdot \partial t_{\mu\nu} \xrightarrow{6.3.1} t_{\mu\nu}^{-1} \cdot A_\mu \cdot t_{\mu\nu} + t_{\mu\nu}^{-1} \cdot \partial t_{\mu\nu}. \\ V_\mu &= t_{\mu\nu} \cdot V_\nu \xrightarrow{6.1.1} t_{\mu\nu}^{-1} \cdot (\partial V_\mu + A_\mu \cdot V_\mu) = \partial V_\nu + (t_{\mu\nu}^{-1} \cdot \partial t_{\mu\nu} + t_{\mu\nu}^{-1} \cdot A_\mu \cdot t_{\mu\nu}) \cdot V_\nu = \partial V_\nu + A_\nu \cdot V_\nu. \\ A_\mu &= H_\mu^{-1} \cdot \partial H_\mu \xrightarrow{6.3.1} H_\mu \cdot A_\mu = H_\mu \cdot H_\mu^{-1} \cdot \partial H_\mu = I_r \cdot \partial H_\mu = \partial H_\mu, \quad A_\mu^* \cdot H_\mu = \bar{\partial} H_\mu \cdot H_\mu^{-1} \cdot H_\mu = \bar{\partial} H_\mu. \\ d(\langle V, W \rangle_H) &\xrightarrow{6.2.3} d(W_\mu^* \cdot H_\mu \cdot V_\mu) = (dW_\mu + A_\mu \cdot W_\mu)^* \cdot H_\mu \cdot V_\mu + W_\mu^* \cdot H_\mu \cdot (dV_\mu + A_\mu \cdot V_\mu). \\ (\tilde{\nabla}^{0,1} = \bar{\partial}_E, \tilde{\nabla}(H) = 0) &\xrightarrow{6.3.1} \partial H_\mu = H_\mu \cdot \tilde{A}_\mu \xrightarrow{6.2.3} \tilde{A}_\mu = H_\mu^{-1} \cdot \partial H_\mu = A_\mu. \end{aligned}$$

Thus the local matrices glue and define the unique Chern connection. $\square$

6.4. Curvature and the second Bianchi identity.

Definition 6.4.1 (Covariant exterior derivative). For $\Psi \in \Omega^p(X, \text{End}(E))$, define

$$D_H(\Psi) = d\Psi + A \wedge \Psi - (-1)^p \cdot \Psi \wedge A.$$

Definition 6.4.2 (Curvature). The curvature of the Chern connection is

$$(6.4.1) \quad F_H = \nabla_H^2 = dA + A \wedge A.$$

For matrix-valued forms B and C ,

$$B \wedge C \xrightarrow{1.5.2} (B \wedge C)^\alpha_\beta = \sum_{\gamma=1}^r B^\alpha_\gamma \wedge C^\gamma_\beta.$$

Proposition 6.4.3 (Local curvature formula). *In a holomorphic frame,*

$$F_H = \bar{\partial}A = \sum_{k=1}^n \sum_{j=1}^n F_{H,\bar{k}j} \cdot d\bar{z}^k \wedge dz^j, \quad F_{H,\bar{k}j} = \frac{\partial A_j}{\partial \bar{z}^k}.$$

Writing

$$F_{H,\bar{k}j} = \begin{bmatrix} F_{H,\bar{k}j}^1{}_1 & \cdots & F_{H,\bar{k}j}^1{}_r \\ \vdots & \ddots & \vdots \\ F_{H,\bar{k}j}^r{}_1 & \cdots & F_{H,\bar{k}j}^r{}_r \end{bmatrix}, \quad R_{H,\bar{k}j} = -F_{H,\bar{k}j},$$

one has

$$(\nabla_{H,j} \nabla_{H,\bar{k}} - \nabla_{H,\bar{k}} \nabla_{H,j})(V) = R_{H,\bar{k}j} \cdot V = -F_{H,\bar{k}j} \cdot V.$$

On overlaps,

$$(6.4.2) \quad F_{H,v} = t_{\mu v}^{-1} \cdot F_{H,\mu} \cdot t_{\mu v}.$$

Proof. By $A = H^{-1} \cdot \partial H$, $F_H = dA + A \wedge A$, and $H^{-1} \cdot H = I_r$,

$$H^{-1} \cdot H = I_r \xrightarrow{6.2.3} 0 = \partial(H^{-1} \cdot H) = (\partial H^{-1}) \cdot H + H^{-1} \cdot \partial H \implies \partial H^{-1} = -H^{-1} \cdot \partial H \cdot H^{-1}.$$

$$A = H^{-1} \cdot \partial H \xrightarrow{6.3.1} \partial A = \partial(H^{-1} \cdot \partial H) = (\partial H^{-1}) \wedge \partial H + H^{-1} \cdot \partial^2 H = -H^{-1} \cdot \partial H \cdot H^{-1} \wedge \partial H + H^{-1} \cdot 0 = -A \wedge A.$$

$$F_H = dA + A \wedge A \xrightarrow{6.4.1} F_H = \partial A + \bar{\partial}A + A \wedge A = -A \wedge A + \bar{\partial}A + A \wedge A = \bar{\partial}A.$$

$$\nabla_{H,j}(V) = \frac{\partial V}{\partial z^j} + A_j \cdot V \xrightarrow{6.3.2} \nabla_{H,j} \nabla_{H,\bar{k}}(V) = \frac{\partial^2 V}{\partial z^j \partial \bar{z}^k} + A_j \cdot \frac{\partial V}{\partial \bar{z}^k},$$

$$\nabla_{H,\bar{k}}(V) = \frac{\partial V}{\partial \bar{z}^k} \xrightarrow{6.3.2} \nabla_{H,\bar{k}} \nabla_{H,j}(V) = \frac{\partial^2 V}{\partial \bar{z}^k \partial z^j} + \frac{\partial A_j}{\partial \bar{z}^k} \cdot V + A_j \cdot \frac{\partial V}{\partial \bar{z}^k}.$$

$$(\nabla_{H,j} \nabla_{H,\bar{k}} - \nabla_{H,\bar{k}} \nabla_{H,j})(V) = -\frac{\partial A_j}{\partial \bar{z}^k} \cdot V = -F_{H,\bar{k}j} \cdot V.$$

$$(\nabla_H(V))_\mu = t_{\mu v} \cdot (\nabla_H(V))_v \xrightarrow{6.3.2} (\nabla_H^2(V))_\mu = t_{\mu v} \cdot (\nabla_H^2(V))_v,$$

$$F_{H,\mu} \cdot t_{\mu v} \cdot V_v = t_{\mu v} \cdot F_{H,v} \cdot V_v \xrightarrow{6.1.1} F_{H,v} = t_{\mu v}^{-1} \cdot F_{H,\mu} \cdot t_{\mu v}.$$

Thus F_H is a global $(1,1)$ -form with values in $\text{End}(E)$. □

Theorem 6.4.4 (Second Bianchi identity). *The Chern curvature satisfies*

$$D_H(F_H) = dF_H + A \wedge F_H - F_H \wedge A = 0.$$

Proof. By $F_H = dA + A \wedge A$, $d(\eta \wedge \xi) = d\eta \wedge \xi + (-1)^{\deg(\eta)} \cdot \eta \wedge d\xi$, and $(A \wedge A) \wedge A = A \wedge (A \wedge A)$,

$$F_H = dA + A \wedge A \xrightarrow{6.4.1} dF_H = d(dA) + d(A \wedge A) = 0 + dA \wedge A - A \wedge dA.$$

$$dA = F_H - A \wedge A \xrightarrow{6.4.1} dF_H = (F_H - A \wedge A) \wedge A - A \wedge (F_H - A \wedge A).$$

$$dF_H = F_H \wedge A - A \wedge A \wedge A - A \wedge F_H + A \wedge A \wedge A = F_H \wedge A - A \wedge F_H.$$

$$dF_H + A \wedge F_H - F_H \wedge A = F_H \wedge A - A \wedge F_H + A \wedge F_H - F_H \wedge A = 0 \xrightarrow{6.4.1} D_H(F_H) = 0.$$

Thus the Chern curvature is covariantly closed. $\square$

6.5. Chern–Weil theory, degree, and slope.

Definition 6.5.1 (Determinant bundle and first Chern class). For a rank- r holomorphic vector bundle E , define

$$\det(E) = \Lambda^r E, \quad e_{v,1} \wedge \cdots \wedge e_{v,r} = \det(t_{\mu\nu}) \cdot e_{\mu,1} \wedge \cdots \wedge e_{\mu,r},$$

$$H_{\det(E),\mu} = \det(H_\mu), \quad H_\nu = t_{\mu\nu}^* \cdot H_\mu \cdot t_{\mu\nu} \xrightarrow{6.2.1} \det(H_\nu) = \det(t_{\mu\nu}^*) \cdot \det(H_\mu) \cdot \det(t_{\mu\nu}) = |\det(t_{\mu\nu})|^2 \cdot \det(H_\mu),$$

$$c_1(E) = c_1(\det(E)).$$

Theorem 6.5.2 (Chern–Weil formula for the first Chern class). *The scalar two-form $\text{Tr}(F_H)$ is global and closed, and its de Rham class is independent of H . Moreover,*

$$F_H^* \cdot H + H \cdot F_H = 0, \quad \overline{\text{Tr}(F_H)} = -\text{Tr}(F_H), \quad \overline{i \cdot \text{Tr}(F_H)} = i \cdot \text{Tr}(F_H),$$

and

$$(6.5.1) \quad c_1(E) = \left[\frac{i}{2 \cdot \pi} \cdot \text{Tr}(F_H) \right] \in H^2(X; \mathbb{R}).$$

In the determinant frame,

$$A_{\det(E),\mu} = \text{Tr}(A_\mu) = \partial \log(\det(H_\mu)),$$

$$F_{\det(E)}|_{X_\mu} = \text{Tr}(F_{H,\mu}) = -\partial \bar{\partial} \log(\det(H_\mu)).$$

Proof. By $F_{H,\nu} = t_{\mu\nu}^{-1} \cdot F_{H,\mu} \cdot t_{\mu\nu}$, $D_H(F_H) = 0$, and $\text{Tr}(P \wedge Q) = (-1)^{p \cdot q} \cdot \text{Tr}(Q \wedge P)$ for $(P, Q) \in \Omega^p(X, \text{End}(E)) \times \Omega^q(X, \text{End}(E))$,

$$F_{H,\nu} = t_{\mu\nu}^{-1} \cdot F_{H,\mu} \cdot t_{\mu\nu} \xrightarrow{6.4.2} \text{Tr}(F_{H,\nu}) = \text{Tr}(t_{\mu\nu}^{-1} \cdot F_{H,\mu} \cdot t_{\mu\nu}) = \text{Tr}(F_{H,\mu} \cdot t_{\mu\nu} \cdot t_{\mu\nu}^{-1}) = \text{Tr}(F_{H,\mu}).$$

$$d\text{Tr}(F_H) = \text{Tr}(dF_H) \xrightarrow{6.4.4} -\text{Tr}(A \wedge F_H) + \text{Tr}(F_H \wedge A) = -\text{Tr}(A \wedge F_H) + \text{Tr}(A \wedge F_H) = 0.$$

$$\nabla_H(H) = 0 \xrightarrow{6.3.1} 0 = d^2(\langle V, W \rangle_H) = W^* \cdot (F_H^* \cdot H + H \cdot F_H) \cdot V \implies F_H^* \cdot H + H \cdot F_H = 0.$$

$$F_H^* = -H \cdot F_H \cdot H^{-1} \implies \overline{\text{Tr}(F_H)} = \text{Tr}(F_H^*) = -\text{Tr}(H \cdot F_H \cdot H^{-1}) = -\text{Tr}(F_H).$$

For metric independence, let $B = \tilde{A} - A \in \Omega^1(X, \text{End}(E))$ and $A_t = A + t \cdot B$.

$$B = \tilde{A} - A \xrightarrow{6.3.2} B_\nu = t_{\mu\nu}^{-1} \cdot B_\mu \cdot t_{\mu\nu} \implies \text{Tr}(B_\nu) = \text{Tr}(B_\mu), \quad F_t = dA_t + A_t \wedge A_t.$$

$$F_t = dA_t + A_t \wedge A_t \xrightarrow{6.4.1} \frac{dF_t}{dt} = dB + B \wedge A_t + A_t \wedge B,$$

$$\frac{d}{dt} \text{Tr}(F_t) = d\text{Tr}(B) + \text{Tr}(B \wedge A_t) + \text{Tr}(A_t \wedge B) = d\text{Tr}(B) + \text{Tr}(B \wedge A_t) - \text{Tr}(B \wedge A_t) = d\text{Tr}(B).$$

$$\text{Tr}(\tilde{F}) - \text{Tr}(F_H) = \int_0^1 \frac{d}{dt} \text{Tr}(F_t) \cdot dt = d \left(\text{Tr}(B) \cdot \int_0^1 1 \cdot dt \right) = d(\text{Tr}(B) \cdot 1) = d\text{Tr}(B).$$

For the determinant bundle, Jacobi's formula and the connection gauge law give

$$A_\mu = H_\mu^{-1} \cdot \partial H_\mu \xrightarrow{6.3.1} \text{Tr}(A_\mu) = \text{Tr}(H_\mu^{-1} \cdot \partial H_\mu) \xrightarrow{5.2.1} \partial \log(\det(H_\mu)),$$

$$A_\nu = t_{\mu\nu}^{-1} \cdot A_\mu \cdot t_{\mu\nu} + t_{\mu\nu}^{-1} \cdot \partial t_{\mu\nu} \xrightarrow{6.3.2} \text{Tr}(A_\nu) = \text{Tr}(A_\mu) + \det(t_{\mu\nu})^{-1} \cdot \partial \det(t_{\mu\nu}) = \text{Tr}(A_\mu) + \partial \log(\det(t_{\mu\nu})).$$

$$\begin{aligned}
A_\mu \wedge A_\mu &\xrightarrow{1.5.2} \text{Tr}(A_\mu \wedge A_\mu) = \sum_{\alpha=1}^r A_\mu^\alpha{}_\alpha \wedge A_\mu^\alpha{}_\alpha \\
&+ \sum_{\alpha=1}^{r-1} \sum_{\beta=\alpha+1}^r (A_\mu^\alpha{}_\beta \wedge A_\mu^\beta{}_\alpha + A_\mu^\beta{}_\alpha \wedge A_\mu^\alpha{}_\beta) = 0 + 0 = 0. \\
(A_{\det(E)} = \text{Tr}(A_\mu), \text{Tr}(A_\mu \wedge A_\mu) = 0) &\xrightarrow{6.4.1} F_{\det(E)} = d\text{Tr}(A_\mu). \\
F_{\det(E)} = d\text{Tr}(A_\mu) = \text{Tr}(dA_\mu + A_\mu \wedge A_\mu) = \text{Tr}(F_{H,\mu}) &\xrightarrow{2.1.3} F_{\det(E)} = -\partial\bar{\partial} \log(\det(H_\mu)). \\
(c_1(E) = c_1(\det(E)), F_{\det(E)} = \text{Tr}(F_H)) &\xrightarrow{2.1.3} c_1(E) = \left[\frac{i}{2 \cdot \pi} \cdot \text{Tr}(F_H) \right].
\end{aligned}$$

Thus $\frac{i}{2 \cdot \pi} \cdot \text{Tr}(F_H)$ is the real, metric-independent Chern–Weil representative. $\square$

Definition 6.5.3 (Degree and slope). Let $E \xrightarrow{\pi} (X^n, \omega)$ be a rank- r holomorphic vector bundle over a compact Kähler manifold. Define

$$\begin{aligned}
\deg_\omega(E) &= \int_X c_1(E) \wedge \frac{\omega^{n-1}}{(n-1)!} \xrightarrow{6.5.2} \frac{i}{2 \cdot \pi} \cdot \int_X \text{Tr}(F_H) \wedge \frac{\omega^{n-1}}{(n-1)!}, \\
\mu_\omega(E) &= \frac{\deg_\omega(E)}{\text{rank}(E)} = \frac{\deg_\omega(E)}{r}. \\
\left(c_1(E) = \left[\frac{i}{2 \cdot \pi} \cdot \text{Tr}(F_H) \right], \deg_\omega(E) = \int_X c_1(E) \wedge \frac{\omega^{n-1}}{(n-1)!} \right) &\xrightarrow{6.5.2} \deg_\omega(E) \text{ is independent of } H.
\end{aligned}$$

6.6. The Hermitian–Einstein equation. For

$$\omega = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n g_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k,$$

the Lefschetz contraction is

$$(6.6.1) \quad \Lambda_\omega = 2 \cdot i \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \iota_{\partial/\partial z^j} \circ \iota_{\partial/\partial \bar{z}^k}.$$

$$F_H = \bar{\partial}A \xrightarrow{6.4.3} F_H = \sum_{k=1}^n \sum_{j=1}^n F_{H,\bar{k}j} \cdot d\bar{z}^k \wedge dz^j \xrightarrow{6.6.1} i \cdot \Lambda_\omega(F_H) = -2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot F_{H,\bar{k}j}.$$

Definition 6.6.1 (Hermitian–Einstein metric). A Hermitian metric H is Hermitian–Einstein when there is $\lambda \in \mathbb{R}$ such that

$$(6.6.2) \quad i \cdot \Lambda_\omega(F_H) = \lambda \cdot I_E.$$

Equivalently,

$$\begin{aligned}
i \cdot \Lambda_\omega(F_H) &= \lambda \cdot I_E \xrightarrow{6.6.1} -2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot F_{H,\bar{k}j}^\alpha{}_\beta = \lambda \cdot \delta^\alpha{}_\beta, \\
-2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \begin{bmatrix} F_{H,\bar{k}j}^{11} & \cdots & F_{H,\bar{k}j}^{1r} \\ \vdots & \ddots & \vdots \\ F_{H,\bar{k}j}^{r1} & \cdots & F_{H,\bar{k}j}^{rr} \end{bmatrix} &= \lambda \cdot \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.
\end{aligned}$$

Example 6.6.2 (A complete rank-two Chern computation). On the trivial rank-two bundle over $\mathbb{C}$, take

$$H = \begin{bmatrix} e^{-2 \cdot |z|^2} & 0 \\ 0 & e^{-2 \cdot |z|^2} \end{bmatrix}, \quad H^{-1} = \begin{bmatrix} e^{2 \cdot |z|^2} & 0 \\ 0 & e^{2 \cdot |z|^2} \end{bmatrix},$$

$$\partial H = \begin{bmatrix} -2 \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} & 0 \\ 0 & -2 \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} \end{bmatrix} \cdot dz.$$

The full product is

$$(H^{-1}, \partial H) \xrightarrow{6.3.1} A = \begin{bmatrix} e^{2 \cdot |z|^2} & 0 \\ 0 & e^{2 \cdot |z|^2} \end{bmatrix} \cdot \begin{bmatrix} -2 \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} & 0 \\ 0 & -2 \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} \end{bmatrix} \cdot dz,$$

$$A = \begin{bmatrix} e^{2 \cdot |z|^2} \cdot (-2) \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} + 0 \cdot 0 & e^{2 \cdot |z|^2} \cdot 0 + 0 \cdot (-2) \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} \\ 0 \cdot (-2) \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} + e^{2 \cdot |z|^2} \cdot 0 & 0 \cdot 0 + e^{2 \cdot |z|^2} \cdot (-2) \cdot \bar{z} \cdot e^{-2 \cdot |z|^2} \end{bmatrix} \cdot dz = \begin{bmatrix} -2 \cdot \bar{z} & 0 \\ 0 & -2 \cdot \bar{z} \end{bmatrix} \cdot dz.$$

Consequently,

$$A = H^{-1} \cdot \partial H \xrightarrow{6.4.3} F_H = \bar{\partial} A = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \cdot d\bar{z} \wedge dz, \quad \text{Tr}(F_H) = (-2 - 2) \cdot d\bar{z} \wedge dz = -4 \cdot d\bar{z} \wedge dz.$$

For $\omega = \frac{i}{2} \cdot dz \wedge d\bar{z}$,

$$F_H = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \cdot d\bar{z} \wedge dz \xrightarrow{6.6.1} i \cdot \Lambda_\omega(F_H) = -2 \cdot \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}.$$

$$-2 \cdot \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} (-2) \cdot (-2) & (-2) \cdot 0 \\ (-2) \cdot 0 & (-2) \cdot (-2) \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = 4 \cdot I_2 \xrightarrow{6.6.1} (H \text{ is Hermitian-Einstein, } \lambda = 4).$$

The determinant and trace calculations give the same scalar curvature

$$\det(H) = e^{-2 \cdot |z|^2} \cdot e^{-2 \cdot |z|^2} - 0 \cdot 0 = e^{-4 \cdot |z|^2}, \quad \log(\det(H)) = -4 \cdot |z|^2,$$

$$(A = H^{-1} \cdot \partial H, \det(H) = e^{-4 \cdot |z|^2}) \xrightarrow{5.2.1} \text{Tr}(A) = ((-2) \cdot \bar{z} + (-2) \cdot \bar{z}) \cdot dz = -4 \cdot \bar{z} \cdot dz = \partial \log(\det(H)),$$

$$F_{\det(E)} = \text{Tr}(F_H) \xrightarrow{6.5.2} \text{Tr}(F_H) = ((-2) + (-2)) \cdot d\bar{z} \wedge dz = -4 \cdot d\bar{z} \wedge dz = -\partial \bar{\partial} \log(\det(H)).$$

Lemma 6.6.3 (Trace-wedge identity). *The Chern curvature satisfies*

$$i \cdot \text{Tr}(F_H) \wedge \frac{\omega^{n-1}}{(n-1)!} = \text{Tr}(i \cdot \Lambda_\omega(F_H)) \cdot \frac{\omega^n}{n!}.$$

Proof. By $p \in X$, $g_{j\bar{k}}(p) = \delta_{jk}$, and $\alpha_j = \frac{i}{2} \cdot dz^j \wedge d\bar{z}^j$,

$$g_{j\bar{k}}(p) = \delta_{jk} \xrightarrow{1.4.2} \omega(p) = \sum_{j=1}^n \alpha_j, \quad \frac{\omega^n}{n!}(p) = \bigwedge_{j=1}^n \alpha_j, \quad \frac{\omega^{n-1}}{(n-1)!}(p) = \sum_{j=1}^n \bigwedge_{\substack{1 \leq \ell \leq n \\ \ell \neq j}} \alpha_\ell.$$

$$d\bar{z}^j \wedge dz^j = -dz^j \wedge d\bar{z}^j = 2 \cdot i \cdot \alpha_j.$$

$$i \cdot \text{Tr}(F_H) \wedge \frac{\omega^{n-1}}{(n-1)!}(p) = i \cdot \sum_{j=1}^n \text{Tr}(F_{H, \bar{j}j}) \cdot 2 \cdot i \cdot \bigwedge_{\ell=1}^n \alpha_\ell = -2 \cdot \sum_{j=1}^n \text{Tr}(F_{H, \bar{j}j}) \cdot \frac{\omega^n}{n!}(p).$$

$$i \cdot \Lambda_\omega(F_H)(p) = -2 \cdot \sum_{j=1}^n F_{H, \bar{j}j} \xrightarrow{6.6.1} \text{Tr}(i \cdot \Lambda_\omega(F_H))(p) = -2 \cdot \sum_{j=1}^n \text{Tr}(F_{H, \bar{j}j}).$$

Thus the two global top-degree forms agree at every $p \in X$. $\square$

Corollary 6.6.4 (The Einstein constant). *If H is Hermitian–Einstein, then*

$$\lambda = \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)}, \quad \text{Vol}_\omega(X) = \int_X \frac{\omega^n}{n!}.$$

Proof. By $i \cdot \Lambda_\omega(F_H) = \lambda \cdot I_E$, $\text{rank}(E) = r$, and $i \cdot \text{Tr}(F_H) \wedge \omega^{n-1}/(n-1)! = \text{Tr}(i \cdot \Lambda_\omega(F_H)) \cdot \omega^n/n!$,

$$\deg_\omega(E) \xrightarrow{6.5.3} \frac{i}{2 \cdot \pi} \cdot \int_X \text{Tr}(F_H) \wedge \frac{\omega^{n-1}}{(n-1)!} \xrightarrow{6.6.3} \frac{1}{2 \cdot \pi} \cdot \int_X \text{Tr}(i \cdot \Lambda_\omega(F_H)) \cdot \frac{\omega^n}{n!}.$$

$$i \cdot \Lambda_\omega(F_H) = \lambda \cdot I_E \xrightarrow{6.6.1} \text{Tr}(i \cdot \Lambda_\omega(F_H)) = \text{Tr}(\lambda \cdot I_r) = \sum_{\alpha=1}^r \lambda \cdot 1 = \lambda \cdot r.$$

$$\deg_\omega(E) = \frac{\lambda \cdot r}{2 \cdot \pi} \cdot \int_X \frac{\omega^n}{n!} = \frac{\lambda \cdot r}{2 \cdot \pi} \cdot \text{Vol}_\omega(X).$$

$$\lambda = \frac{2 \cdot \pi \cdot \deg_\omega(E)}{r \cdot \text{Vol}_\omega(X)} \xrightarrow{6.5.3} \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)}.$$

Thus the Kähler slope and volume determine the Einstein constant. $\square$

7. SLOPE STABILITY AND CURVATURE OF HOLOMORPHIC SUBBUNDLES

Let (X, ω) be a compact connected Kähler manifold and let (E, H) be a holomorphic Hermitian vector bundle. The Hermitian–Einstein equation and its topological constant satisfy, whenever H is Hermitian–Einstein,

$$H \text{ is Hermitian–Einstein} \xrightarrow{6.6.1} i \cdot \Lambda_\omega(F_H) = \lambda \cdot I_E \xrightarrow{6.6.4} \lambda = \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)}.$$

The comparison with a holomorphic subbundle is controlled by the square norm of its second fundamental form.

7.1. Slope stability and the Donaldson–Uhlenbeck–Yau correspondence.

Definition 7.1.1 (Degree and slope of a coherent sheaf). For a torsion-free coherent sheaf $\mathcal{S}$ of rank s , define

$$\det(\mathcal{S}) = \left(\bigwedge^s(\mathcal{S}) \right)^{**}, \quad \deg_\omega(\mathcal{S}) = \int_X c_1(\det(\mathcal{S})) \wedge \frac{\omega^{n-1}}{(n-1)!}, \quad \mu_\omega(\mathcal{S}) = \frac{\deg_\omega(\mathcal{S})}{s}.$$

It is enough to test slope inequalities on saturated coherent subsheaves.

Fact 7.1.2 (Reduction to saturated subsheaves). *Let $\widetilde{\mathcal{S}} \subset E$ be the saturation of a torsion-free coherent subsheaf $\mathcal{S} \subset E$. Then*

$$\text{rank}(\widetilde{\mathcal{S}}) = \text{rank}(\mathcal{S}), \quad \deg_\omega(\widetilde{\mathcal{S}}) \geq \deg_\omega(\mathcal{S}), \quad \mu_\omega(\widetilde{\mathcal{S}}) \geq \mu_\omega(\mathcal{S}).$$

Proof. By $\mathcal{S} \subset \widetilde{\mathcal{S}} \subset E$, $\mathcal{T} = \widetilde{\mathcal{S}}/\mathcal{S}$ torsion, and $D = \sum_{a=1}^N m_a \cdot D_a$ effective,

$$0 \longrightarrow \mathcal{S} \longrightarrow \widetilde{\mathcal{S}} \longrightarrow \mathcal{T} \longrightarrow 0 \implies \text{rank}(\widetilde{\mathcal{S}}) = \text{rank}(\mathcal{S}).$$

$$\det(\widetilde{\mathcal{S}}) = \det(\mathcal{S}) \otimes \mathcal{O}_X(D), \quad D = \sum_{a=1}^N m_a \cdot D_a, \quad m_a \geq 0.$$

$$\left(\det(\widetilde{\mathcal{S}}) = \det(\mathcal{S}) \otimes \mathcal{O}_X(D) \right) \implies c_1(\det(\widetilde{\mathcal{S}})) = c_1(\det(\mathcal{S})) + \sum_{a=1}^N m_a \cdot [D_a].$$

$$\left(c_1(\det(\widetilde{\mathcal{S}})) = c_1(\det(\mathcal{S})) + \sum_{a=1}^N m_a \cdot [D_a] \right) \xrightarrow{7.1.1} \deg_\omega(\widetilde{\mathcal{S}}) = \deg_\omega(\mathcal{S}) + \sum_{a=1}^N m_a \cdot \int_{D_a} \frac{\omega^{n-1}}{(n-1)!} \geq \deg_\omega(\mathcal{S}).$$

$$\left(\text{rank}(\widetilde{\mathcal{S}}) = \text{rank}(\mathcal{S}), \quad \deg_{\omega}(\widetilde{\mathcal{S}}) \geq \deg_{\omega}(\mathcal{S}) \right) \xrightarrow{7.1.1} \mu_{\omega}(\widetilde{\mathcal{S}}) \geq \mu_{\omega}(\mathcal{S}).$$

Thus saturation cannot decrease slope. $\square$

Definition 7.1.3 (Slope stability). A holomorphic vector bundle E is stable when every coherent subsheaf $\mathcal{S} \subset E$ with

$$0 < \text{rank}(\mathcal{S}) < \text{rank}(E)$$

satisfies

$$\mu_{\omega}(\mathcal{S}) < \mu_{\omega}(E), \quad \mu_{\omega}(\mathcal{S}) = \frac{\deg_{\omega}(\mathcal{S})}{\text{rank}(\mathcal{S})}.$$

It is semistable when the strict inequality is replaced by $\mu_{\omega}(\mathcal{S}) \leq \mu_{\omega}(E)$. It is polystable when

$$E = \bigoplus_{a=1}^N E_a, \quad E_a \text{ is stable}, \quad \mu_{\omega}(E_a) = \mu_{\omega}(E) \quad (1 \leq a \leq N).$$

Theorem 7.1.4 (Donaldson–Uhlenbeck–Yau). *The bundle E admits a Hermitian–Einstein metric if and only if E is polystable. If E is stable, its Hermitian–Einstein metric is unique up to multiplication by a positive constant.*

Remark 7.1.5. The two directions are

$$E \text{ admits a Hermitian–Einstein metric} \xrightarrow{7.5.4} E \text{ is polystable.}$$

$$E \text{ is polystable} \xrightarrow{8.5.3} E \text{ admits a Hermitian–Einstein metric.}$$

7.2. Orthogonal splitting and second fundamental forms. Let $S \subset E$ be a holomorphic subbundle, let $T = S^{\perp_H}$, let $S \xrightarrow{is} E$ be the inclusion, and let π_S and π_T denote the orthogonal projections. Thus

$$E = S \oplus T, \quad I_E = \pi_S + \pi_T, \quad \pi_S \cdot \pi_T = \pi_T \cdot \pi_S = 0.$$

Definition 7.2.1 (Induced connections and second fundamental forms). For $s \in \Gamma(X, S)$ and $t \in \Gamma(X, T)$, define

$$\begin{aligned} \nabla^S(s) &= \pi_S(\nabla_H(s)), & \beta(s) &= \pi_T(\nabla_H(s)), \\ \nabla^T(t) &= \pi_T(\nabla_H(t)), & \gamma(t) &= \pi_S(\nabla_H(t)). \end{aligned}$$

The holomorphicity of S and the unitarity of ∇_H give

$$\left(\nabla_H^{0,1} = \bar{\partial}_E, \quad \bar{\partial}_E(\Gamma(X, S)) \subset \Omega^{0,1}(X, S) \right) \xrightarrow{6.3.1} \pi_T \circ \nabla_H^{0,1}|_S = 0 \xrightarrow{7.2.1} \beta \in \Omega^{1,0}(X, \text{Hom}(S, T)),$$

$$\nabla_H(H) = 0 \xrightarrow{6.3.1} \gamma = -\beta^{*H} \xrightarrow{7.2.1} \gamma \in \Omega^{0,1}(X, \text{Hom}(T, S)).$$

Fact 7.2.2 (Leibniz rules). *The operator ∇^S is a connection on S , whereas β is tensorial.*

Proof. By $f \in C^\infty(X)$, $s \in \Gamma(X, S)$, $V \in \Gamma(X, T_{\mathbb{C}}X)$, $\pi_S(s) = s$, and $\pi_T(s) = 0$,

$$\nabla_H(f \cdot s) \xrightarrow{6.3.1} df \otimes s + f \cdot \nabla_H(s).$$

$$\nabla_H(f \cdot s) = df \otimes s + f \cdot \nabla_H(s) \xrightarrow{7.2.1} \nabla_V^S(f \cdot s) = \pi_S(V(f) \cdot s + f \cdot \nabla_{H,V}(s)) = V(f) \cdot s + f \cdot \nabla_V^S(s).$$

$$\nabla_H(f \cdot s) = df \otimes s + f \cdot \nabla_H(s) \xrightarrow{7.2.1} \beta(f \cdot s) = \pi_T(df \otimes s + f \cdot \nabla_H(s)) = df \otimes 0 + f \cdot \beta(s) = f \cdot \beta(s).$$

Thus ∇^S obeys the Leibniz rule and β is $C^\infty(X)$ -linear. $\square$

Fact 7.2.3 (Block form of the connection). *For*

$$\Phi = s + t = \begin{bmatrix} s \\ t \end{bmatrix},$$

the Chern connection has the block form

$$\nabla_H(\Phi) = \begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix} \cdot \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} \nabla^S(s) + \gamma(t) \\ \beta(s) + \nabla^T(t) \end{bmatrix}, \quad \gamma = -\beta^{*H}.$$

Equivalently,

$$\nabla_{H,j} = \begin{bmatrix} \nabla_j^S & 0 \\ \beta_j & \nabla_j^T \end{bmatrix}, \quad \nabla_{H,\bar{k}} = \begin{bmatrix} \nabla_{\bar{k}}^S & \gamma_{\bar{k}} \\ 0 & \nabla_{\bar{k}}^T \end{bmatrix}.$$

Proof. By $\Phi = s + t$, $I_E = \pi_S + \pi_T$, $\nabla^S = \pi_S \circ \nabla_H$, and $\beta = \pi_T \circ \nabla_H|_S$,

$$\nabla_H(\Phi) = \nabla_H(s) + \nabla_H(t) = \nabla^S(s) + \beta(s) + \gamma(t) + \nabla^T(t).$$

$$(\nabla_H(s) = \nabla^S(s) + \beta(s), \quad \nabla_H(t) = \gamma(t) + \nabla^T(t)) \xrightarrow{7.2.1} \nabla_H(\Phi) = \begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix} \cdot \begin{bmatrix} s \\ t \end{bmatrix}.$$

$$\begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix} \cdot \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} \nabla^S(s) + \gamma(t) \\ \beta(s) + \nabla^T(t) \end{bmatrix}.$$

$$\bar{\partial}_E(s) \in \Omega^{0,1}(X, S) \xrightarrow{6.3.1} \pi_T(\nabla_H^{0,1}(s)) = 0 \implies \beta^{0,1} = 0.$$

By $\langle s, t \rangle_H = 0$ and the unitarity of ∇_H ,

$$\nabla_H(H) = 0 \xrightarrow{6.3.1} 0 = d(\langle s, t \rangle_H) = \langle \beta(s), t \rangle_H + \langle s, \gamma(t) \rangle_H = \langle s, (\beta^{*H} + \gamma)(t) \rangle_H.$$

$$\langle s, (\beta^{*H} + \gamma)(t) \rangle_H = 0 \xrightarrow{6.2.3} \gamma = -\beta^{*H}, \quad \beta \in \Omega^{1,0}(X, \text{Hom}(S, T)) \xrightarrow{7.2.1} \gamma \in \Omega^{0,1}(X, \text{Hom}(T, S)).$$

Thus the connection and its two types have the stated block form. $\square$

Fact 7.2.4 (The quotient as the orthogonal complement). *For $Q = E/S$ and the quotient map p , the restriction*

$$T \xrightarrow{p|_T} Q$$

is a smooth vector-bundle isomorphism.

Proof. By $E = S \oplus T$, $T = S^{\perp_H}$, and $e = s + t$,

$$[e] = [s + t] = [t] \implies (p|_T)(t) = [e],$$

so $p|_T$ is surjective. Moreover,

$$(p|_T)(t) = 0 \implies [t] = 0 \implies t \in S \cap T \implies t = 0,$$

so $p|_T$ is injective and hence an isomorphism. $\square$

7.3. The Dolbeault class of an extension.

Theorem 7.3.1 (Dolbeault classification of extensions). *For holomorphic vector bundles S and Q , equivalence classes of holomorphic extensions satisfy*

$$\text{Ext}_{\mathcal{O}_X}^1(Q, S) \cong H_{\bar{\partial}}^{0,1}(X, \text{Hom}(Q, S)).$$

For $0 \longrightarrow S \longrightarrow E \longrightarrow Q \longrightarrow 0$, the corresponding class is $[\gamma]$, and it is independent of the Hermitian metric used to choose $T = S^{\perp_H}$.

Proof. By $E \cong S \oplus Q$ smoothly, $\gamma \in \Omega^{0,1}(X, \text{Hom}(Q, S))$, and $\bar{\partial}_E^2 = 0$,

$$E \cong S \oplus Q \xrightarrow{7.2.4} Q \cong T \xrightarrow{7.2.3} \bar{\partial}_E = \begin{bmatrix} \bar{\partial}_S & \gamma \\ 0 & \bar{\partial}_Q \end{bmatrix}, \quad \bar{\partial}_E^2 = \begin{bmatrix} \bar{\partial}_S^2 & \bar{\partial}_{\text{Hom}(Q,S)}(\gamma) \\ 0 & \bar{\partial}_Q^2 \end{bmatrix}.$$

$$\bar{\partial}_E^2 = 0 \implies \begin{bmatrix} 0 & \bar{\partial}_{\text{Hom}(Q,S)}(\gamma) \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \implies \bar{\partial}_{\text{Hom}(Q,S)}(\gamma) = 0.$$

For a change of smooth splitting by $u \in \Omega^0(X, \text{Hom}(Q, S))$,

$$P_u = \begin{bmatrix} I_S & u \\ 0 & I_Q \end{bmatrix}, \quad P_u^{-1} = \begin{bmatrix} I_S & -u \\ 0 & I_Q \end{bmatrix},$$

$$P_u^{-1} \cdot \bar{\partial}_E \cdot P_u = \begin{bmatrix} I_S & -u \\ 0 & I_Q \end{bmatrix} \cdot \begin{bmatrix} \bar{\partial}_S & \bar{\partial}_S \circ u + \gamma \\ 0 & \bar{\partial}_Q \end{bmatrix} = \begin{bmatrix} \bar{\partial}_S & \gamma + \bar{\partial}_{\text{Hom}(Q,S)}(u) \\ 0 & \bar{\partial}_Q \end{bmatrix}.$$

$$\gamma' = \gamma + \bar{\partial}_{\text{Hom}(Q,S)}(u) \implies [\gamma'] = [\gamma].$$

Conversely,

$$\bar{\partial}_{\text{Hom}(Q,S)}(\gamma) = 0 \implies \bar{\partial}_E^2 = 0,$$

and exact changes give equivalent extensions. □

7.4. Curvature of a holomorphic subbundle.

Proposition 7.4.1 (Curvature block matrix). *With respect to $E = S \oplus T$,*

$$(7.4.1) \quad F_E = \begin{bmatrix} F_S + \gamma \wedge \beta & \partial_{\nabla}(\gamma) \\ \bar{\partial}_{\nabla}(\beta) & F_T + \beta \wedge \gamma \end{bmatrix}, \quad \gamma = -\beta^{*H}.$$

In the established $d\bar{z}^k \wedge dz^j$ coefficient order,

$$(F_S)_{\bar{k}j} = \pi_S \cdot (F_E)_{\bar{k}j} \cdot \iota_S - \gamma_{\bar{k}} \cdot \beta_j = \pi_S \cdot (F_E)_{\bar{k}j} \cdot \iota_S + \beta_k^{*H} \cdot \beta_j.$$

Proof. By $F_E = \nabla_H^2$, $\gamma = -\beta^{*H}$, and $\nabla_H = \begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix}$,

$$\nabla_H = \begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix} \xrightarrow{6.4.2} \nabla_H^2 = \begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix} \wedge \begin{bmatrix} \nabla^S & \gamma \\ \beta & \nabla^T \end{bmatrix}.$$

$$(\nabla_H^2)_{SS} \xrightarrow{1.5.2} \nabla^S \wedge \nabla^S + \gamma \wedge \beta, \quad (\nabla_H^2)_{ST} \xrightarrow{1.5.2} \nabla^S \wedge \gamma + \gamma \wedge \nabla^T.$$

$$(\nabla_H^2)_{TS} \xrightarrow{1.5.2} \beta \wedge \nabla^S + \nabla^T \wedge \beta, \quad (\nabla_H^2)_{TT} \xrightarrow{1.5.2} \beta \wedge \gamma + \nabla^T \wedge \nabla^T.$$

$$\nabla_H^2 = \begin{bmatrix} \nabla^S \wedge \nabla^S + \gamma \wedge \beta & \nabla^S \wedge \gamma + \gamma \wedge \nabla^T \\ \beta \wedge \nabla^S + \nabla^T \wedge \beta & \beta \wedge \gamma + \nabla^T \wedge \nabla^T \end{bmatrix} \xrightarrow{6.4.2} F_E = \begin{bmatrix} F_S + \gamma \wedge \beta & \partial_{\nabla}(\gamma) \\ \bar{\partial}_{\nabla}(\beta) & F_T + \beta \wedge \gamma \end{bmatrix}.$$

$$(F_E)_{SS} = F_S + \gamma \wedge \beta \implies F_S = (F_E)_{SS} - \gamma \wedge \beta.$$

$$(\beta \in \Omega^{1,0}(X, \text{Hom}(S, T)), \quad \gamma \in \Omega^{0,1}(X, \text{Hom}(T, S))) \xrightarrow{7.2.1} \beta = \sum_{j=1}^n \beta_j \cdot dz^j, \quad \gamma = \sum_{k=1}^n \gamma_{\bar{k}} \cdot d\bar{z}^k.$$

$$\left(\beta = \sum_{j=1}^n \beta_j \cdot dz^j, \quad \gamma = \sum_{k=1}^n \gamma_{\bar{k}} \cdot d\bar{z}^k \right) \xrightarrow{1.5.2} \gamma \wedge \beta = \sum_{k=1}^n \sum_{j=1}^n \gamma_{\bar{k}} \cdot \beta_j \cdot d\bar{z}^k \wedge dz^j.$$

$$(F_S)_{\bar{k}j} = (\pi_S \cdot (F_E)_{\bar{k}j} \cdot \iota_S) - \gamma_{\bar{k}} \cdot \beta_j \xrightarrow{7.2.1} \pi_S \cdot (F_E)_{\bar{k}j} \cdot \iota_S + \beta_k^{*H} \cdot \beta_j.$$

Thus the coefficient sign agrees with the curvature convention of the preceding section. □

Example 7.4.2 (Complete rank-two block-curvature computation). Let

$$X = \mathbb{C}, \quad \omega = \frac{i}{2} \cdot dz \wedge d\bar{z}, \quad E = X \times \mathbb{C}^2 = S \oplus T,$$

$$S = \text{span}_{\mathbb{C}}(e_1), \quad T = \text{span}_{\mathbb{C}}(e_2).$$

In an adapted H -unitary frame, take

$$\beta = 2 \cdot dz, \quad \gamma = -2 \cdot d\bar{z}, \quad \mathcal{B} = \begin{bmatrix} 0 & -2 \cdot d\bar{z} \\ 2 \cdot dz & 0 \end{bmatrix}.$$

$$\nabla_E = d + \mathcal{B}, \quad \nabla_E^{0,1}(e_1) = \bar{\partial} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) + \begin{bmatrix} 0 & -2 \cdot d\bar{z} \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

$$\nabla_E^{0,1}(e_1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \cdot 1 + (-2 \cdot d\bar{z}) \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \xrightarrow{6.2.1} S \text{ is holomorphic,}$$

and $d(\mathcal{B}) = 0$. Hence

$$\nabla_E = d + \mathcal{B} \xrightarrow{6.4.2} F_E = d(\mathcal{B}) + \mathcal{B} \wedge \mathcal{B} = \mathcal{B} \wedge \mathcal{B} = \begin{bmatrix} 0 & -2 \cdot d\bar{z} \\ 2 \cdot dz & 0 \end{bmatrix} \wedge \begin{bmatrix} 0 & -2 \cdot d\bar{z} \\ 2 \cdot dz & 0 \end{bmatrix},$$

$$\mathcal{B} \wedge \mathcal{B} \xrightarrow{1.5.2} F_E = \begin{bmatrix} 0 \wedge 0 + (-2 \cdot d\bar{z}) \wedge (2 \cdot dz) & 0 \wedge (-2 \cdot d\bar{z}) + (-2 \cdot d\bar{z}) \wedge 0 \\ (2 \cdot dz) \wedge 0 + 0 \wedge (2 \cdot dz) & (2 \cdot dz) \wedge (-2 \cdot d\bar{z}) + 0 \wedge 0 \end{bmatrix},$$

$$F_E = \begin{bmatrix} (-2) \cdot 2 \cdot d\bar{z} \wedge dz & 0 \\ 0 & 2 \cdot (-2) \cdot dz \wedge d\bar{z} \end{bmatrix} = \begin{bmatrix} -4 \cdot d\bar{z} \wedge dz & 0 \\ 0 & 4 \cdot dz \wedge d\bar{z} \end{bmatrix}.$$

$$F_S = F_T = 0, \quad \gamma_{\bar{z}} \cdot \beta_z = (-2) \cdot 2 = -4,$$

$$\pi_S \cdot (F_E)_{\bar{z}z} \cdot \iota_S = \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -4 & 0 \\ 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot (-4) + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (-4) \cdot 1 + 0 \cdot 0 = -4.$$

$$(\pi_S \cdot (F_E)_{\bar{z}z} \cdot \iota_S = -4, \quad \gamma_{\bar{z}} \cdot \beta_z = -4) \xrightarrow{7.4.1} (F_S)_{\bar{z}z} = \pi_S \cdot (F_E)_{\bar{z}z} \cdot \iota_S - \gamma_{\bar{z}} \cdot \beta_z = -4 - ((-2) \cdot 2) = -4 - (-4) = 0.$$

$$\beta_z = 2 \xrightarrow{7.5.1} |\beta|_{H,\omega}^2 = 2 \cdot g^{z\bar{z}} \cdot \text{Tr}(\beta_z^{*H} \cdot \beta_z) = 2 \cdot 1 \cdot (2 \cdot 2) = 2 \cdot 4 = 8,$$

$$((F_E)_{\bar{z}z,SS} = -4) \xrightarrow{6.6.1} \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) = -2 \cdot (-4) = 8.$$

$$(\text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) = 8, \quad |\beta|_{H,\omega}^2 = 8) \xrightarrow{7.5.2} \text{Tr}(i \cdot \Lambda_\omega(F_S)) = -2 \cdot (-4) - 8 = 8 - 8 = 0.$$

Thus the full block multiplication reproduces the vanishing induced curvature.

7.5. The Hermitian–Einstein slope inequality. Define

$$(7.5.1) \quad |\beta|_{H,\omega}^2 = 2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}(\beta_k^{*H} \cdot \beta_j) \geq 0, \quad \|\beta\|_{L^2(H,\omega)}^2 = \int_X |\beta|_{H,\omega}^2 \cdot \frac{\omega^n}{n!}.$$

$$(7.5.2) \quad ((F_S)_{\bar{k}j} = \pi_S \cdot (F_E)_{\bar{k}j} \cdot \iota_S + \beta_k^{*H} \cdot \beta_j) \xrightarrow{6.6.1} \text{Tr}(i \cdot \Lambda_\omega(F_S)) = \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) - |\beta|_{H,\omega}^2.$$

Corollary 7.5.1 (Hermitian–Einstein slope inequality). *If H is Hermitian–Einstein and $S \subset E$ is a holomorphic subbundle, then*

$$\mu_\omega(S) \leq \mu_\omega(E).$$

Equality holds if and only if $\beta = 0$, in which case

$$E = S \oplus S^{\perp_H}$$

is a holomorphic orthogonal splitting.

Proof. By $s = \text{rank}(S)$, $i \cdot \Lambda_\omega(F_E) = \lambda \cdot I_E$, and $S \hookrightarrow E$,

$$(S \hookrightarrow E, \text{rank}(S) = s) \xrightarrow{6.5.3} 2 \cdot \pi \cdot \deg_\omega(S) = i \cdot \int_X \text{Tr}(F_S) \wedge \frac{\omega^{n-1}}{(n-1)!} \xrightarrow{6.6.3} 2 \cdot \pi \cdot \deg_\omega(S) = \int_X \text{Tr}(i \cdot \Lambda_\omega(F_S)) \cdot \frac{\omega^n}{n!}.$$

$$(i \cdot \Lambda_\omega(F_E) = \lambda \cdot I_E, \text{rank}(S) = s) \xrightarrow{6.6.1} \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) = \text{Tr}(\lambda \cdot I_S) = \lambda \cdot s.$$

$$\text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) = \lambda \cdot s \xrightarrow{7.5.2} \text{Tr}(i \cdot \Lambda_\omega(F_S)) = \lambda \cdot s - \|\beta\|_{H,\omega}^2.$$

$$(7.5.3) \quad 2 \cdot \pi \cdot \deg_\omega(S) = \lambda \cdot s \cdot \text{Vol}_\omega(X) - \|\beta\|_{L^2(H,\omega)}^2.$$

$$(i \cdot \Lambda_\omega(F_E) = \lambda \cdot I_E) \xrightarrow{6.6.4} \lambda = \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)} \xrightarrow{7.5.3} 2 \cdot \pi \cdot \deg_\omega(S) = 2 \cdot \pi \cdot s \cdot \mu_\omega(E) - \|\beta\|_{L^2(H,\omega)}^2 \leq 2 \cdot \pi \cdot s \cdot \mu_\omega(E).$$

$$\deg_\omega(S) \leq s \cdot \mu_\omega(E) \xrightarrow{6.5.3} \mu_\omega(S) = \frac{\deg_\omega(S)}{s} \leq \mu_\omega(E).$$

$$\mu_\omega(S) = \mu_\omega(E) \xleftrightarrow{7.5.3} \|\beta\|_{L^2(H,\omega)}^2 = 0 \xleftrightarrow{7.5.1} \beta = 0 \xrightarrow{7.2.1} \gamma = 0 \xrightarrow{7.2.3} \nabla_H = \begin{bmatrix} \nabla^S & 0 \\ 0 & \nabla^T \end{bmatrix}.$$

Thus ∇_H is block diagonal and the orthogonal complement is holomorphic. $\square$

Example 7.5.2 (Complete rank-four slope-defect computation). Let

$$\text{rank}(E) = 4, \quad \text{rank}(S) = 2, \quad i \cdot \Lambda_\omega(F_E) = 40 \cdot I_E.$$

At $p \in X$, choose an adapted H -orthonormal frame and set

$$P_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad \iota_S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$$M_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 40 & 0 & 0 & 0 \\ 0 & 40 & 0 & 0 \\ 0 & 0 & 40 & 0 \\ 0 & 0 & 0 & 40 \end{bmatrix}.$$

$$M_{S,11} = 1 \cdot 40 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = 40,$$

$$M_{S,12} = 1 \cdot 0 + 0 \cdot 40 + 0 \cdot 0 + 0 \cdot 0 = 0,$$

$$M_{S,13} = 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 40 + 0 \cdot 0 = 0,$$

$$M_{S,14} = 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 40 = 0,$$

$$M_{S,21} = 0 \cdot 40 + 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = 0,$$

$$M_{S,22} = 0 \cdot 0 + 1 \cdot 40 + 0 \cdot 0 + 0 \cdot 0 = 40,$$

$$M_{S,23} = 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 40 + 0 \cdot 0 = 0,$$

$$M_{S,24} = 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 40 = 0.$$

$$\begin{bmatrix} 40 & 0 & 0 & 0 \\ 0 & 40 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 40 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 & 40 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 40 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 & 0 \cdot 0 + 40 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 40 & 0 \\ 0 & 40 \end{bmatrix}.$$

Suppose

$$g^{j\bar{k}}(p) = \delta_{jk} \quad (1 \leq j, k \leq n), \quad \beta_j(p) = 0 \quad (2 \leq j \leq n),$$

and

$$\beta_1(p) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \beta_1^{*H}(p) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}.$$

$$\beta_1^{*H}(p) \cdot \beta_1(p) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 3 \cdot 3 & 1 \cdot 2 + 3 \cdot 4 \\ 2 \cdot 1 + 4 \cdot 3 & 2 \cdot 2 + 4 \cdot 4 \end{bmatrix},$$

$$\beta_1^{*H}(p) \cdot \beta_1(p) = \begin{bmatrix} 1+9 & 2+12 \\ 2+12 & 4+16 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}, \quad \text{Tr}(\beta_1^{*H}(p) \cdot \beta_1(p)) = 10 + 20 = 30.$$

$$\begin{aligned}
\left(g^{j\bar{k}}(p) = \delta_{jk}, \quad \beta_j(p) = 0 \quad (2 \leq j \leq n) \right) &\xrightarrow{7.5.1} |\beta(p)|_{H,\omega}^2 = 2 \cdot 1 \cdot 30 = 2 \cdot 30 = 60. \\
(i \cdot \Lambda_\omega(F_E))(p) = 40 \cdot I_E, \quad \beta_j(p) = 0 \quad (2 \leq j \leq n) &\xrightarrow{7.4.1} i \cdot \Lambda_\omega(F_S)(p) = 40 \cdot I_S - 2 \cdot \beta_1^{*H}(p) \cdot \beta_1(p). \\
i \cdot \Lambda_\omega(F_S)(p) &= \begin{bmatrix} 40 & 0 \\ 0 & 40 \end{bmatrix} - 2 \cdot \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}. \\
i \cdot \Lambda_\omega(F_S)(p) &= \begin{bmatrix} 40 - 2 \cdot 10 & 0 - 2 \cdot 14 \\ 0 - 2 \cdot 14 & 40 - 2 \cdot 20 \end{bmatrix} = \begin{bmatrix} 40 - 20 & 0 - 28 \\ 0 - 28 & 40 - 40 \end{bmatrix}. \\
i \cdot \Lambda_\omega(F_S)(p) &= \begin{bmatrix} 20 & -28 \\ -28 & 0 \end{bmatrix}. \\
\text{Tr}(i \cdot \Lambda_\omega(F_S)(p)) &= 20 + 0 = 20 = 40 \cdot 2 - 60 = 80 - 60.
\end{aligned}$$

Thus the matrix identity reproduces the pointwise slope defect, including the entry $4 \cdot 4 = 16$.

Proposition 7.5.3 (Chern–Weil identity for a saturated subsheaf). *Let $\mathcal{S} \subset E$ be saturated of rank s , let $Z \subset X$ be its singular locus, and set*

$$X^\circ = X \setminus Z, \quad S = \mathcal{S}|_{X^\circ}, \quad \text{codim}_{\mathbb{C}}(Z) \geq 2.$$

Then the orthogonal projection and second fundamental form on X° satisfy

$$(7.5.4) \quad 2 \cdot \pi \cdot \deg_\omega(\mathcal{S}) = \int_{X^\circ} \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) \cdot \frac{\omega^n}{n!} - \int_{X^\circ} |\beta|_{H,\omega}^2 \cdot \frac{\omega^n}{n!}.$$

If H is Hermitian–Einstein, equality of slopes holds exactly when the orthogonal projection extends to a smooth ∇_H -parallel projection on E . In that case $\mathcal{S}$ is locally free and is a holomorphic orthogonal direct summand.

Proof. By $X^\circ = X \setminus Z$, $\text{codim}_{\mathbb{R}}(Z) \geq 4$, and $\chi_\varepsilon \in C_c^\infty(X^\circ)$,

$$0 \leq \chi_\varepsilon \leq 1, \quad \chi_\varepsilon \longrightarrow 1, \quad \int_X |d(\chi_\varepsilon)|_\omega^2 \cdot \frac{\omega^n}{n!} \longrightarrow 0,$$

the determinant metric gives

$$\begin{aligned}
(\det(\mathcal{S})|_{X^\circ} = \det(S), \quad \chi_\varepsilon \longrightarrow 1) &\xrightarrow{6.5.3} 2 \cdot \pi \cdot \deg_\omega(\mathcal{S}) = \lim_{\varepsilon \downarrow 0} i \cdot \int_{X^\circ} \chi_\varepsilon \cdot \text{Tr}(F_S) \wedge \frac{\omega^{n-1}}{(n-1)!}. \\
2 \cdot \pi \cdot \deg_\omega(\mathcal{S}) &= \lim_{\varepsilon \downarrow 0} i \cdot \int_{X^\circ} \chi_\varepsilon \cdot \text{Tr}(F_S) \wedge \frac{\omega^{n-1}}{(n-1)!} \xrightarrow{6.6.3} \lim_{\varepsilon \downarrow 0} \int_{X^\circ} \chi_\varepsilon \cdot \text{Tr}(i \cdot \Lambda_\omega(F_S)) \cdot \frac{\omega^n}{n!}. \\
\text{Tr}(i \cdot \Lambda_\omega(F_S)) &\xrightarrow{7.5.2} \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) - |\beta|_{H,\omega}^2. \\
2 \cdot \pi \cdot \deg_\omega(\mathcal{S}) &= \int_{X^\circ} \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) \cdot \frac{\omega^n}{n!} - \int_{X^\circ} |\beta|_{H,\omega}^2 \cdot \frac{\omega^n}{n!}.
\end{aligned}$$

Suppose now that $i \cdot \Lambda_\omega(F_E) = \lambda \cdot I_E$.

$$\begin{aligned}
(i \cdot \Lambda_\omega(F_E) = \lambda \cdot I_E) &\xrightarrow{6.6.1} \text{Tr}(\pi_S \cdot i \cdot \Lambda_\omega(F_E) \cdot \iota_S) = \text{Tr}(\lambda \cdot I_S) = \lambda \cdot s. \\
(i \cdot \Lambda_\omega(F_E) = \lambda \cdot I_E) &\xrightarrow{6.6.4} \lambda = \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)}.
\end{aligned}$$

$$\mu_\omega(\mathcal{S}) = \mu_\omega(E) \xleftrightarrow{7.5.4} \int_{X^\circ} |\beta|_{H,\omega}^2 \cdot \frac{\omega^n}{n!} = 0 \xleftrightarrow{7.5.1} \beta = 0 \xrightarrow{7.2.1} \gamma = 0, \quad \beta = \gamma = 0 \xrightarrow{7.2.3} \nabla_H(\pi_S) = 0.$$

In a smooth local frame, write $\nabla_H = d + A_H$. Then

$$\nabla_H(\pi_S) = 0 \xrightarrow{6.4.1} d(\pi_S) = -A_H \cdot \pi_S + \pi_S \cdot A_H, \quad \|\pi_S\|_{\text{op}} = 1 \implies |d(\pi_S)| \leq 2 \cdot |A_H|.$$

Thus π_S extends continuously across Z , the displayed differential equation extends distributionally, and repeated differentiation makes the extension smooth and parallel. Its image and kernel give the asserted holomorphic orthogonal splitting. $\square$

Corollary 7.5.4 (The Hermitian–Einstein direction). *If E admits a Hermitian–Einstein metric, then E is polystable.*

Proof. By $\mathcal{S} \subset E$ saturated, $S = \mathcal{S}|_{X \setminus Z}$ locally free, $\text{codim}_{\mathbb{C}}(Z) \geq 2$, and $i \cdot \Lambda_{\omega}(F_E) = \lambda \cdot I_E$,

$$(\mathcal{S} \subset E \text{ saturated, } i \cdot \Lambda_{\omega}(F_E) = \lambda \cdot I_E) \xrightarrow{7.5.3} \mu_{\omega}(\mathcal{S}) = \mu_{\omega}(E) - \frac{1}{2 \cdot \pi \cdot \text{rank}(\mathcal{S})} \cdot \int_{X \setminus Z} |\beta|_{H, \omega}^2 \cdot \frac{\omega^n}{n!} \leq \mu_{\omega}(E).$$

$$(\mu_{\omega}(\mathcal{S}) \leq \mu_{\omega}(E) \text{ for every saturated } \mathcal{S} \subset E) \xrightarrow{7.1.2} (\mu_{\omega}(\mathcal{S}) \leq \mu_{\omega}(E) \text{ for every coherent } \mathcal{S} \subset E).$$

$$(\mu_{\omega}(\mathcal{S}) \leq \mu_{\omega}(E) \text{ for every coherent } \mathcal{S} \subset E) \xrightarrow{7.1.3} E \text{ is semistable.}$$

Suppose that E is not stable.

$$(E \text{ is semistable, } E \text{ is not stable}) \xrightarrow{7.1.3} 0 < \text{rank}(\mathcal{A}) < \text{rank}(E), \quad \mu_{\omega}(\mathcal{A}) = \mu_{\omega}(E).$$

$$\mathcal{A} \subset \widetilde{\mathcal{A}} \xrightarrow{7.1.2} \text{rank}(\widetilde{\mathcal{A}}) = \text{rank}(\mathcal{A}), \quad \mu_{\omega}(\widetilde{\mathcal{A}}) \geq \mu_{\omega}(\mathcal{A}) = \mu_{\omega}(E).$$

$$(E \text{ is semistable, } \mu_{\omega}(\widetilde{\mathcal{A}}) \geq \mu_{\omega}(E)) \xrightarrow{7.1.3} \mu_{\omega}(\widetilde{\mathcal{A}}) = \mu_{\omega}(E).$$

$$\mu_{\omega}(\widetilde{\mathcal{A}}) = \mu_{\omega}(E) \xrightarrow{7.5.3} E = \widetilde{\mathcal{A}} \oplus \widetilde{\mathcal{A}}^{\perp_H} \text{ holomorphically.}$$

$$i \cdot \Lambda_{\omega}(F_E) = \begin{bmatrix} i \cdot \Lambda_{\omega}(F_{\widetilde{\mathcal{A}}}) & 0 \\ 0 & i \cdot \Lambda_{\omega}(F_{\widetilde{\mathcal{A}}^{\perp_H}}) \end{bmatrix} = \lambda \cdot \begin{bmatrix} I_{\widetilde{\mathcal{A}}} & 0 \\ 0 & I_{\widetilde{\mathcal{A}}^{\perp_H}} \end{bmatrix}.$$

$$(E = \widetilde{\mathcal{A}} \oplus \widetilde{\mathcal{A}}^{\perp_H} \text{ holomorphically}) \xrightarrow{7.4.1} i \cdot \Lambda_{\omega}(F_{\widetilde{\mathcal{A}}}) = \lambda \cdot I_{\widetilde{\mathcal{A}}}, \quad i \cdot \Lambda_{\omega}(F_{\widetilde{\mathcal{A}}^{\perp_H}}) = \lambda \cdot I_{\widetilde{\mathcal{A}}^{\perp_H}}.$$

$$(i \cdot \Lambda_{\omega}(F_{\widetilde{\mathcal{A}}}) = \lambda \cdot I_{\widetilde{\mathcal{A}}}, \quad i \cdot \Lambda_{\omega}(F_{\widetilde{\mathcal{A}}^{\perp_H}}) = \lambda \cdot I_{\widetilde{\mathcal{A}}^{\perp_H}}) \xrightarrow{6.6.4} \mu_{\omega}(\widetilde{\mathcal{A}}) = \mu_{\omega}(\widetilde{\mathcal{A}}^{\perp_H}) = \mu_{\omega}(E).$$

$$\text{rank}(\widetilde{\mathcal{A}}) < \text{rank}(E), \quad \text{rank}(\widetilde{\mathcal{A}}^{\perp_H}) < \text{rank}(E) \implies E = \bigoplus_{a=1}^N E_a, \quad \mu_{\omega}(E_a) = \mu_{\omega}(E), \quad E_a \text{ is stable.}$$

$$\left(E = \bigoplus_{a=1}^N E_a, \quad \mu_{\omega}(E_a) = \mu_{\omega}(E), \quad E_a \text{ is stable} \right) \xrightarrow{7.1.3} E \text{ is polystable.}$$

Thus finite induction on $\text{rank}(E)$ terminates. $\square$

8. THE ANALYTIC UHLENBECK–YAU ARGUMENT

Let (X, ω) be a compact connected Kähler manifold and let E be a holomorphic vector bundle of rank r . Write

$$\lambda = \frac{2 \cdot \pi \cdot \mu_{\omega}(E)}{\text{Vol}_{\omega}(X)}.$$

Fix a reference metric $\hat{H}$ whose determinant metric is normalized by

$$\text{Tr}(i \cdot \Lambda_{\omega}(F_{\hat{H}})) = r \cdot \lambda, \quad \text{Tr}(i \cdot \Lambda_{\omega}(F_{\hat{H}}) - \lambda \cdot I_E) = 0.$$

Proposition 8.0.1 (Reference determinant normalization). *Given any Hermitian metric $\hat{H}_0$, there is $\psi \in C^{\infty}(X, \mathbb{R})$ such that*

$$\hat{H} = \exp(\psi) \cdot \hat{H}_0, \quad \text{Tr}(i \cdot \Lambda_{\omega}(F_{\hat{H}})) = r \cdot \lambda.$$

Proof. By $\hat{H}_0$ Hermitian, $\lambda = 2 \cdot \pi \cdot \mu_\omega(E) / \text{Vol}_\omega(X)$, and $c_1(E) = [i \cdot \text{Tr}(F_{\hat{H}_0}) / (2 \cdot \pi)]$,

$$\begin{aligned} \left(c_1(E) = \left[\frac{i}{2 \cdot \pi} \cdot \text{Tr}(F_{\hat{H}_0}) \right] \right) &\xrightarrow{6.5.3} 2 \cdot \pi \cdot \deg_\omega(E) = i \cdot \int_X \text{Tr}(F_{\hat{H}_0}) \wedge \frac{\omega^{n-1}}{(n-1)!} \\ i \cdot \int_X \text{Tr}(F_{\hat{H}_0}) \wedge \frac{\omega^{n-1}}{(n-1)!} &\xrightarrow{6.6.3} \int_X \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}_0})) \cdot \frac{\omega^n}{n!} = 2 \cdot \pi \cdot \deg_\omega(E) = r \cdot \lambda \cdot \text{Vol}_\omega(X). \\ \int_X (\text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}_0})) - r \cdot \lambda) \cdot \frac{\omega^n}{n!} &= 0 \xrightarrow{3.5.5} \exists \psi \in C^\infty(X, \mathbb{R}). \\ \int_X \psi \cdot \frac{\omega^n}{n!} &= 0, \quad 2 \cdot r \cdot \Delta_\omega(\psi) = \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}_0})) - r \cdot \lambda. \\ \hat{H} = \exp(\psi) \cdot \hat{H}_0 &\xrightarrow{6.3.1} A_{\hat{H}} = A_{\hat{H}_0} + \partial(\psi) \cdot I_E \xrightarrow{6.4.3} F_{\hat{H}} = F_{\hat{H}_0} + \bar{\partial}(\partial(\psi)) \cdot I_E. \\ (F_{\hat{H}} - F_{\hat{H}_0} = \bar{\partial}(\partial(\psi)) \cdot I_E) &\xrightarrow{3.4.1} \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}})) = \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}_0})) - 2 \cdot r \cdot \Delta_\omega(\psi) = r \cdot \lambda. \end{aligned}$$

Thus a conformal change of the reference metric gives the stated determinant normalization. $\square$

The positive-parameter continuity construction supplies the perturbed metrics used below. The argument separates uniform metric control from the formation of a weak holomorphic projection.

8.1. Metric deformations and the perturbed equation.

Definition 8.1.1 (Metric endomorphism). For a second Hermitian metric H , its metric endomorphism relative to $\hat{H}$ is the unique section

$$h \in C^\infty(X, \text{End}(E)), \quad H = \hat{H} \cdot h, \quad h = \hat{H}^{-1} \cdot H.$$

For $V, W \in E_x$,

$$H(V, W) = \hat{H}(h \cdot V, W), \quad h^{*\hat{H}} = \hat{H}^{-1} \cdot h^* \cdot \hat{H} = h, \quad h > 0.$$

The functions h^r and $\log(h)$ are defined by the spectral functional calculus.

Example 8.1.2 (A complete rank-two metric endomorphism). Let

$$\hat{H} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix}, \quad \hat{H}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\left(\hat{H} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix} \right) \xrightarrow{8.1.1} h = \hat{H}^{-1} \cdot H = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \cdot 6 + 0 \cdot 2 & \frac{1}{2} \cdot 2 + 0 \cdot 2 \\ 0 \cdot 6 + 1 \cdot 2 & 0 \cdot 2 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}.$$

The weighted adjoint is computed without suppressing either matrix product.

$$\begin{aligned} h &= \hat{H}^{-1} \cdot H \xrightarrow{8.1.1} h^{*\hat{H}} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \\ h^{*\hat{H}} &= \begin{bmatrix} \frac{1}{2} \cdot 3 + 0 \cdot 1 & \frac{1}{2} \cdot 2 + 0 \cdot 2 \\ 0 \cdot 3 + 1 \cdot 1 & 0 \cdot 2 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \\ h^{*\hat{H}} &= \begin{bmatrix} \frac{3}{2} \cdot 2 + 1 \cdot 0 & \frac{3}{2} \cdot 0 + 1 \cdot 1 \\ 1 \cdot 2 + 2 \cdot 0 & 1 \cdot 0 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} = h. \end{aligned}$$

Moreover,

$$H = \hat{H} \cdot h \xrightarrow{8.1.1} \hat{H} \cdot h = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 3 + 0 \cdot 2 & 2 \cdot 1 + 0 \cdot 2 \\ 0 \cdot 3 + 1 \cdot 2 & 0 \cdot 1 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix} = H.$$

Finally,

$$\det(h) = 3 \cdot 2 - 1 \cdot 2 = 6 - 2 = 4, \quad \text{Tr}(h) = 3 + 2 = 5, \quad \det(t \cdot I_2 - h) = t^2 - 5 \cdot t + 4 = (t - 1) \cdot (t - 4).$$

Thus the two $\hat{H}$ -eigenvalues are 1 and 4.

Let $A_{\hat{H}}$ be the Chern connection matrix of $\hat{H}$. On $\text{End}(E)$, define

$$\hat{\nabla}^{1,0}(h) = \partial(h) + A_{\hat{H}} \cdot h - h \cdot A_{\hat{H}}.$$

Fact 8.1.3 (Curvature change under a metric deformation). *If $H = \hat{H} \cdot h$, then*

$$(8.1.1) \quad A_H - A_{\hat{H}} = h^{-1} \cdot \hat{\nabla}^{1,0}(h), \quad F_H - F_{\hat{H}} = \bar{\partial}_{\text{End}(E)}(h^{-1} \cdot \hat{\nabla}^{1,0}(h)).$$

Equivalently,

$$F_H - F_{\hat{H}} = \bar{\partial}_{\text{End}(E)}(h^{-1} \cdot \hat{\nabla}^{1,0}(h)) \xrightarrow{6.4.3} (F_H)_{\bar{k}j} - (F_{\hat{H}})_{\bar{k}j} = \hat{\nabla}_{\bar{k}}(h^{-1} \cdot \hat{\nabla}_j(h)).$$

Proof. By $H = \hat{H} \cdot h$, $H^{-1} = h^{-1} \cdot \hat{H}^{-1}$, and $A_H = H^{-1} \cdot \partial(H)$,

$$\begin{aligned} (H = \hat{H} \cdot h, H^{-1} = h^{-1} \cdot \hat{H}^{-1}) &\xrightarrow{6.3.1} A_H = h^{-1} \cdot \hat{H}^{-1} \cdot \partial(\hat{H} \cdot h) = h^{-1} \cdot \hat{H}^{-1} \cdot \partial(\hat{H}) \cdot h + h^{-1} \cdot \partial(h). \\ \hat{\nabla}^{1,0}(h) = \partial(h) + A_{\hat{H}} \cdot h - h \cdot A_{\hat{H}} &\xrightarrow{6.4.1} A_H - A_{\hat{H}} = h^{-1} \cdot A_{\hat{H}} \cdot h + h^{-1} \cdot \partial(h) - A_{\hat{H}} = h^{-1} \cdot (\partial(h) + A_{\hat{H}} \cdot h - h \cdot A_{\hat{H}}). \\ (A_H - A_{\hat{H}} = h^{-1} \cdot \hat{\nabla}^{1,0}(h)) &\xrightarrow{6.4.3} F_H - F_{\hat{H}} = \bar{\partial}_{\text{End}(E)}(h^{-1} \cdot \hat{\nabla}^{1,0}(h)). \end{aligned}$$

Thus the curvature-change sign agrees with the $d\bar{z}^k \wedge dz^j$ coefficient convention. $\square$

Definition 8.1.4 (Perturbed Hermitian–Einstein equation). For $0 < \varepsilon \leq 1$, a perturbed metric is

$$H_\varepsilon = \hat{H} \cdot h_\varepsilon, \quad h_\varepsilon = h_\varepsilon^{*\hat{H}} > 0,$$

and it solves

$$(8.1.2) \quad i \cdot \Lambda_\omega(F_{H_\varepsilon}) - \lambda \cdot I_E = -\varepsilon \cdot \log(h_\varepsilon).$$

Using the curvature coefficients fixed in the preceding sections, the complete local system is

$$\begin{aligned} H_\varepsilon = \hat{H} \cdot h_\varepsilon &\xrightarrow{8.1.1} F_{H_\varepsilon} = F_{\hat{H}} + \bar{\partial}_{\text{End}(E)}(h_\varepsilon^{-1} \cdot \hat{\nabla}^{1,0}(h_\varepsilon)). \\ (F_{H_\varepsilon})_{\bar{k}j} &\xrightarrow{6.6.1} i \cdot \Lambda_\omega(F_{H_\varepsilon}) = -2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot (F_{H_\varepsilon})_{\bar{k}j}. \\ (F_{H_\varepsilon})_{\bar{k}j} &= (F_{\hat{H}})_{\bar{k}j} + \hat{\nabla}_{\bar{k}}(h_\varepsilon^{-1} \cdot \hat{\nabla}_j(h_\varepsilon)). \\ -2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot &\left[(F_{\hat{H}})_{\bar{k}j} + \hat{\nabla}_{\bar{k}}(h_\varepsilon^{-1} \cdot \hat{\nabla}_j(h_\varepsilon)) \right] - \lambda \cdot I_E + \varepsilon \cdot \log(h_\varepsilon) = 0. \end{aligned}$$

Equivalently,

$$(8.1.3) \quad 2 \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \hat{\nabla}_{\bar{k}}(h_\varepsilon^{-1} \cdot \hat{\nabla}_j(h_\varepsilon)) = i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E + \varepsilon \cdot \log(h_\varepsilon).$$

Proposition 8.1.5 (Fixed determinant normalization). *Suppose $h_\varepsilon = h_\varepsilon^{*\hat{H}} > 0$ solves the perturbed Hermitian–Einstein equation and $\text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}})) = r \cdot \lambda$. Then*

$$\det(h_\varepsilon) = 1, \quad \text{Tr}(\log(h_\varepsilon)) = 0.$$

Proof. By $u_\varepsilon = \log(\det(h_\varepsilon))$, $\text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}})) = r \cdot \lambda$, and $i \cdot \Lambda_\omega(F_{H_\varepsilon}) - \lambda \cdot I_E = -\varepsilon \cdot \log(h_\varepsilon)$,

$$(u_\varepsilon = \log(\det(h_\varepsilon)), \quad h_\varepsilon = h_\varepsilon^{*\hat{H}} > 0) \xrightarrow{5.2.1} \text{Tr}(h_\varepsilon^{-1} \cdot \hat{\nabla}^{1,0}(h_\varepsilon)) = \text{Tr}(h_\varepsilon^{-1} \cdot \partial(h_\varepsilon)) = \partial(u_\varepsilon).$$

$$h_\varepsilon = h_\varepsilon^{*\hat{H}} > 0 \xrightarrow{8.1.1} \text{Tr}(\log(h_\varepsilon)) = \log(\det(h_\varepsilon)) = u_\varepsilon.$$

$$(F_{H_\varepsilon} - F_{\hat{H}} = \bar{\partial}_{\text{End}(E)}(h_\varepsilon^{-1} \cdot \hat{\nabla}^{1,0}(h_\varepsilon))) \xrightarrow{8.1.3} \text{Tr}(F_{H_\varepsilon} - F_{\hat{H}}) = \bar{\partial}(\partial(u_\varepsilon)).$$

$$(i \cdot \Lambda_\omega(F_{H_\varepsilon}) - \lambda \cdot I_E = -\varepsilon \cdot \log(h_\varepsilon), \quad \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}})) = r \cdot \lambda) \xrightarrow{8.1.2} \text{Tr}(i \cdot \Lambda_\omega(F_{H_\varepsilon}) - i \cdot \Lambda_\omega(F_{\hat{H}})) = -\varepsilon \cdot u_\varepsilon.$$

$$\text{Tr}(i \cdot \Lambda_\omega(F_{H_\varepsilon} - F_{\hat{H}})) = i \cdot \Lambda_\omega(\bar{\partial}(\partial(u_\varepsilon))) \xrightarrow{3.4.1} -2 \cdot \Delta_\omega(u_\varepsilon) = -\varepsilon \cdot u_\varepsilon \implies 2 \cdot \Delta_\omega(u_\varepsilon) = \varepsilon \cdot u_\varepsilon.$$

$$\begin{aligned} u_\varepsilon(x_{\max}) > 0 &\implies 2 \cdot \Delta_\omega(u_\varepsilon)(x_{\max}) \leq 0 < \varepsilon \cdot u_\varepsilon(x_{\max}), \\ u_\varepsilon(x_{\min}) < 0 &\implies 2 \cdot \Delta_\omega(u_\varepsilon)(x_{\min}) \geq 0 > \varepsilon \cdot u_\varepsilon(x_{\min}). \end{aligned}$$

Hence $u_\varepsilon = 0$, which gives both asserted identities. $\square$

Theorem 8.1.6 (Positive-parameter solvability input). *For every $0 < \varepsilon \leq 1$, the positive-parameter continuity method produces a smooth solution $H_\varepsilon = \hat{H} \cdot h_\varepsilon$ of*

$$i \cdot \Lambda_\omega(F_{H_\varepsilon}) - \lambda \cdot I_E = -\varepsilon \cdot \log(h_\varepsilon), \quad h_\varepsilon = h_\varepsilon^{*\hat{H}} > 0 \xrightarrow{8.1.5} \det(h_\varepsilon) = 1.$$

This is the first analytic input in the Uhlenbeck–Yau continuity argument. The remaining issue is the behavior of these solutions as $\varepsilon \downarrow 0$.

8.2. The bounded branch.

Theorem 8.2.1 (Elliptic compactness for a bounded perturbed branch). *Suppose $h_j = h_j^{*\hat{H}} > 0$ satisfies*

$$2 \cdot \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \hat{\nabla}_{\bar{q}}(h_j^{-1} \cdot \hat{\nabla}_p(h_j)) = i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E + \varepsilon_j \cdot \log(h_j), \quad \varepsilon_j \downarrow 0,$$

and assume that constants $0 < m \leq M < \infty$ satisfy

$$m \cdot I_E \leq h_j \leq M \cdot I_E \quad (j \geq 1).$$

Then, after passing to a subsequence,

$$\sup_{j \geq 1} \left(\|h_j\|_{C^k(X, \hat{H})} + \|h_j^{-1}\|_{C^k(X, \hat{H})} \right) < \infty \quad (k \geq 0), \quad h_j \longrightarrow h_\infty \quad \text{in } C^\infty.$$

This is the standard bundle-valued elliptic compactness input for the uniformly positive perturbed system.

Proposition 8.2.2 (Convergence of a uniformly bounded branch). *Let $\varepsilon_j \downarrow 0$, let $h_j = h_{\varepsilon_j}$, and assume*

$$\sup_{j \geq 1} \sup_{x \in X} \text{Tr}(h_j(x)) \leq M < \infty.$$

Then a subsequence of $H_j = \hat{H} \cdot h_j$ converges smoothly to a Hermitian–Einstein metric H_∞ satisfying

$$i \cdot \Lambda_\omega(F_{H_\infty}) = \lambda \cdot I_E.$$

Proof. By $h_j = h_j^{*\hat{H}} > 0$, $\det(h_j) = 1$, and $\text{Tr}(h_j) \leq M$, let $\mu_{j,1}, \dots, \mu_{j,r} > 0$ be the eigenvalues of h_j .

$$\left(h_j = h_j^{*\hat{H}} > 0, \quad h_j \cdot e_{j,a} = \mu_{j,a} \cdot e_{j,a} \right) \xrightarrow{8.1.1} \sum_{a=1}^r \mu_{j,a} = \text{Tr}(h_j) \leq M.$$

$$(i \cdot \Lambda_\omega(F_{H_j}) - \lambda \cdot I_E = -\varepsilon_j \cdot \log(h_j), \quad \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}})) = r \cdot \lambda) \xrightarrow{8.1.5} \prod_{a=1}^r \mu_{j,a} = \det(h_j) = 1.$$

$$0 < \mu_{j,a} \leq M, \quad \mu_{j,a}^{-1} = \prod_{\substack{b=1 \\ b \neq a}}^r \mu_{j,b} \leq \prod_{\substack{b=1 \\ b \neq a}}^r M = M^{r-1}, \quad \text{Tr}(h_j^{-1}) \leq r \cdot M^{r-1}.$$

$$(M^{1-r} \leq \mu_{j,a} \leq M \quad (1 \leq a \leq r)) \xrightarrow{8.1.1} M^{1-r} \cdot I_E \leq h_j \leq M \cdot I_E, \quad M^{1-r} \cdot \hat{H} \leq H_j \leq M \cdot \hat{H}.$$

The complete elliptic system remains

$$(i \cdot \Lambda_\omega(F_{H_j}) - \lambda \cdot I_E = -\varepsilon_j \cdot \log(h_j)) \xrightarrow{8.1.3} 2 \cdot \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \hat{\nabla}_{\bar{q}}(h_j^{-1} \cdot \hat{\nabla}_p(h_j)) = i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E + \varepsilon_j \cdot \log(h_j).$$

$$(M^{1-r} \cdot I_E \leq h_j \leq M \cdot I_E, \quad \varepsilon_j \downarrow 0) \xrightarrow{8.2.1} \sup_{j \geq 1} \left(\|h_j\|_{C^m(X, \hat{H})} + \|h_j^{-1}\|_{C^m(X, \hat{H})} \right) \leq C_m \quad (m \geq 0).$$

$$\begin{aligned}
h_j &\longrightarrow h_\infty \quad \text{in } C^\infty. \\
H_\infty &= \hat{H} \cdot h_\infty, \quad \varepsilon_j \cdot \log(h_j) \longrightarrow 0 \implies i \cdot \Lambda_\omega(F_{\hat{H} \cdot h_\infty}) - \lambda \cdot I_E = 0. \\
i \cdot \Lambda_\omega(F_{H_\infty}) &= \lambda \cdot I_E \xrightarrow{6.6.1} H_\infty \text{ is Hermitian-Einstein.}
\end{aligned}$$

Thus the bounded branch converges to the required metric. $\square$

8.3. Spectral calculus and Sobolev control.

Fact 8.3.1 (Moving-frame computation). *At a fixed point choose a $\hat{H}$ -orthonormal eigenframe $e_1, \dots, e_r$ and its dual frame $e^1, \dots, e^r$ such that*

$$h \cdot e_a = \mu_a \cdot e_a, \quad \hat{H}(e_a, e_b) = \delta_{ab}, \quad e^a(e_b) = \delta_b^a.$$

Then

$$h \cdot e_a = \mu_a \cdot e_a \xrightarrow{8.1.1} h = \sum_{a=1}^r \mu_a \cdot e_a \otimes e^a, \quad h^\sigma = \sum_{a=1}^r \mu_a^\sigma \cdot e_a \otimes e^a.$$

If

$$\hat{\nabla}(e_a) = \sum_{c=1}^r e_c \cdot A^c{}_a, \quad \hat{\nabla}(e^a) = - \sum_{d=1}^r A^a{}_d \cdot e^d,$$

then

$$(8.3.1) \quad \hat{\nabla}(h) = \sum_{a=1}^r d(\mu_a) \cdot e_a \otimes e^a + \sum_{a=1}^r \sum_{c=1}^r (\mu_a - \mu_c) \cdot A^c{}_a \cdot e_c \otimes e^a.$$

Proof. By $h = \sum_{a=1}^r \mu_a \cdot e_a \otimes e^a$, $\hat{\nabla}(e_a) = \sum_{c=1}^r e_c \cdot A^c{}_a$, $\hat{\nabla}(e^a) = - \sum_{d=1}^r A^a{}_d \cdot e^d$, and $\hat{\nabla}(u \otimes \eta) = \hat{\nabla}(u) \otimes \eta + u \otimes \hat{\nabla}(\eta)$,

$$\begin{aligned}
\hat{\nabla}(h) &= \sum_{a=1}^r d(\mu_a) \cdot e_a \otimes e^a + \sum_{a=1}^r \sum_{c=1}^r \mu_a \cdot A^c{}_a \cdot e_c \otimes e^a - \sum_{a=1}^r \sum_{d=1}^r \mu_a \cdot A^a{}_d \cdot e_a \otimes e^d \\
&\quad - \sum_{a=1}^r \sum_{d=1}^r \mu_a \cdot A^a{}_d \cdot e_a \otimes e^d = - \sum_{a=1}^r \sum_{c=1}^r \mu_c \cdot A^c{}_a \cdot e_c \otimes e^a.
\end{aligned}$$

Combining the two connection terms gives the asserted formula. $\square$

Example 8.3.2 (Multiplication of frame endomorphisms). For

$$E_{12} = e_1 \otimes e^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad E_{21} = e_2 \otimes e^1 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

the complete product is

$$E_{12} \cdot E_{21} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 0 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = E_{11}.$$

In general,

$$(e_a \otimes e^b) \cdot (e_c \otimes e^d) = \delta_c^b \cdot (e_a \otimes e^d).$$

Example 8.3.3 (Complete computation of $h^{-1} \cdot \hat{\nabla}(h)$). For $r = 2$, the moving-frame matrices are

$$\begin{aligned}
h \cdot e_a &= \mu_a \cdot e_a \xrightarrow{8.3.1} h = \begin{bmatrix} \mu_1 & 0 \\ 0 & \mu_2 \end{bmatrix}, \quad h^{-1} = \begin{bmatrix} \mu_1^{-1} & 0 \\ 0 & \mu_2^{-1} \end{bmatrix}, \\
\hat{\nabla}(h) &= \begin{bmatrix} d(\mu_1) & (\mu_2 - \mu_1) \cdot A^1{}_2 \\ (\mu_1 - \mu_2) \cdot A^2{}_1 & d(\mu_2) \end{bmatrix}.
\end{aligned}$$

Therefore

$$(h^{-1}, \hat{\nabla}(h)) \xrightarrow{8.3.1} h^{-1} \cdot \hat{\nabla}(h) = \begin{bmatrix} \mu_1^{-1} & 0 \\ 0 & \mu_2^{-1} \end{bmatrix} \cdot \begin{bmatrix} d(\mu_1) & (\mu_2 - \mu_1) \cdot A^1{}_2 \\ (\mu_1 - \mu_2) \cdot A^2{}_1 & d(\mu_2) \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} \mu_1^{-1} \cdot d(\mu_1) + 0 \cdot (\mu_1 - \mu_2) \cdot A^2_1 & \mu_1^{-1} \cdot (\mu_2 - \mu_1) \cdot A^1_2 + 0 \cdot d(\mu_2) \\ 0 \cdot d(\mu_1) + \mu_2^{-1} \cdot (\mu_1 - \mu_2) \cdot A^2_1 & 0 \cdot (\mu_2 - \mu_1) \cdot A^1_2 + \mu_2^{-1} \cdot d(\mu_2) \end{bmatrix} \\
&= \begin{bmatrix} \mu_1^{-1} \cdot d(\mu_1) & \mu_1^{-1} \cdot (\mu_2 - \mu_1) \cdot A^1_2 \\ \mu_2^{-1} \cdot (\mu_1 - \mu_2) \cdot A^2_1 & \mu_2^{-1} \cdot d(\mu_2) \end{bmatrix}.
\end{aligned}$$

The positive quantity is the adjoint-contracted norm

$$|h^{-1} \cdot \hat{\nabla}(h)|_{\hat{H}, \omega}^2 = \sum_{a=1}^r \mu_a^{-2} \cdot |d(\mu_a)|_\omega^2 + \sum_{a=1}^r \sum_{b=1}^r \mu_a^{-2} \cdot |\mu_b - \mu_a|^2 \cdot |A^a_b|_\omega^2.$$

Fact 8.3.4 (Energy estimate for a blow-up sequence). *Let $\varepsilon_j \downarrow 0$, let $h_j = h_{\varepsilon_j}$, and suppose*

$$M_j = \sup_{x \in X} \text{Tr}(h_j(x)) \longrightarrow \infty, \quad \tilde{h}_j = M_j^{-1} \cdot h_j.$$

For every fixed $0 < \sigma \leq 1$,

$$(8.3.2) \quad \left\| \tilde{h}_j^\sigma \right\|_{W^{1,2}(X, \text{End}(E))} \leq C\sigma.$$

Proof. By $2 \cdot \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \hat{\nabla}_{\bar{q}}(h_j^{-1} \cdot \hat{\nabla}_p(h_j)) = i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E + \varepsilon_j \cdot \log(h_j)$, $M_j = \sup_{x \in X} \text{Tr}(h_j(x))$, and $\tilde{h}_j = M_j^{-1} \cdot h_j$,

$$\begin{aligned}
&\left(h_j = h_j^{*\hat{H}} > 0, \quad M_j = \sup_{x \in X} \text{Tr}(h_j(x)) \right) \xrightarrow{8.1.1} 0 < \tilde{h}_j \leq I_E, \quad \sup_{x \in X} \text{Tr}(\tilde{h}_j(x)) = 1. \\
&(h_j = h_{\varepsilon_j}, \quad \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}})) = r \cdot \lambda) \xrightarrow{8.1.5} \det(h_j) = 1, \quad \det(\tilde{h}_j) = M_j^{-r}.
\end{aligned}$$

Testing the complete system by $\tilde{h}_j^{2\sigma}$ and integrating by parts gives

$$\begin{aligned}
&(h_j, \tilde{h}_j^{2\sigma}) \xrightarrow{8.1.3} 2 \cdot \int_X \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \text{Tr}(\tilde{h}_j^{-1} \cdot \hat{\nabla}_p(\tilde{h}_j) \cdot \hat{\nabla}_{\bar{q}}(\tilde{h}_j^{2\sigma})) \cdot \frac{\omega^n}{n!} \\
&= - \int_X \text{Tr}((i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E + \varepsilon_j \cdot \log(h_j)) \cdot \tilde{h}_j^{2\sigma}) \cdot \frac{\omega^n}{n!}.
\end{aligned}$$

For $0 < x \leq 1$,

$$\begin{aligned}
f_\sigma(x) &= -x^{2\sigma} \cdot \log(x), \quad f'_\sigma(x) = -x^{2\sigma-1} \cdot (2 \cdot \sigma \cdot \log(x) + 1), \\
f'_\sigma(x) &= 0 \iff x = \exp\left(-\frac{1}{2 \cdot \sigma}\right), \quad 0 \leq f_\sigma(x) \leq \frac{1}{2 \cdot e \cdot \sigma}. \\
-\varepsilon_j \cdot \text{Tr}(\log(h_j) \cdot \tilde{h}_j^{2\sigma}) &= -\varepsilon_j \cdot \log(M_j) \cdot \text{Tr}(\tilde{h}_j^{2\sigma}) - \varepsilon_j \cdot \text{Tr}(\log(\tilde{h}_j) \cdot \tilde{h}_j^{2\sigma}) \leq \frac{r}{2 \cdot e \cdot \sigma}.
\end{aligned}$$

At the chosen point let $\tilde{h}_j \cdot e_a = \tilde{\mu}_{j,a} \cdot e_a$.

$$\begin{aligned}
(\tilde{h}_j, \tilde{h}_j^\tau) &\xrightarrow{8.3.1} (\hat{\nabla}(\tilde{h}_j^\tau))^a_b = \frac{\tilde{\mu}_{j,a}^\tau - \tilde{\mu}_{j,b}^\tau}{\tilde{\mu}_{j,a} - \tilde{\mu}_{j,b}} \cdot (\hat{\nabla}(\tilde{h}_j))^a_b \quad (\tilde{\mu}_{j,a} \neq \tilde{\mu}_{j,b}). \\
\tilde{\mu}_{j,a} &= \tilde{\mu}_{j,b} \implies (\hat{\nabla}(\tilde{h}_j^\tau))^a_b = \tau \cdot \tilde{\mu}_{j,a}^{\tau-1} \cdot (\hat{\nabla}(\tilde{h}_j))^a_b.
\end{aligned}$$

For eigenvalues $x, y > 0$ and $0 < \sigma \leq 1$, the divided differences satisfy

$$x^{-1} \cdot \frac{x^{2\sigma} - y^{2\sigma}}{x - y} \geq \left(\frac{x^\sigma - y^\sigma}{x - y} \right)^2, \quad x = y \implies 2 \cdot \sigma \cdot x^{2\sigma-2} \geq \sigma^2 \cdot x^{2\sigma-2}.$$

Consequently,

$$\left(x^{-1} \cdot \frac{x^{2\sigma} - y^{2\sigma}}{x - y} \geq \left(\frac{x^\sigma - y^\sigma}{x - y} \right)^2 \right) \implies \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \text{Tr}(\tilde{h}_j^{-1} \cdot \hat{\nabla}_p(\tilde{h}_j) \cdot \hat{\nabla}_{\bar{q}}(\tilde{h}_j^{2\sigma})) \geq \left| \hat{\nabla}(\tilde{h}_j^\sigma) \right|_{\hat{H}, \omega}^2.$$

$$\int_X \left| \hat{\nabla}(\tilde{h}_j^\sigma) \right|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!} \leq C_\sigma, \quad \int_X \left| \tilde{h}_j^\sigma \right|_{\hat{H}}^2 \cdot \frac{\omega^n}{n!} \leq r \cdot \text{Vol}_\omega(X).$$

This proves the scale-independent Sobolev estimate. $\square$

Fact 8.3.5 (Banach–Alaoglu and Rellich compactness). *If $v_j \in W^{1,2}(X, \text{End}(E))$ and $\|v_j\|_{W^{1,2}} \leq C$, then a subsequence satisfies*

$$v_{j_\ell} \rightharpoonup v_\infty \quad \text{weakly in } W^{1,2}(X, \text{End}(E)), \quad v_{j_\ell} \longrightarrow v_\infty \quad \text{strongly in } L^2(X, \text{End}(E)).$$

Proof. By $\|v_j\|_{W^{1,2}} \leq C$, $W^{1,2}(X, \text{End}(E))$ reflexive, and $W^{1,2}(X, \text{End}(E)) \Subset L^2(X, \text{End}(E))$,

$$\|v_j\|_{W^{1,2}} \leq C \implies v_{j_\ell} \rightharpoonup v_\infty \quad \text{in } W^{1,2}.$$

$$W^{1,2}(X, \text{End}(E)) \Subset L^2(X, \text{End}(E)) \implies v_{j_\ell} \longrightarrow v_\infty \quad \text{in } L^2.$$

Thus weak Sobolev compactness and strong L^2 compactness hold along one subsequence. $\square$

Fact 8.3.6 (Adjoint-order trace–gradient inequality). *For every $h = h^{*\hat{H}} > 0$,*

$$\frac{\sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \text{Tr}((\hat{\nabla}_{\bar{q}}(h)) \cdot h^{-1} \cdot \hat{\nabla}_p(h))}{\text{Tr}(h)} - \frac{\sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \partial_p(\text{Tr}(h)) \cdot \partial_{\bar{q}}(\text{Tr}(h))}{(\text{Tr}(h))^2} \geq 0.$$

Proof. By $h = h^{*\hat{H}} > 0$, $g^{p\bar{q}} = \delta_{pq}$, and $h = \text{diag}(\mu_1, \dots, \mu_r)$ at the chosen point,

$$(h = \text{diag}(\mu_1, \dots, \mu_r), \quad g^{p\bar{q}} = \delta_{pq}) \xrightarrow{8.3.1} \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \text{Tr}((\hat{\nabla}_{\bar{q}}(h)) \cdot h^{-1} \cdot \hat{\nabla}_p(h)) = \sum_{p=1}^n \sum_{a=1}^r \sum_{b=1}^r \mu_a^{-1} \cdot |\hat{\nabla}_p(h)^a_b|^2.$$

$$\sum_{p=1}^n |\partial_p(\text{Tr}(h))|^2 = \sum_{p=1}^n \left| \sum_{a=1}^r \hat{\nabla}_p(h)^a_a \right|^2 \leq \sum_{p=1}^n \left(\sum_{a=1}^r \mu_a \right) \left(\sum_{a=1}^r \mu_a^{-1} \cdot |\hat{\nabla}_p(h)^a_a|^2 \right).$$

$$\sum_{p=1}^n |\partial_p(\text{Tr}(h))|^2 \leq \text{Tr}(h) \cdot \sum_{p=1}^n \sum_{a=1}^r \sum_{b=1}^r \mu_a^{-1} \cdot |\hat{\nabla}_p(h)^a_b|^2.$$

Division by $(\text{Tr}(h))^2$ gives the asserted inequality. $\square$

Fact 8.3.7 (Trace nonconcentration). *For the normalized blow-up sequence, set*

$$u_j = \log(\text{Tr}(\tilde{h}_j)) = \log(\text{Tr}(h_j)) - \log(M_j).$$

Then

$$u_j \leq 0, \quad \sup_{x \in X} u_j(x) = 0, \quad \Delta_\omega(u_j) \geq -\frac{1}{2} \cdot \left(\|i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E\|_{\text{op}} + \frac{1}{e} \right), \quad \int_X u_j \cdot \frac{\omega^n}{n!} \geq -C.$$

In particular, every strong L^2 limit of $\tilde{h}_j^\sigma$ supplied by the energy estimate is nonzero.

Proof. By $\det(h_j) = 1$, $2 \cdot \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \hat{\nabla}_{\bar{q}}(h_j^{-1}) \cdot \hat{\nabla}_p(h_j) = i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E + \varepsilon_j \cdot \log(h_j)$, and $\text{Tr}(h_j) \geq r$, multiply the full endomorphism equation by h_j .

$$h_j \cdot \hat{\nabla}_{\bar{q}}(h_j^{-1}) \cdot \hat{\nabla}_p(h_j) = \hat{\nabla}_{\bar{q}}(\hat{\nabla}_p(h_j)) - (\hat{\nabla}_{\bar{q}}(h_j)) \cdot h_j^{-1} \cdot \hat{\nabla}_p(h_j).$$

$$(h_j, \hat{\nabla}(h_j)) \xrightarrow{8.1.3} \Delta_\omega(\log(\text{Tr}(h_j))) = \frac{\text{Tr}(h_j \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) + \varepsilon_j \cdot \text{Tr}(h_j \cdot \log(h_j))}{2 \cdot \text{Tr}(h_j)}$$

$$+ \frac{\sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \text{Tr}((\hat{\nabla}_{\bar{q}}(h_j)) \cdot h_j^{-1} \cdot \hat{\nabla}_p(h_j))}{\text{Tr}(h_j)} - \frac{\sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \partial_p(\text{Tr}(h_j)) \cdot \partial_{\bar{q}}(\text{Tr}(h_j))}{(\text{Tr}(h_j))^2}.$$

$$\left(h_j = h_j^{*\hat{H}} > 0, \quad g^{p\bar{q}} > 0 \right) \xrightarrow{8.3.6} \frac{\sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \text{Tr}((\hat{\nabla}_{\bar{q}}(h_j)) \cdot h_j^{-1} \cdot \hat{\nabla}_p(h_j))}{\text{Tr}(h_j)} - \frac{|\partial(\text{Tr}(h_j))|_\omega^2}{(\text{Tr}(h_j))^2} \geq 0.$$

$$\begin{aligned} \mathrm{Tr}(h_j \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) &\geq -\|i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E\|_{\mathrm{op}} \cdot \mathrm{Tr}(h_j), \quad \mathrm{Tr}(h_j \cdot \log(h_j)) = \sum_{a=1}^r \mu_{j,a} \cdot \log(\mu_{j,a}) \geq -\frac{r}{e}. \\ \Delta_\omega(u_j) &= \Delta_\omega(\log(\mathrm{Tr}(h_j))) \geq -\frac{1}{2} \cdot \left(\|i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E\|_{\mathrm{op}} + \frac{1}{e} \right). \end{aligned}$$

The Green estimate for a function with $\sup_X(u_j) = 0$ gives $\int_X u_j \cdot \frac{\omega^n}{n!} \geq -C$. Jensen's inequality then yields

$$\begin{aligned} \left(0 < \tilde{h}_j \leq I_E, \quad 0 < \sigma \leq 1 \right) &\xrightarrow{8.1.1} \int_X \mathrm{Tr}(\tilde{h}_j^\sigma) \cdot \frac{\omega^n}{n!} \geq \int_X \mathrm{Tr}(\tilde{h}_j) \cdot \frac{\omega^n}{n!} = \int_X \exp(u_j) \cdot \frac{\omega^n}{n!}. \\ \int_X \exp(u_j) \cdot \frac{\omega^n}{n!} &\geq \mathrm{Vol}_\omega(X) \cdot \exp\left(\frac{1}{\mathrm{Vol}_\omega(X)} \int_X u_j \cdot \frac{\omega^n}{n!} \right) > 0. \end{aligned}$$

Thus no strong L^2 limit can vanish. □

Theorem 8.3.8 (Uhlenbeck–Yau blow-up extraction input). *Assume $\varepsilon_j \downarrow 0$ and*

$$h_j = h_{\varepsilon_j} = h_j^{*\hat{H}} > 0, \quad H_j = \hat{H} \cdot h_j, \quad \det(h_j) = 1.$$

Assume further that the complete perturbed system holds

$$-2 \cdot \sum_{p=1}^n \sum_{q=1}^n g^{p\bar{q}} \cdot \left[(F_{\hat{H}})_{\bar{q}p} + \hat{\nabla}_{\bar{q}}(h_j^{-1} \cdot \hat{\nabla}_p(h_j)) \right] - \lambda \cdot I_E + \varepsilon_j \cdot \log(h_j) = 0.$$

If

$$M_j = \sup_{x \in X} \mathrm{Tr}(h_j(x)) \longrightarrow \infty, \quad \tilde{h}_j = M_j^{-1} \cdot h_j,$$

then, after one diagonal subsequence, the normalized powers have simultaneous limits

$$\tilde{h}_j^\sigma \rightharpoonup q_\sigma \quad \text{in } W^{1,2}, \quad \tilde{h}_j^\sigma \longrightarrow q_\sigma \quad \text{in } L^2 \quad (\sigma \in \mathbb{Q} \cap (0, 1)).$$

The monotone limit

$$q_0 = \lim_{\sigma \downarrow 0} q_\sigma, \quad \pi = I_E - q_0$$

satisfies

$$q_0 \in W^{1,2}(X, \mathrm{End}(E)), \quad q_0 = q_0^{*\hat{H}} = q_0^2, \quad q_0 \cdot \bar{\partial}_E(q_0) = 0, \quad 0 < \mathrm{Tr}(\pi) < r \quad \text{almost everywhere.}$$

It also satisfies the cutoff-limit inequality

$$(8.3.3) \quad \int_X \mathrm{Tr}(\pi \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) \cdot \frac{\omega^n}{n!} - \int_X |\bar{\partial}_E(\pi)|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!} \geq 0.$$

This is the second deep analytic input. The normalized energy estimate and trace nonconcentration calculation above are its explicit preparation. The additional weighted cutoff estimate supplies the σ -uniform weak derivative control and the limiting inequality.

8.4. Weak projections and coherent subsheaves.

Theorem 8.4.1 (Weak projection from blow-up). *Assume*

$$\varepsilon_j \downarrow 0, \quad M_j = \sup_{x \in X} \mathrm{Tr}(h_{\varepsilon_j}(x)) \longrightarrow \infty.$$

Then there exists a nonzero proper projection

$$\pi \in W^{1,2}(X, \mathrm{End}(E)), \quad \pi = \pi^{*\hat{H}} = \pi^2, \quad 0 < \mathrm{rank}(\pi) < r,$$

satisfying

$$(8.4.1) \quad (I_E - \pi) \cdot \bar{\partial}_E(\pi) = 0 \quad \text{almost everywhere.}$$

Proof. By $\varepsilon_j \downarrow 0$, $M_j = \sup_{x \in X} \text{Tr}(h_{\varepsilon_j}(x)) \rightarrow \infty$, and $\tilde{h}_j = M_j^{-1} \cdot h_{\varepsilon_j}$, choose one diagonal subsequence for every rational $\sigma \in (0, 1)$.

$$\begin{aligned} (M_j \rightarrow \infty, \quad \tilde{h}_j = M_j^{-1} \cdot h_{\varepsilon_j}) &\xrightarrow{8.3.4} \|\tilde{h}_j^\sigma\|_{W^{1,2}(X, \text{End}(E))} \leq C_\sigma. \\ (M_j \rightarrow \infty, \quad \tilde{h}_j = M_j^{-1} \cdot h_{\varepsilon_j}) &\xrightarrow{8.3.7} \int_X \text{Tr}(\tilde{h}_j^\sigma) \cdot \frac{\omega^n}{n!} \geq c_\sigma > 0. \\ \|\tilde{h}_j^\sigma\|_{W^{1,2}(X, \text{End}(E))} &\leq C_\sigma \xrightarrow{8.3.5} \tilde{h}_j^\sigma \rightarrow q_\sigma \quad \text{in } W^{1,2}, \quad \tilde{h}_j^\sigma \rightarrow q_\sigma \quad \text{in } L^2. \\ (0 < \tilde{h}_j^\sigma \leq I_E, \quad \int_X \text{Tr}(\tilde{h}_j^\sigma) \cdot \frac{\omega^n}{n!} &\geq c_\sigma > 0) \implies 0 \leq q_\sigma \leq I_E, \quad q_\sigma^{*\hat{H}} = q_\sigma, \quad q_\sigma \neq 0. \end{aligned}$$

For rational $\sigma_1, \sigma_2 > 0$ with $\sigma_1 + \sigma_2 < 1$,

$$\begin{aligned} \tilde{h}_j &= \tilde{h}_j^{*\hat{H}} > 0 \xrightarrow{8.1.1} \tilde{h}_j^{\sigma_1} \cdot \tilde{h}_j^{\sigma_2} = \tilde{h}_j^{\sigma_1 + \sigma_2}. \\ \|\tilde{h}_j^{\sigma_1} \cdot \tilde{h}_j^{\sigma_2} - q_{\sigma_1} \cdot q_{\sigma_2}\|_{L^1} &\leq \|\tilde{h}_j^{\sigma_1}\|_{L^2} \cdot \|\tilde{h}_j^{\sigma_2} - q_{\sigma_2}\|_{L^2} + \|\tilde{h}_j^{\sigma_1} - q_{\sigma_1}\|_{L^2} \cdot \|q_{\sigma_2}\|_{L^2} \rightarrow 0. \\ q_{\sigma_1} \cdot q_{\sigma_2} &= q_{\sigma_1 + \sigma_2}. \end{aligned}$$

For $0 < \tau < \sigma < 1$, the scalar inequality $0 < x^\sigma \leq x^\tau \leq 1$ on $0 < x \leq 1$ gives

$$(0 < \tilde{h}_j \leq I_E, \quad 0 < \tau < \sigma < 1) \xrightarrow{8.1.1} 0 \leq \tilde{h}_j^\sigma \leq \tilde{h}_j^\tau \leq I_E \implies 0 \leq q_\sigma \leq q_\tau \leq I_E \implies q_0 = \lim_{\sigma \downarrow 0} q_\sigma, \quad \pi = I_E - q_0.$$

$$q_\sigma^2 = q_{2\sigma} \implies q_0^2 = q_0 \implies \pi^2 = (I_E - q_0)^2 = I_E - 2 \cdot q_0 + q_0^2 = I_E - q_0 = \pi.$$

For a fixed rational $\sigma_0 \in (0, 1)$, strong L^2 convergence and an almost-everywhere subsequence give

$$\begin{aligned} (h_j = h_{\varepsilon_j}, \quad \text{Tr}(i \cdot \Lambda_\omega(F_{\hat{H}}))) &\xrightarrow{8.1.5} \det(h_j) = 1, \quad \det(\tilde{h}_j) = M_j^{-r}. \\ \det(q_{\sigma_0}) &= \lim_{j \rightarrow \infty} \det(\tilde{h}_j^{\sigma_0}) = \lim_{j \rightarrow \infty} M_j^{-r \cdot \sigma_0} = 0. \\ q_{\sigma_0/2^m}^{2^m} = q_{\sigma_0} &\implies \ker(q_{\sigma_0/2^m}) = \ker(q_{\sigma_0}) \neq 0 \implies \ker(q_0) \neq 0 \implies q_0 \neq I_E. \\ \left(\int_X \text{Tr}(\tilde{h}_j^{\sigma_0}) \cdot \frac{\omega^n}{n!} \geq c_{\sigma_0} > 0 \right) &\xrightarrow{8.3.7} q_{\sigma_0} \neq 0, \quad q_0 \geq q_{\sigma_0} \neq 0. \\ (q_\sigma = \lim_{j \rightarrow \infty} \tilde{h}_j^\sigma, \quad q_0 = \lim_{\sigma \downarrow 0} q_\sigma) &\xrightarrow{8.3.8} q_0 \in W^{1,2}(X, \text{End}(E)), \quad q_0 \cdot \bar{\partial}_E(q_0) = 0. \\ (I_E - \pi) \cdot \bar{\partial}_E(\pi) &= q_0 \cdot \bar{\partial}_E(I_E - q_0) = -q_0 \cdot \bar{\partial}_E(q_0) = 0. \\ \text{Tr}(\pi) \in W^{1,2}(X, \mathbb{Z}), \quad X \text{ connected} &\implies d(\text{Tr}(\pi)) = 0, \quad \text{Tr}(\pi) = s, \quad 0 < s < r. \end{aligned}$$

Hence π is a nonzero proper weak holomorphic projection. $\square$

Example 8.4.2 (A complete rank-two projection limit). On a coordinate disk set

$$D = 1 + |z|^2, \quad s(z) = \begin{bmatrix} 1 \\ z \end{bmatrix}, \quad \pi(z) = \frac{1}{D} \cdot \begin{bmatrix} 1 & \bar{z} \\ z & |z|^2 \end{bmatrix}.$$

The projection identity is the full multiplication

$$\begin{aligned} \pi(z)^2 &= \frac{1}{D^2} \cdot \begin{bmatrix} 1 & \bar{z} \\ z & |z|^2 \end{bmatrix} \cdot \begin{bmatrix} 1 & \bar{z} \\ z & |z|^2 \end{bmatrix} \\ &= \frac{1}{D^2} \cdot \begin{bmatrix} 1 \cdot 1 + \bar{z} \cdot z & 1 \cdot \bar{z} + \bar{z} \cdot |z|^2 \\ z \cdot 1 + |z|^2 \cdot z & z \cdot \bar{z} + |z|^2 \cdot |z|^2 \end{bmatrix} = \frac{1}{D^2} \cdot \begin{bmatrix} D & \bar{z} \cdot D \\ z \cdot D & |z|^2 \cdot D \end{bmatrix} = \pi(z). \end{aligned}$$

Moreover,

$$\bar{\partial}_E(\pi) = \frac{d\bar{z}}{D^2} \cdot \begin{bmatrix} -z & 1 \\ -z^2 & z \end{bmatrix}, \quad I_E - \pi = \frac{1}{D} \cdot \begin{bmatrix} |z|^2 & -\bar{z} \\ -z & 1 \end{bmatrix}.$$

$$\begin{aligned}
(I_E - \pi) \cdot \bar{\partial}_E(\pi) &= \frac{d\bar{z}}{D^3} \cdot \begin{bmatrix} |z|^2 & -\bar{z} \\ -z & 1 \end{bmatrix} \cdot \begin{bmatrix} -z & 1 \\ -z^2 & z \end{bmatrix} \\
&= \frac{d\bar{z}}{D^3} \cdot \begin{bmatrix} -z^2 \cdot \bar{z} + \bar{z} \cdot z^2 & z \cdot \bar{z} - z \cdot \bar{z} \\ z^2 - z^2 & -z + z \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot d\bar{z}.
\end{aligned}$$

At $z = 2$,

$$\begin{aligned}
\pi(2) &= \frac{1}{5} \cdot \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad \pi(2)^2 = \frac{1}{25} \cdot \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 & 1 \cdot 2 + 2 \cdot 4 \\ 2 \cdot 1 + 4 \cdot 2 & 2 \cdot 2 + 4 \cdot 4 \end{bmatrix} \\
&= \frac{1}{25} \cdot \begin{bmatrix} 1+4 & 2+8 \\ 2+8 & 4+16 \end{bmatrix} = \frac{1}{5} \cdot \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \pi(2).
\end{aligned}$$

At $z = 2$, set $\pi = \pi(2)$. Then

$$I_E - \pi = \frac{1}{5} \cdot \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix},$$

and the determinant-one family has the full matrix

$$h_m = m^2 \cdot (I_E - \pi) + m^{-2} \cdot \pi = \frac{1}{5} \cdot \begin{bmatrix} 4 \cdot m^2 + m^{-2} & -2 \cdot m^2 + 2 \cdot m^{-2} \\ -2 \cdot m^2 + 2 \cdot m^{-2} & m^2 + 4 \cdot m^{-2} \end{bmatrix}.$$

$$\begin{aligned}
\det(h_m) &= \frac{1}{25} \cdot [(4 \cdot m^2 + m^{-2}) \cdot (m^2 + 4 \cdot m^{-2}) - (-2 \cdot m^2 + 2 \cdot m^{-2})^2] \\
&= \frac{1}{25} \cdot [4 \cdot m^4 + 17 + 4 \cdot m^{-4} - (4 \cdot m^4 - 8 + 4 \cdot m^{-4})] = \frac{25}{25} = 1.
\end{aligned}$$

$$\text{Tr}(h_m) = \frac{1}{5} \cdot [(4 \cdot m^2 + m^{-2}) + (m^2 + 4 \cdot m^{-2})] = \frac{1}{5} \cdot (5 \cdot m^2 + 5 \cdot m^{-2}) = m^2 + m^{-2}.$$

For every fixed $\sigma > 0$,

$$\left(\frac{h_m}{\text{Tr}(h_m)} \right)^\sigma \longrightarrow I_E - \pi, \quad I_E - (I_E - \pi) = \pi.$$

Thus the numerical projection is exactly the projection extracted by the normalized-power construction.

Theorem 8.4.3 (Compactness alternative). *Let $\varepsilon_j \downarrow 0$ and let H_{ε_j} solve the fixed-determinant perturbed equation. After passing to a subsequence, exactly one of the following occurs.*

$$\sup_{j \geq 1} \sup_{x \in X} \text{Tr}(h_{\varepsilon_j}(x)) < \infty \implies H_{\varepsilon_j} \longrightarrow H_\infty \text{ smoothly, } i \cdot \Lambda_\omega(F_{H_\infty}) = \lambda \cdot I_E,$$

or

$$\sup_{x \in X} \text{Tr}(h_{\varepsilon_j}(x)) \longrightarrow \infty \implies \pi = \pi^{*\hat{n}} = \pi^2, \quad 0 < \text{rank}(\pi) < r, \quad (I_E - \pi) \cdot \bar{\partial}_E(\pi) = 0.$$

Proof. By $M_j = \sup_{x \in X} \text{Tr}(h_{\varepsilon_j}(x))$, $\det(h_{\varepsilon_j}) = 1$, and $M_j \in (0, \infty)$,

$$(M_j)_{j \geq 1} \text{ bounded or } \exists (j_\ell)_{\ell \geq 1} \text{ with } M_{j_\ell} \longrightarrow \infty.$$

$$(M_j)_{j \geq 1} \text{ bounded} \xrightarrow{8.2.2} H_{\varepsilon_j} \longrightarrow H_\infty, \quad i \cdot \Lambda_\omega(F_{H_\infty}) = \lambda \cdot I_E,$$

$$(M_j)_{j \geq 1} \text{ unbounded} \implies M_{j_\ell} \longrightarrow \infty \xrightarrow{8.4.1} \pi = \pi^{*\hat{n}} = \pi^2, \quad 0 < \text{rank}(\pi) < r, \quad (I_E - \pi) \cdot \bar{\partial}_E(\pi) = 0.$$

These alternatives are mutually exclusive and exhaust every sequence. $\square$

Theorem 8.4.4 (Uhlenbeck–Yau regularity for weak projections). *Let $\pi \in W^{1,2}(X, \text{End}(E))$ be a nonzero proper projection satisfying*

$$\pi = \pi^{*\hat{n}} = \pi^2, \quad (I_E - \pi) \cdot \bar{\partial}_E(\pi) = 0 \text{ almost everywhere.}$$

Then there is an analytic subset $Z \subset X$ with $\text{codim}_{\mathbb{C}}(Z) \geq 2$ such that π is smooth on $X \setminus Z$ and $\pi(E)|_{X \setminus Z}$ is a holomorphic subbundle. Moreover,

$$\mathcal{F}(U) = \{s \in \mathcal{O}(E)(U) \mid \pi \cdot s = s \text{ almost everywhere}\}$$

defines a saturated coherent subsheaf $\mathcal{F} \subset \mathcal{O}(E)$ with torsion-free quotient and

$$\mathcal{F}|_{X \setminus Z} = \pi(E), \quad \text{rank}(\mathcal{F}) = \text{rank}(\pi).$$

This theorem is the deep weak-holomorphic-map regularity input. The preceding matrix and Sobolev computations establish its hypotheses.

8.5. Exclusion of blow-up by stability.

Fact 8.5.1 (Weak projection degree inequality). *The projection produced by a blow-up sequence satisfies*

$$(8.5.1) \quad \int_X \text{Tr}(\pi \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) \cdot \frac{\omega^n}{n!} - \int_X |\bar{\partial}_E(\pi)|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!} \geq 0.$$

Proof. By $M_j = \sup_{x \in X} \text{Tr}(h_j(x)) \rightarrow \infty$, $\tilde{h}_j = M_j^{-1} \cdot h_j$, and $\pi = I_E - \lim_{\sigma \downarrow 0} \lim_{j \rightarrow \infty} \tilde{h}_j^\sigma$,

$$(M_j \rightarrow \infty, \quad \pi = I_E - q_0) \xrightarrow{8.3.8} \int_X \text{Tr}(\pi \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) \cdot \frac{\omega^n}{n!} - \int_X |\bar{\partial}_E(\pi)|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!} \geq 0.$$

Thus the cutoff-limit input gives the asserted inequality. $\square$

Theorem 8.5.2 (Destabilizing subsheaf from blow-up). *If the blow-up branch occurs, the saturated coherent subsheaf supplied by weak-projection regularity satisfies*

$$0 < \text{rank}(\mathcal{F}) < \text{rank}(E), \quad \mu_\omega(\mathcal{F}) \geq \mu_\omega(E).$$

Proof. By $\mathcal{F}|_{X \setminus Z} = \pi(E)$, $0 < \text{rank}(\pi) < r$, and $\text{codim}_{\mathbb{C}}(Z) \geq 2$,

$$(\pi = \pi^{*\hat{H}} = \pi^2, \quad (I_E - \pi) \cdot \bar{\partial}_E(\pi) = 0) \xrightarrow{8.4.4} \mathcal{F}|_{X \setminus Z} = \pi(E), \quad \text{codim}_{\mathbb{C}}(Z) \geq 2.$$

$$0 < \text{rank}(\mathcal{F}) = \text{Tr}(\pi) < r.$$

The fixed-metric weak Chern–Weil identity is

$$\begin{aligned} (\mathcal{F} \subset E \text{ saturated}, \quad \mathcal{F}|_{X \setminus Z} = \pi(E)) &\xrightarrow{7.5.3} 2 \cdot \pi \cdot \deg_\omega(\mathcal{F}) = \int_X \text{Tr}(\pi \cdot i \cdot \Lambda_\omega(F_{\hat{H}})) \cdot \frac{\omega^n}{n!} - \int_X |\bar{\partial}_E(\pi)|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!}. \\ i \cdot \Lambda_\omega(F_{\hat{H}}) &= \lambda \cdot I_E + (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E) \implies 2 \cdot \pi \cdot \deg_\omega(\mathcal{F}) = \lambda \cdot \text{rank}(\mathcal{F}) \cdot \text{Vol}_\omega(X) \\ &\quad + \int_X \text{Tr}(\pi \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) \cdot \frac{\omega^n}{n!} - \int_X |\bar{\partial}_E(\pi)|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!}. \\ \left(\int_X \text{Tr}(\pi \cdot (i \cdot \Lambda_\omega(F_{\hat{H}}) - \lambda \cdot I_E)) \cdot \frac{\omega^n}{n!} - \int_X |\bar{\partial}_E(\pi)|_{\hat{H}, \omega}^2 \cdot \frac{\omega^n}{n!} \geq 0 \right) &\xrightarrow{8.5.1} 2 \cdot \pi \cdot \deg_\omega(\mathcal{F}) \geq \lambda \cdot \text{rank}(\mathcal{F}) \cdot \text{Vol}_\omega(X). \\ \lambda = \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)} \implies \deg_\omega(\mathcal{F}) &\geq \text{rank}(\mathcal{F}) \cdot \mu_\omega(E) \xrightarrow{7.1.1} \mu_\omega(\mathcal{F}) \geq \mu_\omega(E). \end{aligned}$$

Thus every blow-up sequence produces a proper destabilizing subsheaf. $\square$

Theorem 8.5.3 (The Uhlenbeck–Yau converse). *If E is stable, then E admits a Hermitian–Einstein metric. If E is polystable, the stable-factor metrics combine to a Hermitian–Einstein metric on E .*

Proof. By E stable, $0 < \varepsilon \leq 1$, and $\varepsilon_j \downarrow 0$,

$$(E \text{ stable}, \quad 0 < \varepsilon \leq 1) \xrightarrow{8.1.6} H_\varepsilon = \hat{H} \cdot h_\varepsilon, \quad i \cdot \Lambda_\omega(F_{H_\varepsilon}) - \lambda \cdot I_E = -\varepsilon \cdot \log(h_\varepsilon), \quad \det(h_\varepsilon) = 1.$$

$$(E \text{ stable}, \quad H_{\varepsilon_j} \text{ solves the perturbed equation}) \xrightarrow{8.4.3} (\text{the branch is bounded or a weak projection forms}).$$

Suppose first that the blow-up branch occurs.

$$\left(\sup_{x \in X} \text{Tr}(h_{\varepsilon_j}(x)) \rightarrow \infty \right) \xrightarrow{8.5.2} 0 < \text{rank}(\mathcal{F}) < \text{rank}(E), \quad \mu_\omega(\mathcal{F}) \geq \mu_\omega(E).$$

$$(0 < \text{rank}(\mathcal{F}) < \text{rank}(E), \quad \mu_\omega(\mathcal{F}) \geq \mu_\omega(E)) \xrightarrow{7.1.3} \text{a contradiction.}$$

$$\sup_{j \geq 1} \sup_{x \in X} \text{Tr}(h_{\varepsilon_j}(x)) < \infty \xrightarrow{8.2.2} H_{\varepsilon_j} \longrightarrow H_\infty, \quad i \cdot \Lambda_\omega(F_{H_\infty}) = \lambda \cdot I_E.$$

Suppose now that

$$E \text{ is polystable} \xrightarrow{7.1.3} E = \bigoplus_{a=1}^N E_a, \quad E_a \text{ stable}, \quad \mu_\omega(E_a) = \mu_\omega(E) \quad (1 \leq a \leq N).$$

Apply the stable case above to every E_a .

$$(H_a \text{ is Hermitian-Einstein}, \quad \mu_\omega(E_a) = \mu_\omega(E)) \xrightarrow{6.6.4} i \cdot \Lambda_\omega(F_{H_a}) = \frac{2 \cdot \pi \cdot \mu_\omega(E_a)}{\text{Vol}_\omega(X)} \cdot I_{E_a} = \frac{2 \cdot \pi \cdot \mu_\omega(E)}{\text{Vol}_\omega(X)} \cdot I_{E_a} = \lambda \cdot I_{E_a}.$$

$$\left(E = \bigoplus_{a=1}^N E_a, \quad H = \bigoplus_{a=1}^N H_a \right) \xrightarrow{7.2.3} F_H = \begin{bmatrix} F_{H_1} & 0 & \cdots & 0 \\ 0 & F_{H_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & F_{H_N} \end{bmatrix}.$$

$$i \cdot \Lambda_\omega(F_H) = \begin{bmatrix} \lambda \cdot I_{E_1} & 0 & \cdots & 0 \\ 0 & \lambda \cdot I_{E_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda \cdot I_{E_N} \end{bmatrix} = \lambda \cdot I_E \xrightarrow{6.6.1} H \text{ is Hermitian-Einstein}.$$

Thus stability gives existence and equal-slope direct sums give the polystable case. $\square$

If H_0 and H_1 are Hermitian-Einstein metrics on a stable bundle, the standard convexity identity for $h = H_0^{-1} \cdot H_1$ gives

$$0 = 2 \cdot \int_X \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\nabla_{H_0, \bar{k}}(h)) \cdot h^{-1} \cdot \nabla_{H_0, j}(h)) \cdot \frac{\omega^n}{n!} \implies \nabla_{H_0}^{1,0}(h) = 0 \implies \nabla_{H_0}(h) = 0.$$

Stability makes every parallel holomorphic endomorphism scalar, so $h = c \cdot I_E$ for some $c > 0$. The two directions are

$$(E \text{ admits a Hermitian-Einstein metric}) \xrightarrow{7.5.4} (E \text{ is polystable}).$$

$$(E \text{ is polystable}) \xrightarrow{8.5.3} (E \text{ admits a Hermitian-Einstein metric}).$$

Hence

$$(E \text{ admits a Hermitian-Einstein metric}) \iff (E \text{ is polystable}).$$

9. HEISENBERG GROUP

Definition 9.0.1 (Heisenberg Lie algebra). Let n be a positive integer, we denote the Heisenberg Lie algebra $\mathfrak{h}_n$ as the real Lie algebra of dimension $2n + 1$ generated by elements $P_1, \dots, P_n, Q_1, \dots, Q_n$ and S satisfy the following structure equation,

$$[P_j, Q_k] = \delta_{jk} S, \quad (j, k = 1, \dots, n),$$

$$[P_j, P_k] = [Q_j, Q_k] = [P_j, S] = [Q_j, S] = 0, \quad (j, k = 1, \dots, n),$$

where δ_{jk} is the usual Kronecker symbol such that $\delta_{jk} = 1$ if $j = k$, $\delta_{jk} = 0$ otherwise. Thus, $\mathfrak{h}_n$ is nilpotent of step 2 (all third-order brackets $[[X, Y], Z]$ vanish), and the linear span of S is both the center and the commutator subalgebra of $\mathfrak{h}_n$.

Definition 9.0.2. Define

$$\mathfrak{h}_n^{\text{mat}} = \{m(p, q, s) : p, q \in \mathbb{R}^n, s \in \mathbb{R}\}, \quad m(p, q, s) = \begin{bmatrix} 0 & p_1 & \cdots & p_n & s \\ 0 & 0 & \cdots & 0 & q_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & q_n \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix}.$$

Fact 9.0.3. $\mathfrak{h}_n^{\text{mat}}$ is a Lie subalgebra of $\mathfrak{gl}(n+2, \mathbb{R})$, and the map $(p, q, s) \mapsto m(p, q, s)$ is an isomorphism $\mathfrak{h}_n \xrightarrow{\sim} \mathfrak{h}_n^{\text{mat}}$.

Proof. Let $\alpha, \beta \in \mathbb{R}$ be constants, and to prove linearity of the map to show it is a linear subspace of $\mathfrak{gl}(n+2, \mathbb{R})$. Let $x = (p, q, s), y = (p', q', s')$ and $\phi(x) = m(x)$ such that

$$\begin{aligned} \alpha x + \beta y &= (\alpha p + \beta p', \alpha q + \beta q', \alpha s + \beta s'), \\ \implies \phi(\alpha x + \beta y) &= m(\alpha x + \beta y) = \begin{bmatrix} 0 & (\alpha p + \beta p')^T & s \\ 0 & 0_{n \times n} & (\alpha q + \beta q') \\ 0 & 0 & 0 \end{bmatrix}, \\ &= \alpha \begin{bmatrix} 0 & p^T & s \\ 0 & 0_{n \times n} & q \\ 0 & 0 & 0 \end{bmatrix} + \beta \begin{bmatrix} 0 & (p')^T & s' \\ 0 & 0_{n \times n} & q' \\ 0 & 0 & 0 \end{bmatrix} = \alpha \phi(x) + \beta \phi(y) = \alpha m(x) + \beta m(y), \end{aligned}$$

proving linearity. Hence, $\mathfrak{h}_n^{\text{mat}} \subset \mathfrak{gl}(n+2, \mathbb{R})$ is a linear subspace. To show it is closed, it suffice to show that it is closed under the Lie bracket of form $[X, Y] = XY - YX$. Let

$$\begin{aligned} X &= m(p, q, s) = \begin{bmatrix} 0 & p^T & s \\ 0 & 0_{n \times n} & q \\ 0 & 0 & 0 \end{bmatrix}, \quad Y = m(p', q', s') = \begin{bmatrix} 0 & (p')^T & s' \\ 0 & 0_{n \times n} & q' \\ 0 & 0 & 0 \end{bmatrix}, \\ \implies [X, Y] &= XY - YX = \begin{bmatrix} 0 & p^T & s \\ 0 & 0_{n \times n} & q \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & (p')^T & s' \\ 0 & 0_{n \times n} & q' \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & (p')^T & s' \\ 0 & 0_{n \times n} & q' \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & p^T & s \\ 0 & 0_{n \times n} & q \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \cdot 0 + p^T \cdot 0 + s \cdot 0 & 0 \cdot (p')^T + p^T \cdot 0_{n \times n} + s \cdot 0 & 0 \cdot s' + p^T \cdot q' + s \cdot 0 \\ 0 \cdot 0 + 0_{n \times n} \cdot 0 + q \cdot 0 & 0 \cdot (p')^T + 0_{n \times n} \cdot 0_{n \times n} + q \cdot 0 & 0 \cdot s' + 0_{n \times n} \cdot q' + q \cdot 0 \\ 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 & 0 \cdot (p')^T + 0 \cdot 0_{n \times n} + 0 \cdot 0 & 0 \cdot s' + 0 \cdot q' + 0 \cdot 0 \end{bmatrix} \\ &\quad - \begin{bmatrix} 0 \cdot 0 + (p')^T \cdot 0 + s' \cdot 0 & 0 \cdot p^T + (p')^T \cdot 0_{n \times n} + s' \cdot 0 & 0 \cdot s + (p')^T \cdot q + s' \cdot 0 \\ 0 \cdot 0 + 0_{n \times n} \cdot 0 + q' \cdot 0 & 0 \cdot p^T + 0_{n \times n} \cdot 0_{n \times n} + q' \cdot 0 & 0 \cdot s + 0_{n \times n} \cdot q + q' \cdot 0 \\ 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 & 0 \cdot p^T + 0 \cdot 0_{n \times n} + 0 \cdot 0 & 0 \cdot s + 0 \cdot q + 0 \cdot 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & (0_n)^T & p^T \cdot q' \\ 0_n & 0_{n \times n} & 0_n \\ 0 & (0_n)^T & 0 \end{bmatrix} - \begin{bmatrix} 0 & (0_n)^T & (p')^T \cdot q \\ 0_n & 0_{n \times n} & 0_n \\ 0 & (0_n)^T & 0 \end{bmatrix} = \begin{bmatrix} 0 & (0_n)^T & p^T \cdot q' - (p')^T \cdot q \\ 0_n & 0_{n \times n} & 0_n \\ 0 & (0_n)^T & 0 \end{bmatrix} \\ &= m(0, 0, p^T \cdot q' - (p')^T \cdot q) \in \mathfrak{h}_n^{\text{mat}} \subset \mathfrak{gl}(n+2, \mathbb{R}) \text{ is a closed linear subspace.} \end{aligned}$$

□

Part 3. Seminar talks⁶**10. SPINORIAL PERELMAN ENTROPY AND GRADIENT FLOW**

Let (M^n, g) be a closed Riemannian spin manifold, let $f \in C^\infty(M, \mathbb{R})$, let $\tau > 0$, and let ΣM denote the complex spin bundle.

Definition 10.0.1 (Weighted spin geometry). Define

$$d\Omega_f = \frac{1}{(4 \cdot \pi \cdot \tau)^{n/2}} \cdot \exp(-f) \cdot d\mu_g, \quad \operatorname{div}_f(\alpha) = \exp(f) \cdot \operatorname{div}(\exp(-f) \cdot \alpha).$$

For $u \in C^\infty(M, \mathbb{R})$ and $\psi \in \Gamma(M, \Sigma M)$, define

$$\Delta_f(u) = \Delta(u) - \langle \nabla f, \nabla u \rangle_g, \quad D_f(\psi) = \exp(f/2) \cdot D(\exp(-f/2) \cdot \psi).$$

$$(10.0.1) \quad D_f(\psi) = D(\psi) - \frac{1}{2} \cdot c(\nabla f) \cdot \psi.$$

The weighted spinor Laplacian and weighted curvatures are

$$\Delta_f^\Sigma(\psi) = -\nabla^*(\nabla(\psi)) - \nabla_{\nabla f}(\psi), \quad \operatorname{Ric}_f = \operatorname{Ric}(g) + \operatorname{Hess}(f),$$

$$(10.0.2) \quad R_f = R(g) + 2 \cdot \Delta(f) - |\nabla f|_g^2 = R(g) + |\nabla f|_g^2 - 2 \cdot \Delta_f(f).$$

Proposition 10.0.2 (Weighted integration by parts). For $u, v \in C^\infty(M, \mathbb{R})$,

$$\int_M \Delta_f(u) \cdot v \cdot \exp(-f) \cdot d\mu_g = - \int_M \langle \nabla u, \nabla v \rangle_g \cdot \exp(-f) \cdot d\mu_g.$$

Consequently Δ_f is self-adjoint on $L^2(M, \exp(-f)d\mu_g)$. For $\psi, \chi \in \Gamma(M, \Sigma M)$, one also has

$$\int_M \operatorname{Re}(\langle -\Delta_f^\Sigma(\psi), \chi \rangle) \cdot \exp(-f) \cdot d\mu_g = \int_M \operatorname{Re}(\langle \nabla(\psi), \nabla(\chi) \rangle_g) \cdot \exp(-f) \cdot d\mu_g.$$

Proof. By $u, v \in C^\infty(M, \mathbb{R})$, $\partial M = \emptyset$, and div_f as in the displayed definition,

$$\operatorname{div}(v \cdot \exp(-f) \cdot \nabla u) = v \cdot \Delta_f(u) \cdot \exp(-f) + \langle \nabla u, \nabla v \rangle_g \cdot \exp(-f).$$

$$\partial M = \emptyset \implies \int_M \operatorname{div}(v \cdot \exp(-f) \cdot \nabla u) \cdot d\mu_g = 0 \xrightarrow{10.0.1} \int_M \Delta_f(u) \cdot v \cdot \exp(-f) \cdot d\mu_g = - \int_M \langle \nabla u, \nabla v \rangle_g \cdot \exp(-f) \cdot d\mu_g.$$

Interchanging u and v proves self-adjointness. The same divergence computation applied componentwise in a synchronous spin frame gives the spinor identity. $\square$

Definition 10.0.3 (Perelman functional). Perelman's $\mathcal{F}$ -functional is

$$\mathcal{F}(g, f) = \int_M \left(R(g) + |\nabla f|_g^2 \right) \cdot \exp(-f) \cdot d\mu_g.$$

This is Perelman's original weighted curvature functional [2].

Proposition 10.0.4 (Weighted scalar-curvature form). The functional satisfies

$$(10.0.3) \quad \mathcal{F}(g, f) = \int_M R_f \cdot \exp(-f) \cdot d\mu_g.$$

⁶Based on lectures by Mu-Tao Wang, Aaron Chow [4] [6].

Proof. By $R_f = R(g) + 2\Delta(f) - |\nabla f|_g^2$, $\partial M = \emptyset$, and weighted integration by parts,

$$\begin{aligned} \int_M \Delta(f) \cdot \exp(-f) \cdot d\mu_g &= - \int_M \langle \nabla f, \nabla(\exp(-f)) \rangle_g \cdot d\mu_g = \int_M |\nabla f|_g^2 \cdot \exp(-f) \cdot d\mu_g. \\ \int_M R_f \cdot \exp(-f) \cdot d\mu_g &= \int_M (R(g) + 2\Delta(f) - |\nabla f|_g^2) \cdot \exp(-f) \cdot d\mu_g \xrightarrow{10.0.2} \int_M (R(g) + |\nabla f|_g^2) \cdot \exp(-f) \cdot d\mu_g. \\ \left(\int_M R_f \cdot \exp(-f) \cdot d\mu_g = \int_M (R(g) + |\nabla f|_g^2) \cdot \exp(-f) \cdot d\mu_g \right) &\xrightarrow{10.0.3} \int_M R_f \cdot \exp(-f) \cdot d\mu_g = \mathcal{F}(g, f). \end{aligned}$$

Thus the two forms of Perelman's functional agree. $\square$

Proposition 10.0.5 (Weighted Weitzenböck identity). *For every smooth spinor ψ ,*

$$(10.0.4) \quad D_f^2(\psi) = -\Delta_f^\Sigma(\psi) + \frac{1}{4} \cdot R_f \cdot \psi.$$

Proof. By $D_f(\psi) = \exp(f/2) \cdot D(\exp(-f/2) \cdot \psi)$, $D^2(\psi) = \nabla^*(\nabla(\psi)) + R(g)\psi/4$, and the weighted definitions,

$$\begin{aligned} D_f^2(\psi) &= \exp(f/2) \cdot D^2(\exp(-f/2) \cdot \psi). \\ \exp(f/2) \cdot \nabla^* \nabla(\exp(-f/2) \cdot \psi) &= -\Delta_f^\Sigma(\psi) + \frac{1}{4} \cdot (2\Delta(f) - |\nabla f|_g^2) \cdot \psi. \\ D_f^2(\psi) &= -\Delta_f^\Sigma(\psi) + \frac{1}{4} \cdot (R(g) + 2\Delta(f) - |\nabla f|_g^2) \cdot \psi \xrightarrow{10.0.2} D_f^2(\psi) = -\Delta_f^\Sigma(\psi) + \frac{1}{4} \cdot R_f \cdot \psi. \end{aligned}$$

This is the weighted Weitzenböck identity. $\square$

Theorem 10.0.6 (Spinorial form of Perelman's functional). *For every smooth spinor ψ ,*

$$(10.0.5) \quad 4 \cdot \int_M |D_f(\psi)|^2 \cdot \exp(-f) \cdot d\mu_g - \mathcal{F}(g, f) = \int_M (4|\nabla(\psi)|_g^2 + R_f(|\psi|^2 - 1)) \cdot \exp(-f) \cdot d\mu_g.$$

This is the spinorial identity of Chow and Fong [1, Eq. (1.1)].

Proof. By $\psi \in \Gamma(M, \Sigma M)$, $\partial M = \emptyset$, and the weighted Weitzenböck identity,

$$\begin{aligned} U_f(\psi) &= \exp(-f/2) \cdot \psi, \quad D_f(\psi) = U_f^{-1}(D(U_f(\psi))). \\ \int_M \langle D_f(\psi), \chi \rangle \cdot \exp(-f) \cdot d\mu_g &= \int_M \langle D(U_f(\psi)), U_f(\chi) \rangle \cdot d\mu_g. \\ D = D^* \implies \int_M \langle D_f(\psi), \chi \rangle \cdot \exp(-f) \cdot d\mu_g &= \int_M \langle \psi, D_f(\chi) \rangle \cdot \exp(-f) \cdot d\mu_g. \\ 4 \cdot \int_M |D_f(\psi)|^2 \cdot \exp(-f) \cdot d\mu_g &= 4 \cdot \int_M \operatorname{Re}(\langle D_f^2(\psi), \psi \rangle) \cdot \exp(-f) \cdot d\mu_g. \\ D_f^2(\psi) &= -\Delta_f^\Sigma(\psi) + \frac{1}{4} R_f \psi \xrightarrow{10.0.2} 4 \cdot \int_M |D_f(\psi)|^2 \cdot \exp(-f) \cdot d\mu_g = \int_M (4|\nabla(\psi)|_g^2 + R_f |\psi|^2) \cdot \exp(-f) \cdot d\mu_g. \\ \mathcal{F}(g, f) &\xrightarrow{10.0.3} \mathcal{F}(g, f) = \int_M R_f \cdot \exp(-f) \cdot d\mu_g. \end{aligned}$$

Subtracting the last equality gives the stated spinorial identity. $\square$

Example 10.0.7 (A complete Clifford-matrix calculation). In an oriented Euclidean two-plane take

$$c(e_1) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad c(e_2) = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \quad v = 3 \cdot e_1 + 4 \cdot e_2.$$

Then

$$c(v) = 3 \cdot c(e_1) + 4 \cdot c(e_2) = \begin{bmatrix} 0 & 3+4i \\ -3+4i & 0 \end{bmatrix}.$$

The full adjoint product is

$$c(v)^* \cdot c(v) = \begin{bmatrix} 0 & -3-4i \\ 3-4i & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 3+4i \\ -3+4i & 0 \end{bmatrix}.$$

$$c(v)^* \cdot c(v) = \begin{bmatrix} (-3-4i) \cdot (-3+4i) & 0 \cdot (3+4i) + (-3-4i) \cdot 0 \\ (3-4i) \cdot 0 + 0 \cdot (-3+4i) & (3-4i) \cdot (3+4i) \end{bmatrix} = \begin{bmatrix} 9+16 & 0 \\ 0 & 9+16 \end{bmatrix} = 25 \cdot I_2.$$

Thus

$$c(v)^* \cdot c(v) = (3 \cdot 3 + 4 \cdot 4) \cdot I_2 = (9 + 16) \cdot I_2 = |v|^2 \cdot I_2.$$

Definition 10.0.8 (Perelman lambda functional). Define

$$\lambda(g) = \inf \left\{ \mathcal{F}(g, f) \mid f \in C^\infty(M, \mathbb{R}), \quad \int_M \exp(-f) \cdot d\mu_g = 1 \right\}.$$

For the weighted rough spinor Laplacian define

$$v_1(-\Delta_f^\Sigma) = \inf_{\psi \neq 0} \frac{\int_M |\nabla(\psi)|_g^2 \cdot \exp(-f) \cdot d\mu_g}{\int_M |\psi|^2 \cdot \exp(-f) \cdot d\mu_g}.$$

Fact 10.0.9 (Perelman minimizer). *The infimum defining $\lambda(g)$ is achieved by a smooth function $\hat{f}$, normalized by $\int_M \exp(-\hat{f}) d\mu_g = 1$, and*

$$R_{\hat{f}} = \lambda(g).$$

This is the variational characterization underlying Perelman's entropy [2].

Theorem 10.0.10 (Dirac and weighted-Laplacian identity). *Let $\lambda_1(D)^2$ be the lowest eigenvalue of D^2 . At the Perelman minimizer $\hat{f}$,*

$$(10.0.6) \quad \lambda(g) = 4 \cdot \lambda_1(D)^2 - 4 \cdot v_1(-\Delta_{\hat{f}}^\Sigma).$$

The identity is the spectral comparison in [1, Section 1.1].

Proof. By $\int_M |\psi|^2 \exp(-f) d\mu_g = 1$, $\eta = \exp(-f/2)\psi$, and the weighted Dirac conjugation,

$$\int_M |\eta|^2 \cdot d\mu_g = \int_M |\psi|^2 \cdot \exp(-f) \cdot d\mu_g = 1.$$

$$D_f(\psi) = \exp(f/2) \cdot D(\eta) \xrightarrow{10.0.1} \int_M |D_f(\psi)|^2 \cdot \exp(-f) \cdot d\mu_g = \int_M |D(\eta)|^2 \cdot d\mu_g.$$

$$\inf_{\int_M |\psi|^2 \exp(-f) d\mu_g = 1} 4 \cdot \int_M |D_f(\psi)|^2 \cdot \exp(-f) \cdot d\mu_g = 4 \cdot \lambda_1(D)^2.$$

$$R_{\hat{f}} = \lambda(g) \xrightarrow{10.0.5} 4 \cdot \lambda_1(D)^2 - \lambda(g) = \inf_{\int_M |\psi|^2 \exp(-\hat{f}) d\mu_g = 1} 4 \cdot \int_M |\nabla(\psi)|_g^2 \cdot \exp(-\hat{f}) \cdot d\mu_g.$$

$$\inf_{\int_M |\psi|^2 \exp(-\hat{f}) d\mu_g = 1} \int_M |\nabla(\psi)|_g^2 \cdot \exp(-\hat{f}) \cdot d\mu_g \xrightarrow{10.0.8} 4 \cdot \lambda_1(D)^2 - \lambda(g) = 4 \cdot v_1(-\Delta_{\hat{f}}^\Sigma).$$

Rearrangement gives the eigenvalue identity. □

Definition 10.0.11 (Spinorial Perelman entropy). For $\lambda \in \mathbb{R}$, define

$$(10.0.7) \quad \mathbb{W}_\lambda(g, f, \psi, \tau) = \int_M (4|\nabla(\psi)|_g^2 + R_f(|\psi|^2 - \tau) - \lambda(f - n)) \cdot d\Omega_f.$$

Perelman's $\mathcal{W}$ -functional is

$$\mathcal{W}(g, f, \tau) = \int_M (\tau(R(g) + |\nabla f|_g^2) + f - n) \cdot d\Omega_f.$$

This is the Chow–Fong spinorial entropy [1, Definition 1.1].

Proposition 10.0.12 (Specializations of the spinorial entropy). *The weighted Weitzenböck identity gives*

$$(10.0.8) \quad \mathbb{W}_\lambda(g, f, \psi, \tau) = \int_M (4|D_f(\psi)|^2 - \tau(R(g) + |\nabla f|_g^2) - \lambda(f - n)) \cdot d\Omega_f.$$

If $D_f(\psi) = 0$, then

$$\lambda = 1 \implies \mathbb{W}_1(g, f, \psi, \tau) = -\mathcal{W}(g, f, \tau), \quad \lambda = 0 \implies \mathbb{W}_0(g, f, \psi, \tau) = -\frac{\tau}{(4 \cdot \pi \cdot \tau)^{n/2}} \cdot \mathcal{F}(g, f).$$

At $\lambda = 0$ and $\tau = 1$, $\mathbb{W}_0$ is the normalized Baldauf–Ozuch spinorial energy.

Proof. By $\mathbb{W}_\lambda$ as displayed, $D_f^2(\psi) = -\Delta_f^\Sigma(\psi) + R_f\psi/4$, and $\partial M = \emptyset$,

$$\left(D_f^2(\psi) = -\Delta_f^\Sigma(\psi) + \frac{1}{4}R_f\psi, \quad \partial M = \emptyset \right) \xrightarrow{10.0.4} 4 \cdot \int_M |D_f(\psi)|^2 \cdot d\Omega_f = \int_M (4|\nabla(\psi)|_g^2 + R_f|\psi|^2) \cdot d\Omega_f.$$

$$\left(d\Omega_f = (4 \cdot \pi \cdot \tau)^{-n/2} \exp(-f) d\mu_g \right) \xrightarrow{10.0.3} \int_M R_f \cdot d\Omega_f = \frac{1}{(4 \cdot \pi \cdot \tau)^{n/2}} \cdot \mathcal{F}(g, f).$$

$$\mathbb{W}_\lambda(g, f, \psi, \tau) = \int_M (4|D_f(\psi)|^2 - \tau(R(g) + |\nabla f|_g^2) - \lambda(f - n)) \cdot d\Omega_f.$$

$$D_f(\psi) = 0, \quad \lambda = 1 \xrightarrow{10.0.11} \mathbb{W}_1(g, f, \psi, \tau) = -\mathcal{W}(g, f, \tau).$$

$$D_f(\psi) = 0, \quad \lambda = 0 \xrightarrow{10.0.3} \mathbb{W}_0(g, f, \psi, \tau) = -\frac{\tau}{(4 \cdot \pi \cdot \tau)^{n/2}} \cdot \mathcal{F}(g, f).$$

Setting $\tau = 1$ gives the normalized Baldauf–Ozuch energy. □

Definition 10.0.13 (Spinorial tensors). For a local orthonormal frame $e_1, \dots, e_n$, define

$$V_f = \sum_{j=1}^n \operatorname{Re}(\langle \psi, c(e_j) \cdot D_f(\psi) \rangle) \cdot e_j.$$

For vector fields X, Y , define

$$S_{g,f,\psi}(X, Y) = \operatorname{Re}(\langle D_f(\psi), c(X) \cdot \nabla_Y(\psi) + c(Y) \cdot \nabla_X(\psi) \rangle).$$

Fact 10.0.14 (Chow–Fong critical-point equations). Fix $\tau > 0$. Under the constraints

$$\int_M d\Omega_f = 1, \quad \int_M |\psi|^2 \cdot d\Omega_f = c,$$

a constrained critical point satisfies, for constants $\alpha, \beta \in \mathbb{R}$,

$$\begin{aligned} \operatorname{Ric}(g) + \mathcal{L}_{\frac{1}{2}\nabla f - \frac{1}{\tau}V_f}(g) - \frac{2}{\tau} \cdot S_{g,f,\psi} - \frac{\lambda}{2\tau} \cdot g &= 0, & D_f^2(\psi) &= \frac{\beta}{4} \cdot \psi, \\ \tau \cdot R_f + \lambda(f - n) &= \lambda - \alpha. \end{aligned}$$

Thus the metric is a twisted Ricci soliton and the spinor is an eigen-spinor of D_f^2 . These are the constrained critical-point equations in [1, Proposition 3.1].

Definition 10.0.15 (Spinorial Ricci flow). Assume $|\psi_0|^2 = c > 0$, $\int_M d\Omega_{f_0} = 1$, and fix $\tau > 0$. The Chow–Fong flow is

$$(10.0.9) \quad \frac{\partial g}{\partial t} = -2 \left(\operatorname{Ric}(g) + \mathcal{L}_{\frac{1}{2}\nabla f - \frac{1}{\tau}V_f}(g) - \frac{2}{\tau} \cdot S_{g,f,\psi} - \frac{\lambda}{2\tau} \cdot g \right).$$

$$(10.0.10) \quad \frac{\partial f}{\partial t} = -\Delta(f) - R(g) + \frac{\lambda \cdot n}{2\tau} + \frac{4}{\tau} \cdot \operatorname{Tr}_g(S_{g,f,\psi}) + \frac{2}{\tau} \cdot \operatorname{div}(V_f).$$

$$(10.0.11) \quad \frac{\partial \psi}{\partial t} = \Delta_f^\Sigma(\psi) + \frac{|\nabla(\psi)|_g^2}{|\psi|^2} \cdot \psi.$$

This is the coupled flow introduced in [1, Definition 3.5] and discussed in the chalkboard lectures [?].

Proposition 10.0.16 (Pointwise spinor-norm preservation). *Every smooth solution with $|\psi_0|^2 = c > 0$ satisfies*

$$(10.0.12) \quad \frac{\partial}{\partial t} |\psi|^2 = \Delta_f(|\psi|^2), \quad |\psi(t)|^2 = c.$$

Proof. By $\partial_t(\psi) = \Delta_f^\Sigma(\psi) + |\nabla(\psi)|_g^2 \psi / |\psi|^2$, $|\psi_0|^2 = c$, and $\nabla \langle \psi, \psi \rangle = 2 \operatorname{Re}(\langle \nabla(\psi), \psi \rangle)$,

$$\frac{\partial}{\partial t} |\psi|^2 = 2 \operatorname{Re} \left(\left\langle \Delta_f^\Sigma(\psi) + \frac{|\nabla(\psi)|_g^2}{|\psi|^2} \psi, \psi \right\rangle \right).$$

$$2 \operatorname{Re}(\langle \Delta_f^\Sigma(\psi), \psi \rangle) + 2 |\nabla(\psi)|_g^2 = \Delta_f(|\psi|^2).$$

$$\frac{\partial}{\partial t} |\psi|^2 = \Delta_f(|\psi|^2), \quad |\psi_0|^2 = c \implies |\psi(t)|^2 = c.$$

The last implication is the scalar maximum principle on the closed manifold. $\square$

Fact 10.0.17 (Chow–Fong constraint preservation and dissipation). *Along every smooth solution with $\int_M d\Omega_{f_0} = 1$ and $|\psi_0|^2 = c$,*

$$\int_M d\Omega_f = 1, \quad |\psi(t)|^2 = c.$$

Moreover

$$(10.0.13) \quad \frac{d}{dt} \mathbb{W}_\lambda(g, f, \psi, \tau) = - \int_M (2\tau \cdot |\mathcal{A}|_g^2 + 8 \cdot |\mathcal{B}|^2) \cdot d\Omega_f \leq 0,$$

where the two displayed terms are written completely as

$$\mathcal{A} = \operatorname{Ric}(g) + \mathcal{L}_{\frac{1}{2}\nabla_f - \frac{1}{\tau}V_f}(g) - \frac{2}{\tau} \cdot S_{g,f,\psi} - \frac{\lambda}{2\tau} \cdot g, \quad \mathcal{B} = \Delta_f^\Sigma(\psi) + \frac{|\nabla(\psi)|_g^2}{|\psi|^2} \cdot \psi.$$

Equality holds precisely when $\mathcal{A} = 0$ and $\mathcal{B} = 0$. The complete first variation includes the variation of $d\Omega_f$; its cancellation with the three coupled flow equations is retained here as the principal analytic input from Chow and Fong. The complete argument is [1, Proposition 3.6].

Fact 10.0.18 (Chow–Fong short-time existence). *Let (g_0, f_0, ψ_0) be smooth with $|\psi_0|^2 = c$. If $c > \tau$, the DeTurck-gauged system is uniformly forward parabolic in (g, f, ψ) and has a unique short-time solution with initial data (g_0, f_0, ψ_0) . If $c < \tau$, the (g, ψ) -equations are forward parabolic while the f -equation is backward parabolic, and the short-time solution uses*

$$g(0) = g_0, \quad \psi(0) = \psi_0, \quad f(\varepsilon) = f_0.$$

The case $c = \tau$ is degenerate at the principal-symbol level. This is the short-time theory in [1, Theorem 4.1].

Lemma 10.0.19 (Stationary integral balance). *At a stationary solution with $|\psi|^2 = c$ and $\int_M d\Omega_f = 1$, set*

$$A = \int_M |D_f(\psi)|^2 \cdot d\Omega_f, \quad B = \int_M |\nabla(\psi)|_g^2 \cdot d\Omega_f.$$

Then

$$(10.0.14) \quad \left(1 - \frac{c}{\tau}\right) \cdot A = \frac{n \cdot c \cdot \lambda}{8\tau} + B.$$

This balance is [1, Lemma 3.7].

Proof. By $\partial_t(g) = 0$, $\partial_t(\psi) = 0$, $|\psi|^2 = c$, and $\int_M d\Omega_f = 1$,

$$\partial_t(\psi) = 0 \xrightarrow{10.0.11} \Delta_f^\Sigma(\psi) = -\frac{|\nabla(\psi)|_g^2}{c} \cdot \psi.$$

$$\left(D_f^2(\psi) = -\Delta_f^\Sigma(\psi) + \frac{1}{4}R_f\psi, \quad |\psi|^2 = c \right) \xrightarrow{10.0.4} A = B + \frac{c}{4} \cdot \int_M R_f \cdot d\Omega_f.$$

The tensor definitions and weighted integration by parts give

$$\begin{aligned} \int_M (\operatorname{div}(V_f) + \operatorname{Tr}_g(S_{g,f,\psi})) \cdot d\Omega_f &\xrightarrow{10.0.13} 2 \cdot \int_M |D_f(\psi)|^2 \cdot d\Omega_f = 2A. \\ \partial_t(g) = 0 &\xrightarrow{10.0.9} 0 = R(g) + \Delta(f) - \frac{2}{\tau} \operatorname{div}(V_f) - \frac{2}{\tau} \operatorname{Tr}_g(S_{g,f,\psi}) - \frac{n\lambda}{2\tau}. \\ \int_M (R(g) + \Delta(f)) \cdot d\Omega_f &\xrightarrow{10.0.2} \int_M (R(g) + |\nabla f|_g^2) \cdot d\Omega_f = \int_M R_f \cdot d\Omega_f. \\ \left(\int_M (\operatorname{div}(V_f) + \operatorname{Tr}_g(S_{g,f,\psi})) \cdot d\Omega_f = 2A \right) &\implies \int_M R_f \cdot d\Omega_f = \frac{4}{\tau} \cdot A + \frac{n \cdot \lambda}{2\tau}. \\ A = B + \frac{c}{4} \left(\frac{4}{\tau} A + \frac{n\lambda}{2\tau} \right) &\implies \left(1 - \frac{c}{\tau} \right) A = B + \frac{n \cdot c \cdot \lambda}{8\tau}. \end{aligned}$$

This is the stationary balance. $\square$

Corollary 10.0.20 (Stationary points in the forward regime). *At a stationary point with $c > \tau$, $|\psi|^2 = c$, and $\int_M d\Omega_f = 1$, one has $\lambda \leq 0$. If $\lambda = 0$, then*

$$\nabla(\psi) = 0, \quad D_f(\psi) = 0, \quad \nabla f = 0, \quad \operatorname{Ric}(g) = 0.$$

This is the forward-regime conclusion in [1, Corollary 3.8].

Proof. By $c > \tau$, $\partial_t(g) = \partial_t(f) = \partial_t(\psi) = 0$, $|\psi|^2 = c$, and $\int_M d\Omega_f = 1$,

$$(\partial_t(g) = \partial_t(f) = \partial_t(\psi) = 0, \quad |\psi|^2 = c) \xrightarrow{10.0.19} \left(1 - \frac{c}{\tau} \right) \cdot \int_M |D_f(\psi)|^2 \cdot d\Omega_f = \frac{n \cdot c \cdot \lambda}{8\tau} + \int_M |\nabla(\psi)|_g^2 \cdot d\Omega_f.$$

$$c > \tau \implies 1 - \frac{c}{\tau} < 0, \quad \int_M |D_f(\psi)|^2 d\Omega_f \geq 0, \quad \int_M |\nabla(\psi)|_g^2 d\Omega_f \geq 0 \implies \lambda \leq 0.$$

$$\lambda = 0 \implies \left(1 - \frac{c}{\tau} \right) \int_M |D_f(\psi)|^2 d\Omega_f = \int_M |\nabla(\psi)|_g^2 d\Omega_f \implies D_f(\psi) = 0, \quad \nabla(\psi) = 0.$$

$$D_f(\psi) = D(\psi) - \frac{1}{2}c(\nabla f)\psi, \quad D(\psi) = 0, \quad |\psi|^2 = c > 0 \xrightarrow{10.0.1} \nabla f = 0.$$

$$\partial_t(g) = 0 \xrightarrow{10.0.9} \mathcal{A} = 0, \quad D_f(\psi) = 0 \xrightarrow{10.0.13} V_f = 0, \quad S_{g,f,\psi} = 0.$$

$$(\mathcal{A} = 0, \quad \nabla f = 0, \quad V_f = 0, \quad S_{g,f,\psi} = 0, \quad \lambda = 0) \implies 0 = \mathcal{A} = \operatorname{Ric}(g) + \mathcal{L}_0(g) - 0 - 0 = \operatorname{Ric}(g).$$

Thus a zero-sign stationary point in the forward regime is Ricci-flat with a parallel spinor. $\square$

11. CALIBRATING FORMS FOR MINIMAL GRAPHS

Let $\Omega \subset \mathbb{R}^n$ be a domain and let $\Omega \xrightarrow{F} \mathbb{R}^m$ be smooth. Write

$$F = (f_1, \dots, f_m), \quad \iota_F(x) = (x, F(x)), \quad \Sigma_F = \iota_F(\Omega) \subset \mathbb{R}^{n+m}.$$

Definition 11.0.1 (Minimal graph). The induced metric and volume form of the graph are

$$g_{jk} = \delta_{jk} + \sum_{\alpha=1}^m \partial_j(f_\alpha) \cdot \partial_k(f_\alpha), \quad G = [g_{jk}]_{j,k=1}^n, \quad g = \det(G),$$

$$\text{vol}_{\Sigma_F} = \sqrt{g} \cdot dx^1 \wedge \dots \wedge dx^n.$$

The graph is minimal precisely when every component satisfies

$$(11.0.1) \quad \sum_{j=1}^n \sum_{k=1}^n \partial_j \left(\sqrt{g} \cdot (G^{-1})^{jk} \cdot \partial_k(f_\alpha) \right) = 0, \quad 1 \leq \alpha \leq m.$$

Definition 11.0.2 (Calibration and comass). For an n -form Θ on Euclidean space, define

$$\text{comass}(\Theta) = \sup \{ |\Theta(e_1, \dots, e_n)| \mid e_1, \dots, e_n \text{ is an oriented orthonormal frame} \}.$$

The form Θ is a calibration when

$$d(\Theta) = 0, \quad \text{comass}(\Theta) \leq 1.$$

An oriented n -submanifold Σ is calibrated by Θ when

$$\Theta|_\Sigma = \text{vol}_\Sigma.$$

This is the calibration framework of Harvey and Lawson [5].

Proposition 11.0.3 (Calibration implies area minimization). *If Σ is calibrated by Θ , then Σ minimizes area in its homology class. This is the fundamental calibration inequality [5].*

Proof. By $d(\Theta) = 0$, $\text{comass}(\Theta) \leq 1$, and $\Theta|_\Sigma = \text{vol}_\Sigma$, let $\Sigma' - \Sigma = \partial C$.

$$\begin{aligned} \int_{\Sigma'} \Theta - \int_\Sigma \Theta &= \int_{\partial C} \Theta = \int_C d(\Theta) = 0. \\ \text{Area}(\Sigma) &= \int_\Sigma \text{vol}_\Sigma \xrightarrow{11.0.2} \text{Area}(\Sigma) = \int_\Sigma \Theta = \int_{\Sigma'} \Theta. \\ \text{comass}(\Theta) \leq 1 &\xrightarrow{11.0.2} \int_{\Sigma'} \Theta \leq \int_{\Sigma'} \text{vol}_{\Sigma'} = \text{Area}(\Sigma'). \end{aligned}$$

Thus every homologous competitor has at least the area of Σ . □

Example 11.0.4 (Classical calibrations). On $\mathbb{C}^N$, the Kähler form

$$\omega = \frac{i}{2} \cdot \sum_{j=1}^N dz^j \wedge d\bar{z}^j$$

satisfies the Wirtinger inequality, so $\omega^k/k!$ calibrates each complex k -fold. For $\vartheta \in \mathbb{R}$,

$$\text{Re}(\exp(-i \cdot \vartheta) \cdot dz^1 \wedge \dots \wedge dz^N)$$

calibrates the special Lagrangian N -folds of phase ϑ .

For a hypersurface graph $y = f(x)$, set

$$\Theta(f) = \frac{dx^1 \wedge \dots \wedge dx^n + dy \wedge *d(f)}{\sqrt{1 + |\nabla f|^2}}.$$

Then every multiplication in the pullback is explicit

$$\iota_f^*(dy \wedge *d(f)) = d(f) \wedge *d(f) = |\nabla f|^2 \cdot dx^1 \wedge \dots \wedge dx^n,$$

$$\iota_f^*(\Theta(f)) = \frac{1 + |\nabla f|^2}{\sqrt{1 + |\nabla f|^2}} \cdot dx^1 \wedge \cdots \wedge dx^n = \sqrt{1 + |\nabla f|^2} \cdot dx^1 \wedge \cdots \wedge dx^n.$$

Moreover

$$d(\Theta(f)) = 0 \iff \operatorname{div} \left(\frac{\nabla f}{\sqrt{1 + |\nabla f|^2}} \right) = 0.$$

Thus the usual minimal-hypersurface equation is the first case of the calibration construction.

Definition 11.0.5 (Graph linear algebra). At $p = \iota_F(x)$, write

$$A_p = d(F)|_x \in \mathbb{R}^{m \times n}, \quad G_p = I_n + A_p^* \cdot A_p, \quad H_p = I_m + A_p \cdot A_p^*.$$

Define isometries

$$\begin{aligned} L_p(v) &= (G_p^{-1/2} \cdot v, A_p \cdot G_p^{-1/2} \cdot v) \in T_p \Sigma_F, \\ L_p^\perp(w) &= (-A_p^* \cdot H_p^{-1/2} \cdot w, H_p^{-1/2} \cdot w) \in N_p \Sigma_F. \end{aligned}$$

The graph-normal map is

$$(11.0.2) \quad J_p = L_p^\perp \circ A_p \circ L_p^{-1} \in \operatorname{Hom}(T_p \Sigma_F, N_p \Sigma_F).$$

Lemma 11.0.6 (Sylvester identities). For every $A \in \mathbb{R}^{m \times n}$,

$$\begin{aligned} \det(I_n + A^* \cdot A) &= \det(I_m + A \cdot A^*), \\ (I_m + A \cdot A^*)^{-1} &= I_m - A \cdot (I_n + A^* \cdot A)^{-1} \cdot A^*, \\ (11.0.3) \quad (I_m + A \cdot A^*)^{-1} \cdot A &= A \cdot (I_n + A^* \cdot A)^{-1}. \end{aligned}$$

Proof. By $A \in \mathbb{R}^{m \times n}$, $G = I_n + A^* \cdot A$, and $H = I_m + A \cdot A^*$,

$$H \cdot A = (I_m + A \cdot A^*) \cdot A = A + A \cdot A^* \cdot A = A \cdot (I_n + A^* \cdot A) = A \cdot G.$$

$$H \cdot A = A \cdot G \implies H^{-1} \cdot A = A \cdot G^{-1}.$$

$$H \cdot (I_m - A \cdot G^{-1} \cdot A^*) = I_m + A \cdot A^* - A \cdot G^{-1} \cdot A^* - A \cdot A^* \cdot A \cdot G^{-1} \cdot A^*.$$

$$H \cdot (I_m - A \cdot G^{-1} \cdot A^*) = I_m + A \cdot (I_n - G^{-1} - (A^* \cdot A) \cdot G^{-1}) \cdot A^* = I_m + A \cdot (I_n - G \cdot G^{-1}) \cdot A^* = I_m.$$

The determinant identity follows by multiplying the nonzero singular-value factors $1 + \lambda_j^2$ on both sides. $\square$

Example 11.0.7 (A complete graph-matrix calculation). Take $n = m = 2$ and

$$F(x^1, x^2) = \left(x^1, \frac{1}{2}x^2 \right), \quad A = d(F) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.$$

Then

$$A^* \cdot A = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot \frac{1}{2} \\ 0 \cdot 1 + \frac{1}{2} \cdot 0 & 0 \cdot 0 + \frac{1}{2} \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix}.$$

$$A \cdot A^* = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix}.$$

$$G = I_2 + A^* \cdot A = \begin{bmatrix} 2 & 0 \\ 0 & \frac{5}{4} \end{bmatrix}, \quad H = I_2 + A \cdot A^* = \begin{bmatrix} 2 & 0 \\ 0 & \frac{5}{4} \end{bmatrix}.$$

$$\det(G) = 2 \cdot \frac{5}{4} - 0 \cdot 0 = \frac{10}{4} = \frac{5}{2}, \quad \det(H) = 2 \cdot \frac{5}{4} - 0 \cdot 0 = \frac{5}{2}.$$

This verifies the determinant identity and will give an explicit calibrated graph below.

Definition 11.0.8 (Tsai–Wang form). Fix $p \in \Sigma_F$, choose an oriented orthonormal basis $e_1, \dots, e_n$ of $T_p \Sigma_F$, and define an operator on forms by

$$\Psi(\xi) = \sum_{j=1}^n (J_p(e_j))^b \wedge \iota_{e_j}(\xi).$$

Set

$$\Psi^0(F) = \text{vol}_{\Sigma_F}, \quad \Psi^\ell(F) = \Psi^\ell(\text{vol}_{\Sigma_F}) \quad (1 \leq \ell \leq \min(n, m)).$$

The Tsai–Wang n -form is

$$(11.0.4) \quad \Theta(F) = \Psi^0(F) - \sum_{\ell=2}^{\min(n, m)} (-1)^\ell \cdot \frac{\ell-1}{\ell!} \cdot \Psi^\ell(F).$$

Each $\Psi^\ell(F)$ is extended from Σ_F to $\Omega \times \mathbb{R}^m$ by parallel translation in the $\mathbb{R}^m$ -direction. Thus $\Theta(F)$ is an ambient form whose exterior derivative and comass are defined on the whole graphical cylinder. Thus

$$\Theta(F) = \Psi^0(F) - \frac{1}{2} \Psi^2(F) + \frac{2}{3!} \Psi^3(F) - \frac{3}{4!} \Psi^4(F) + \dots.$$

This is the form introduced in [4, Definition 2.4] and presented in the chalkboard lecture [?].

Proposition 11.0.9 (Singular-value form). At $p \in \Sigma_F$, let $r = \text{rank}(d(F)|_p)$, choose singular frames with

$$J_p(e_j) = \lambda_j \cdot e_{n+j} \quad (1 \leq j \leq r), \quad \lambda_1 \geq \dots \geq \lambda_r > 0,$$

and let $\omega^1, \dots, \omega^{n+m}$ be the dual ambient coframe. For $\mathcal{J} \subset \{1, \dots, r\}$, define

$$\eta_{\mathcal{J}}^j = \omega^{n+j} \quad (j \in \mathcal{J}), \quad \eta_{\mathcal{J}}^j = \omega^j \quad (j \notin \mathcal{J}),$$

$$\Delta_{\mathcal{J}} = \eta_{\mathcal{J}}^1 \wedge \dots \wedge \eta_{\mathcal{J}}^n.$$

Then

$$(11.0.5) \quad \Theta(F) = \omega^1 \wedge \dots \wedge \omega^n - \sum_{\ell=2}^r (-1)^\ell (\ell-1) \sum_{\substack{\mathcal{J} \subset \{1, \dots, r\} \\ |\mathcal{J}|=\ell}} \left(\prod_{j \in \mathcal{J}} \lambda_j \right) \cdot \Delta_{\mathcal{J}}.$$

This is the singular-frame expansion in [4, Eq. (2.10)].

Proof. By $J_p(e_j) = \lambda_j \cdot e_{n+j}$, $\Psi(\xi) = \sum_{j=1}^n (J_p(e_j))^b \wedge \iota_{e_j}(\xi)$, and $\Psi^0(F) = \omega^1 \wedge \dots \wedge \omega^n$,

$$\Psi^\ell(F) = \ell! \cdot \sum_{\substack{\mathcal{J} \subset \{1, \dots, r\} \\ |\mathcal{J}|=\ell}} \left(\prod_{j \in \mathcal{J}} \lambda_j \right) \cdot \Delta_{\mathcal{J}}.$$

$$\left(\Psi^\ell(F) = \ell! \cdot \sum_{\substack{\mathcal{J} \subset \{1, \dots, r\} \\ |\mathcal{J}|=\ell}} \left(\prod_{j \in \mathcal{J}} \lambda_j \right) \cdot \Delta_{\mathcal{J}} \right) \xrightarrow{11.0.4} \Theta(F) = \omega^1 \wedge \dots \wedge \omega^n - \sum_{\ell=2}^r (-1)^\ell (\ell-1) \sum_{\substack{\mathcal{J} \subset \{1, \dots, r\} \\ |\mathcal{J}|=\ell}} \left(\prod_{j \in \mathcal{J}} \lambda_j \right) \cdot \Delta_{\mathcal{J}}.$$

This is the singular-value expansion. □

Proposition 11.0.10 (Restriction to the graph). The Tsai–Wang form satisfies

$$\Theta(F)|_{\Sigma_F} = \text{vol}_{\Sigma_F}.$$

Proof. By $J_p(T_p\Sigma_F) \subset N_p\Sigma_F$, $(J_p(e_j))^\flat|_{T_p\Sigma_F} = 0$, and the definition of Ψ ,

$$\Psi(\xi)|_{T_p\Sigma_F} = \sum_{j=1}^n (J_p(e_j))^\flat|_{T_p\Sigma_F} \wedge \iota_{e_j}(\xi) = 0.$$

$$1 \leq \ell \leq \min(n, m) \implies \Psi^\ell(F)|_{T_p\Sigma_F} = 0.$$

$$\left(\Psi^\ell(F)|_{T_p\Sigma_F} = 0 \text{ for } 1 \leq \ell \leq \min(n, m) \right) \xrightarrow{11.0.4} \Theta(F)|_{T_p\Sigma_F} = \Psi^0(F)|_{T_p\Sigma_F} = \text{vol}_{\Sigma_F}|_p.$$

Since p is arbitrary, the restriction identity holds globally. $\square$

Proposition 11.0.11 (Coordinate form). *Let*

$$h_{\alpha\beta} = \delta_{\alpha\beta} + \sum_{j=1}^n \partial_j(f_\alpha) \cdot \partial_j(f_\beta), \quad H = [h_{\alpha\beta}]_{\alpha,\beta=1}^m, \quad h = \det(H).$$

Then

$$(11.0.6) \quad \Theta(F) = \sqrt{h} \cdot (\text{Tr}(H^{-1}) - (m-1)) \cdot dx^1 \wedge \cdots \wedge dx^n + \sum_{\alpha=1}^m \sum_{\beta=1}^m dy^\alpha \wedge \sqrt{h} \cdot (H^{-1})^{\alpha\beta} \cdot *d(f_\beta).$$

Here $$ is the Euclidean Hodge star on $\mathbb{R}^n$. This is the coordinate representation in [4, Proposition 3.1].*

Proof. By a singular-value decomposition of $d(F)$, the singular-value form of $\Theta(F)$, and the Euclidean Hodge star, set $q = \min(n, m)$, put $\lambda_j = 0$ after the rank of $d(F)$, and choose orthonormal coordinates with

$$d(F)(a_j) = \lambda_j \cdot b_j \quad (1 \leq j \leq q), \quad \partial_j(f_\alpha) = \lambda_j \cdot \delta_{\alpha j}.$$

When $n > m$, introduce dummy b_j and y^j for $j > m$, with $\lambda_j = 0$. The paper's oriented tangent and normal frames are

$$e_j = \frac{a_j + \lambda_j \cdot b_j}{\sqrt{1 + \lambda_j^2}}, \quad e_{n+j} = \frac{-\lambda_j \cdot a_j + b_j}{\sqrt{1 + \lambda_j^2}} \quad (1 \leq j \leq n).$$

For

$$\widehat{dx}^k = dx^1 \wedge \cdots \wedge dx^{k-1} \wedge dx^{k+1} \wedge \cdots \wedge dx^n, \quad *(dx^k) = (-1)^{k-1} \cdot \widehat{dx}^k,$$

the expansion of the defining finite sum gives

$$(11.0.7) \quad (\partial_j(f_\alpha) = \lambda_j \cdot \delta_{\alpha j}, \quad \lambda_j = 0 \text{ after rank}(d(F))) \xrightarrow{11.0.5} \Theta(F) = \Theta_0(F) + \Theta_1(F).$$

$$\Theta_0(F) = \sqrt{\prod_{j=1}^n (1 + \lambda_j^2)} \cdot \left(1 - \sum_{j=1}^n \frac{\lambda_j^2}{1 + \lambda_j^2} \right) \cdot dx^1 \wedge \cdots \wedge dx^n,$$

$$\Theta_1(F) = \sqrt{\prod_{j=1}^n (1 + \lambda_j^2)} \cdot \sum_{k=1}^q \frac{(-1)^{k-1} \cdot \lambda_k}{1 + \lambda_k^2} \cdot dy^k \wedge \widehat{dx}^k.$$

On the oriented tangent frame, every factor is visible

$$(\Theta_0(F) + \Theta_1(F))(e_1, \dots, e_n) = 1 - \sum_{j=1}^n \frac{\lambda_j^2}{1 + \lambda_j^2} + \sum_{k=1}^n \frac{\lambda_k^2}{1 + \lambda_k^2} = 1.$$

For $2 \leq \ell \leq \text{rank}(d(F))$, evaluation on the frame with its first ℓ tangent vectors replaced by normal vectors gives

$$\begin{aligned} & (\Theta_0(F) + \Theta_1(F))(e_{n+1}, \dots, e_{n+\ell}, e_{\ell+1}, \dots, e_n) \\ &= \left(1 - \sum_{j=1}^n \frac{\lambda_j^2}{1 + \lambda_j^2} \right) \cdot \prod_{k=1}^{\ell} (-\lambda_k) + \sum_{j=1}^{\ell} \frac{\lambda_j}{1 + \lambda_j^2} \cdot \frac{\prod_{k=1}^{\ell} (-\lambda_k)}{-\lambda_j} + \sum_{j=\ell+1}^n \frac{\lambda_j^2}{1 + \lambda_j^2} \cdot \prod_{k=1}^{\ell} (-\lambda_k), \end{aligned}$$

$$= (-1)^\ell \cdot \prod_{k=1}^\ell \lambda_k \cdot \left(1 - \sum_{j=1}^\ell \frac{\lambda_j^2}{1 + \lambda_j^2} - \sum_{j=1}^\ell \frac{1}{1 + \lambda_j^2} \right) = (\ell - 1)(-1)^{\ell-1} \cdot \prod_{k=1}^\ell \lambda_k.$$

These evaluations agree with every coefficient in the singular-value form. The two diagonal graph matrices are

$$G = \begin{bmatrix} 1 + \lambda_1^2 & 0 & \cdots & 0 \\ 0 & 1 + \lambda_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 + \lambda_n^2 \end{bmatrix}, \quad H = \begin{bmatrix} 1 + \lambda_1^2 & 0 & \cdots & 0 \\ 0 & 1 + \lambda_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 + \lambda_m^2 \end{bmatrix}.$$

$$\det(H) = \prod_{\alpha=1}^m (1 + \lambda_\alpha^2) = \prod_{j=1}^n (1 + \lambda_j^2) = \det(G), \quad H^{-1} = \begin{bmatrix} (1 + \lambda_1^2)^{-1} & 0 & \cdots & 0 \\ 0 & (1 + \lambda_2^2)^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & (1 + \lambda_m^2)^{-1} \end{bmatrix}.$$

$$(\det(H) = h, \quad \lambda_\alpha = 0 \text{ for } \alpha > q) \implies \sqrt{h} \cdot (\text{Tr}(H^{-1}) - (m - 1)) = \sqrt{\prod_{j=1}^n (1 + \lambda_j^2)} \cdot \left(1 - \sum_{j=1}^n \frac{\lambda_j^2}{1 + \lambda_j^2} \right).$$

$$\sum_{\alpha=1}^m \sum_{\beta=1}^m dy^\alpha \wedge \sqrt{h} \cdot (H^{-1})^{\alpha\beta} \cdot *d(f_\beta) = \sqrt{\prod_{j=1}^n (1 + \lambda_j^2)} \cdot \sum_{k=1}^q \frac{(-1)^{k-1} \cdot \lambda_k}{1 + \lambda_k^2} \cdot dy^k \wedge \widehat{dx}^k.$$

The last two displays reproduce $\Theta_0(F)$ and $\Theta_1(F)$, so orthogonal invariance gives the coordinate formula in the original frame. $\square$

Theorem 11.0.12 (Closedness and the minimal graph system). *The form $\Theta(F)$ is closed if and only if Σ_F is minimal. This equivalence is [4, Theorem 3.3].*

Proof. By the coordinate form of $\Theta(F)$, the Sylvester identities, and the definition of the induced metric,

$$\sqrt{h} = \sqrt{g}, \quad \sum_{\beta=1}^m (H^{-1})^{\alpha\beta} \cdot \partial_j(f_\beta) = \sum_{k=1}^n (G^{-1})^{jk} \cdot \partial_k(f_\alpha).$$

$$d\left(\sqrt{h}(\text{Tr}(H^{-1}) - (m - 1)) \cdot dx^1 \wedge \cdots \wedge dx^n\right) = 0.$$

$$d(\Theta(F)) \xrightarrow{11.0.6} d(\Theta(F)) = - \sum_{\alpha=1}^m \sum_{\beta=1}^m dy^\alpha \wedge d\left(\sqrt{h} \cdot (H^{-1})^{\alpha\beta} \cdot *d(f_\beta)\right).$$

$$d(\Theta(F)) = 0 \xLeftrightarrow{11.0.6} \sum_{j=1}^n \sum_{k=1}^n \partial_j \left(\sqrt{g} \cdot (G^{-1})^{jk} \cdot \partial_k(f_\alpha) \right) = 0 \quad (1 \leq \alpha \leq m).$$

$$\left(\sum_{j=1}^n \sum_{k=1}^n \partial_j (\sqrt{g} (G^{-1})^{jk} \partial_k(f_\alpha)) = 0 \right) \xLeftrightarrow{11.0.1} \Sigma_F \text{ is minimal.}$$

Thus closedness is exactly the minimal graph system. $\square$

Definition 11.0.13 (Two-dilation). Let $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ be the singular values of $d(F)$. Define

$$\|\wedge^2 d(F)\| = \lambda_1 \cdot \lambda_2,$$

with value zero when $\text{rank}(d(F)) \leq 1$. The map F is area-non-increasing when $\|\wedge^2 d(F)\| \leq 1$.

Lemma 11.0.14 (Comass of a block two-form). *For*

$$\alpha = a \cdot \omega^1 \wedge \omega^2 + b \cdot \omega^3 \wedge \omega^4$$

on an oriented Euclidean four-space,

$$\text{comass}(\alpha) = \max(|a|, |b|).$$

Proof. By $\omega^1, \dots, \omega^4$ orthonormal, $a, b \in \mathbb{R}$, and the skew matrix of α ,

$$K = \begin{bmatrix} 0 & a & 0 & 0 \\ -a & 0 & 0 & 0 \\ 0 & 0 & 0 & b \\ 0 & 0 & -b & 0 \end{bmatrix}, \quad K^\top = \begin{bmatrix} 0 & -a & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & 0 & 0 & -b \\ 0 & 0 & b & 0 \end{bmatrix}.$$

$$K \cdot K^\top = \begin{bmatrix} 0 & a & 0 & 0 \\ -a & 0 & 0 & 0 \\ 0 & 0 & 0 & b \\ 0 & 0 & -b & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & -a & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & 0 & 0 & -b \\ 0 & 0 & b & 0 \end{bmatrix} = \begin{bmatrix} a^2 & 0 & 0 & 0 \\ 0 & a^2 & 0 & 0 \\ 0 & 0 & b^2 & 0 \\ 0 & 0 & 0 & b^2 \end{bmatrix}.$$

$$\text{comass}(\alpha) = \|K\|_{\text{op}} = \sqrt{\max(a^2, b^2)} = \max(|a|, |b|).$$

This proves the comass formula. $\square$

Theorem 11.0.15 (Codimension-two calibration). *Let $\Omega \xrightarrow{F} \mathbb{R}^2$ have a minimal graph. If*

$$\|\wedge^2 d(F)\| = \lambda_1 \cdot \lambda_2 \leq 1,$$

then $\Theta(F)$ calibrates Σ_F , and Σ_F is area-minimizing. This is the codimension-two criterion in [4, Theorem 1.4].

Proof. By Σ_F minimal, $\lambda_1 \lambda_2 \leq 1$, and the SVD frame $\omega^1, \dots, \omega^{n+2}$,

$$\Sigma_F \text{ is minimal} \xrightarrow{11.0.12} d(\Theta(F)) = 0.$$

$$n = 1 \implies \text{rank}(d(F)) \leq 1 \xrightarrow{11.0.5} \Theta(F) = \omega^1, \quad \text{comass}(\Theta(F)) = 1.$$

$$(n = 1, \quad d(\Theta(F)) = 0, \quad \Theta(F)|_{\Sigma_F} = \text{vol}_{\Sigma_F}, \quad \text{comass}(\Theta(F)) = 1) \xrightarrow{11.0.2} \Theta(F) \text{ calibrates } \Sigma_F.$$

$$\Theta(F) \text{ calibrates } \Sigma_F \xrightarrow{11.0.3} \Sigma_F \text{ is area-minimizing.}$$

Assume henceforth that $n \geq 2$. Then

$$\Theta(F) \xrightarrow{11.0.5} \Theta(F) = (\omega^1 \wedge \omega^2 - \lambda_1 \lambda_2 \cdot \omega^{n+1} \wedge \omega^{n+2}) \wedge \omega^3 \wedge \dots \wedge \omega^n.$$

The terminal wedge is interpreted as 1 when $n = 2$.

$$\text{comass}(\Theta(F)) \xrightarrow{11.0.14} \text{comass}(\Theta(F)) = \max(1, \lambda_1 \lambda_2) = 1.$$

$$\Theta(F)|_{\Sigma_F} = \text{vol}_{\Sigma_F}, \quad d(\Theta(F)) = 0, \quad \text{comass}(\Theta(F)) = 1 \xrightarrow{11.0.2} \Theta(F) \text{ calibrates } \Sigma_F.$$

$$\Theta(F) \text{ calibrates } \Sigma_F \xrightarrow{11.0.3} \Sigma_F \text{ is area-minimizing.}$$

Thus the two-dilation bound promotes minimality to area minimization. $\square$

Proposition 11.0.16 (Explicit codimension-two form). *Write $F = (f, g)$ and set*

$$\Xi = (1 + |\nabla f|^2)(1 + |\nabla g|^2) - \langle \nabla f, \nabla g \rangle^2.$$

Define

$$\begin{aligned} A_0 &= 1 - |\nabla f|^2 |\nabla g|^2 + \langle \nabla f, \nabla g \rangle^2, \\ B_1 &= (1 + |\nabla g|^2) \cdot *d(f) - \langle \nabla f, \nabla g \rangle \cdot *d(g), \\ B_2 &= -\langle \nabla f, \nabla g \rangle \cdot *d(f) + (1 + |\nabla f|^2) \cdot *d(g). \end{aligned}$$

Then

$$(11.0.8) \quad \Theta(F) = \frac{1}{\sqrt{\Xi}} (A_0 \cdot dx^1 \wedge \dots \wedge dx^n + dy^1 \wedge B_1 + dy^2 \wedge B_2).$$

This is the explicit formula in [4, Eq. (1.3)].

Proof. By $m = 2$, the coordinate formula for $\Theta(F)$, and

$$\begin{aligned} H &= \begin{bmatrix} 1 + |\nabla f|^2 & \langle \nabla f, \nabla g \rangle \\ \langle \nabla f, \nabla g \rangle & 1 + |\nabla g|^2 \end{bmatrix}, \\ \det(H) &= (1 + |\nabla f|^2)(1 + |\nabla g|^2) - \langle \nabla f, \nabla g \rangle^2 = \Xi. \\ H^{-1} &= \frac{1}{\Xi} \begin{bmatrix} 1 + |\nabla g|^2 & -\langle \nabla f, \nabla g \rangle \\ -\langle \nabla f, \nabla g \rangle & 1 + |\nabla f|^2 \end{bmatrix}. \\ \sqrt{\Xi} \cdot (\text{Tr}(H^{-1}) - 1) &= \frac{1 + |\nabla f|^2 + 1 + |\nabla g|^2 - \Xi}{\sqrt{\Xi}} = \frac{A_0}{\sqrt{\Xi}}. \end{aligned}$$

Substitution in the coordinate form gives the displayed expression. $\square$

Example 11.0.17 (An explicit calibrated plane). Continue the complete matrix example with

$$f(x^1, x^2) = x^1, \quad g(x^1, x^2) = \frac{1}{2}x^2, \quad *dx^1 = dx^2, \quad *dx^2 = -dx^1.$$

Then

$$\begin{aligned} |\nabla f|^2 &= 1, \quad |\nabla g|^2 = \frac{1}{4}, \quad \langle \nabla f, \nabla g \rangle = 0, \quad \Xi = (1 + 1) \left(1 + \frac{1}{4}\right) = 2 \cdot \frac{5}{4} = \frac{5}{2}. \\ A_0 &= 1 - 1 \cdot \frac{1}{4} + 0 = \frac{3}{4}, \quad B_1 = \left(1 + \frac{1}{4}\right) dx^2 = \frac{5}{4} dx^2, \quad B_2 = (1 + 1) \cdot * \left(\frac{1}{2} dx^2\right) = -dx^1. \\ \Theta(F) &\xrightarrow{11.0.8} \Theta(F) = \sqrt{\frac{2}{5}} \left(\frac{3}{4} dx^1 \wedge dx^2 + \frac{5}{4} dy^1 \wedge dx^2 - dy^2 \wedge dx^1 \right). \end{aligned}$$

Along the graph,

$$\begin{aligned} \iota_F^*(dy^1) &= dx^1, \quad \iota_F^*(dy^2) = \frac{1}{2} dx^2. \\ \iota_F^*(\Theta(F)) &= \sqrt{\frac{2}{5}} \left(\frac{3}{4} + \frac{5}{4} + \frac{1}{2} \right) dx^1 \wedge dx^2 = \sqrt{\frac{2}{5}} \cdot \frac{5}{2} \cdot dx^1 \wedge dx^2 = \sqrt{\frac{5}{2}} \cdot dx^1 \wedge dx^2. \\ \det(G) &= \frac{5}{2} \xrightarrow{11.0.1} \text{vol}_{\Sigma_F} = \sqrt{\frac{5}{2}} \cdot dx^1 \wedge dx^2 = \iota_F^*(\Theta(F)). \end{aligned}$$

Moreover F is linear, so every second derivative vanishes and the minimal graph system is zero term by term. Its singular values are 1 and $1/2$, hence

$$\|\wedge^2 d(F)\| = 1 \cdot \frac{1}{2} = \frac{1}{2} \leq 1 \xrightarrow{11.0.15} \Sigma_F \text{ is area-minimizing.}$$

Fact 11.0.18 (Tsai–Wang comass estimate). Let $r = \text{rank}(d(F)) \geq 2$, let $\lambda_1 \geq \dots \geq \lambda_r > 0$, and define

$$\Lambda_\ell = \max_{\substack{\mathcal{J} \subset \{1, \dots, r\} \\ |\mathcal{J}| = \ell}} \left(\prod_{j \in \mathcal{J}} \lambda_j \right) \quad (2 \leq \ell \leq r).$$

Then

$$\text{comass}(\Theta(F)) \leq \max_{0 \leq \vartheta \leq \pi/2} \left((\cos \vartheta)^r + \sum_{\ell=2}^r \Lambda_\ell (\ell-1) \binom{r}{\ell} (\cos \vartheta)^{r-\ell} (\sin \vartheta)^\ell \right).$$

If $\lambda_j \lambda_k \leq (r-1)^{-2}$ for all $j < k$, the right-hand side is at most one. This is the quantitative comass bound in [4, Proposition 4.3].

Theorem 11.0.19 (Arbitrary-codimension calibration). *Let $\Sigma_F \subset \mathbb{R}^{n+m}$ be a minimal graph and set*

$$r = \sup_{x \in \Omega} (\text{rank}(d(F)|_x)) \geq 2.$$

If every pair of singular values satisfies

$$\lambda_j \cdot \lambda_k \leq \frac{1}{(r-1)^2} \quad (1 \leq j < k \leq r),$$

then $\Theta(F)$ calibrates Σ_F , so Σ_F is area-minimizing. This is the arbitrary-codimension theorem in [4, Theorem 4.4].

Proof. By Σ_F minimal, the pairwise singular-value bound, and the Tsai–Wang comass estimate,

$$\Sigma_F \text{ is minimal} \xrightarrow{11.0.12} d(\Theta(F)) = 0.$$

$$p = \iota_F(x), \quad s = \text{rank}(d(F)|_x) \leq r, \quad s \leq 1 \xrightarrow{11.0.5} \text{comass}(\Theta(F)|_p) = 1.$$

Suppose $s \geq 2$. Then

$$\lambda_j \lambda_k \leq \frac{1}{(r-1)^2} \leq \frac{1}{(s-1)^2} \implies \Lambda_\ell \leq \frac{1}{(s-1)^\ell} \quad (2 \leq \ell \leq s).$$

Indeed, multiplying all $\binom{\ell}{2}$ pairwise bounds gives

$$\left(\prod_{j \in \mathcal{J}} \lambda_j \right)^{\ell-1} \leq \left(\frac{1}{s-1} \right)^{\ell(\ell-1)} \implies \prod_{j \in \mathcal{J}} \lambda_j \leq \left(\frac{1}{s-1} \right)^\ell.$$

The bounded binomial sum is

$$\sum_{\ell=2}^s \frac{\ell-1}{(s-1)^\ell} \binom{s}{\ell} = 1.$$

For $x = \cos(\vartheta)$ and $y = \sin(\vartheta)$,

$$0 \leq x, y \leq 1 \implies x^s \leq x^2, \quad x^{s-\ell} y^\ell \leq y^2 \quad (2 \leq \ell \leq s).$$

$$\left(\Lambda_\ell \leq (s-1)^{-\ell}, \quad x^2 + y^2 = 1 \right) \xrightarrow{11.0.18} \text{comass}(\Theta(F)|_p) \leq x^2 + y^2 = 1.$$

Since x is arbitrary, $\text{comass}(\Theta(F)) \leq 1$.

$$(F \in C^\infty(\Omega, \mathbb{R}^m), \quad \Theta(F) \text{ is the Tsai–Wang form}) \xrightarrow{11.0.10} \Theta(F)|_{\Sigma_F} = \text{vol}_{\Sigma_F}.$$

$$(d(\Theta(F)) = 0, \quad \text{comass}(\Theta(F)) \leq 1, \quad \Theta(F)|_{\Sigma_F} = \text{vol}_{\Sigma_F}) \xrightarrow{11.0.2} \Theta(F) \text{ calibrates } \Sigma_F \xrightarrow{11.0.3} \Sigma_F \text{ is area-minimizing.}$$

This proves the arbitrary-codimension criterion. $\square$

Fact 11.0.20 (Weak Lawson–Osserman consequence). *Let $F = (f_1, \dots, f_m)$ be locally Lipschitz. Write*

$$g_{jk} = \delta_{jk} + \sum_{\alpha=1}^m \partial_j(f_\alpha) \cdot \partial_k(f_\alpha), \quad G = [g_{jk}]_{j,k=1}^n, \quad g = \det(G).$$

It is a weak solution of the minimal graph system when, for every $\varphi \in C_c^\infty(\Omega, \mathbb{R})$ and $1 \leq \alpha \leq m$,

$$(11.0.9) \quad \int_{\Omega} \sum_{j=1}^n \sum_{k=1}^n \sqrt{g} \cdot (G^{-1})^{jk} \cdot \partial_k(f_\alpha) \cdot \partial_j(\varphi) \cdot dx^1 \wedge \dots \wedge dx^n = 0.$$

Set

$$r = \text{ess sup}_{x \in \Omega} (\text{rank}(d(F)|_x)).$$

If $r \leq 1$, if $m = 2$ and $\|\wedge^2 d(F)\| \leq 1$ almost everywhere, or if $r \geq 2$ and

$$\lambda_j \cdot \lambda_k \leq (r-1)^{-2} \quad (1 \leq j < k \leq r)$$

almost everywhere, then the graph current of F is locally mass minimizing. It is therefore stationary, and the Tsai–Wang regularity theorem implies

$$F \in C^\infty(\Omega, \mathbb{R}^m).$$

The proof uses the locally real flat-cochain representation of the weakly closed L^∞ calibration, a higher-codimension Bernstein theorem, and Allard regularity. These three results are analytic inputs here. The weak formulation and regularity conclusion are [4, Proposition 5.1 and Corollary 5.2].

Part 4. Appendix

APPENDIX A. FOUNDATIONAL TOOLS

A.1. Scale and Shape in the Trace–Gradient Defect.

Definition A.1.1 (Trace-normalized shape). Let $E \rightarrow X$ be a rank- r Hermitian bundle with unitary connection D , and let $h = h^* > 0$ be a positive endomorphism. Define

$$(A.1.1) \quad s = \text{Tr}(h), \quad q = s^{-1} \cdot h, \quad q^{-1} = s \cdot h^{-1}.$$

Then

$$\text{Tr}(q) = 1, \quad \det(q) = \frac{\det(h)}{(\text{Tr}(h))^r}, \quad h = e^f \cdot \tilde{h} \implies \frac{h}{\text{Tr}(h)} = \frac{\tilde{h}}{\text{Tr}(\tilde{h})}.$$

Theorem A.1.2 (Projectivized trace–gradient identities). Let $g^{j\bar{k}}$ be the inverse coefficients of a Hermitian metric on the base. For $h = h^* > 0$, $s = \text{Tr}(h)$, and $q = h/s$, one has

$$(A.1.2) \quad \begin{aligned} \mathcal{A}_{\rightarrow}(h) &= \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((D_j(h)) \cdot h^{-1} \cdot D_{\bar{k}}(h)), \\ \mathcal{A}_{\leftarrow}(h) &= \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((D_{\bar{k}}(h)) \cdot h^{-1} \cdot D_j(h)), \quad \mathcal{G}(h) = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \partial_j(\text{Tr}(h)) \cdot \partial_{\bar{k}}(\text{Tr}(h)). \\ \mathcal{D}_{\rightarrow}(h) &= \frac{\mathcal{A}_{\rightarrow}(h)}{\text{Tr}(h)} - \frac{\mathcal{G}(h)}{(\text{Tr}(h))^2} = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((D_j(q)) \cdot q^{-1} \cdot D_{\bar{k}}(q)) \geq 0. \end{aligned}$$

The reversed adjoint order satisfies

$$(A.1.3) \quad \mathcal{D}_{\leftarrow}(h) = \frac{\mathcal{A}_{\leftarrow}(h)}{\text{Tr}(h)} - \frac{\mathcal{G}(h)}{(\text{Tr}(h))^2} = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((D_{\bar{k}}(q)) \cdot q^{-1} \cdot D_j(q)) \geq 0.$$

Both defects are invariant under $h \mapsto e^f \cdot h$.

Proof. By $h = s \cdot q$, $h^{-1} = s^{-1} \cdot q^{-1}$, and $\text{Tr}(q) = 1$,

$$D_j(h) = \partial_j(s) \cdot q + s \cdot D_j(q), \quad D_{\bar{k}}(h) = \partial_{\bar{k}}(s) \cdot q + s \cdot D_{\bar{k}}(q), \quad \text{Tr}(D_j(q)) = \text{Tr}(D_{\bar{k}}(q)) = 0.$$

The complete first ordered product is

$$\begin{aligned} (D_j(h)) \cdot h^{-1} \cdot D_{\bar{k}}(h) &= (\partial_j(s) \cdot q + s \cdot D_j(q)) \cdot s^{-1} \cdot q^{-1} \cdot (\partial_{\bar{k}}(s) \cdot q + s \cdot D_{\bar{k}}(q)). \\ (D_j(h)) \cdot h^{-1} \cdot D_{\bar{k}}(h) &= \frac{\partial_j(s) \cdot \partial_{\bar{k}}(s)}{s} \cdot q + \partial_j(s) \cdot D_{\bar{k}}(q) + \partial_{\bar{k}}(s) \cdot D_j(q) + s \cdot D_j(q) \cdot q^{-1} \cdot D_{\bar{k}}(q). \end{aligned}$$

Taking the trace removes both mixed terms and gives

$$\begin{aligned} \text{Tr}((D_j(h)) \cdot h^{-1} \cdot D_{\bar{k}}(h)) &= \frac{\partial_j(s) \cdot \partial_{\bar{k}}(s)}{s} + s \cdot \text{Tr}((D_j(q)) \cdot q^{-1} \cdot D_{\bar{k}}(q)). \\ s = \text{Tr}(h) &\xrightarrow{A.1.1} \frac{\text{Tr}((D_j(h)) \cdot h^{-1} \cdot D_{\bar{k}}(h))}{\text{Tr}(h)} - \frac{\partial_j(\text{Tr}(h)) \cdot \partial_{\bar{k}}(\text{Tr}(h))}{(\text{Tr}(h))^2} = \text{Tr}((D_j(q)) \cdot q^{-1} \cdot D_{\bar{k}}(q)). \end{aligned}$$

At a fixed point choose coordinates and a unitary frame satisfying

$$\begin{aligned} g^{j\bar{k}} &= \delta_{jk}, \quad q = \text{diag}(v_1, \dots, v_r), \quad v_a > 0, \quad B_j = D_j(q), \quad D_{\bar{j}}(q) = B_j^*. \\ \sum_{j=1}^n \text{Tr}(B_j \cdot q^{-1} \cdot B_j^*) &= \sum_{j=1}^n \sum_{a=1}^r \sum_{b=1}^r v_b^{-1} \cdot |(B_j)^a_b|^2 \geq 0, \end{aligned}$$

which proves the forward identity. Reversing the first and last factors gives

$$\sum_{j=1}^n \text{Tr}(B_j^* \cdot q^{-1} \cdot B_j) = \sum_{j=1}^n \sum_{a=1}^r \sum_{b=1}^r v_a^{-1} \cdot |(B_j)^a_b|^2 \geq 0,$$

and the same four-term expansion proves the reverse identity. Finally $q(e^f \cdot h) = q(h)$, so both defects ignore scalar rescaling. $\square$

Corollary A.1.3 (Exact spectral remainder and equality). *At a point with*

$$g^{j\bar{k}} = \delta_{jk}, \quad h = \text{diag}(\mu_1, \dots, \mu_r), \quad B_j = D_j(h), \quad s = \sum_{a=1}^r \mu_a,$$

the forward defect equals

$$\begin{aligned} \mathcal{O}_{\rightarrow}(h) &= \sum_{j=1}^n \frac{1}{s} \cdot \sum_{\substack{a,b=1 \\ a \neq b}}^r \mu_b^{-1} \cdot |(B_j)^a_b|^2, \\ \mathcal{V}(h) &= \sum_{j=1}^n \frac{1}{s^2} \cdot \sum_{1 \leq a < b \leq r} \left| \sqrt{\frac{\mu_b}{\mu_a}} \cdot (B_j)^a_a - \sqrt{\frac{\mu_a}{\mu_b}} \cdot (B_j)^b_b \right|^2. \\ \mathcal{D}_{\rightarrow}(h) &= \mathcal{O}_{\rightarrow}(h) + \mathcal{V}(h). \end{aligned} \tag{A.1.4}$$

The reverse defect replaces μ_b^{-1} by μ_a^{-1} in the off-diagonal sum. Equality holds if and only if $D(q) = 0$, equivalently

$$D(h) = d(\log(s)) \cdot h.$$

If $\det(h)$ is constant, equality holds if and only if $D(h) = 0$.

Proof. By $s = \sum_{a=1}^r \mu_a$, $B_j = ((B_j)^a_b)_{a,b=1}^r$, and $h^{-1} = \text{diag}(\mu_1^{-1}, \dots, \mu_r^{-1})$,

$$s \cdot \sum_{a=1}^r \mu_a^{-1} \cdot |x_a|^2 - \left| \sum_{a=1}^r x_a \right|^2 = \sum_{1 \leq a < b \leq r} \left| \sqrt{\frac{\mu_b}{\mu_a}} \cdot x_a - \sqrt{\frac{\mu_a}{\mu_b}} \cdot x_b \right|^2.$$

Indeed, the coefficient of $|x_a|^2$ on the right is

$$\sum_{\substack{b=1 \\ b \neq a}}^r \frac{\mu_b}{\mu_a} = \frac{s - \mu_a}{\mu_a} = s \cdot \mu_a^{-1} - 1,$$

and the cross terms sum to $-2 \cdot \text{Re}(\sum_{1 \leq a < b \leq r} x_a \bar{x}_b)$. Apply this identity to $x_a = (B_j)^a_a$, separate the terms with $a \neq b$, and use

$$\begin{aligned} (x_a = (B_j)^a_a, a \neq b) &\xrightarrow{\text{A.1.2}} \mathcal{D}_{\rightarrow}(h) = \mathcal{O}_{\rightarrow}(h) + \mathcal{V}(h). \\ \text{defect} = 0 &\xrightarrow{\text{A.1.2}} D(q) = 0 \xrightarrow{\text{A.1.1}} D(h) = d(\log(s)) \cdot h. \end{aligned}$$

$\det(h)$ constant, $D(h) = d(\log(s)) \cdot h \implies 0 = d(\log(\det(h))) = \text{Tr}(h^{-1} \cdot D(h)) = r \cdot d(\log(s)) \implies D(h) = 0$.

The converse is immediate. $\square$

Corollary A.1.4 (Kähler rigidity of equality). *Let $\hat{\omega}$ and ω be Kähler metrics and $h = \hat{G}^{-1} \cdot G$. Compute the forward defect with $D = \hat{\nabla}$. If $n \geq 2$ and it vanishes at p , then $\hat{\nabla}^{1,0}(h)(p) = 0$. If $n = 1$, the defect vanishes identically.*

Proof. By $D = \hat{\mathbf{V}}$, $\hat{G}(p) = I_n$, $\partial(\hat{G})(p) = 0$, and $G(p) = \text{diag}(\lambda_1, \dots, \lambda_n)$,

$$\mathcal{D}_{\rightarrow}(h)(p) = 0 \xrightarrow{\text{A.1.3}} \partial_j(g_{a\bar{b}})(p) = c_j \cdot g_{a\bar{b}}(p) \quad (1 \leq j, a, b \leq n).$$

For fixed j , choose $a \neq j$ and set $b = a$. Kähler symmetry gives

$$c_j \cdot \lambda_a = c_j \cdot g_{a\bar{a}} = \partial_j(g_{a\bar{a}}) \xrightarrow{1.4.3} \partial_a(g_{j\bar{a}}) = c_a \cdot g_{j\bar{a}} = 0.$$

Thus $c_j = 0$ for every j , and $\hat{\mathbf{V}}^{1,0}(h)(p) = \partial(h)(p) = 0$. If $n = 1$, then $q = h/\text{Tr}(h) = 1$, so the projectivized identity gives zero directly. $\square$

Example A.1.5 (A complete rank-two defect computation). Take

$$h = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \quad h^{-1} = \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}, \quad X^* = \begin{bmatrix} 4 & 3 \\ 2 & -1 \end{bmatrix}.$$

The complete multiplication is

$$\begin{aligned} X \cdot h^{-1} &= \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 \cdot \frac{1}{4} + 2 \cdot 0 & 4 \cdot 0 + 2 \cdot 1 \\ 3 \cdot \frac{1}{4} + (-1) \cdot 0 & 3 \cdot 0 + (-1) \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ \frac{3}{4} & -1 \end{bmatrix}, \\ X \cdot h^{-1} \cdot X^* &= \begin{bmatrix} 1 & 2 \\ \frac{3}{4} & -1 \end{bmatrix} \cdot \begin{bmatrix} 4 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 2 \cdot 2 & 1 \cdot 3 + 2 \cdot (-1) \\ \frac{3}{4} \cdot 4 + (-1) \cdot 2 & \frac{3}{4} \cdot 3 + (-1) \cdot (-1) \end{bmatrix} = \begin{bmatrix} 8 & 1 \\ 1 & \frac{13}{4} \end{bmatrix}. \end{aligned}$$

Since $\text{Tr}(h) = 5$, $\text{Tr}(X) = 3$, and $\text{Tr}(X \cdot h^{-1} \cdot X^*) = 45/4$,

$$\frac{\text{Tr}(X \cdot h^{-1} \cdot X^*)}{\text{Tr}(h)} - \frac{|\text{Tr}(X)|^2}{(\text{Tr}(h))^2} = \frac{45}{4 \cdot 5} - \frac{3 \cdot 3}{5 \cdot 5} = \frac{225 - 36}{100} = \frac{189}{100}.$$

The spectral remainder reproduces the same number entry by entry,

$$\frac{1}{5} \cdot \frac{|2|^2}{1} + \frac{1}{5} \cdot \frac{|3|^2}{4} + \frac{1}{5^2} \cdot \left| \sqrt{\frac{1}{4}} \cdot 4 - \sqrt{\frac{4}{1}} \cdot (-1) \right|^2 = \frac{4}{5} + \frac{9}{20} + \frac{16}{25} = \frac{80 + 45 + 64}{100} = \frac{189}{100}.$$

Corollary A.1.6 (The positive remainder in the Aubin–Yau computation). For $h = \hat{G}^{-1} \cdot G$ and $q = h/\text{Tr}(h)$,

$$\mathcal{B}_{\hat{\omega}}(h, F) = \Delta_{\hat{\omega}}(F) - \hat{S} + \sum_{j=1}^n \sum_{k=1}^n \sum_{p=1}^n \sum_{m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}j p}^m \cdot h^p_m.$$

$$(A.1.5) \quad \Delta_{\omega_{\phi}}(\log(\text{Tr}(h))) = \frac{\mathcal{B}_{\hat{\omega}}(h, F)}{\text{Tr}(h)} + \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\hat{\mathbf{V}}_j(q)) \cdot q^{-1} \cdot \hat{\mathbf{V}}_{\bar{k}}(q)).$$

Thus the trace-gradient term measures only variation of the normalized shape q , not scalar variation of h .

Proof. By $p \in X$, $\hat{G}(p) = I_n$, $\partial(\hat{G})(p) = 0$, $T = \text{Tr}(h) > 0$, and $\Delta_{\omega_{\phi}}(\log(T)) = T^{-1} \Delta_{\omega_{\phi}}(T) - T^{-2} |\partial(T)|_{\omega_{\phi}}^2$, $q = h/T$,

$$\hat{\mathbf{V}}_j(q)(p) = \partial_j(q)(p), \quad \hat{\mathbf{V}}_{\bar{k}}(q)(p) = \partial_{\bar{k}}(q)(p).$$

$$\Delta_{\omega_{\phi}}(T) \xrightarrow{5.5.1} \Delta_{\hat{\omega}}(F) - \hat{S} + \sum_{j=1}^n \sum_{k=1}^n \sum_{p=1}^n \sum_{m=1}^n g^{j\bar{k}} \cdot \hat{R}_{\bar{k}j p}^m \cdot h^p_m + \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j(h)) \cdot h^{-1} \cdot \partial_{\bar{k}}(h)).$$

$$\frac{1}{T} \cdot \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\partial_j(h)) \cdot h^{-1} \cdot \partial_{\bar{k}}(h)) - \frac{|\partial(T)|_{\omega_{\phi}}^2}{T^2} \xrightarrow{A.1.2} \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot \text{Tr}((\hat{\mathbf{V}}_j(q)) \cdot q^{-1} \cdot \hat{\mathbf{V}}_{\bar{k}}(q)).$$

Substitution gives the claimed projective-remainder formula. $\square$

A.2. Exact Real $(1, 1)$ -Forms. Only the real $(1, 1)$ -case of the $\partial\bar{\partial}$ -lemma is used in the main notes. In that case a Poisson equation and one primitive-form identity give a short proof with every normalization visible.

Lemma A.2.1 (Trace–wedge identity for a real $(1, 1)$ -form). *Let*

$$\alpha = \frac{i}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n a_{j\bar{k}} \cdot dz^j \wedge d\bar{z}^k, \quad \Lambda_\omega(\alpha) = \sum_{j=1}^n \sum_{k=1}^n g^{j\bar{k}} \cdot a_{j\bar{k}}.$$

Then

$$(A.2.1) \quad \alpha \wedge \frac{\omega^{n-1}}{(n-1)!} = \Lambda_\omega(\alpha) \cdot \frac{\omega^n}{n!}.$$

Proof. By $p \in X$, $G(p) = I_n$, and $\alpha_m = (i/2) \cdot dz^m \wedge d\bar{z}^m$,

$$\frac{\omega^{n-1}(p)}{(n-1)!} = \sum_{m=1}^n \alpha_1 \wedge \cdots \wedge \widehat{\alpha_m} \wedge \cdots \wedge \alpha_n.$$

Every off-diagonal term of α repeats one holomorphic or antiholomorphic differential and vanishes after wedging. Therefore

$$\alpha \wedge \frac{\omega^{n-1}}{(n-1)!}(p) = \sum_{m=1}^n a_{m\bar{m}} \cdot \alpha_1 \wedge \cdots \wedge \alpha_n = \Lambda_\omega(\alpha)(p) \cdot \frac{\omega^n(p)}{n!}.$$

Since p was arbitrary, the identity is global. $\square$

Lemma A.2.2 (Primitive $(1, 1)$ -star identity). *If $\beta \in \Omega^{1,1}(X, \mathbb{R})$ and $\Lambda_\omega(\beta) = 0$, then for $n \geq 2$,*

$$(A.2.2) \quad *\beta = -\beta \wedge \frac{\omega^{n-2}}{(n-2)!}.$$

Proof. By $p \in X$, $G(p) = I_n$, $\beta(p) = \sum_{a=1}^n \mu_a \cdot \alpha_a$ in an ω -unitary frame diagonalizing $\beta(p)$, and $\sum_{a=1}^n \mu_a = \Lambda_\omega(\beta)(p) = 0$,

$$\widehat{\alpha}_b = \alpha_1 \wedge \cdots \wedge \widehat{\alpha_b} \wedge \cdots \wedge \alpha_n, \quad *\beta(p) = \sum_{b=1}^n \mu_b \cdot \widehat{\alpha}_b.$$

$$\beta \wedge \frac{\omega^{n-2}}{(n-2)!}(p) = \sum_{b=1}^n \left(\sum_{\substack{a=1 \\ a \neq b}}^n \mu_a \right) \cdot \widehat{\alpha}_b = - \sum_{b=1}^n \mu_b \cdot \widehat{\alpha}_b = -*\beta(p).$$

The pointwise calculation proves the star identity. $\square$

Theorem A.2.3 (Real $\partial\bar{\partial}$ -lemma). *Let (X^n, ω) be a connected compact Kähler manifold. If $\alpha \in \Omega^{1,1}(X, \mathbb{R})$ is exact, then there is $u \in C^\infty(X, \mathbb{R})$, unique modulo constants, such that*

$$(A.2.3) \quad \alpha = \frac{i}{2} \cdot \partial(\bar{\partial}(u)).$$

The condition

$$\int_X u \cdot \frac{\omega^n}{n!} = 0$$

makes u unique.

Proof. By $\alpha = d(\eta)$ for some $\eta \in \Omega^1(X, \mathbb{R})$, $d(\omega) = 0$, and X compact without boundary,

$$\int_X \Lambda_\omega(\alpha) \cdot \frac{\omega^n}{n!} \xrightarrow{A.2.1} \int_X d(\eta) \wedge \frac{\omega^{n-1}}{(n-1)!} = \int_X d\left(\eta \wedge \frac{\omega^{n-1}}{(n-1)!}\right) = 0.$$

$$\int_X \Lambda_\omega(\alpha) \cdot \frac{\omega^n}{n!} = 0 \xrightarrow{3.5.5} \exists! u \in C^\infty(X, \mathbb{R}), \quad \Delta_\omega(u) = \Lambda_\omega(\alpha), \quad \int_X u \cdot \frac{\omega^n}{n!} = 0.$$

Set

$$\beta = \alpha - \frac{i}{2} \cdot \partial(\bar{\partial}(u)).$$

Then

$$d(\beta) = 0, \quad \beta = d\left(\eta - \frac{i}{2} \cdot \bar{\partial}(u)\right), \quad \Lambda_\omega(\beta) = \Lambda_\omega(\alpha) - \Delta_\omega(u) = 0.$$

If $n = 1$, every primitive real $(1, 1)$ -form vanishes, so $\beta = 0$. If $n \geq 2$, then

$$\|\beta\|_{L^2}^2 = \int_X \beta \wedge * \beta \xrightarrow{A.2.2} - \int_X \beta \wedge \beta \wedge \frac{\omega^{n-2}}{(n-2)!}.$$

Writing $\beta = d(\zeta)$ and using $d(\beta) = d(\omega) = 0$,

$$\|\beta\|_{L^2}^2 = - \int_X d\left(\zeta \wedge \beta \wedge \frac{\omega^{n-2}}{(n-2)!}\right) = 0 \implies \beta = 0.$$

Hence $\alpha = (i/2) \cdot \partial(\bar{\partial}(u))$. Finally

$$\frac{i}{2} \cdot \partial(\bar{\partial}(u - v)) = 0 \xrightarrow{3.4.1} \Delta_\omega(u - v) = 0 \xrightarrow{3.5.2} u - v \in \mathbb{R},$$

and the mean-zero condition removes that constant. $\square$

Corollary A.2.4 (Quantitative potential estimate). *For $k \in \mathbb{N} \cup \{0\}$, $0 < \gamma < 1$, and the mean-zero potential above,*

$$\|u\|_{C^{k+2,\gamma}(X)} \leq C(X, \omega, k, \gamma) \cdot \|\alpha\|_{C^{k,\gamma}(X)}.$$

Proof. By $\Delta_\omega(u) = \Lambda_\omega(\alpha)$, $\int_X u \cdot \omega^n / n! = 0$, and $\Lambda_\omega(\alpha) \in C^{k,\gamma}(X)$,

$$(\Delta_\omega(u) = \Lambda_\omega(\alpha), u \in L^2(X)) \xrightarrow{A.3.2} \|u\|_{C^{k+2,\gamma}} \leq C(X, \omega, k, \gamma) \cdot (\|\Lambda_\omega(\alpha)\|_{C^{k,\gamma}} + \|u\|_{L^2}).$$

$$\Delta_\omega(u) = \Lambda_\omega(\alpha) \xrightarrow{3.5.1} \|d(u)\|_{L^2}^2 \leq C(X, \omega) \cdot \|\Lambda_\omega(\alpha)\|_{L^2} \cdot \|u\|_{L^2}.$$

$$\int_X u \cdot \frac{\omega^n}{n!} = 0 \xrightarrow{3.5.3} \|u\|_{L^2}^2 \leq C(X, \omega) \cdot \|d(u)\|_{L^2}^2.$$

Combining the two inequalities and $\|\Lambda_\omega(\alpha)\|_{C^{k,\gamma}} \leq C(X, \omega, k, \gamma) \cdot \|\alpha\|_{C^{k,\gamma}}$ proves the estimate. $\square$

A.3. Elliptic Upgrade on Compact Manifolds. This section isolates what is genuinely imported from elliptic theory, writes the complex Laplacian as a real uniformly elliptic operator, and then gives the complete difference-quotient equation that bootstraps a $C^{2,\gamma}$ Monge–Ampère solution.

Proposition A.3.1 (Real form of the complex Laplacian). *Write $z^j = x^j + i \cdot y^j$ and*

$$G^{-1} = A + i \cdot B, \quad A^\top = A, \quad B^\top = -B.$$

For real u ,

$$\mathcal{P}_A(u) = \frac{1}{4} \cdot \sum_{j=1}^n \sum_{k=1}^n A^{jk} \cdot (\partial_{x^j}(\partial_{x^k}(u)) + \partial_{y^j}(\partial_{y^k}(u))),$$

$$\mathcal{M}_B(u) = \frac{1}{2} \cdot \sum_{j=1}^n \sum_{k=1}^n B^{jk} \cdot \partial_{x^j}(\partial_{y^k}(u)).$$

$$(A.3.1) \quad \Delta_\omega(u) = \mathcal{P}_A(u) - \mathcal{M}_B(u).$$

Its real principal matrix is

$$(A.3.2) \quad \mathcal{A} = \frac{1}{4} \cdot \begin{bmatrix} A & -B \\ B & A \end{bmatrix}.$$

If $m \cdot I_n \preceq G \preceq M \cdot I_n$, then

$$\frac{1}{4 \cdot M} \cdot |\xi|^2 \leq \xi^\top \cdot \mathcal{A} \cdot \xi \leq \frac{1}{4 \cdot m} \cdot |\xi|^2 \quad (\xi \in \mathbb{R}^{2n}).$$

Proof. By $\partial_j = (\partial_{x^j} - i \cdot \partial_{y^j})/2$, $\partial_{\bar{k}} = (\partial_{x^k} + i \cdot \partial_{y^k})/2$, and $g^{j\bar{k}} = A^{jk} + i \cdot B^{jk}$,

$$\partial_j(\partial_{\bar{k}}(u)) = \frac{1}{4} \cdot (\partial_{x^j}(\partial_{x^k}(u)) + \partial_{y^j}(\partial_{y^k}(u)) + i \cdot \partial_{x^j}(\partial_{y^k}(u)) - i \cdot \partial_{y^j}(\partial_{x^k}(u))).$$

The symmetric matrix A annihilates the antisymmetric mixed bracket, while the antisymmetric matrix B annihilates the symmetric pure bracket. Hence

$$\sum_{j=1}^n \sum_{k=1}^n B^{jk} \cdot (\partial_{x^j}(\partial_{y^k}(u)) - \partial_{y^j}(\partial_{x^k}(u))) = 2 \cdot \sum_{j=1}^n \sum_{k=1}^n B^{jk} \cdot \partial_{x^j}(\partial_{y^k}(u)),$$

which proves the real-form and principal-matrix identities. For $\xi = [p^\top \ q^\top]^\top$,

$$\xi^\top \cdot \mathcal{A} \cdot \xi = \frac{1}{4} \cdot [p^\top \ q^\top] \cdot \begin{bmatrix} A & -B \\ B & A \end{bmatrix} \cdot \begin{bmatrix} p \\ q \end{bmatrix} = \frac{1}{4} \cdot (p + i \cdot q)^* \cdot G^{-1} \cdot (p + i \cdot q).$$

$$m \cdot I_n \preceq G \preceq M \cdot I_n \implies M^{-1} \cdot I_n \preceq G^{-1} \preceq m^{-1} \cdot I_n \implies \frac{|\xi|^2}{4 \cdot M} \leq \xi^\top \cdot \mathcal{A} \cdot \xi \leq \frac{|\xi|^2}{4 \cdot m}.$$

□

Theorem A.3.2 (Global Schauder estimate and elliptic regularity). *Let M be compact, $k \in \mathbb{N} \cup \{0\}$, $0 < \gamma < 1$, and suppose that in every real coordinate chart*

$$P(u) = \sum_{a=1}^d \sum_{b=1}^d a^{ab}(x) \cdot \partial_a(\partial_b(u)) + \sum_{a=1}^d b^a(x) \cdot \partial_a(u) + c(x) \cdot u,$$

where the coefficients are smooth, $a^{ab} = a^{ba}$, and

$$\lambda \cdot |\xi|^2 \leq \sum_{a=1}^d \sum_{b=1}^d a^{ab}(x) \cdot \xi_a \cdot \xi_b \leq \Lambda \cdot |\xi|^2.$$

If $u \in L^2(M)$, $P(u) = f$ distributionally, and $f \in C^{k,\gamma}(M)$, then $u \in C^{k+2,\gamma}(M)$ and

$$(A.3.3) \quad \|u\|_{C^{k+2,\gamma}(M)} \leq C(M, P, k, \gamma, \lambda, \Lambda) \cdot \left(\|f\|_{C^{k,\gamma}(M)} + \|u\|_{L^2(M)} \right).$$

In particular, smooth f gives smooth u .

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